BORIS SHEIKO

POWERLIFTING

FOUNDATIONS AND METHODS

WITH MIKE ISRAETEL, PhD AND DEREK WILCOX, PhDc

BORIS SHEIKO

Powerlifting:

Foundations and Methods

with Mike Israetel, PhD and Derek Wilcox, PhDc

UFA
2018

УДК 796.8
ББК 75.712
Ш 39

Reviewers:
P. S. Gorulev - PhD, Professor, accomplished coach of Russia, Head of Bashkir institute of Physical Education.
G. M. Yulamanova - PhD, Assistant Professor, head of the department of theory and methodology of physical education and sports at M. Akmullah of Bashkir State Pedagogical University.

Boris Sheiko
Powerlifting: Foundations and Methods. – Ufa, 2018. – 377 p.

with Dr. Mike Israetel and Derek Wilcox.

The monograph “Powerlifting: Foundations and Methods” is a powerlifting program manual written by Professor B.I. Sheiko. It contains the basics of modern theories and methods of powerlifting. The book describes a modern systematic approach to teaching and perfecting the techniques of competition exercises in powerlifting. The book is written for coaches, athletes, and fans of powerlifting alike.

Written by: Professor Boris Sheiko

Project manager: Dmitry Spiridonov

Editors: Mike Israetel, PhD and Derek Wilcox, PhDc

Consultant: Robert Frederick

Translations by: Anna Tkaczuk, Jacob Shamailov, Zach Sokol, and Gabriel Naspinski

Proofread by: Mike Israetel, PhD, Derek Wilcox, PhDc, Christophe Rebreyend, PhD and Malgorzata Krakowian

Chapter 4 written by Leonid Ostapenko

Copyright © 2018 by Boris Sheiko.

All rights reserved. No part of this publication may be reproduced, distributed, translated, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without the prior written permission of the author.

ISBN 978-5-906299-05-5

Contacts:
Boris' email - sheikoprogram@gmail.com
(for the seminars inquiries and other questions)
Website - www.sheiko-program.ru
Forum - www.sheiko-program.ru/forum
Instagram - www.instagram.com/borissheiko
Facebook - www.facebook.com/BorisSheikoOfficial
YouTube - https://www.youtube.com/user/MrBorisSheiko

Official Sheiko apps:
iOS: https://itunes.apple.com/us/app/sheiko/id807352223
Android: https://play.google.com/store/apps/details?id=me.pushapp.sheiko

Check the personal website of the iOS app developer Robert Frederick. Contains tons of unique insights about strength training:
http://www.strongur.io/
Sheiko t-shirts: https://shop.spreadshirt.com/sheiko/

Boris's manager Dmitry Spiridonov's email: fprvideo@mail.ru
Dmitry Spiridonov's Social Media:
Instagram: www.instagram.com/dmitry.spiridonov/
Facebook: https://www.facebook.com/dmitry.spiridonov.5
YouTube: https://www.youtube.com/channel/UCShc1tD-mPMdg6el6ofQMpQ

Boris Sheiko's programs written for the Kizen Training project:
Sheiko Intermediate Offseason Program https://kizentraining.teachable.com/courses/sheiko-intermediate-offseason?affcode=89295_ofjafdrp
Sheiko Intermediate Competitive Program https://kizentraining.teachable.com/courses/sheiko-intermediate-competitive?affcode=89295_ofjafdrp
3 Day Maximum Strength Program
https://kizentraining.teachable.com/courses/sheiko-intermediate-express?affcode=89295_ofjafdrp

Renaissance Periodization Website
https://renaissanceperiodization.com/
Renaissance Periodization Expert Forum:
https://renaissanceperiodization.com/rp-plus
Renaissance Periodization Instagram:
https://www.instagram.com/rpstrength/?hl=en

TABLE OF CONTENTS

Foreword 9

Chapter 1. Classification of Powerlifting Exercises 11

1.1. Historical overview 11

1.2. Principles of physical exercise classification 12

1.3. Classification of basic and additional exercises in powerlifting 13

1.4. Barbell squats exercises 16

1.4.1. Group 1 – competitive exercise 18

1.4.2. Group 2 – exercises for specific purposes 18

1.4.3. Group 3 – general developmental exercises 22

1.5. Bench press exercises 28

1.5.1. Group 1 – competitive exercise 30

1.5.2. Group 2 – exercises for specific purposes 30

1.5.3. Group 3 – general exercises for bench press 36

1.5.3.1. Basic exercises for the chest muscles 36

1.5.3.2. Basic exercises for the development of the shoulder girdle 41

1.5.3.3. Basic exercises for the development of the triceps muscles 43

1.5.3.4. Basic exercises for the development of the biceps muscles 46

1.5.3.5. Basic exercises for the latissimus dorsi muscle 47

1.6. Deadlift exercises 50

1.6.1. Group 1 – competitive exercise 52

1.6.2. Group 2 – exercises for specific purposes 52

1.6.3. Group 3 – general and developmental exercises 60

1.6.3.1. Exercises for the back muscles development 60

1.6.3.2. General exercises for the abdominal muscles development 64

Chapter 2. Basic Concepts of Biomechanics and Technique in Powerlifting 66

2.1. Technical structure of competitive exercises 67

2.1.1. Period and phase structure of exercises 69

2.2. Biomechanical video analysis of complex movements in powerlifting 70

2.2.1. Hardware-software for biomechanical analysis of motor actions in powerlifters 71

2.3. Squat Technique 76
2.3.1. Rules and regulations for the squat 76
2.3.2. Squat phase structure 77
2.3.3. Technique and structure of squats 81
2.3.4. Squat techniques of advanced athletes 90
2.3.4.1. K. Pavlov's squat technique 90
2.3.4.2. E. Koval'kov's squat technique 94
2.3.4.3. Alexey Serov's squat technique 100
2.4. Bench Press Technique 105
2.4.1. Bench press rules and regulations 105
2.4.2. The phase structure of the bench press 106
2.4.3. Technique and motor structure of the bench press 109
2.4.4. Biomechanical analysis of the bench press 115
2.4.5. The influence of the arch on bench press 120
2.4.6. The transfer of the mechanical impulse from the legs to the bar 122
2.4.7. Bench press techniques of advanced athletes 123
2.4.7.1. K. Pavlov's bench press technique 123
2.4.7.2. Y. Chistyakova's bench press technique 127
2.4.7.3. A. Babin's bench press technique 130
2.5. Deadlift technique 133
2.5.1. Deadlift rules and regulations 133
2.5.2. Deadlift phases structure 133
2.5.3. Technique and structure of deadlift 136
2.5.4. Comparative analysis of deadlift styles 142
2.5.5. Deadlift techniques of advanced athletes 144
2.5.5.1. A.Belyaev's deadlift technique 144
2.5.5.2. M. Baruzdin's deadlift technique 149
2.5.5.3. V.Sheglova's deadlift technique 154
2.5.5.4. Deadlift mistakes of Stephen Prichard 157

Chapter 3. Methods of Teaching Competitive Exercises Techniques 159

3.1. General Training Principles 159

3.1.1. Methods of Teaching 161

3.1.2. Formation of motor skills 164

3.1.3. The Structure of Teaching Motor Activities 165

3.1.4. Typical Mistakes in Exercises 167

3.1.5. Motor Error Factors 168

3.2. Fundamentals of teaching competitive exercises techniques 169

3.2.1. Squat Technique Training 170

3.2.2. Common mistakes in competitive bar squat technique 175

3.2.3. Bench Press Technique Training 181

3.2.4. Back Arching in the Bench Press Technique 185

3.2.5. Common Competitive Bench Press Techniques Mistakes 190

3.2.6. Deadlift Technique Training 195

3.2.7. Most common deadlift errors 202

Chapter 4. Powerlifting Nutrition 208

4.1. Physiological basis for the development of muscle strength 208

4.1.1. Biochemical basis for the development of muscle strength 209

4.1.2. Anatomical characteristics of powerlifting 209

4.2. Importance of adequacy and balanced diet 210

4.3. Algorithms for calculating an adequate diet and its components 214

4.3.1. Short-term and long-term requirements 214

4.3.2. Calculation and adjustment of energy requirements 215

4.3.3. Tools for conclusions about the adjustments 215

4.3.4. Balancing the daily diet 216

4.3.5. Protein balance 216

4.3.6. Carbohydrate balance 217

4.3.7. Fat balance 218

4.3.8. Vitamin and mineral balance 218

4.4. Food for a balanced diet 219

4.4.1. Food groups and their properties 219
4.4.2. Optimum daily nutrition scheme 221
4.5. Methods of objective control over diet quality 224
4.5.1. Nutrition in preparatory period 225
4.5.2. Nutrition in pre-contest period 227
4.5.3. Competition day nutrition 229
4.6. Fundamentals of body weight regulation technique 229

Chapter 5. The earliest studies in the field of muscle strength training 233
5.1. Modern Methods and Means for Strength Training Athletes 238
5.2. Training methods of Foreign and Russian Powerlifting Specialists 243
5.2.1. Training Methods of Foreign Specialists 243
5.2.2. Training methods of Russian specialists 247

Chapter 6. Powerlifting training structure 250
6.1. Single training session and micro cycles' structure 250
6.2. Distribution of training loads to weekly cycles and training sessions 252
6.3. Mesocycle Training Structure 261
6.4. Long cycles (macrocycles) training structure 265
6.5. Principles of long-term (multi-year) planning 269
6.6. Training volume and intensity planning 270
6.7. Load distribution by zones of intensity 271

Chapter 7. Powerlifting Programs 274
7.1. Three lifts: squat emphasis 278
7.2. Three lifts: bench press emphasis 294
7.3. Three lifts: deadlift emphasis 310
7.4. Bench press only 326
7.5. Three lifts. Medium volume for experienced athletes 344

Bibliography 364

FOREWORD

Mike Israetel, PhD

Derek Wilcox, PhDc

Few of us think “great powerlifters” without at some point thinking of “the Russians.” We used to joke back in the day that no matter how good of a 110kg competitor you thought you were, there were 10 Russians at 90kg that could out-total you, while never breaking perfect technique. Of course, in the modern age, amazing competitors come from all over the world, but to this day, Russian powerlifting has a special presence and influence in our sport. The core of that influence was and is, of course, the amazing abilities and record-setting performances of Russian lifters. But something else adds to the allure; mystique. Due originally to the Iron Curtain of the Cold War, and due today to the language barrier, Russian training methods have always been shrouded in mystery. Inklings about childhood selection programs, advanced sport science, and meticulous programming make their way into our discussions, further increasing our wonder about what’s really going on in Russia that turns out so many amazing lifters. Especially in the last 15 years, one of the common features of these speculative discussion has been “Sheiko.” Boris Sheiko, to be specific, Russia’s most decorated powerlifting coach. For the longest time, and even mostly to this day, we in the west learned scarcely little about this man and his training methods; the very same methods that rocketed Russian powerlifting teams to so many championships that most people stopped counting. Much of what we do know about his methods has come from his programs. A simple google search of “Sheiko powerlifting program” will lead to you to dozens of routines, some of them indeed authored by Sheiko, and some of them just copycats. These programs gave western lifters the first real glimpses into what Russian powerlifting training was like, but the scope of that vision was very limited. For one, nearly all of these programs, up until very recently, were never meant to apply to all lifters. They were custom-made programs

for specific lifters, and their specific strengths, weaknesses, and training ages, that simply leaked online and were used by thousands. Thousands of people that saw lots of benefits, but were also left wondering what would happen if they actually used a program tailored to them. Another big limitation to the “try a random program” introduction to Sheiko’s training that many people had was that the principles and methods behind the program remained largely mysterious. In a time when seemingly every powerlifting coach has their own method and descriptive article or book about it, the methods of Sheiko, mostly because of the language barrier, remained elusive. But that was then, and this is now. Now, right now, you hold the complete guide to those very methods in your hands. This book is a full description of the principles and methods behind the way Boris Sheiko, distinguished professor of Russian sport science and Russia’s greatest powerlifting coach, designs programs. Included in this book is a list of nearly all of the exercises he recommends, instructional manuals on how to perform them, details on program design, and even descriptions of the different elements of program design appropriate for lifters for different training ages and abilities. This book was no easy task to translate, but we finally got it done. And as sport scientists and longtime powerlifting competitors and coaches ourselves, we learned an unbelievable amount just from getting a chance to work on this very special book. We have no doubt that you’ll get a huge amount of knowledge out of this book; knowledge you can take to the gym right away and begin using to do the very important task of making you stronger. So take a shot of vodka, punch a bear in the face, and get ready, because the wait is over; Boris Sheiko’s powerlifting manual is finally here!

Mike Israetel, PhD and Derek Wilcox, PhDc Renaissance Periodization, December 2018.

Chapter 1. Classification of Powerlifting Exercises

1.1 Historical overview

Novikov A., one of the founders of the physical education theory, claimed that the general taxonomy (organization) of physical exercises should be the same for all units of the physical education system, otherwise it loses its scientific and practical value. Classification of physical exercises, being an essential condition of their pedagogical (educational) use, is one of the main elements of physical education.

The scientific and practical value of the classification of physical exercises is determined by the features the exercises are based on, their importance for the assessment of an exercise and its application. If the feature of which the classification is based on has an important pedagogical value, it helps to navigate the extraordinary variety of physical exercises, as well as select the appropriate ones and use them for the purpose of physical education (Matveev L., Novikov A., 1976).

The classification of physical exercises has long been developed by weightlifting experts. One of the first attempts to classify physical exercises was made by Luchkin N. in "Weightlifting," a textbook written for sports universities (1956). Ever since he published the book, all weightlifting textbooks have included the materials regarding the classification of physical exercises as well as weightlifting terminology: Falameyev A. 1962, 1981; Mihayluk M, 1965; Samusevich A., 1967; Vorobyov A., 1967, 1972, 1981, 1988; Medvedev A., 1981, 1985, 1986.

In his book "Special Weightlifting Exercises" (1973), Laputin N. systematized and analyzed special and supplementary weightlifting movements.

Dvorkin L. (2001, 2005) and Vinogradov G. (2009) were the first to provide a classification of weightlifting sports.

Medvedev A. and Yakubenko Y. (1997) were the first to attempt classifying the exercises used in powerlifting training and they provided a layout of the basic and supplementary powerlifting exercises.

Sheiko B (2001, 2005) and Gorulev P. (2004) have presented the most thorough classification of powerlifting exercises.

As a result of studying the physical exercise classification system, general principles have been defined, including the following:

1. Physical qualities being developed: strength, speed, endurance, agility, etc. (Ozolin N. 1949, Dyachkov V. 1950, 1955, 1967, Kuznetsov V. 1975, etc.).
2. Resemblance between an exercise and its competitive version (Farfel V. 1969, Kuznetsov 1975).
3. Exercises' effect on the nervous system (Ivanova L. Parshiganin Y. 1973).
4. Exercises' classification based on their intensity and the energy needed for their performance (Saksonov N. 1969; Kuznetsov V. 1975).
5. Classification according to exercise intensity (regarding cyclic physical exercises – Farfel 1949).
6. Exercise arrangement according to three principles: exercise method, aim and movement (Natalov G. 1964, 1968).
7. Computer analysis of training. Chernyak A. and Gisin M. (1977), Chernyak A. and

Kachaev S. (1978) attempted to group physical exercises based on the most informative training load analysis and divided them into training cycles. As a result of the experiment they concluded that computers may serve as an additional tool to monitor a training process.

1.2 Principles of physical exercise classification

In order to perform an exercise, one should first visualize it in a rational sequence. In a training system there are specific principles of exercise classification. Classification is a system of grouping similar objects into classes, groups, etc.; they are grouped based on common features and attributes.

The classified categories are first separated into the largest groups (classes) based on their common features. Sub-categories are then separated into smaller groups based on more specific features they have in common. This grouping system is usually presented as a scheme (classification) and is used as visual means of establishing a connection between the classified items.

Matveev L. (1977) notes that one of the main features of exercises classification used in sports training, is their similarities (or differences) with the sports chosen as a specialization. It is assumed that each physical exercise has more or less constant principles, this is why according to the physical education theory there are three categories of physical exercises in all sports:

Group 1 – competitive exercises;
Group 2 – exercises for specific purposes;
Group 3 – general exercises.

Competitive exercises are holistic physical activities (including complex set of activities) that serve as means for specific training and are performed in the same way as during competition conditions in a chosen sport. The term “competitive exercises” in this case is identical to the concept “sport”. From a methodological point of view, one should distinguish between the actual competition exercises and their training form. The former are done during actual competitions, in full compliance with the competition rules laid down for this type of sport. The latter would be more general in nature, but still similar to competition exercises. Specific and often subtle alterations to technique in these exercises are intended to emphasize particular training adaptations.

Exercises for specific purposes belong to the second group of physical exercises. The latter would be more general in nature, but still similar to competition exercises. Specific and often subtle alterations to technique in these exercises are intended to emphasize particular training adaptations. The structure of these exercises is mainly determined by the character of a selected sport, they are performed with heavy weights, which allows for training with larger loads. In this way, this exercise group appears to be the most important when preparing athletes, these exercises help to develop specific physical qualities, as well as athletes’ technical mastery of competitive exercises.

General exercises are used to increase the level of general physical preparation of an athlete as well as to involve specific muscle groups (exercises with weights, gymnastic and acrobatic exercises, swimming, sports games, etc.).

Thus, group 1 and 2 provide basic load for an athlete, while group 3 provides additional load.

1.3 Classification of basic and additional exercises in powerlifting

The number of physical exercises used in training is currently quite large, many of which differ from each other in technique (structure and organization), as well as content (in terms of psychology, physiology, biochemistry, and biomechanics). The nature of an exercise determines its technique, which is its internal and external structure (Matveev L. 1977).

As mentioned above, the classification of exercises in powerlifting training has not yet been described in depth, so we offer a somewhat ordered set of powerlifting exercises divided into groups and subgroups based on certain characteristics, as well as their technique and content.

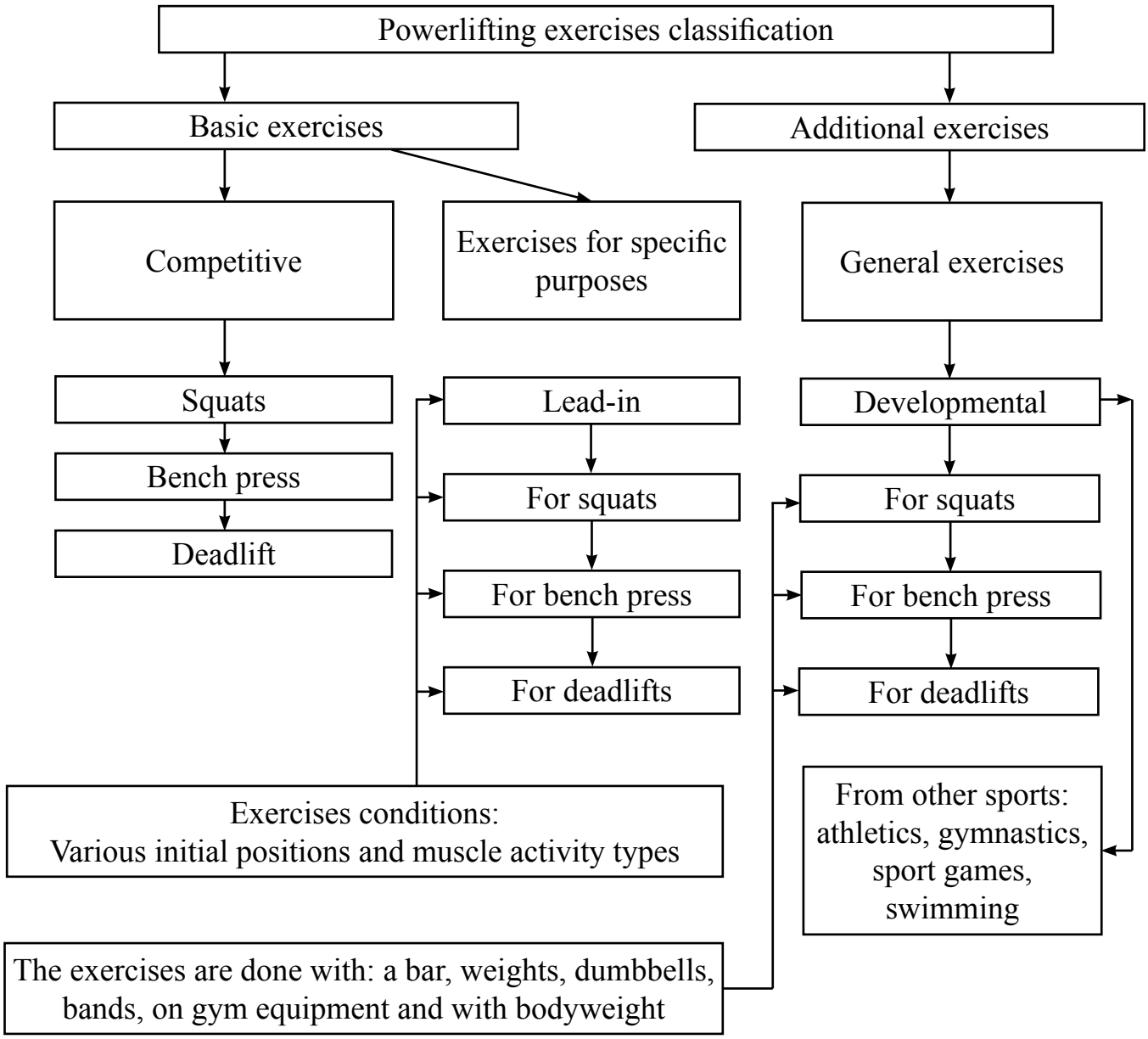


Fig. 1.1. Arrangement of primary and additional powerlifting exercises

In powerlifting, competitive exercises belong to the basic exercises group. They include barbell squats, bench press on a horizontal bench, and deadlift.

The second group includes exercises for specific purposes. One should note that an exercise can be regarded as preparatory only when it has something in common with a competitive exercise.

Depending on the primary focus of exercises, they can be divided into lead-in and developmental. ***Lead-in exercises*** help master the technique as well as movement qualities, whereas ***developmental exercises*** are mainly aimed at developing physical qualities (strength, speed, endurance, etc.) and most of them have a localized impact (see Fig. 1.1.)

Within a training process, lead-in exercises are divided into three exercise groups: squats, bench press, and deadlift.

Lead-in exercises for **barbell squats** include:

- Squat to boxes of various heights or from platforms or pins of various heights
- Front squat
- Squats with one or two pauses
- Slow eccentric squat with rapid upward movement
- Barbell squat against a wall
- Squat negatives (slow control of a heavy squat on the way down, help up by training partners)
- Squat with chains
- Pyramid power rack squat
- Barbell partial squat
- Narrow stance squat

Lead-in exercises for **bench press** on a horizontal bench:

- Wide grip bench press
- Medium grip bench press
- Narrow grip bench press
- Bench press with a foam roller under the lower back
- Bench press with an extended pause
- Explosive bench press
- Speed bench press
- Bench press with pauses
- Slow negatives
- Cambered bar press
- Bench press from a board on the chest (5-25 cm)
- Bench press with chains

Lead-in exercises for **deadlift**:

- Deadlift to the knees
- Deadlift with one pause, bar above knee joints
- Deadlift with two pauses, bar below and above knee joints
- Deadlift on a stand (deficit deadlifts)
- Deficit deadlift with one or two pauses
- Deadlift with chains
- Deadlift from blocks, e.g. bar below the knees
- Deadlift from blocks, e.g. bar above the knees
- Deadlift from below the knees to lockout
- Deadlift from pins (starting at lockout and lowering down slowly)

- Deadlift with slow return of the bar to the platform
- Competitive deadlift + deadlift from below the knees to lockout

General developmental exercises are performed with a bar, kettlebells, dumbbells, bands or gym equipment. They also include gymnastic exercises, swimming and sports games. The abovementioned exercises promote versatile physical development of an athlete. In powerlifting, the exercises from this group are used to increase the level of an athlete's general physical preparation as well as to develop individual muscle groups. These particular exercises affect individual muscles, their technical parameters differ from those of competitive exercises. They are performed with relatively small weights and serve as an additional means of an athlete's training. Therefore, they should be accounted for and analyzed separately (i.e. squats, bench press and deadlift).

Developmental squat exercises:

- Cross-legged barbell squat
- Deep squats, standing on blocks, with weights (kettlebell, etc.) in one's hands
- Leg press on a leg press machine, seated
- Hack squat machine
- Leg extensions machine, seated
- Hamstrings curl machine, lying face down
- Barbell jump squat
- Deep squat jumps

Developmental bench press exercises:

- Incline and decline bench press
- Overhead barbell press (seated or standing)
- Overhead barbell press, wide grip (seated or standing)
- Dumbbell flyes (lying on a horizontal bench)
- Parallel bar dips (with leg or waist weights)
- Pushups with weight on the upper back (chest pushup style)
- Pushups with weight on the upper back (triceps pushup style)
- Pec deck (seated)
- Dumbbell bench press (lying on a horizontal bench)
- French press (bending the arms lying down aka Skull Crushers)
- Bench dips (seated)
- Barbell overhead extensions (standing)
- Pull-ups (wide grip)

Developmental exercises for deadlift:

- Deficit deadlifts
- Bent-over barbell rows
- Shoulder shrugs
- Hyperextensions
- Reverse hyperextensions
- Barbell good mornings (seated, standing)
- Barbell good mornings + squat (standing)
- Deep good mornings

1.4. Barbell squats exercises

Experts consider barbell squat to be a primary exercise, as it is one of the most important exercise for your leg muscles, moreover, it affects the back and upper body muscles.

Below, we present the ranking chart of basic and additional exercises used in powerlifting barbell squat (Fig.1.2).

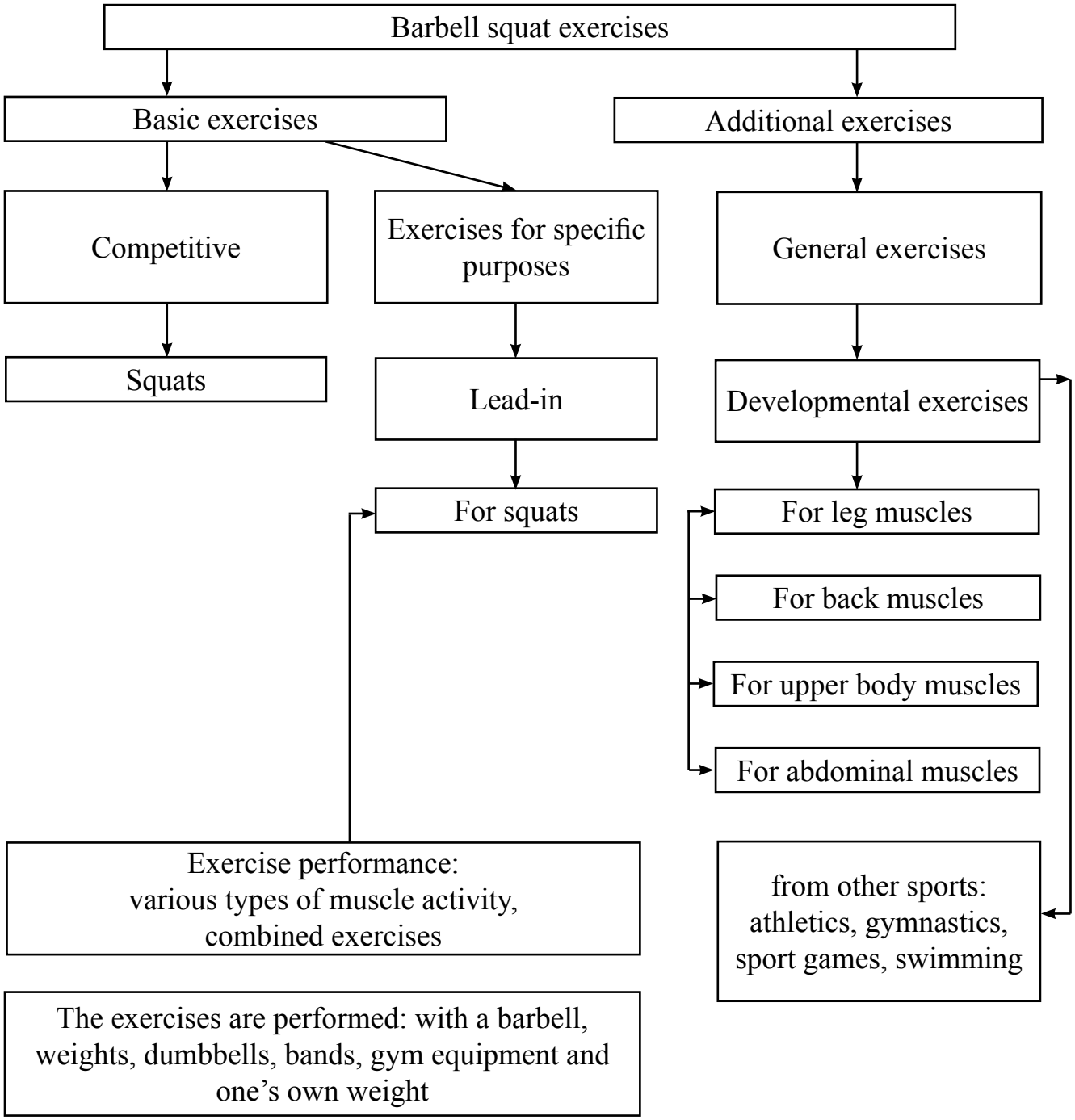


Fig.1.2. Arrangement of basic and additional barbell squat exercises

In his thesis on the basis of the existing classification of powerlifting exercises, Kostryukov V. (2011) attempted to develop an exercise tree that describes the motor structure of the exercises as well as their types.

In order to facilitate the formalization of the exercises, an alpha-numeric code has been assigned to each exercise, where letter "S" stands for squat, "B" for bench press, and "D" for

deadlift. The letters “AS”, “AB” and “AD” stand for additional squat, bench press and deadlift exercises while the number next to the letters represents the sequence number within an exercise group. Figure 1.3. shows a squat exercise tree consisting of three levels.

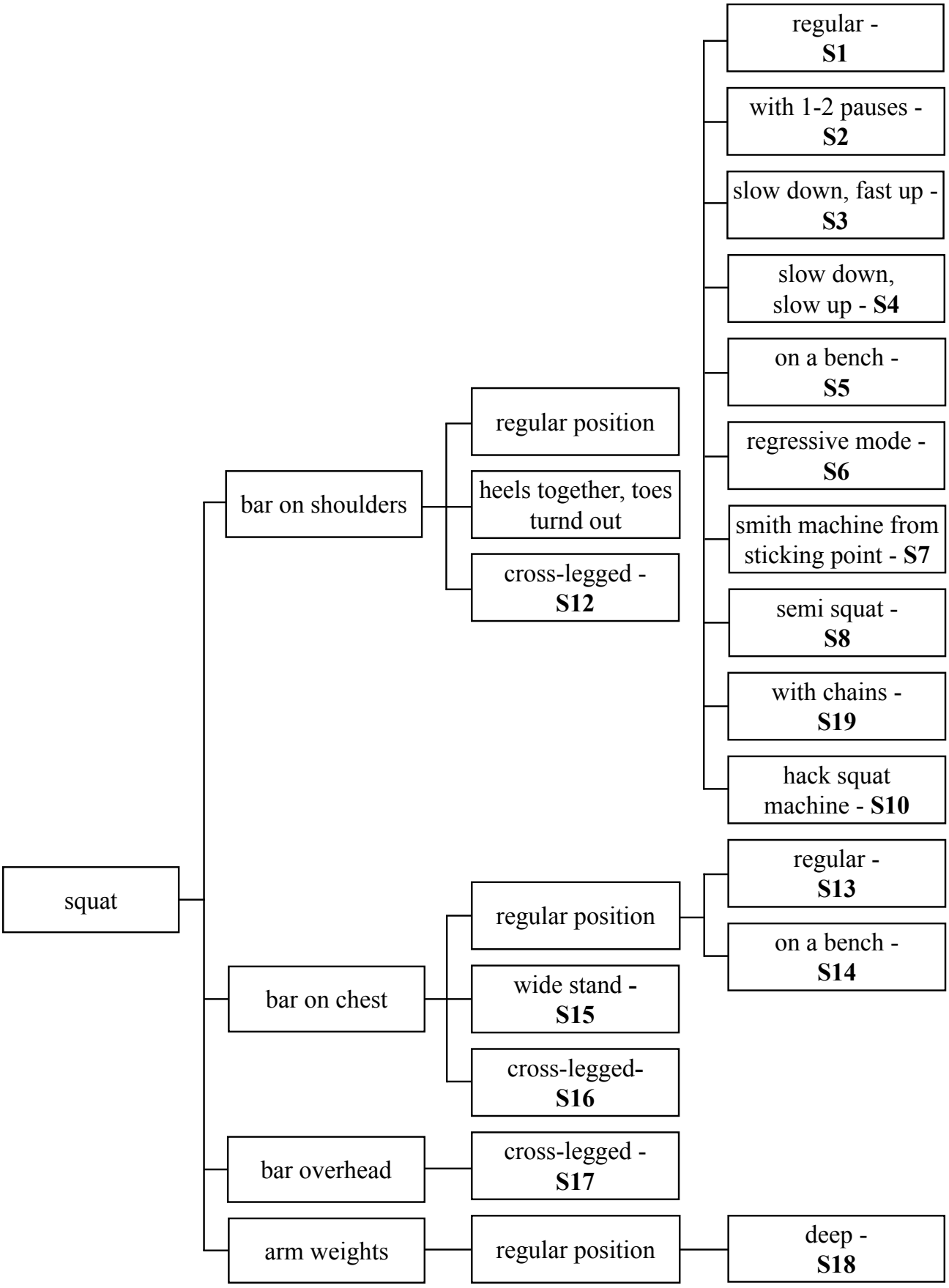


Fig. 1.3. Squat exercises tree (Kostryukov V. 2011)

Competition barbell squat is indicated in the scheme as “S1”. The first level shows that there are 4 types of weights used in squat exercises: a barbell on the shoulders, on the chest, or overhead, and with weights held in the hands.

The second level presents 4 stance types during squats: regular, heels together with toes turned out, wide, cross-legged (lunge position).

The third level describes weight types (plates, chains, Smith machine, hack machine), type of downward and upward movement (slow, fast), as well as squat depth (partial squat, to a box, regular, deep).

Barbell squat exercises are divided into three groups.

1.4.1. Group 1 – competitive exercise – barbell squat.

The exercise is carried out in accordance with the standards and requirements of the competition rules of the Russian Powerlifting Federation (RPF) and the International Powerlifting Federation (IPF).

Depending on the foot placement in the starting position, athletes compete in various styles such as “sumo” (wide foot placement), “conventional” (narrow foot placement), as well as many other foot placement styles (photo 1.1. – 1.3.).

Photo 1.1. Gadiyev R.

Photo 1.2. Dedylya V.

Photo 1.3. Suslov N.

1.4.2. Group 2 – exercises for specific purposes

This group includes the following lead-in as well as developmental exercises:

1. Squats to boxes of various height

SP: standing with a barbell on your shoulders. Squat until you touch the box with your hips. The box is placed at 15-20 cm behind the athlete. Once you touch the box, immediately return to the starting position. The exercise facilitates fast mastery of the squat technique, it also helps control hip and knee movements (photo 1.4 – 1.6).

2. Front barbell squat to a box

SP: standing with a barbell on the chest. Squat until you touch the box with your hips. The box is placed at 15-20 cm behind the athlete. Once you touch the box, immediately return to the starting position. It is recommended to use these exercises when learning squat technique. It helps keep the back straight (upright) and not rounded (photo 1.7 – 1.9).

Photo 1.4

Photo 1.5

Photo 1.6

Photo 1.7

Photo 1.8

Photo 1.9

3. Barbell squat with one or two pauses

SP: standing with a barbell on your shoulders. While squatting, you should make one or two pauses for 2-3 seconds.

This exercise helps master the technique of downward movement (photo 1.10 – 1.12).

Photo 1.10 Starting position

Photo 1.11 first pause

Photo 1.12 Second pause

4. Slow downward and upward squat

SP: standing with a barbell on your shoulders, feet shoulder-width apart. Perform a competitive squat style in a slow manner.

This exercises helps master the competitive squat technique. It is recommended to perform the exercises with 60-70% of maximum weight.

5. Barbell squat against the wall

SP: standing against the wall with a bar on the shoulders. Your toes should be touching the wall.

Photo 1.13

Photo 1.14

Photo 1.15

Photo 1.16

Photo 1.17

Photo 1.18

This exercise is presented by Ruslan Gareyev, master of sport of the Russian Federation, multiple champion of Russia among students.

When performing this exercise against the wall, the athlete's knees are pressed against it and do not go over the toes.

6. Front barbell squat

SP: standing, feet shoulder-width apart, the bar should be placed on the deltoid muscles. Squat and return to the starting position.

This weight-lifting exercise develops the front thigh muscles (quadriceps) and helps improve the back arch (staying upright) (photo 1.19-1.21).

Photo 1.19

Photo 1.20

Photo 1.21

7. Squat negatives

SP: standing with a barbell on your shoulders, feet shoulder width apart. Squat slowly with maximum weight (80-90% of personal record) for 6-10 seconds, then return to the starting position with assistants' help.

8. Barbell squat with narrow stance

SP: standing with a barbell on your shoulders, distance between the heels – foot-width, toes to the sides.

The recommended weight is 30-40% of maximum weight.

9. Barbell squat with chains

SP: standing with a barbell on your shoulders, feet shoulder-width apart. The chain is attached to the bar. As the athlete is ascending the chains are rising gradually increasing the load that reaches its maximum at the top (photo 1.22-1.23).

Photo 1.22

Photo 1.23

Photo 1.24

The exercise is presented by Sergey Mor, 4-time world champion

10. Barbell squat in "pyramid" power rack from the sticking point

This particular machine gives an athlete the opportunity to learn and practice all the phases of squat technique. It allows to change the starting position of the bar: the higher it is, the bigger the weight the athlete trains with should be.

This exercise is aimed at the improvement of the upward movement and increase of leg muscles strength when passing the sticking point (photo 1.25-1.27).

11. Barbell half squats

This exercise is performed with maximum load (100-120% of the maximum squat results) and has a good effect on the athletes who "shake" during squats at competitions (photo 1.28-1.30).

Photo 1.25

Photo 1.26

Photo 1.27

The exercise is presented by the 10-time, world champion, Pavlov, K.

Photo 1.28

Photo 1.29

Photo 1.30

1.4.3. Group 3 – general developmental exercises

1. Barbell squat with slow downward and fast upward movement

SP: standing with a barbell on your shoulders, feet shoulder-width apart. Squat slowly and get up fast. The recommended weight for the exercises is 50-60% of the maximum.

Engaged muscles:

- *Primary muscle group - quadriceps, gluteal muscles*
- *Secondary muscle group - hamstrings, adductors, sacrum and lumbar muscles, abdominal muscles.*

2. Split squat

SP: Standing with a barbell on your shoulders, right foot forward, shoulders back, torso upright. Bend the knee of the right leg, and slightly bend the left one. Return to the starting position. The depth of a squat depends on the degree of mobility in the hip, knee and ankle joints of the right leg.

The main idea of the exercise is not the weight but stretching of the quadriceps. The leg that is in front works the most (photo 1.31-1.34). The exercise improves the flexibility of the hip and ankle joints.

Engaged muscles:

- *Primary muscle group - quadriceps, gluteal muscles*
- *Secondary muscle group - posterior group of thigh muscles, adductors.*

Photo 1.31

Photo 1.32

Photo 1.33

Photo 1.34

3. Deep squat, standing on boxes with weights (kettlebells, plates, etc,) in one's hands or hung on a belt

SP: standing on a box with your feet wide apart, toes to the sides, back is straight, slightly lean forward, hold the weights (dumbbells, kettlebell, etc.) in your hands positioned between your legs parallel to the ground .Bend your knees until your hips are below the knee joints level. Return to the starting position (photo 1.35-1.37)

Engaged muscles:

- *Primary muscle group – quadriceps, gluteal muscles*
- *Secondary muscle group - hamstrings, adductor muscles, erector spinae muscles, abdominal muscles*

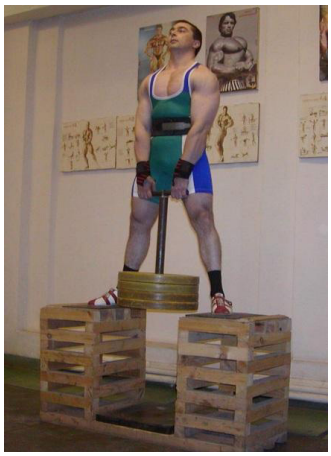


Photo 1.35

Photo 1.36

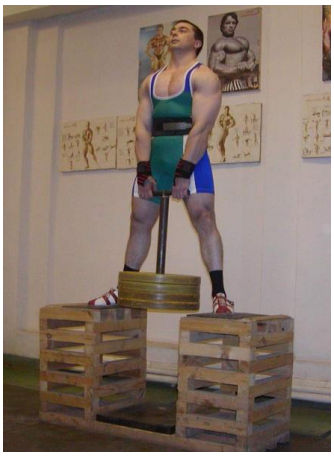


Photo 1.37

This exercises is presented by the 4-time world champion, Sergey Mor.

4. Leg press on a leg press machine in a seated position

SP: Seated with your back on the leg press machine, the back is pressed against the bench set at 45-degree angle. The feet are placed on the platform at shoulder-width, the toes are slightly to the sides and the hands hold the machine's handles.. Slowly bend the legs at your knees and hip joints to lower the platform as slowly as possible moving the knees to the shoulders. The knees should be away from each other. Press the weight up until your legs are completely straight.

It is recommended to do the exercises with the weight larger than the maximum barbell squat weight. By using a significantly larger weight than the barbell squat weight, an athlete reduces the spinal load of heavy squatting but can still develop some of the strength qualities needed for heavy squatting (photo 1.38 – 1.40).

Engaged muscles:

- *Primary muscle group – quadriceps*
- *Secondary muscle group - gluteal muscles, hamstrings, adductors.*

5. Hack machine squat

SP: standing with your back pressed against the back pad of the machine, place your shoulders under the shoulder pads, the feet are shoulder-width apart, the toes are on the platform pointed out. The weight is attached to the stands. Squat by bending your knee and hip joints, then straightening the legs return to the starting position. The exercises should be done slowly (photo 1.41 – 1.43).

Engaged muscles:

- *Primary muscle group – quadriceps*
- *Secondary muscle group - gluteal muscles, hamstrings, adductors.*

Photo 1.38

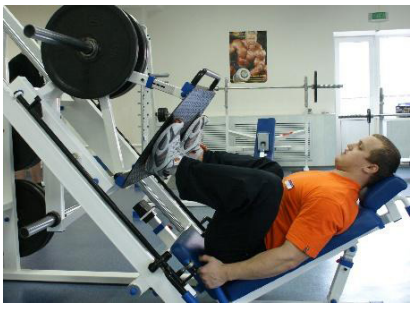


Photo 1.39

Photo 1.40

Photo 1.41

Photo 1.42

Photo 1.43

6. Leg extensions on a leg-extension machine (seated)

SP: seated on a leg-extension machine with your back straight. Place your legs under the pad, hold the side bars with your hands and slowly straighten your legs. Pause at the top of the movement. Bend your knees at a 90-degree angle when returning to the starting position. The exercise is performed slowly without sharp movements or swinging (photo 1.44-1.46).

Engaged muscles:

- *Primary muscle group – quadriceps*
- *Secondary muscle group - tibialis anterior muscle.*

7. Hamstrings curl machine, lying face down

SP: lying face down on the machine's bench, hold the bars or the edge of the bench. Place your legs behind the pad, your knees should be below the bench level. Bend the knees and then straighten them returning to the starting position. To achieve the maximum hamstrings contraction, your legs should be bent at the knee joints to an acute angle (less than 90-degree knee joint angle), pause at the top of the contraction (photo 1.47-1.49).

Engaged muscles:

- *Primary muscle group – hamstrings*
- *Secondary muscle group - gluteal muscles, lower leg muscles.*

Photo 1.44

Photo 1.45

Photo 1.46

Photo 1.47

Photo 1.48

Photo 1.49

8. Barbell jump squats

SP: stand with a barbell on your shoulders with your legs hip-width apart. Squat deeply by bending your knees. Jump up by straightening your legs. We recommend you perform this exercise with light weight (10-20% of the maximum weight). The exercise develops explosive speed and trains calf muscles, mobility in the hip, knee and ankle joints. When jumping you need to ensure that the barbell is not separated from the trapezius muscles (bar doesn't float off of your neck). Performing this exercise with heavy weight may lead to injuries, as it can cause damage to the intervertebral discs (photo 1.50-1.55).

Photo 1.50 – 1.55

9. Depth jumps (plyo jumps)

SP: standing on a 40-50 cm tall box. Jump off the box, and jump again once you touch the ground. This exercise develops explosive strength of leg muscles (photo 1.56 – 1.60).

Photo 1.56 – 1.60

10. Jumps onto a box

This exercise develops jumping skills as well as strengthens leg extensors. It also improves the performance of quick extension of the trunk and legs(photo 1.61 - 1.65).

Photo 1.61 – 1.65

11. Seated calf raises

SP: seated on a bench, place your feet on a box. Hold dumbbells in your hands, place them on your knees, lower your heels to the maximum, then raise heels as high as possible. Pause at the very top and return to the starting position.

Engaged muscles: triceps surae.

12.Barbell calf raises

SP: standing with a barbell on your shoulders, place your toes on a block, lower the heels to the maximum and raise on the balls of your feet as high as possible. Pause at the very top and return to the starting position (photo 1.66 – 1.68).

Engaged muscles: triceps surae.

Photo 1.66 **Photo 1.67** **Photo 1.68**

1.5. Bench press exercises

Bench press s the second competitive exercise. It's important for an athlete to develop the bench press along with their squat. Pressing exercises are considered to be the most effective movements when it comes to developing upper bod muscles. They also lead to an increase of strength indicators, muscle mass and they strengthen connective tissues.

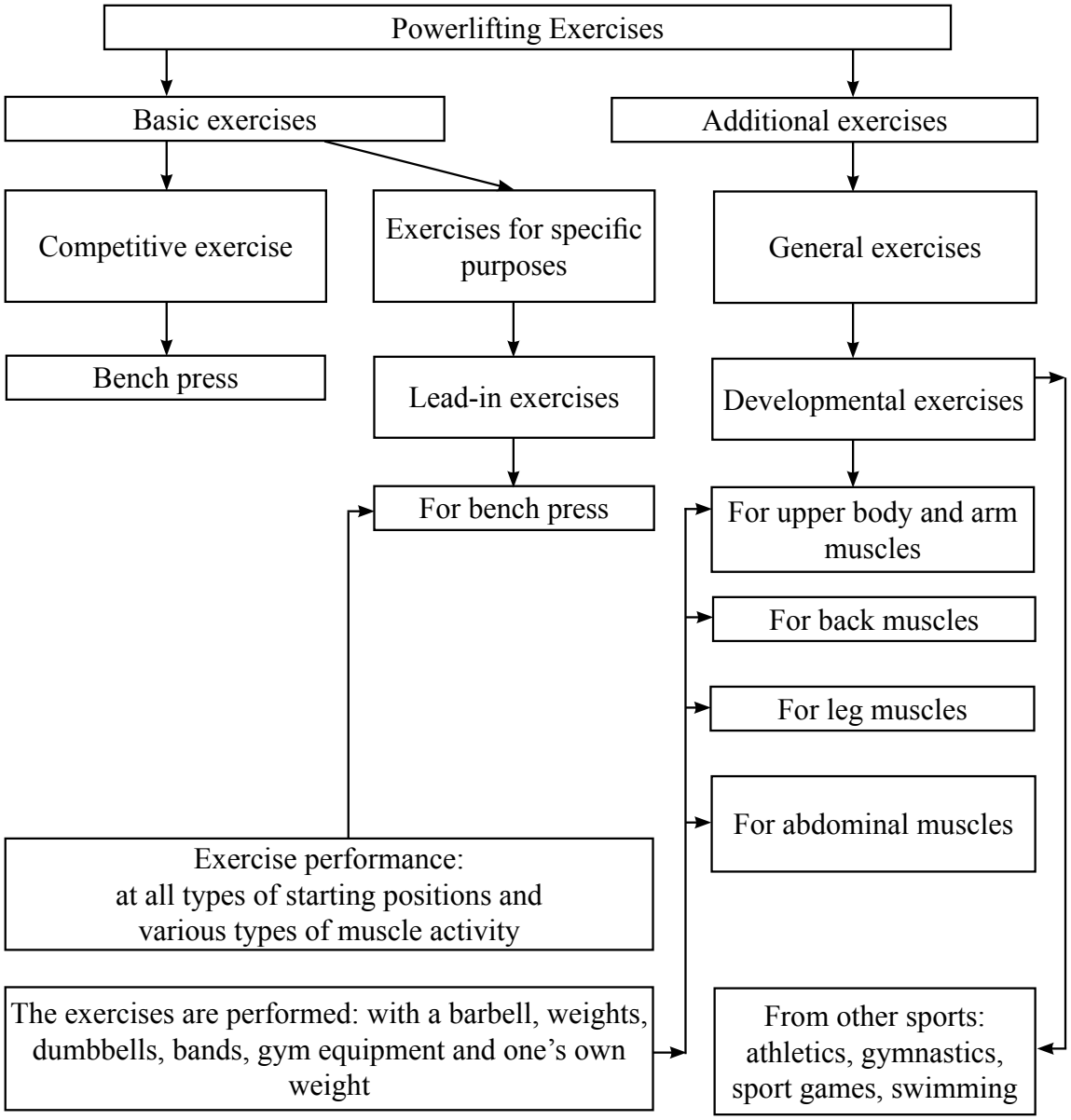


Fig.1.4. Arrangement of basic and additional bench press exercises

Fig. 1.4 represents a bench press tree (V.V. Kostryukov 2011) consisting of five levels.

Competitive bench press is indicated by "B1" code on the scheme.

The first specification level indicates the type of weight used: barbell, weights.

The second level describes the initial position of the press: form the chest or behind the head.

The third level shows the body position while performing the bench press: lying down, sitting down, standing, or push press.

The fourth level describes the following: grip type (straight wrist, reversed), torso angle, space-rhythm parameter (simultaneous, alternate, French press).

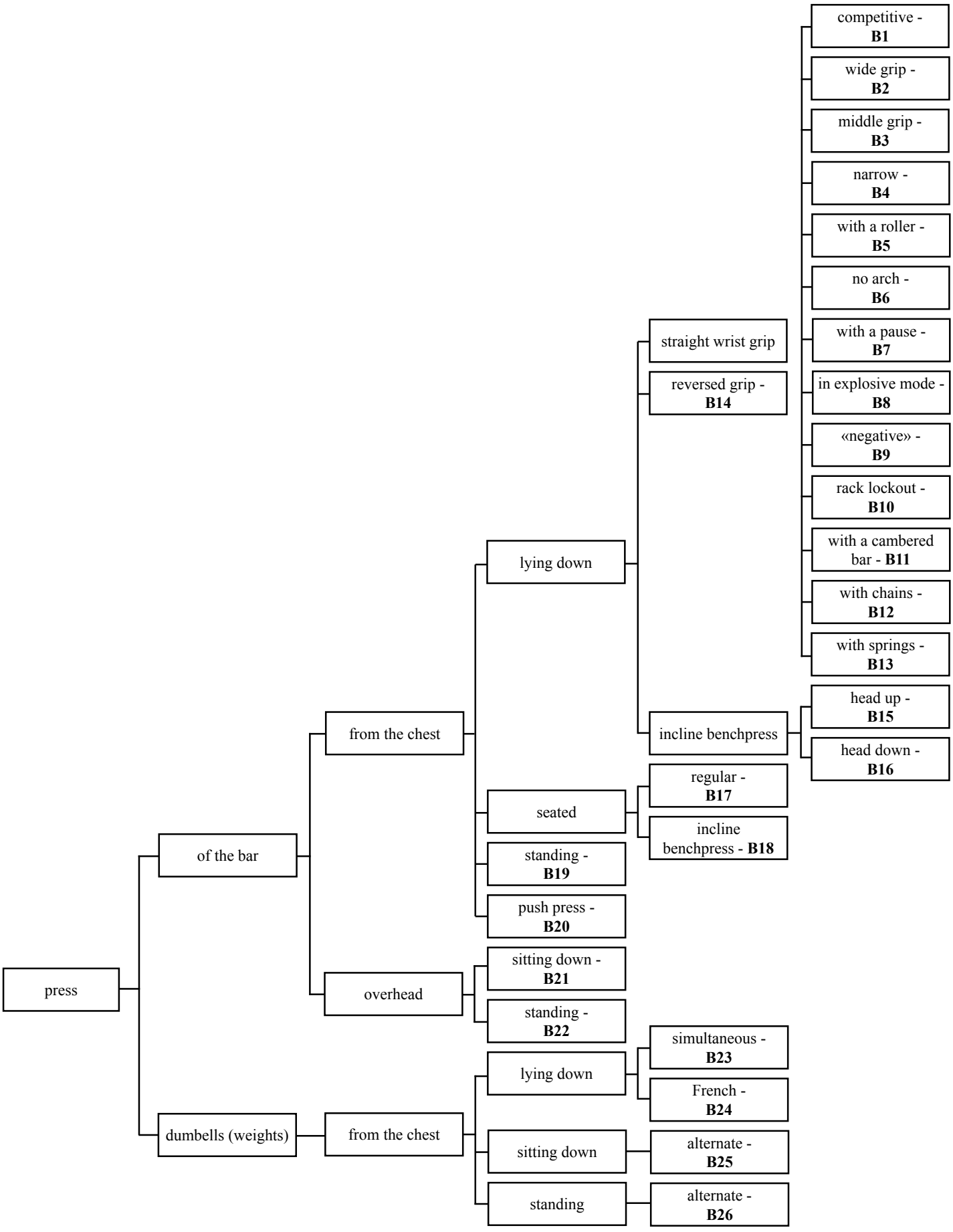


Fig. 1.5. Bench press exercises tree (V.V. Kostryukov 2011)

The fifth level refers exclusively to bench press and describes: grip width (wide, mid, narrow); mode (with pause, explosive, negative), range (with a roller, lockout), details in torso position (no arch), type of weight (cambered bar, chains, bands), and bench inclination (head up or down).

1.5.1. Group 1 – competitive exercises bench press.

Bench press on a horizontal bench. The exercise is carried out in accordance with the standards and requirements of the competition rules of the Russian Powerlifting Federation (RPF) and the International Powerlifting Federation (IPF).

Photo 1.69. Competitive bench press

1.5.2. Group 2 Exercises for specific purposes.

Exercises are performed with a bar from the starting position (SP): lying down on a horizontal bench.

1. Wide grip bench press

SP: grip width is 85-95 cm. With this particular grip width the biggest load goes on the pectoral muscles. The lifter must lower the bar slowly and closer to his throat (photo 1.70-1.72).

Photo 1.70

Photo 1.71

Photo 1.72

The exercise is presented by the three-time world champion, Irina Abramova.

2. Medium grip

SP: grip width is 50-60 cm. In this exercise, all the upper body muscles are engaged on the same level (photo 1.73-1.75).

Photo 1.73

Photo 1.74

Photo 1.75

Photo 1.76

Photo 1.77

Photo 1.78

3. Narrow grip

SP: grip width is 30-40 cm. While performing this exercise, two fingers should be placed on the smooth part of the bar. This exercise brings a strong focus to the triceps (photo 1.76-1.78).

Photo 1.79

Photo 1.80

Photo 1.81

The exercise is presented by the two-time world champion, Tatyana Yeltsova

4. Bench press with a roller under the back

SP: with a roller placed under the lower back. The roller helps assume the proper torso position while performing this particular exercise, which allows for a rapid mastering of the arch technique. The height of the roller is from eight up to fifteen cm (photo 1.79-1.81).

5. Bench press with an extended pause

This exercise is performed in a competitive style with a three-five seconds pause on the chest. Due to this delay, the stress on the pectoral and deltoid muscles is increased.

6. Explosive bench press

The exercise is performed with a slow lowering of the bar to the chest (on the count of one-two-three) and a rapid press (on the count of one). The exercise helps increase the speed of bench press.

7. Bench press with a pause

While lowering the bar to the chest or pressing it off the chest, the athlete should make a one to three seconds pause in the sticking point. If an athlete experiences difficulties while performing this exercise in other points of his range of motion, he may make additional pauses.

8. Reverse grip bench press

SP: grip with the palms placed towards the athlete's face. In this exercise, the weight used is not big, triceps and deltoids are more engaged than pecs. The RPF and IPF rules forbid this particular grip from being used during competitions (photo 1.82-1.84).

Photo 1.82

Photo 1.83

Photo 1.84

9. "Negative" bench press

SP: the grip width is the same as in the competitive bench press technique. The bar should be lowered to the chest very slowly (from five to ten seconds). As soon as the bar touches the chest, two spotters return the bar to the initial position. The exercise is performed with 90-100% weight. Not only does the athlete strengthen his connective tissues, he also gets mentally used to bigger weights. However, this exercise heavily taxes the nervous system, thus it should be used with caution.

10. Board press

SP: the athlete unracks the bar with one of his spotter's help, the second spotter puts a board on the athlete's chest (the thickness of the board can vary from five to fifteen cm). The taller the board, the bigger the weight used in training should be (photo 1.85-1.87). This exercise increases lockout strength as well as strength at the sticking point, depending on the size of the board.

Photo 1.85

Photo 1.86

Photo 1.87

11. Bench press with chains

SP: chain locks are put on the barbell sleeves. As the bar is being pressed, the chains gradually come off the floor, the weight becomes heavier, which increases the load. The chains allow for a steady, gradual press (photo 1.88-1.90). The exercise helps strengthen the lockout of the bench press.

Photo 1.88

Photo 1.89

Photo 1.90

The exercise is presented by the two-time world bench press champion, F. Muhamatyanov.

12. Bench press with bands

SP: lying down on the bench. When lifting the bar, the bands are stretched, thus increasing the load. Load reaches it's peak at the top (photo 1.91-1.93). The bands are used to create dynamic resistance, and allow for a steady press in which resistance is added smoothly as the weight moves up.

Photo 1.91

Photo 1.92

Photo 1.93

13. Bench press with springs

SP: lying down on a flat bench. Locks with springs are placed on the bar's sleeves. As the bar is pressed, the springs stretch, thus increasing the load. The springs allow for a steady and gradual press. The exercise helps strengthen the arm muscles at the final stage of the bench press (photo 1.94 -1.96)

Photo 1.94

Photo 1.95

Photo 1.96

Bench press with Sling Shot, RAM, Catapult

SP: lying down. Sling shot and its alternatives (RAM, Catapult) provide additional support in the eccentric and concentric phases of pressing the bar. They mimic the natural movement of the muscles due to elongation and contraction. These tools help press the bar off the chest, which allows for work with greater weight and training to overcome the sticking point, as well as the lockouts. Sling Shot allows for training with 5-10% heavier weights than normal in the bench press, which will help increase workout intensity. (Note: some Sling Shot variants can provide even greater assistance than 10%, but that much excess load may only be more for its own sake, and no longer logical for the purpose of increasing your raw bench press strength.) (photo 1.97-1.102).

Photo 1.97

Photo 1.98

Photo 1.99

Sling shot bench press is presented by the world and European champion, Kirill Sarychev

Photo 1.100

Photo 1.101

Photo 1.102

The bench press with RAM are presented by the world junior bench press champion, Yan Urusov

14. Bench press with elevated legs

In this exercise, the legs are not placed on the floor but held in the air, there is no arch in the back. All this allows to disengage the leg and back muscles, thereby giving the upper body muscles a bigger proportion of the load. The weights used in this exercise are generally 10% - 20% lighter than during competition pressing with feet down and back arched (photo 1.103-1.105).

Photo 1.103

Photo 1.104

Photo 1.105

15. Floor press

SP: laying on the floor, a spotter helps to place the bar in the starting position. The exercise is aimed at improving the lockout in competitive bench press. When performing this exercise one should take into account injury risks – the weight of 90% and higher can lead to injury of the elbow joints. The recommended number of repetitions is 3-6 in 4-5 sets (photo 1.106-1.108).

Photo 1.106

Photo 1.107

Photo 1.108

Floor press disengages the leg muscles, which puts more pressure on the pectoral muscles, deltoid and triceps muscles.

Floor press develops mass and strength of the pectoral muscles.

16. Rack lockout

SP: laying on the bench. The higher the bar is on the racks, the bigger the weight should be (photo 1.109-1.111). Depending on the racks' height, this exercise increases the arms strength in the initial stage of the press as well as the sticking point.

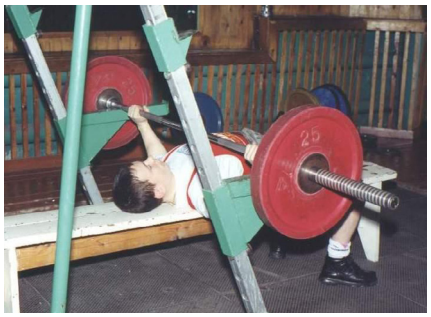


Photo 1.109

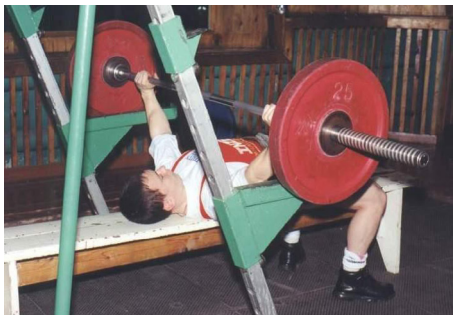


Photo 1.110

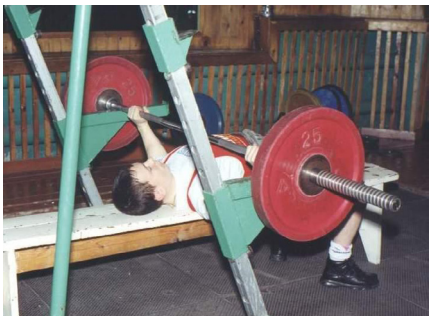


Photo 1.111

17. Cambered bar press

SP: laying on the bench. Due to the shape of the bar, it can be lowered below the chest level. When the range of motion is increased, the stress on the chest and arm muscles is bigger as well (photo 1.112-1.114). Warning: the bar should be lowered slowly, otherwise, when it is lowered abruptly, there is a risk of a chest muscle injury.

Photo 1.112

Photo 1.113

Photo 1.114

The exercise is presented by the five-time European bench press champion, F. Muhamatyanov

1.5.3. Group 3 – general exercises for bench press.

This group includes exercises for the development of the upper body and arm muscles, as well as leg and back muscles, and abdominals. The exercises used are not only from gymnastics, but also from track and field, gymnastics, swimming, and others.

1.5.3.1. Basic exercises for the chest muscles

1. Incline press

SP: laying on the incline bench at 30-45 degrees angle with the head up, the back is pressed to the bench, the feet on the floor. Grip should be outside shoulder width. Lower the bar to the top of the pectoral muscles. When performing the exercise, one must ensure that the elbows are turned outwards, away from the body. The bigger the angle of the bench, the more the deltoid muscles will be engaged. By changing the angle of the movement, the athlete additionally engages the upper pectoral muscles, and activates the deltoid muscles. The optimal variant of the angle is 30 degrees (photo 1.115-1.117).

Engaged muscles:

Primary muscle group - upper fibers of the pectoralis major muscle.

Secondary muscle group – front of the deltoid, tricep.

Photo 1.115

Photo 1.116

Photo 1.117

Photo 1.118

Photo 1.119

Photo 1.120

2. Decline bench press

SP: laying on the bench head down, the incline is 30-40 degrees, the back is pressed to the bench, and the feet are pressed to the stand. Overhand grip, the arms wider than the shoulders. The bar touches the body three-five centimeters below the pectoral muscles. Due to the shortened range of motion and mechanical advantage provided by the position of the elbows close to the body, it is possible to lift more weight than in the horizontal position (photo 1.118-1.120).

Engaged muscles:

Primary muscle group - medium fibers of the pectoralis major muscle.

Secondary muscle group - triceps, anterior fibers of the deltoid muscle.

3. Dumbbell bench press

SP: laying down on a horizontal bench, the feet are on the floor. When performing this exercise with especially heavy loads, two spotters should pass the dumbbells while the athlete is laying on the bench. He takes the dumbbells at the chest level, the palms are positioned inward (the wrists should be placed so that the palms assume the bench press position). Then, the lifter presses the dumbbells in a straight line from the shoulders not allowing them to touch at the top. The main advantage of the exercise is that it provides a greater range of motion and, consequently, a greater impact on the muscles (photo 1.121-1.123).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid, triceps.

Photo 1.121

Photo 1.122

Photo 1.123

Photo 1.124

Photo 1.125

Photo 1.126

4. Incline dumbbell flyes

SP: laying down on a horizontal bench, the feet are on the floor, the dumbbells are above the head on the extended arms with palms facing each other. Move the arms away from each other until the chest level or below. Return to the starting position. Throughout the whole exercise the arms should be slightly bent. In order to avoid injuries, the weight should not be too heavy as the exercise is aimed at stretching the chest muscles (photo 1.124-1.126).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid

5. Flat dumbbell flyes

SP: laying down on a horizontal bench, the feet are on the floor, dumbbells are on extended arms above he upper part of the chest touching, the palms are facing each other. The arms movement is the same as in the exercise above. The arms should be slightly bent in the elbows throughout the whole exercise. (photo 1.127-1.129).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid

Photo 1.127

Photo 1.128

Photo 1.129

6. Pec deck fly

SP: sitting down on the machine with the back pressed to the vertical pad. The arms are pulled back and bent at a 90 degree angle. The forearms rest on the pads. On the exhale, pull the arms to each other as close as possible. Hold this position before returning to the starting position on the inhale. Try to achieve the biggest possible range of motion by stretching the muscles to the maximum when the arms are pulled back, and creating an additional muscle contraction when the arms are pulled to each other (photo 1.130-1.132).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid

Photo 1.130

Photo 1.131

Photo 1.132

7. Dumbbell pullover with extended arms

SP: laying down across the bench, feet on the floor. The hips do not touch the bench. Take the dumbbell with both hands and hold it in front of your chest, pressing the palms to the inner plate. The elbows are bent at a 90 degrees angle. After lowering the dumbbell to the lowest point possible, return it to the starting position. The hips should be lowered throughout the whole exercise as it allows for maximum stretching and expanding of the chest (photo 1.133-1.135).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid, latissimus.

Photo 1.133

Photo 1.134

Photo 1.135

8. Parallel bar dips with added weight on the waist or legs

SP: parallel bar dip on fully extended arms, the head slightly forward. The knees are locked, the torso is upright. When bending your arms, slightly turn your elbows out, lean forward. The elbows are bent at a 90 degrees angle. It is not recommended to make a deep dip, in order not to injure the shoulder joint. Fully extend the arms, return to the starting position (photo 1.136-1.138). The smaller the distance between the bars, the bigger engagement of the triceps, whereas the bigger the distance is, the more the chest muscles will be engaged.

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid, triceps, forearm muscles.

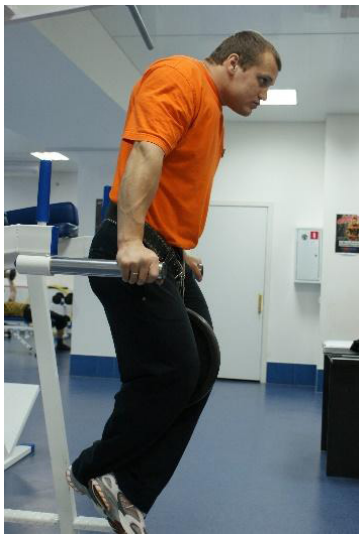


Photo 1.136

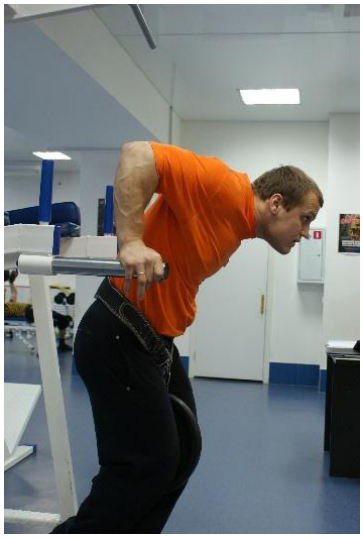


Photo 1.137

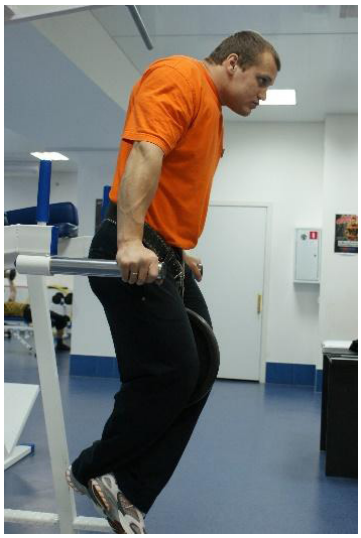


Photo 1.138

9. Pushups with added weight on the back (chest emphasis).

SP: Do a plank on extended arms, the arms are wider than the shoulders (the torso and legs should be straight, no arch in the back). Bend the elbows until the chest reaches the floor (move the elbows to the side). Extend the arms and return to the starting position. In order to increase the load, put additional weight on your back, the feet should be placed on a bench or a stand (photo 1.139-1.141).

Engaged muscles:

Primary muscle group - the pectoralis major muscle.

Secondary muscle group - anterior fibers of the deltoid, latissimus, triceps.

Photo 1.139

Photo 1.140

Photo 1.141

10. Pushups with weight on the back (triceps emphasis).

SP: Do a plank on extended arms, the arms are shoulder-width apart. Elbows are pressed into the torso. Bend the elbows until the chest reaches the floor. Extend the arms and return to the starting position. In order to increase the load, put additional weight on your back, the feet should be placed on a bench or a stand (photo 1.142-1.144).

Engaged muscles:

Primary muscle group - triceps muscle.

Secondary muscle group - anterior fibers of the deltoid, medium fibers of the pectoralis major muscle, forearm muscles.

Photo 1.142

Photo 1.143

Photo 1.144

1.5.3.2.Basic exercises for the development of the shoulder girdle

1. Overhead barbell press from the chest (seated position)

SP: seated on a bench, the barbell is on the chest. Press the barbell up by fully extending the arms, return the bar to the starting position by bending the arms (photo 1.145-1.47).

Engaged muscles:

Primary muscle group – front and side fibers of the pectoralis major muscle

Secondary muscle group – triceps, trapezius muscle, upper fibers of the pectoralis major muscle.

Photo 1.145

Photo 1.146

Photo 1.147

2. Overhead barbell press behind the neck (seated or standing)

SP: sitting on the bench with a barbell behind the neck. Press the barbell from the overhead position. The abdominal and back muscles are stable, which means that they maintain the torso in the upright position (photo 1.148-1.150).

Engaged muscles:

Primary muscle group – front and side fibers of the pectoralis major muscle.

Secondary muscle group – rear fibers of the deltoid muscle, triceps.

Photo 1.148

Photo 1.149

Photo 1.150

Photo 1.151

Photo 1.152

Photo 1.153

3. Alternating dumbbell press (standing or seated)

SP: seated on the bench, arms with the dumbbells are at the shoulders level. Lift the right arm with the dumbbell until it is fully extended. Then bend it and return to the starting position. Lift the left arm. When performing this exercise, it is crucial to lower the dumbbells as slowly as possible (photo 1.151-1.153).

Engaged muscles:

Primary muscle group – front and side fibers of the pectoralis major muscle.

Secondary muscle group – side fibers of the pectoralis major muscle, triceps.

4. Side dumbbell raises (standing or seated)

SP: standing, the arms with the dumbbells are down, the legs are shoulder-width apart. Lift the dumbbell up to the sides, then slowly lower them to the starting position by creating resistance. Do not swing the dumbbells, lift them only with the strength of the deltoid muscles (photo 1.154-1.156).

Engaged muscles:

Primary muscle group –side fibers of the pectoralis major muscle.

Secondary muscle group – front and rear fibers of the pectoralis major muscle, triceps.

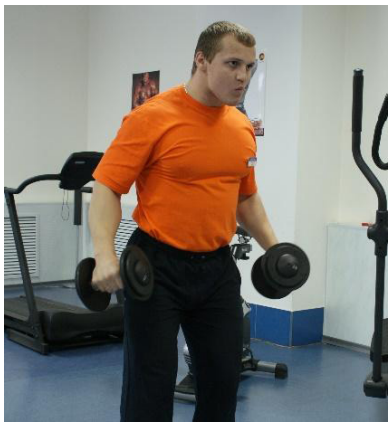


Photo 1.154

Photo 1.155

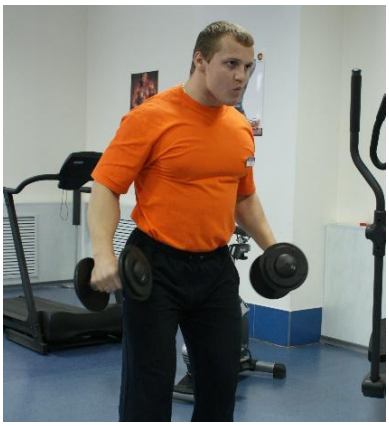


Photo 1.156

Photo 1.157

Photo 1.158

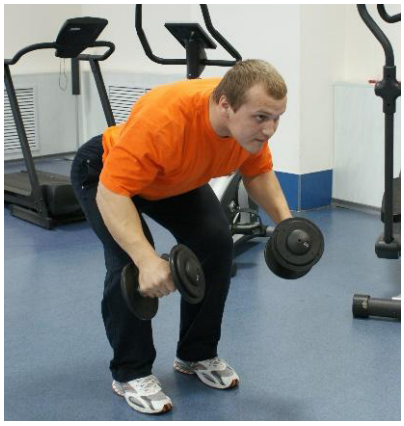


Photo 1.159

5. Rear lateral raise (standing)

SP: standing, the legs are slightly bent, lean forward by bending the body in the hip joints until the torso is almost parallel to the floor. The arms with the dumbbells are down. Raise the dumbbells out to your sides, then lower them down to the starting position creating resistance throughout the whole exercise (photo 1.157-1.159).

Engaged muscles:

Primary muscle group –rear fibers of the pectoralis major muscle.

Secondary muscle group – side fibers of the pectoralis major muscle, trapezius and infraspinatus muscles, rhomboid muscle, teres minor and teres major muscle.

6. Front dumbbell raise (standing or seated)

SP: standing with the dumbbells in the arms. With the arm fully extended, alternate the lift to the front and lift each dumbbell up until it's parallel to the floor (photo 1.160 – 1.162).

Engaged muscles:

Primary muscle group –anterior fibers of the pectoralis major muscle.

Secondary muscle group –pectoralis major muscle, trapezius and serratus muscles.

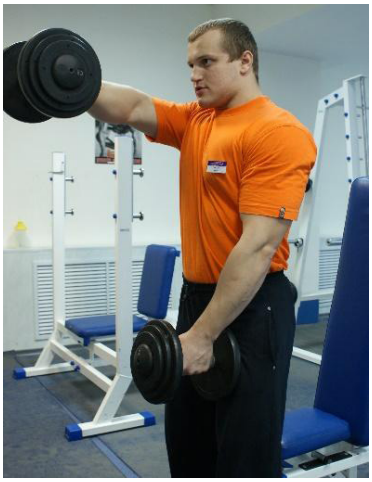


Photo 1.160

Photo 1.161

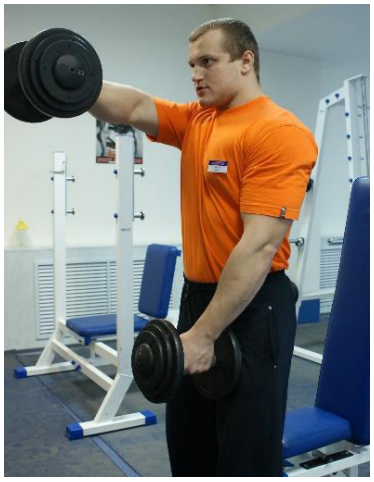


Photo 1.162

1.5.3.3.Basic exercises for the development of the triceps muscles

1. French press (bending the arms lying down)

SP: lying down on the horizontal bench, the hands hold the bar with the narrow grip, the palms are on top. The legs are to the sides pressed to the floor. By bending the arms in the elbows lower the bar to your forehead or overhead. By extending the arms return the bar to the starting position. Do not flare the elbows out (photo 1.163-1.165).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group –chest muscles, deltoid muscle, forearm muscle.

Photo 1.163

Photo 1.164

Photo 1.165

Photo 1.166

Photo 1.167

Photo 1.168

2. Barbell triceps extension (standing or seated)

SP: seated on the bench, the barbell is on extended arms, overhand narrow grip. Lower the bar behind your head, raise the bar in a semicircular arc motion to the initial position. A dumbbell can be used instead of a barbell (photo 1.166-1.168).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group – deltoid muscle, forearm muscle.

3. Bench dips (seated)

SP: hold on to the edge of a box or platform behind you, put your legs on another bench. Bending the arms, slowly lower the torso to the maximum. Then, by extending the arms return to the starting position. It is recommended to increase the load by putting a plate on your lap (photo 1.169-1.171).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group – rear fibers of the pectoralis major muscle.

Photo 1.169

Photo 1.170

Photo 1.171

Photo 1.172

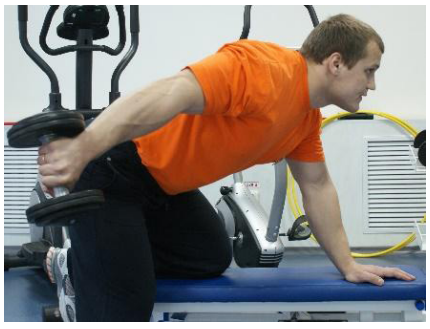


Photo 1.173

Photo 1.174

4. Dumbbell kickbacks

SP: standing bent forward on your right leg, put your left arm and leg on the bench, hold the dumbbell in your right hand with the elbow bent, the elbow should be directed upwards. Fully extend the arm. Hold the final position for one-two seconds and return the dumbbell to the initial position (photo 1.172-1.174).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group –rear deltoids, latissimus.

5. Dumbbell triceps extension

SP: seated on the bench, the dumbbell is in the extended arm above the head, the elbow is close to the head. Lower the dumbbell behind the head until the forearm is parallel to the floor. Return the arm to the starting position (photo 1.175-1.177).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group – deltoid muscles, forearm muscles.

Photo 1.175

Photo 1.176

Photo 1.177

6. Triceps pushdowns (standing)

SP: facing the pulley machine, legs shoulder width apart, back straight. Grip the bar at the chest level, overhand grip, arms bent and pressed to the torso. By extending the forearms, slowly pull the bar until the arms are fully extended. Hold the position for one-two seconds in the lowest point of the movement, then, by bending the arms return to the starting position (photo 1.178-1.180).

Engaged muscles:

Primary muscle group – triceps.

Secondary muscle group – deltoid muscles, forearm muscles.

The wide grip mostly engages the long head of the triceps muscle, while the narrow grip engages the lateral head.

Photo 1.178

Photo 1.179

Photo 1.180

1.5.3.4.Basic exercises for the development of the biceps muscles

1. Barbell curls (standing)

SP: standing, legs shoulder width apart, bar on extended arms, underhand grip. Raise the bar as high as possible along an arc (the elbows are stationary and pressed to the torso). Pause at the top. Lower the bar along the same path until the arms are fully extended. If the fingers are spread throughout the whole move, all of the biceps is equally engaged. If the thumb is placed upwards, the lower part of the biceps is engaged. This exercise can be performed with dumbbells by bending both arms at the same time or alternating (photo 1.181-1.183).

Engaged muscles:

Primary muscle group – biceps.

Secondary muscle group – shoulder muscles, brachioradialis muscle, anterior fibers of the deltoid muscle, forearm muscles.

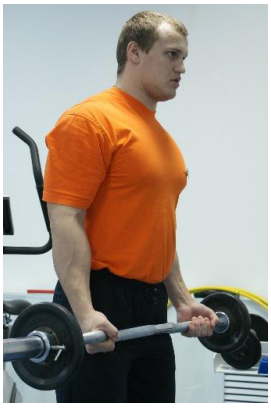


Photo 1.181

Photo 1.182

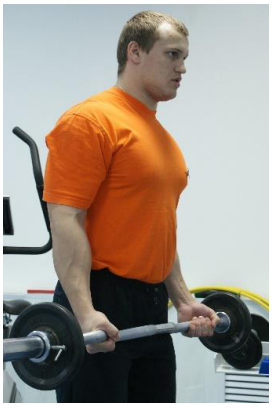


Photo 1.183

Photo 1.184

Photo 1.185

Photo 1.186

2. Barbell curl seated over incline bench

SP: the exercise is performed seated on Scott curl machine with elbows on the incline pad. The grip is shoulder width, palms down. The arms are fully extended while the triceps muscle is pressed to the surface of the pad. Bend the arms in the elbows. Lower the barbell to the starting position. The triceps muscle remains on the pad surface throughout the whole exercise. It is recommended to perform the exercise with an EZ curl bar to avoid the excessive load on the forearms (photo 1.184-1.186).

Engaged muscles:

Primary muscle group – biceps.

Secondary muscle group – shoulder muscles, brachioradialis muscle, anterior fibers of the deltoid muscle, forearm muscles.

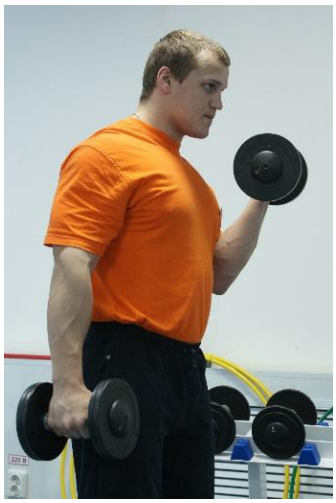


Photo 1.187

Photo 1.188

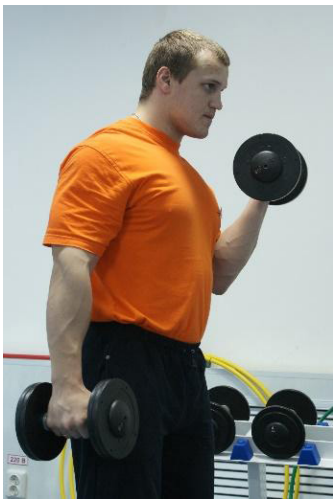


Photo 1.189

3. Dumbbell alternate bicep curl (standing).

SP: standing, arms with dumbbells fully extended and placed alongside the torso. The back is straight, the feet are shoulder width apart. Palms on the bottom. Bend the right arm at the elbow and raise the dumbbell to the shoulder. Slowly lower the dumbbell to the starting position. Repeat the exercise with the other arm (photo 1.187-1.189).

Engaged muscles:

Primary muscle group – biceps.

Secondary muscle group – shoulder muscles, brachioradialis muscle, anterior fibers of the deltoid muscle, forearm muscles.

1.5.3.5.Basic exercises for the latissimus dorsi muscle

1. Pull-ups (wide grip)

SP: Hanging from the bar, wide grip, palms on top. Bend the arms, pull up the torso so that the chin is above the crossbar (photo 1.190-1.192).

Engaged muscles:

Primary muscle group – medium and lower part of the latissimus dorsi.

Secondary muscle group – biceps, forearm muscles, serratus anterior muscle.

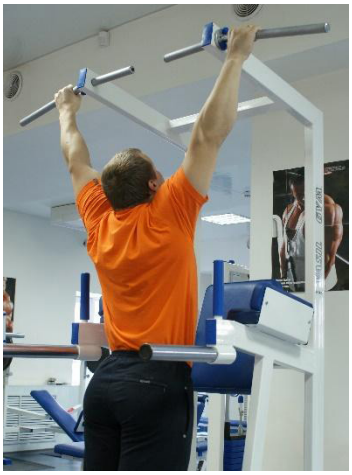


Photo 1.190

Photo 1.191

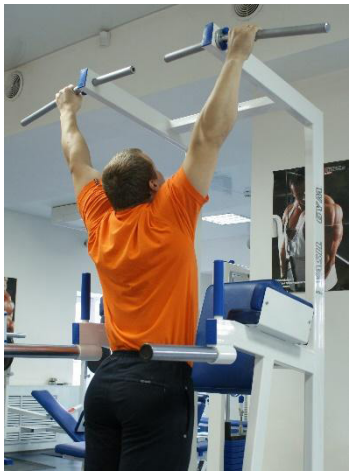


Photo 1.192

Photo 1.193

Photo 1.194

Photo 1.195

2. Crossbar pull-ups with the back of the head touching the crossbar (wide grip)

SP: hanging on the bar, wide grip. Bend the arms, pull your torso up until the back of your head touches the crossbar. The movements are slow with one-two seconds pause at the top (photo 1.193-1.195).

Engaged muscles:

Primary muscle group – upper part of the latissimus dorsi.

Secondary muscle group – biceps, shoulder blade adductor muscles.

The wider the grip, the more the upper lats are engaged.

3. Lat pulldown (seated)

SP: seated on the machine's bench, arms up to the sides. The grip is wider than the shoulders, overhand grip, torso upright. The lat bar should be above the head. By bending the arms at the elbows, pull the bar down over the head until it touches the lower neck. Hold the bar in this position. By extending the arms, return the bar to the starting position (photo 1.196-1.198).

Engaged muscles:

Primary muscle group – latissimus dorsi muscle.

Secondary muscle group – rear fibers of the deltoid muscle, lower fibers of the trapezius muscle, rhomboid muscles.

Photo 1.196

Photo 1.197

Photo 1.198

Photo 1.199

Photo 1.200

Photo 1.201

4. Seated cable row

SP: seated, feet pressed to the machine's platform, legs slightly bent in the knees. Torso upright, arms straight in front of you, the grip is shoulder width, overhand grip. By bending the arms, pull the bar down to your abdominals. Hold the position for one-two seconds and return to the starting position (photo 1.199-1.201).

Engaged muscles:

Primary muscle group – middle and lower fibers of the trapezius muscle, latissimus dorsi muscle.

Secondary muscle group – rhomboid muscles, rear fibers of the deltoid muscle, biceps muscle, forearm muscles.

Photo 1.202

Photo 1.203

Photo 1.204

5. Bent over barbell row

SP: standing, legs slightly bent in the knees, shoulder-width apart, torso bent forward. The grip is wide, the head is slightly raised, focus your gaze in front of you, the back is straight. Lift the bar until it touches the torso and return to the starting position (photo 1.202-1.204).

Engaged muscles:

Primary muscle group –, latissimus dorsi muscle.

Secondary muscle group – trapezius muscle, rhomboid muscles, rear fibers of the deltoid muscle.

1.6. Deadlift exercises

Deadlift is the final exercise in the power triathlon. The exercise has its own peculiarities: all the large leg and back muscles are engaged when the legs and back are straightened.

The following table presents the arrangement of powerlifting deadlift exercises.

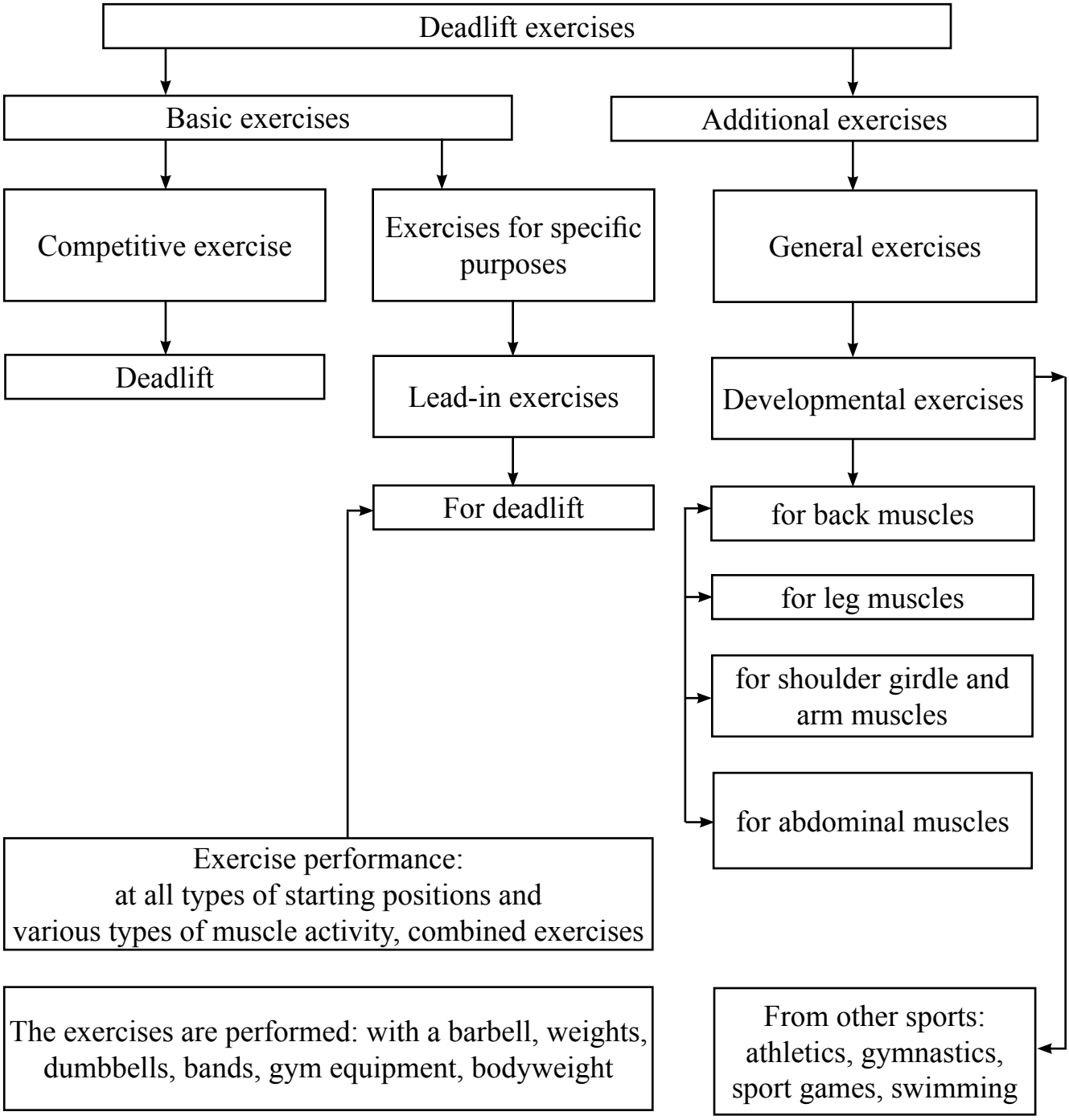


Fig. 1.6. Deadlift exercises arrangement scheme

Fig. 1.6. represents a deadlift tree consisting of three levels.
Competitive deadlift is indicated by "D1" code on the scheme.

The first specification level describes the starting position (standing, from blocks, on a stand, with Smith machine from the knees, seated), and the movement (full, just shrugs).

The second level describes the starting position of the bar (below, lower, or at the knees level), and the final bar position (competitive, to the knees), grip width (regular, wide), weight type (plates, chains, springs), work rate (with pauses, slow return to the platform).

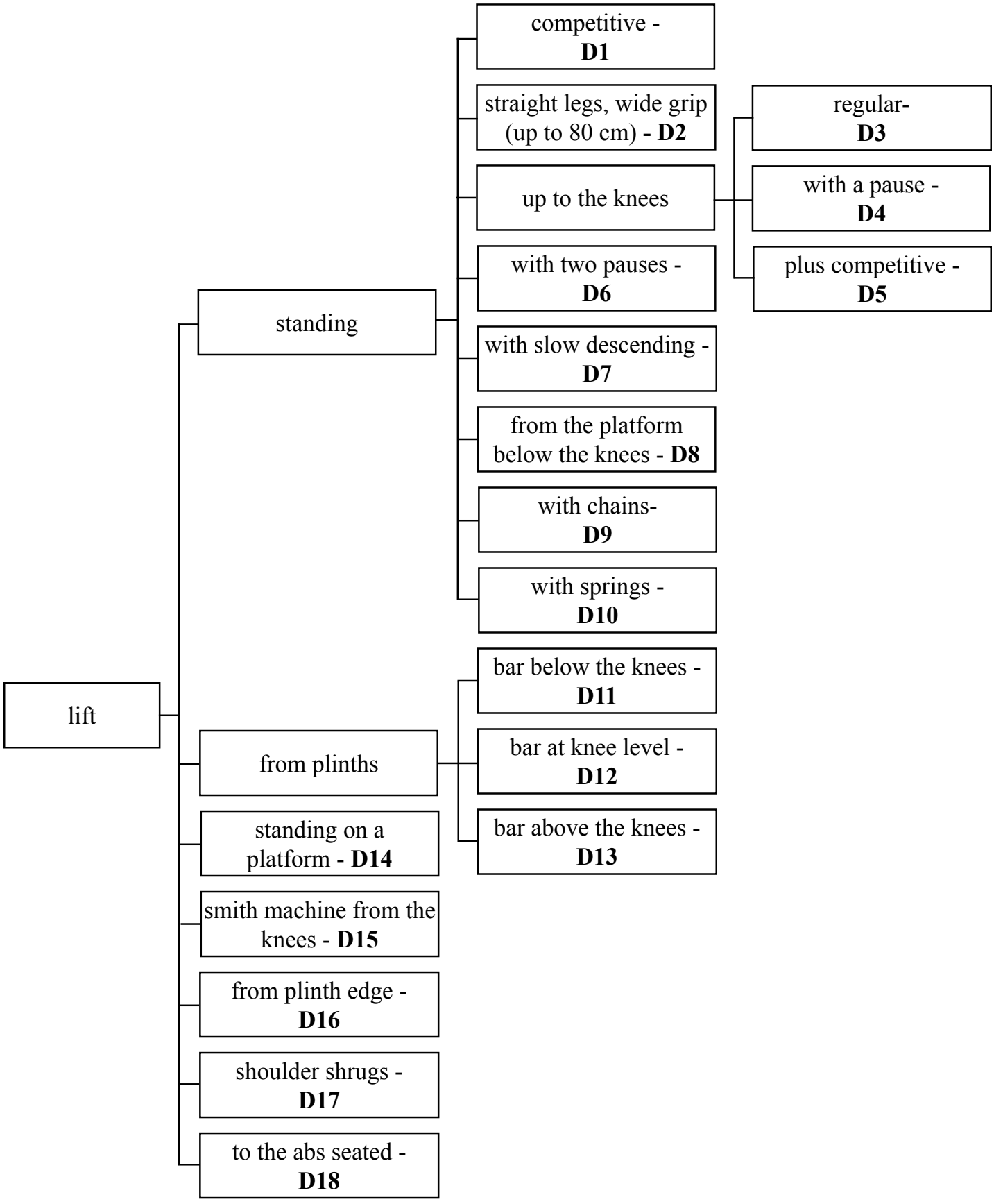


Fig. 1.7. Deadlift exercises tree (V.V. Kostryukov 2011)

The third specification level refers to the knee level deadlift, and indicates additional parameters of lifting the weight (regular lift, with a pause, to the knees plus competitive).

1.6.1. Group 1 - of the deadlift exercises includes a competitive exercise, **similar to squats and bench press**. It should follow all the norms and requirements of the international competitions rules. It is performed in “sumo” style (photo 1.205) as well as “conventional” style (photo 1.207), and in a combination of both (photo 1.206).

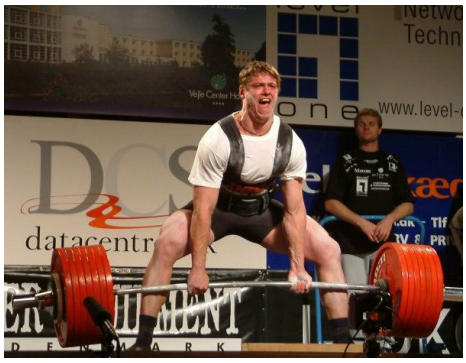


Photo 1.205

Photo 1.206

Photo 1.207

1.6.2.Group 2 - of the deadlift exercises includes exercises for specific purposes, divided into lead-in and developmental exercises.

1. Deadlift to the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Lift the bar to the knee joints level, pause for two-three seconds. After the pause, return the bar to the platform. (photo 1.208-1.210).

Photo 1.208

Photo 1.209

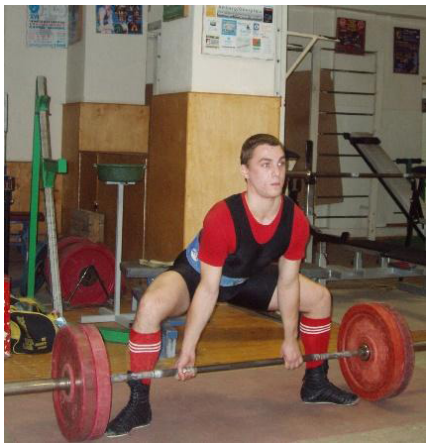


Photo 1.210

2. Deadlift with one pause below the knees

This exercise is a more complicated version of the previous one.

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Lift the bar until it is five-eight centimeters below the knee joints, pause for two-three seconds, then lift the bar to the lockout position (photo 1.211-1.213).

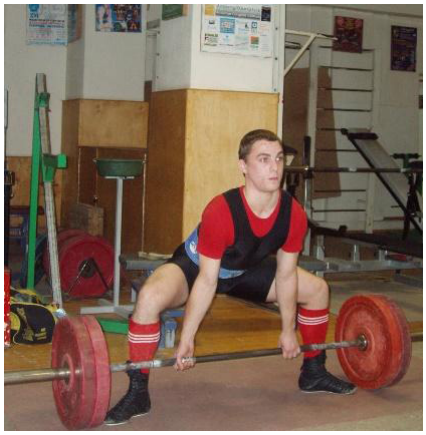


Photo 1.211

Photo 1.212

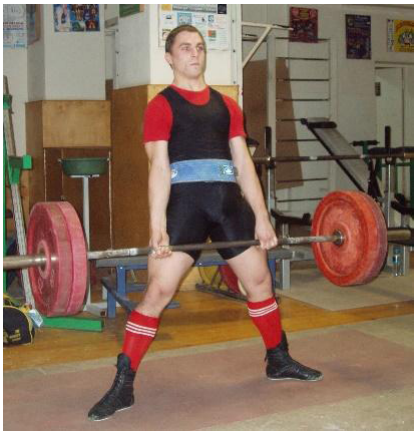


Photo 1.213

3. Deadlift with one pause above the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Lift the bar until it is five-eight centimeters above the knee joints, pause for two-three seconds, then lift the bar to the lockout position (photo 1.214-1.216).

Photo 1.214

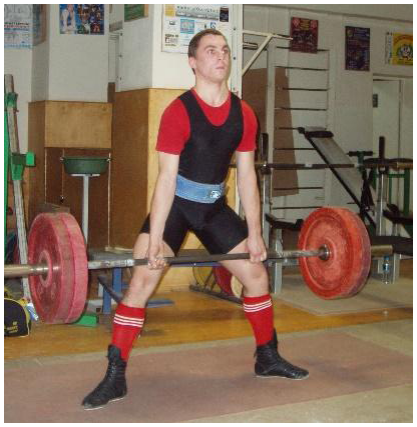


Photo 1.215

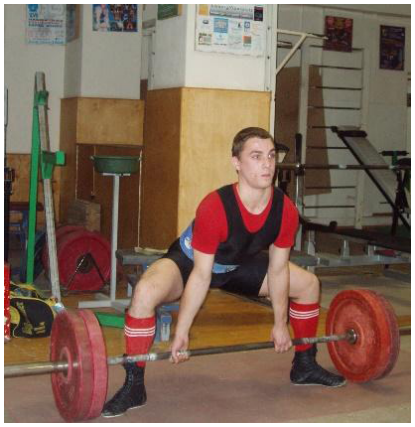


Photo 1.216

4. Deadlift with two pauses below and above the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Lift the bar until it is five-eight centimeters below the knee joints, make the first pause for one-two seconds. Then, lift the bar until it is five-eight centimeters above the knee joints, make the second pause for one-two seconds. After the second pause, lift the bar to the lockout position (photo 1.217-1.220).

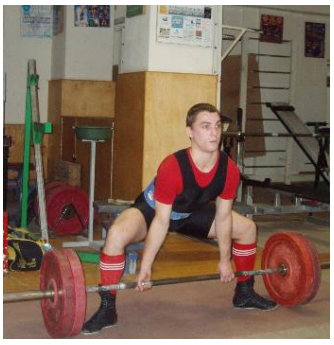


Photo 1.217

Photo 1.218

Photo 1.219

Photo 1.220

5. Competitive deadlift + deadlift from below the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Do the competitive deadlift, then, without returning the bar to the platform, do the hang clean lift below the knee joints. The number of the lifts depends on the bar’s weight. This particular exercise helps improve the deadlift technique, develop the strength of the leg and torso extensor muscles, as well as the speed in the lockout stage (photo 1.221-1.224).

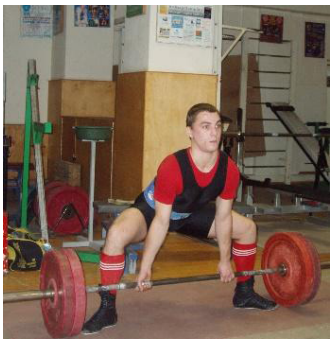


Photo 1.221

Photo 1.222

Photo 1.223

Photo 1.224

6. Competitive deadlift + deadlift from above the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Do the competitive deadlift, then, without returning the bar to the platform, do the hang clean lift above the knee joints. The number of the lifts depends on the bar’s weight. This particular exercise helps improve the deadlift technique, develop the strength of the leg and torso extensor muscles as well as the speed in the lockout stage (photo 1.225-1.228).

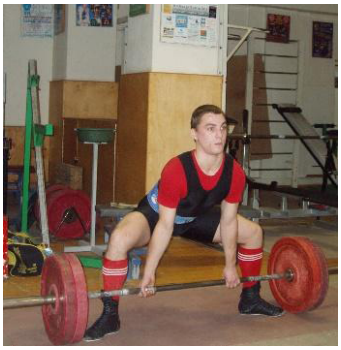


Photo 1.225

Photo 1.226

Photo 1.227

Photo 1.228

7. Deadlift to the knees plus competitive deadlift

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style.

From the starting position, an athlete first performs the lift to the knee joints level (once or twice), followed by the competitive deadlift.

The exercise is aimed at the improvement of the competitive deadlift technique. It helps develop the strength of the leg and torso extensor muscles. (photo 1.229-1.232).

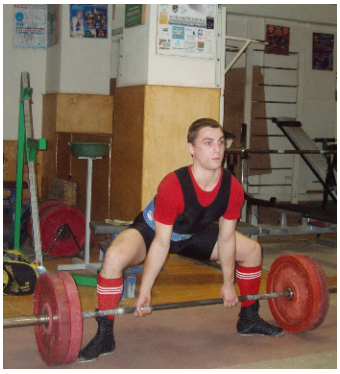
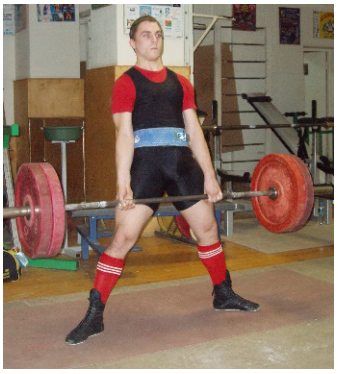



Photo 1.229 **Photo 1.230** **Photo 1.231** **Photo 1.232**

8. Deficit deadlift

SP: standing on a seven-ten centimeters tall stand. Feet spread apart in “sumo” or “conventional” style. It is recommended to spread the feet as it would be done at competitions: “sumo” style (photo 1.233-1.234), “conventional” style (photo 1.235-1.236). When performing the deadlift from a stand, the bar’s path is longer, which increases the load on the leg and torso extensor muscles.

The exercise helps increase strength when lifting the bar off the platform. It is not recommended to use this exercise when training beginners, as the starting position of this exercise is different from the regular one and can lead to the acquisition of the wrong technique.

Photo 1.233 **Photo 1.234** **Photo 1.235** **Photo 1.236**

9. Deficit deadlift with one pause below the knees

SP: standing on a seven-ten centimeters tall stand. Feet spread apart in “sumo” or “conventional” style. Lift the bar to five-eight centimeters below the knee joints and pause for two-three seconds, then lift the bar to the final position (photo 1.237-1.239).

Photo 1.237 **Photo 1.238** **Photo 1.239**

10. Deficit deadlift with one pause above the knees

SP: standing on a seven-ten centimeters tall stand. Feet spread apart in “sumo” or “conventional” style. Lift the bar to five-eight centimeters above the knee joints and pause for two-three seconds, then lift the bar to the final position (photo 1.240-1.242).

Photo 1.240

Photo 1.241

Photo 1.242

11. Deadlift with chains

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Attach the chains to the sleeves of the bar, which will increase the bar weight when lifted. Consequently, the load on leg and back muscle also increases. The load will reach its maximum at the lockout.

The exercise helps increase leg and back muscle strength at the lockout (photo 1.243-1.244).

Photo 1.243

Photo 1.244

12. Deadlift with chains with one pause below the knees

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Perform the exercise in the same way as exercise 3 (photo 1.245-1.247).

Photo 1.245

Photo 1.246

Photo 1.247

Photo 1.248

Photo 1.249

Photo 1.250

13. Deadlift with chains with one pause above the knees

The starting position and the exercise are the same as in exercise 4 (photo 1.248-1.250).

14. Deadlift with chains with two pauses below and above the knees

The starting position and the exercise are the same as in exercise 5 (photo 1.251-1.254).

Photo 1.251

Photo 1.252

Photo 1.253

Photo 1.254

15. Deficit deadlift with chains

The starting position and the exercise are the same as in exercise 6 (photo 1.255-1.256).

Photo 1.255

Photo 1.256

16. Deficit deadlift with chains with one pause below the knees

The starting position and the exercise are the same as in exercise 7 (photo 1.257-1.259).

Photo 1.257

Photo 1.258

Photo 1.259

Photo 1.260

Photo 1.261

Photo 1.262

17. Deficit deadlift with chains with one pause above the knees

The starting position and the exercise are the same as in exercise 7 (photo 1.260-1.262).

18. Deadlift from blocks, bar below the knees

SP: bar on blocks below the level of knee joints. Feet spread apart in “sumo” or “conventional” style (photo 1.263-1.264).

The exercise develops the strength of the leg and torso extensor muscles as well as trapezius muscles. It also helps improve the technique of the top half of the deadlift.

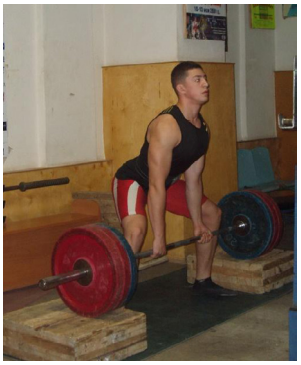


Photo 1.263

Photo 1.264

19. Deadlift from blocks with one pause

SP: bar on blocks below the level of knee joints. Feet spread apart in “sumo” or “conventional” style.

The exercise is performed with one pause at five-ten centimeters above the knee joints (photo 1.265-1.267).

The exercise develops the strength of the leg and torso extensor muscles as well as trapezius muscles. It also helps improve the technique of the top half of the deadlift.

Photo 1.265

Photo 1.266

Photo 1.267

20. Deadlift from blocks with chains

SP: bar on blocks below the level of knee joints. Feet spread apart in “sumo” or “conventional” style (photo 1.268-1.269).

The exercise develops the strength of the leg and torso extensor muscles as well as trapezius muscles. It also helps improve the technique of the top half of the deadlift.

Photo 1.268

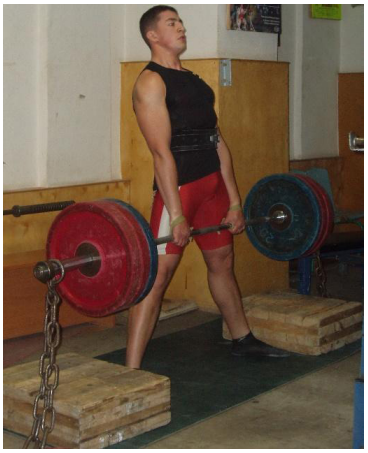


Photo 1.269

21. Deadlift from blocks with chains with one pause above the knees

SP: bar on blocks below the level of knee joints. Feet spread apart in “sumo” or “conventional” style. The exercise is performed with one pause above the knee joints level (photo 1.270-1.272).

The exercise develops the strength of the leg and torso extensor muscles as well as trapezius muscles. It also helps improve the technique of the top half of deadlift.

Photo 1.270

Photo 1.271

Photo 1.272

22. Deadlift from blocks above the knees

SP: bar on blocks above the level of knee joints.

The exercise is aimed at the improvement of the lockout stage. It is recommended to perform it with maximum weights of 90%-110%.

The exercise develops the strength of the leg and torso extensor muscles as well as trapezius muscles. It also helps improve the technique of the top half of deadlift. The high position of the bar (on blocks) allows to focus on the strength development at the lockout (photo 1.273-1.274).

Photo 1.273

Photo 1.274

23. Deadlift with slow descending

SP: bar on the platform. Feet spread apart in “sumo” or “conventional” style. Slow lowering of the bar creates additional load on the back muscles.

1.6.3. Group 3 – general and developmental exercises

General and developmental exercises for powerlifting deadlift consist of the exercises for back, leg, and abdominal muscle development.

1.6.3.1. Exercises for back muscle development

1. Deadlift from the edge of the block (from a “rib”)

SP: put the center of the bar on a narrow block. The grip is wide. You should straighten the body as fast as possible while lowering the bar as slow as possible. When lowering the bar, the legs should be slightly bent (photo 1.275-1.277).

Engaged muscles:

Primary muscle group - posterior group of thigh muscles, gluteal muscles.

Secondary muscle group - extensor muscles of the spine (m. erector spinae), which is divided into three parts: a) bristlecone muscle; b) longissimus muscle; c) the iliac rib muscle.

Photo 1.275

Photo 1.276

Photo 1.277

2. Shoulder shrugs

SP: standing, legs hip-width apart, dumbbells in extended arms.

Raise the shoulders as high as possible, return to the starting position (photo 1.278-1.280).

Engaged muscles:

Primary muscle group – front and medium of the trapezius muscles.

Secondary muscle group - levator scapulae muscles, deltoid muscles.

Photo 1.278

Photo 1.279

Photo 1.280

3. Hyperextensions

SP: lay down on the machine (face down), stabilize the feet. Arms behind the head. Raise the torso until it is parallel to the floor. After lowering the torso, slowly raise it up until fully extended. The exercise can be performed with and without additional weight (barbell, plates, etc.) behind the head or in lowered arms (photo 1.281-1.283).

Engaged muscles:

Primary muscle group - extensor muscles of the spine, bristlecone muscle, longissimus muscle, iliac rib muscle

Secondary muscle group - longissimus dorsi, gluteal muscles, biceps femoris.

Photo 1.281

Photo 1.282

Photo 1.283

4. Reverse hyperextensions

SP: lay with your face down on the machine, grip the handles. Keeping your back straight lift your legs up, slowly return to the position in which the body will be in a straight line (the torso line is the extension of the legs line), hold this position for one-two seconds. You must avoid arching your back (photo 1.284-1.285).

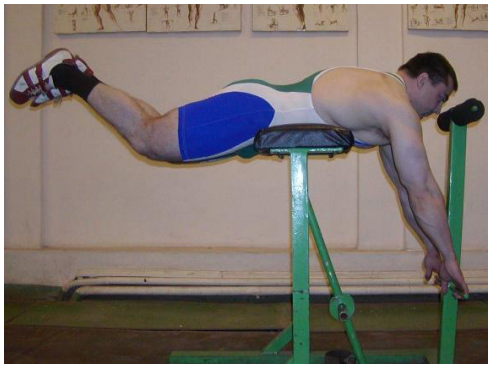


Photo 1.284

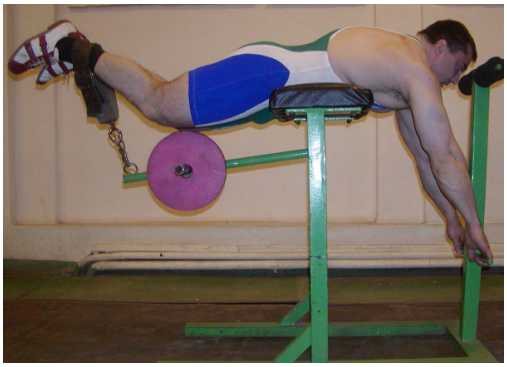


Photo 1.285

5. Barbell good morning

SP: standing, feet apart, with the bar on your shoulders. Slightly bending your legs and keeping your back straight, lean forward until parallel to the floor, return to the starting position (photo 1.286-1.288).

Engaged muscles:

Primary muscle group - extensor muscles of the spine, bristlecone muscle, longissimus muscle, iliac rib muscle

Secondary muscle group – latissimus dorsi, gluteal muscles, rear thigh muscles.

Photo 1.286

Photo 1.287

Photo 1.288

6. Seated barbell good morning

SP: seated on the bench, feet apart, with a barbell your shoulders. Keeping your back straight, lean forward and return to the starting position. The angle of the movement depends on the athlete's hip flexibility. The lower the lean, the bigger the load on the extensor muscles. It is crucial to remember that leaning forward rapidly is dangerous as the athlete can injure the intervertebral ligaments (photo 1.289-1.291).

Engaged muscles:

Primary muscle group - extensor muscles of the spine, longissimus muscle, iliac rib muscle

Secondary muscle group – latissimus dorsi, gluteal muscles, rear thigh muscles.

Photo 1.289

Photo 1.290

Photo 1.291

7. Deep good mornings (standing)

SP: feet shoulder-width apart, weight (kettlebell, plates, etc.) in extended arms. Lean forward. The exercise is performed with your back straight and legs slightly bent in the knees (photo 1.292-1.294).

Engaged muscles:

Primary muscle group - extensor muscles of the spine, bristlecone muscle, longissimus muscle, iliac rib muscle

Secondary muscle group – gluteal muscles, rear thigh muscles.

Photo 1.292

Photo 1.293

Photo 1.294

8. Barbell good morning plus squat (standing)

The barbell and foot position are the same as in squats. On count “one” lean forward until the body is parallel to the floor. On count “two” squat from this position. The shoulders and barbell are stabilized, the hips are lowered. On count “three” lean forward from the squat. On count “four” straighten the back to the starting position (photo 1.295-1.299).

Photo 1.295 **Photo 1.296** **Photo 1.297** **Photo 1.298** **Photo 1.299**

1.6.3.2. General exercises for abdominal muscle development

1. Hanging pike

SP: hanging from a crossbar or wall bars. Raise the slightly bent legs until your shins touch the bar. The movement is primarily done by the hip flexors (photo 1.300-1.302).

Engaged muscles:

Primary muscle group – rectus abdominis.

Secondary muscle group – obliques, iliac rib muscle, rectus femoris, adductor longus muscle.

Photo 1.300 **Photo 1.301** **Photo 1.302**

2. Hanging leg raise

SP: torso hanging upright from the machine, arms bent. Raise the legs up until they are parallel to the floor. To make the exercise easier if needed, the legs can be slightly bent in the knees (photo 1.303-1.305).

Engaged muscles:

Primary muscle group – rectus abdominis.

Secondary muscle group – obliques, iliac rib muscle, rectus femoris, adductor longus muscle.

Photo 1.303

Photo 1.304

Photo 1.305

3. Flat bench sit-ups

SP: laying on a flat abdominal bench. Arms behind your head. Legs bent in the knees (it will decrease the load on the lumbar spine region). Raise the torso (photo 1.306-1.309).

Engaged muscles:

Primary muscle group – rectus abdominis, especially the upper part.

Secondary muscle group – obliques, serratus anterior muscle, iliac rib muscle, rectus femoris.

Photo 1.306

Photo 1.307

Photo 1.308

4. Incline sit-ups

SP: laying with your head down on an incline ab bench. The exercise is performed as the previous one. The load can be increased by adding weight to the arms (photo 1.309-2.311).

Engaged muscles:

Primary muscle group – rectus abdominis and obliques.

Secondary muscle group – serratus anterior muscle, iliac rib muscle, hip flexors.

Photo 1.309

Photo 1.310

Photo 1.311

Chapter 2. Basic Concepts of Biomechanics and Technique in Powerlifting

The Biomechanics of motor actions studies the properties and functions of the musculoskeletal system and human motor actions on the basis of concepts, principles, and laws of classical mechanics.

A sports technique (as in the technique used in a particular sport) is a set of methods and actions that provide the most effective solution to motor tasks arising from a specific sport, its discipline and the type of competition. The best choice of methods and actions is the one when an athlete lifting the barbell at maximum weight (without breaking the rules of the competition) uses his physical and mental capabilities in the most effective way. Formation and improvement of motor actions is a multi-step process, inseparable from the training process. This can be defined as the technical preparation of an athlete.

The technical proficiency of an athlete is the degree of mastering the movements performed by the athlete, i.e. sports technique. This corresponds to the characteristics of an athletic discipline and is aimed at achieving success in competition. Technical training is aimed at teaching the movements technique as well as bringing them to perfection by an athlete (L.P. Matveev, 1982: N.G.Ozolin, 2002; V.N.Platonov, 2004).

Technical skills occupy a special place among other components of sports preparedness. In the execution of anyphysical exercise an athlete demonstrates the results of their physical, tactical, and theoretical training. Currently, many experts consider technical preparation as the strategic direction of modern sports training.

Due to the fact that any activity related to the demonstration of physical qualities (specifically in the barbell exercises with strength, speed, and flexibility) the movement control, which is carried out on the basis of the firmly engrained skills, must evolve along the development of these qualities. Motor skills in strength sports, where sports achievements are associated with heavy lifting, can be described as an ability to perform the movements of a physical exercise automatically.

The level of the development of strength, speed, and flexibility effects the form of an exercise and its techniques. Consequently, performing barbell exercises in powerlifting occurs under specific conditions, and is characterized by a number of athlete's actions:

1. During an exercise with small and medium-sized weights as well as maximum and supra maximum weights, the muscular system of an athlete undergoes varying degrees of dynamic and static stresses. The heavier the weight, the more muscles are involved in the movement.
2. The body of an athlete is an open kinematic chain with a large number of links where almost all the links have three degrees of freedom, ultimately providing subtle and varied movement coordination. With this in mind, for a successful exercise with great power output, it is necessary to put the joints and links in the body in an optimal position in order to provide the maximum synchronization of the muscle groups involved and allow for each muscle's strength to be manifested.
3. In the course of an exercise with weights, one can observe an alternation of superior, relatively calm, inferior, and static forces. There are short-term movements by inertia, as well as a manifestation of varied simultaneous efforts in different body parts. When one

muscle group is actively moving and performing dynamic work, others maintain a certain position in the joints, thus performing static work. There are continuous changes and diverse alternations of dynamic and static operation modes of the body as a whole as well as its separate parts.

4. Unusual balance conditions occur in the performance of exercises with a barbell. The general center of mass of the «athlete-barbell unit» changes continuously until the very end of the exercise by moving upwards and downwards, e.g. squat, over a limited area of support. Therefore, when studying the powerlifting exercises technique, an athlete and a barbell must be considered as a single complete mechanical unit with a common fulcrum.

An athlete must learn to move bars of varying weights with the same technique. Only under this condition during training is an increase in the level of development of physical qualities as well as technological improvement made automatic. A number of studies in this area have shown that first of all, the characteristics of the trained sportsmen's kinematic, dynamic and rhythmic movements vary with increasing the barbell weight during exercise. When the barbell weight increases along with the changes in the spatial and temporal characteristics of the motion, one can also observe changes in the nature of muscle tension and effort. These changes are determined by a kind of «muscular sense.» This feeling must be formed in order to work with different weights successfully.

Secondly, the consideration of what occurs in competition must be made. When an athlete lifts a weight significantly exceeding what has been lifted in training, there is an extrapolation of motion control in the new structure, i.e. the ability of the nervous system to adequately address emerging motor tasks based on prior experience. The athlete's body, which is learning different skills when lifting barbells of different weights, gains the ability to correctly perform exercises with a greater bar weight.

The stability of the technique is related to its interference immunity, i.e. the technique remains consistent and stable regardless of the competition conditions or the functional state of an athlete.

2.1 Technical structure of competitive exercises

The technical structure of competitive exercises can be divided into three categories: the basis of the movements' technique, their sequence, and details.

The basis of the technique can be defined as a combination of parts and features of the dynamic, kinematic and rhythmic structures of the movement, which are necessary to execute a motor task in a certain way, i.e. a particular sequence in the demonstration of muscular strength; the right combination of movements coordinated in space and time, etc. Loss or violation of at least one or part of the elements makes the completion of a motor task impossible. The following requirements should be considered as the basis of the powerlifting movements' technique:

1. Establishment of optimal joint angle, particularly in the most difficult sections of the path (e.g. in «sticking points»), when it is not possible to use inertia for the upward movement.

2. Consistent activation of certain muscle groups, starting from the strongest and moving on to the weakest ones.

3. Providing the most efficient direction of barbell movement with the optimal speed at every step of the process.

4. Creation of the necessary conditions to ensure the effectiveness of the implementation of the final path of the bar.

5. Creating the conditions necessary for supporting an athlete's body (especially for their constituent body parts) to allow longer and more efficient transmission of muscular effort to the barbell throughout the execution of a competitive exercise.

The motor actions technique as any other system has its own structure with a defining element. The execution of the main element usually occurs in a relatively short period of time and requires a lot of muscular effort.

The main element of the technique is the most important and crucial part of a particular method of motor task performance. For example, in the high jump the main element is propulsion, coupled with fast and high gait; in throwing sports - the final effort; arching over the gymnastics apparatus - timely and rapid extension in the hip joints, followed by stopping. Performing basic movements in the execution usually occurs in a relatively short period of time and requires a lot of muscular effort.

Technique's details are secondary movement features that do not disrupt its underlying mechanism. They depend on the morphological and functional characteristics of an athlete. An individual technique, which appears to be the most appropriate one for a particular person, is characterized by the proper use of the so-called technique's details.

In the theory and practice of physical education, the structure of a movement, which is a relatively stable element of a motor activity, is considered as a whole.

The motor structure is the connection between movements in space and time (defined as kinematic structure) as well as the power and energy of interaction (defined as dynamic structure).

When learning an exercise (e.g. lifting the bar), the kinematic structure is determined as the first step. Video shooting followed by its biomechanical analysis is used to study the structure. This makes it possible to determine the trajectory of the barbell, the center of mass of the athlete, the center of mass of the bar, a common center of mass of the athlete in relation to the barbell, and the location of the individual units of the athlete's body. Additionally, the duration of the exercise and the various phases of motion, velocity and acceleration of the body in space, the rate of rising of the bar, etc. can be determined through biomechanical analysis.

The dynamic structure is a natural force (dynamic) interaction of human body parts between each other and external bodies (surroundings, support, equipment, partners, and opponents).

Analysis of the dynamic structure is carried out through various dynamic and strain gauge devices, which allows one to obtain objective data regarding efforts when lifting the equipment. For example, you can see parts of the body interact with each other and external bodies (suspension, partners, and opponents). This defines the application of force in an athlete lifting weights, inertial resistance, and the reasons for a particular motion.

The anatomical structure determines the interaction of the osseous-ligament and muscular apparatus, the mode of operation of the muscles and the coordination of muscle contractions.

Rhythmic structure is a pattern of the interconnection between movements over time, the ratio of movement duration, all motor act or its single actions. The speed and duration of subsequent movements depend on how which effort is emphasized in a particular time. Parts of the movement differ in direction, speed, acceleration, and force. The rhythmic ratio is measured exclusively by means of time indicators. The rhythmic structure serves

as a particularly distinct indicator of an exercise mastery, reflects a peculiar combination of temporal and spatial characteristics of the motion and shows the execution order of the individual phases.

2.1.1 Period and phase structure of exercises

A Period is a time frame during which something occurs (starts and ends). In powerlifting, a period can be defined as a somewhat independent part of an exercise, during which rational conditions for the effectiveness of the subsequent movements are formed. In each period there are significant changes in the spatial, temporal, kinematic, rhythmic movement characteristics. A relative completion of some actions can be observed as well. Period consists of phases with common features.

A motor task can be defined as generalized requirements for motor actions, which are determined by the nature of the impending action and the overall consistency of its stages. Each motor action of an individual carries out certain motor tasks. The task can be e.g. to achieve a particular end goal (such as a squat), or in the performance of a given movement. The completion of a motor task is the goal of motion control. A motor task appears to be a model of something that has not yet occurred («model of a future», according to Bernstein).

Phase (in Greek - manifestation) is a smaller constituent part of an exercise that provides a solution to a motor task. A phase is a change in the muscle contraction form in major muscle groups that participate in a motor action. Previous phases create optimal conditions for the solution of the subsequent phase. According to NA Bernstein «... the end of any movement is the beginning of the subsequent one». Phase characteristics have a certain timing; their duration depends on an athlete's sports qualification, his height, and the barbell weight. If a period is divided into two phases, it solves a more general problem, the result of which extends to the subsequent, more significant part of a motor task and even to the whole exercise.

The analysis of every single movement phase of the competition exercises technique allows for a better understanding of the kinematic and dynamic characteristics of movements, and helps to determine variability and stability for each particular phase. One can better understand and assess the quality of performance to determine the role of each phase of the whole exercise by knowing the requirements for each phase, establishing how they fit together, and how they are to be used as a part of movements for the general results of the exercise. Phase structure, considering the uniqueness of the individual elements and the sequence of motion, is also imperative for proper analysis.

Elements are the phase components. If periods and phases are a general integral part of an exercise, accompanied by the technique of any skilled individual, the elements indicate an individual technique. The elements of the technique can be attributed to its detail.

The process of implementation of all three competitive powerlifting exercises is conventionally divided into the preparatory, primary and final parts. The preparatory part is the starting position. It creates the optimal conditions for the implementation of the main part (the squat, bench press or deadlift), aimed at addressing the main motor task of an exercise. The final part (in all three cases) is the final position. It provides the conditions necessary for the effective completion of the main motor task.

2.2 Biomechanical video analysis of complex movements in powerlifting

Biomechanical analysis is a process that analyzes the biomechanics of sports training. This form of analysis is needed to address the issues of movement formation as well as to determine the biomechanical patterns of motor actions. This analysis will look into the causes and consequences of technical errors in performing sports exercises, it will also find ways to help overcome them, and choose options for effective individual interpretation of the form of exercise and at the same time maintain the integrity of the technique. Basic techniques are required and objectively necessary for any athlete. Sound knowledge of the basic components of a sports exercise present an exceptional value in mastering the technique, which will solve its main tasks. Therefore, it is important to develop the basic requirements to be followed by all elite athletes and used on the basis of biomechanical structural phases.

Detailed knowledge of the characteristics of competitive movements greatly aids in the development of the recommendations on how to improve the technique of powerlifters, as well as in the construction of a modern training program. Efficiency in obtaining and visualizing information is easily achievable with the use of modern technologies, in particular, video analysis. The effect of video analysis is largely determined by the method of analysis and the way in which this information is relayed to the athlete or coach.

The study of the kinematics in the movements of humans and animals using single-frame analysis of the images obtained through photo and film has been done since the end of the 19th century. One of the first studies of the kinematics of locomotion in humans and animals was conducted by J. Marey (1830-1904). The application of motion capturing allowed the analysis of the kinematics of human locomotion performed at high speed, e.g. «the backflip».

The greatest development of this method, known as kymocyclography, was seen through the work of NA Bernstein. Its focus is the registration of moving joints of the human body with a fixed camera. Electric bulbs were attached to the joints of a test subject. The test subject then moved with the bulbs perpendicular to the optical axis of the camera. An obturator with a cut out window rotated in front of the open camera lens. The window of the obturator, occasionally appearing in front of the camera, made the trajectory of the joints movement intermittent. The actual coordinates of the trajectory were received by manual processing on a stereoscope. According to prof. DD Don (1971), this option of recording locomotion has obvious drawbacks: long duration (one cyclograph took two weeks of highly skilled laboratory technicians) and the complexity of the experiment.

The registration part of the kymocyclography was subjected to modernization, e.g. electric bulbs were exchanged with strobing ones, or removed entirely, and reflecting markers were used instead. Kymocyclography is a non-contact method. Using a strobe with the pulse frequency of 100 Hz enabled the determination of the kinematic characteristics of locomotion with a large stride length and speed of movement, such as running, or triple jump.

The technique that eliminates the manual work of determining the coordinates was developed in the 1970s and 1980s in Switzerland and Italy. The essence of the registration was in the active markers, working in the infrared invisible spectrum, which were attached to the body of the test subject. The CCD cameras recorded the infrared image into a digital number and the markers' coordinates in the RAM of a computer. Unfortunately, despite the significant rate of obtaining of the kinematic data, this technique was done in a laboratory and had a number of shortcomings, such as:

a) when the limb was rotated, the source of light disappeared from the camera's field of vision;

b) a part of the trajectory disappeared when markers were covered by a body part;

c) in bright sunlight (additional infrared light) registration accuracy was significantly reduced.

Mechanical recorders of movement were used in both weightlifting and powerlifting for a long time, but gradually they began to use film and video cameras in order to accurately correlate the trajectories of singular points with the quality of exercise performance. A great achievement in this area was the creation of high-speed digital cameras that can shoot at up to 500-1000 frames/sec. Being completely invisible and harmless to the human eye, the infrared radiation directed at the light markers attached to a test subject's body does not distract them during the study. The reflected infrared light from the markers, in turn, is registered by the camera CCD-sensor that detects real-time two-dimensional coordinates of the centers of each reflective marker. When using multiple cameras, the computer builds a three-dimensional model of the test subject's movement. Typically, software tools have abundant opportunities for visualization and quantitative analysis of various parameters of motion.

In some cases, analysis can be done without cameras. The basis of the device is a set of photographic sensors arranged in a semicircle of 30 cm radius. A similar device has been used in the rehabilitation of patients with limb trauma. In this case, angular limb movements alternately cause shading of the photographic sensors, which are later recorded and processed by a computer. The obtained data is used to create graphs of angular movements, velocity, and acceleration.

Specialists of the SPC «Videoscan» (Moscow) have developed a hardware-software system for biomechanical studies based on synchronized high-speed cameras (with a frame rate of 100 to 500 frames / s). The system allows to automatically and semi-automatically track the reflecting markers on an athlete's body and build the trajectory of the markers' movement in space, calculate the visual changes over time, velocities, markers' accelerations, and the current position of the mass centers of selected body parts.

Contemporary foreign measurement systems of kinematic and dynamic characteristics of locomotion such as ELITE, SELSPOT, VICON, COSTEL, ARIEL, PEAK PERFORMANCE, due to their cost, often hundreds of thousands of euros, are unavailable for most Russian research centers.

2.2.1 Hardware-software for biomechanical analysis of motor actions in powerlifters

Computer video analysis of motor movement is based on the algorithmic processing of the primary image data obtained by capturing an object with one or two cameras. The resulting video, if necessary, is processed by software (trimming video clips, change brightness, contrast, compression settings, etc.), and then analyzed in a specially designed program.

The main source requirements for the development of a hardware-software are:

- Ability to use any cameras.
- Lack of contrast markers on an athlete's body or equipment. This is important, for example, when recording in competition where pre-tagging is impossible.
- Mandatory visualization of the trajectories of the points of interest and obtaining the graphs of their movements, velocities, and accelerations.

• Ability to save results in an accessible format for playback in other software environments.

The scientific group, led by a doctor of technical sciences, professor of information and measuring equipment Ufa State Aviation Technical University Vladimir S. Fetisov, in 2006 created a hardware and software system designed to study the kinematic characteristics of athletes' movements in powerlifting and weightlifting. It consists of hardware, software and a database.

The hardware includes:

1. Two units of SONY HDR-HC3E camcorder, one of which is used as the main one, and the second one as an auxiliary. A second camcorder is optional. Although it may be useful, especially at competitions, when some interesting points are not seen or seen badly because of the spotters or equipment. The type of camera is also not important: it can be both digital and analog. Digital cameras with built-in video mode should be used. SONY HDR-HC3E camera allows you to capture images at 200 frames / s for 3 seconds. In normal mode, they will shoot at 25 frames / s, which is sufficient for sports such as weightlifting, or powerlifting.

2. Sync flash, or a beacon. This can simplify the synchronization of video sequences obtained from two cameras. Special forced synchronization of the two cameras is not provided, but the video, obtained by different cameras at the preliminary stage of video processing, may be aligned along the length and synchronized. An anchor (the reference frame for synchronization) is selected from a frame with a characteristic part of the movement, visible in both cameras. The presence of the flash or periodical beacon signal (which should be seen in both cutscenes) greatly simplifies the synchronization and makes it considerably more accurate.

3. Computer for video processing. An IBM PC is recommended, or compatible computer running Windows 98/2000 / NT / XP with at least 1 GB of RAM. It is highly desirable to have a special card for video editing (or digital video interface IEEE 1394) and related software, for example, Pinnacle Studio 10. It is also desirable to have a DVD optical drive. A laptop can be used for the same purposes.

Using the Motion Trace program, coaches will be able to quickly identify weaknesses when performing athletic movements and effectively organize the training process. The main features of the developed software allows the following:

• Visualization of the movement of the interest points (markers);

• Obtaining of movement graphs, velocities, accelerations, expansions in rectangular coordinates, points of interest, which are marked by the user on the first frame. There are up to 16 analyzed points. The auto tracking of one point in a video fragment with up to 100 frames lasts no more than 80 seconds;

• View of the right combination of video segments and motion graphs on the screen at the same time;

• One can to enter vertical and horizontal lines marking (graphs) with different colors of labels attached to movable or fixed points;

• One can calculate and monitor changes in the current provisions of the athlete's center of mass, the center of mass of the barbell and the common center of mass. For convenience, a video frame in this analysis can be replaced by an animated skeleton model;

• The user can apply a special tool to differentiate characteristic reference frames and name them, which makes the analysis of various exercise phases convenient.

These tables also store the information not only about the coordinates of points in the graphs but also the position of the selected reference points that can be used to organize personal automated calculations in Excel;

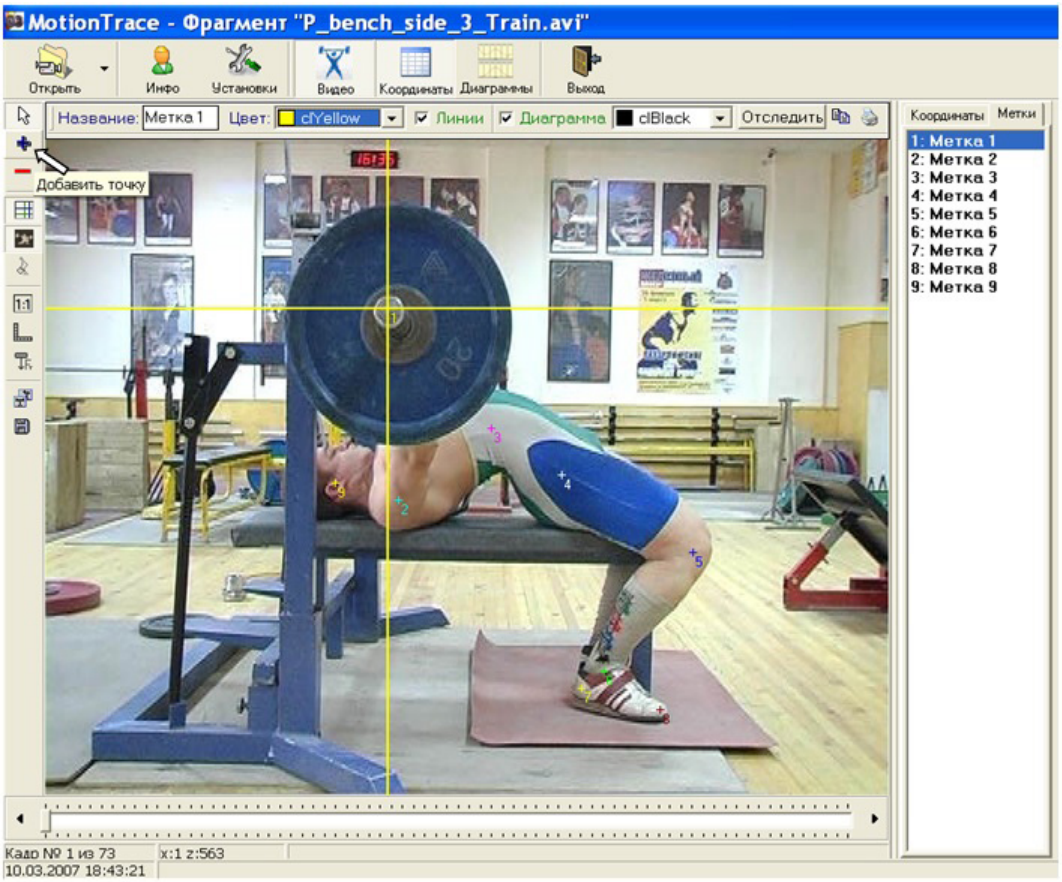


Photo 2.1. Simultaneous marking of points

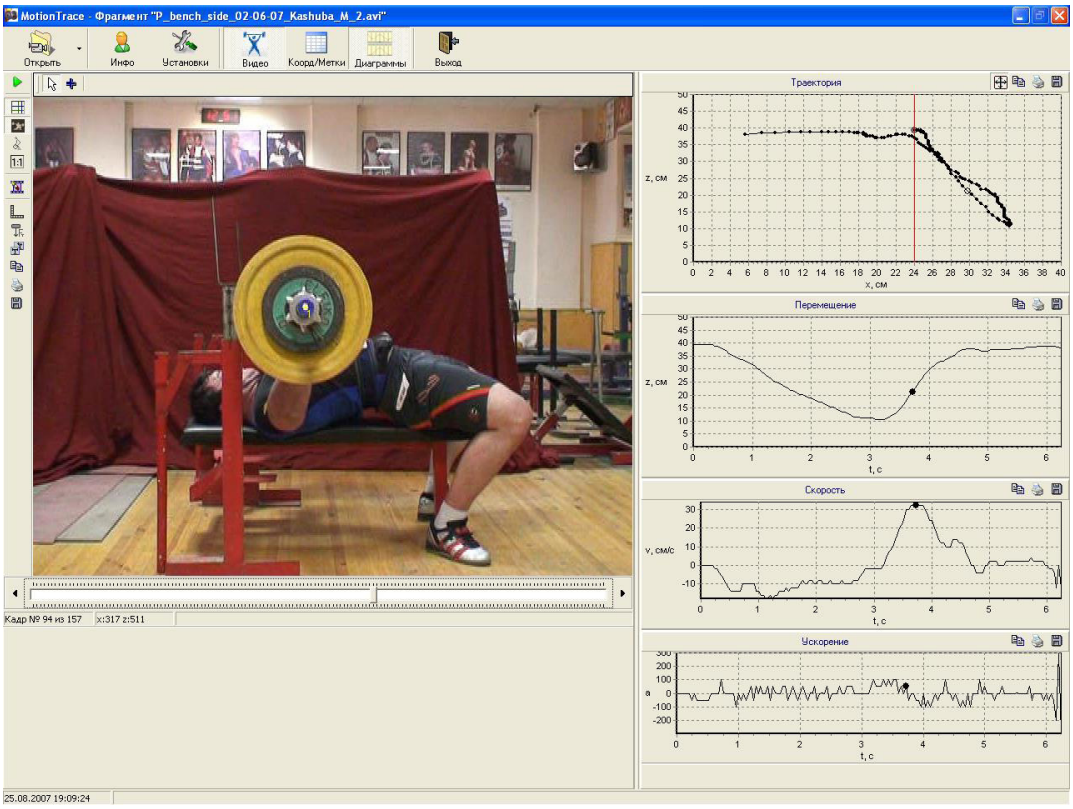


Photo 2.2. Typical graphs of the trajectory of the center of the bar and timing of its vertical movement, velocity, and acceleration (a black dot on the graph marks the position of the observed current frame)

Due to the fact that all barbell points move in relation to an athlete in planes parallel to his sagittal plane, for the analysis of any point of the biokinematic “athlete-barbell”system, it is crucial to know the two components of its motion: vertical and horizontal, the main indicator of which is the trajectory of the movement of any barbell point.

One of the main characteristics of the quality of the competition exercises technique in powerlifting, which provides the corresponding representation of the vertical and horizontal spatial movements of the whole system, is the trajectory of the center of mass of the bar (CTS).

To calculate the center of mass of an athlete one needs to know the coordinates of the ten specific points on their body. They are listed in the lower part of the screen in a special dialogue box. After entering the corresponding markers on an athlete’s body and marking the center of the barbell they are labeled. Then, when using frame position, tracking labels are automatically assigned and the center of mass calculation is computed.

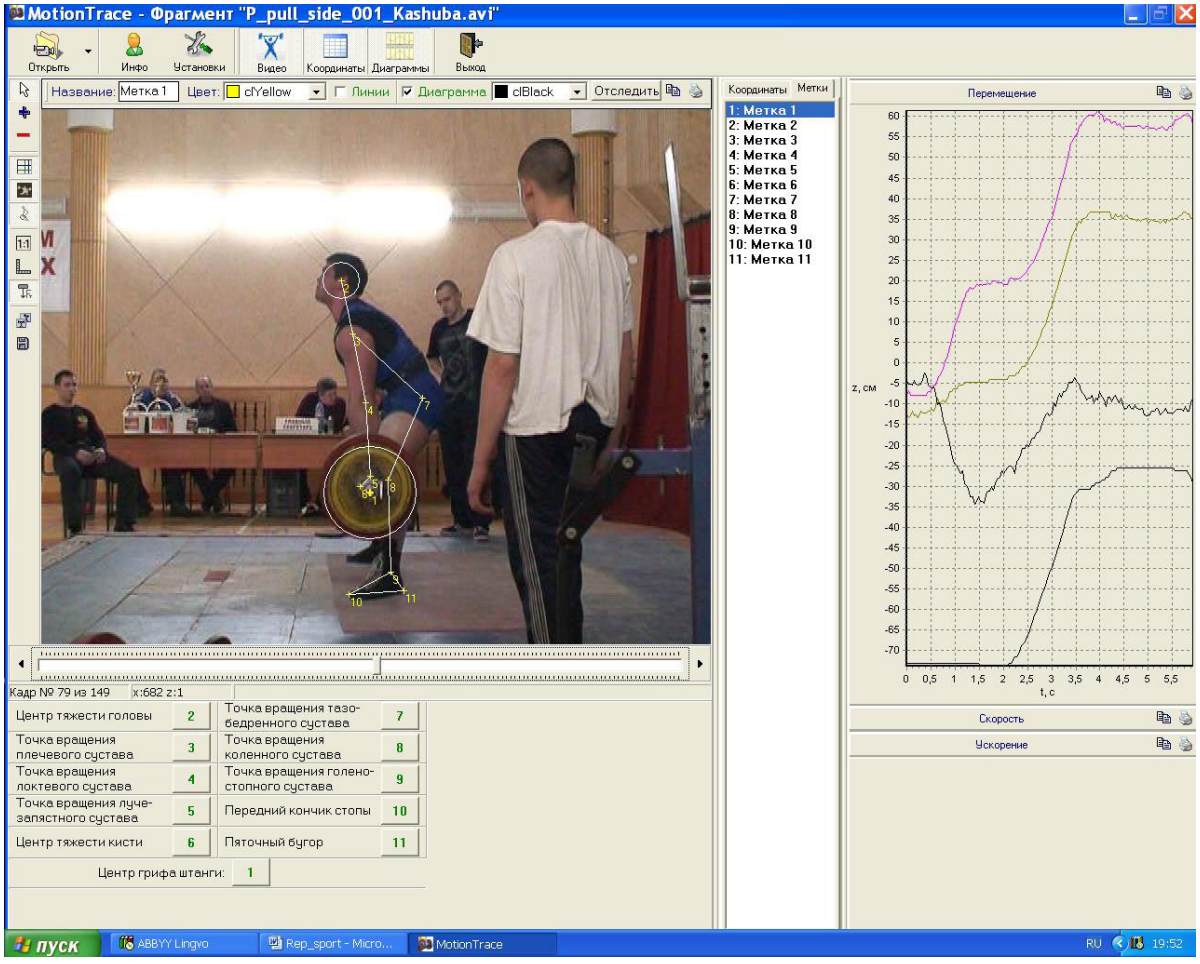


Photo 2.3. Tags for determining the center of mass of an athlete

Photos 2.1-2.3 show the typical window types in Motion Trace for different kinds and stages of powerlifters movements’ analysis.

In addition to the above possibilities, Motion Trace software allows for the recording and analysis of the biomechanical characteristics of the exercises as well as using the information in the online management of the training process of athletes. The information about movement parameters, which is stored in the database, allows to create customized

models for exercise technique and to get the reference characteristics of the competitive exercises technique in strength sports.

In order to improve the readability of the data, the program offers an option for entering a wireframe model, in which the main image of the video frame is turned off, and only the geometric construction is visible.

An athlete's center of mass is shown with a small dotted line, and the common center of mass - with the help of a large dash. This image helps in teaching and learning the exercises technique and eliminating errors in their implementation, as well as in the planning of the training process and the choice of exercise used for the correction of the competitive lifts technique.

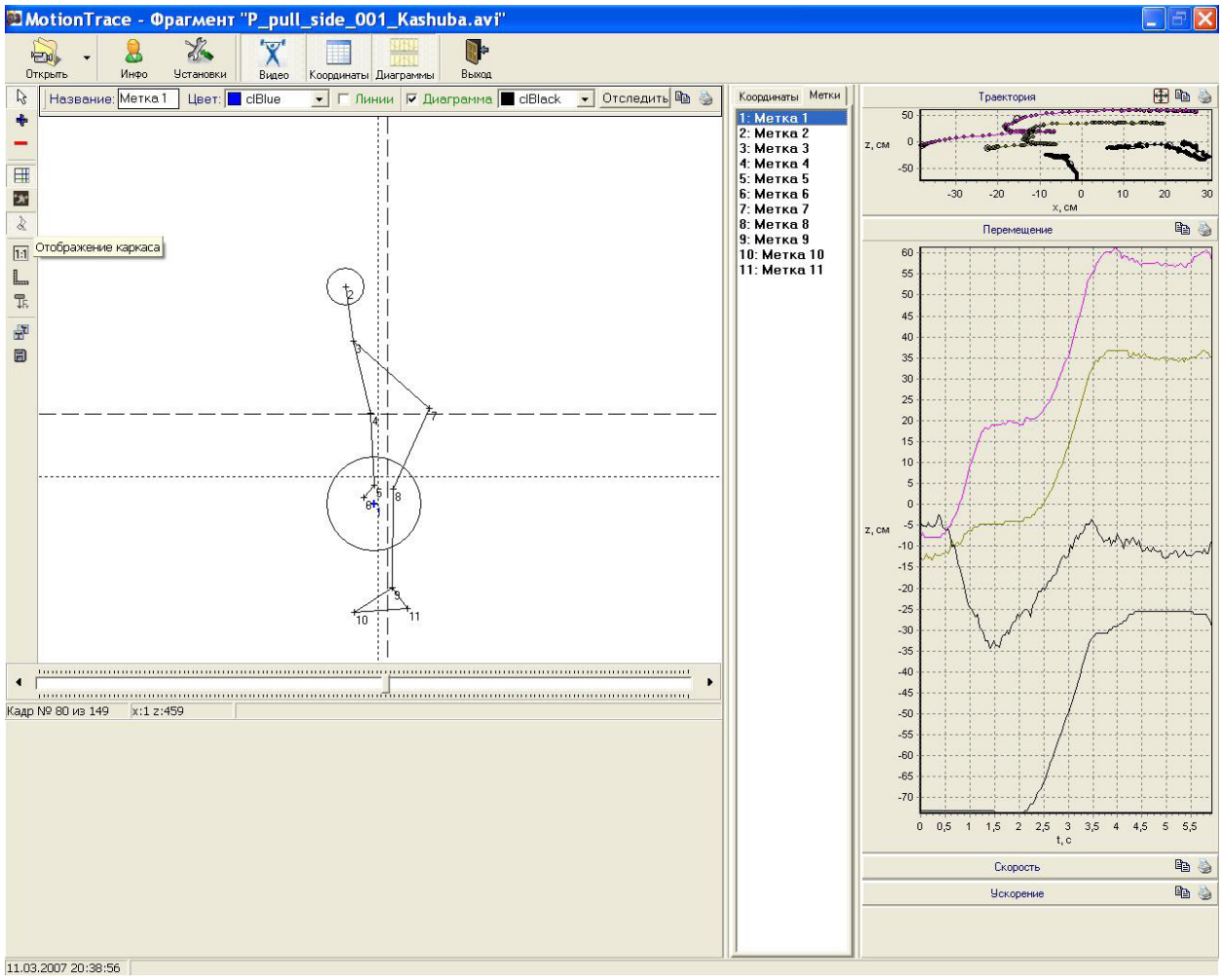


Photo 2.4. Wireframe mode

The video hardware and software developed by professor VS Fetisov and his group performed well in the preparation of athletes for the Bashkortostan Powerlifting Championships in Russia, Europe, and the world. The implementation of this system allowed for identification of specific motor errors preventing athletes from achieving maximum results based on their individual characteristics.

The availability and the possibility of using video analysis of the dynamic, kinematic and temporal structure of athletes movement makes it possible to use the developed software and hardware not only in the training process of Russian powerlifting and weightlifting national teams but also in the educational process of sports institutions and clubs.

2.3 Squat Technique

The Squat is one of the most efficient exercises in strength training in terms of gaining muscle mass. The movement is polyarticular and involves a great number of large and small muscles. Practically all the muscles of the lower body are engaged during this exercise. Different movement types shift the load on different muscle groups.

The effectiveness of squats is indisputable, and the best proof of this is the huge role that this exercise plays in training athletes in different sports. Squats are used in the training of a number of athletes. These include skiers, throwers in track and field (shot put, hammer, etc.), sprinters, jumpers, wrestlers, boxers, football players (both international and North American variants), as well as various other sports.

The squat, which is the first competitive exercise in powerlifting, sets a proper mental attitude of an athlete toward the rest of the competition. It is impossible to achieve high results in the squat without effective completion of motor tasks. This pertains to the need for proper technique, which mainly depends on the technical rules and requirements.

2.3.1 Rules and regulations for the squat

Technical rules and order of the powerlifting squat are approved by the Presidium of the Russian Federation, which comply with the rules of the International Powerlifting Federation (IPF):

1. The athlete shall face the front of the platform with the bar held horizontally across the shoulders. Hands shall hold the bar in any position in the bar inside or in contact with the inner collars. After removing the bar from the rack, the athlete must move backward to establish starting position. The spotter on the platform can assist the athlete in the removal of the bar.

2. Once the athlete has taken a fixed position with a vertical torso and the knees locked, the Chief Referee will give the signal to begin. The signal is a movement of the hand down along with the audible command «squat». Prior to receiving the signal, the lifter is allowed to make any movement which does not involve a breach of the rules in assuming the starting position.

3. After receiving the Chief Referee's signal to begin, the lifter must bend the knees and lower the body so that the upper part of the legs at the hip joint (proximal[1]) are lower than the top of the knee (distal*). An attempt is considered as performed if the athlete's legs were bent at the knees. The athlete is allowed only one decent attempt.

4. After the athlete completes the movement by recovering to an upright position, the Chief Referee gives the signal to place the bar back to the rack by the wrist movement backward and audible command «rack". After this signal, the athlete returns the bar to the rack.

5. After the "rack" command, any motion of the feet is permitted. When returning the bar to the racks (for safety reasons), the athlete can be aided by the spotters.

6. During the runtime of the exercise on the platform shall be not more than five and not less than two spotters.

[1] Proximal part - the closer part of the thigh to the torso, and the distal - the further end of the thigh

2.3.2 Squat phase structure

This section of the book provides comparison of squat phases defined by several foreign and domestic experts. The information has been gathered from specialized journals and articles and summarized in the table below. Although the squat phase structure and its individual elements have been considered by foreign and local experts in various magazine articles, until present they have not been systematized, analyzed or summarized.

L. Ostapenko (1998) and B. Sheiko (2003, 2010) have summarized the experts' views regarding the phase structure of the technique of all three competitive powerlifting exercises.

Table 2.1

The phase structure of squats,
as suggested by foreign experts
(L. Ostapenko, B. Sheiko 2003, 2010)

Author / Phase	John Lear (Canada)	Bill Jamison (USA)	Thomas Harrier (USA)	A. Stetsenko (Ukraine)
Phase 1	Remove the bar from the rack and assume the starting position	Remove the bar from the rack	Remove the bar from the rack	Approach the bar, place the hands, and place the bar on the back
Phase 2	Lowering in the squat	Assume the starting position	Assume the starting position	Move away from the rack and assume the starting position
Phase 3	Rising from the squat	Lowering in the squat	Preparation and beginning the descent	Lowering in the squat
Phase 4		Rising in the squat	Lowering of the squat	Rising in the squat
Phase 5			Rising in the squat	Secure the end position
Phase 6				Returning the bar to the rack and leaving the platform

Many foreign experts do not attach much importance to the starting position and completely ignore the phase of securing the barbell in the final position and returning it on the rack. An exception is the squat phase structure described by the honored coach of Ukraine, A. Stetsenko (Table 2.1).

Table 2.2 shows the variation of the squat phase structure used by Russian experts.

Table 2.2

The phase structure of squats, proposed by Russian specialists
(B. Sheiko 2005, 2010)

Author / Phase	I. Belsky	Y. Nazarenko S. Tye	B. Sheiko	L. Ostapelko	R. Tsedov	B. Schetina
Phase 1	Begin	Begin	Grip, and placement of the barbell on the shoulders, remove barbell from the racks	Approach the barbell, grip and place the barbell on the shoulders	Withdraw the bar from the rack	Approach the barbell
Phase 2	Approaching the bar	Lowering in the squat	Backing up and assuming the starting position	Remove the bar from the rack	Backing up and ssuming the starting position	Assuming the starting position
Phase 3	Squat	Lowering phase	Lowering in the squat and attaining proper depth	Step back and assume the starting position	Start command	Lowering in the squat at proper speed
Phase 4	Securing the final position	Rising from the squat	Rising in the squat	Preparing for the squat	Squat	Achieving maximum depth in the squat
Phase 5		Assuming the final position	Attaining the final position	Lowering in the squat	Achieving sufficient depth in the squat	Rising in the squat
Phase 6			Returning the bar to rack	Rising from the squat	Rising in the squat	Receiving the rack command
Phase 7				Secure the final position	Receiving the rack command	Returning the bar to the rack
Phase 8				Returning the bar to the rack	Lowering the bar into the rack	Leaving the platform

B. Schetina describes six main phases and two auxiliary phases: Phase 1 - approach to the bar and phase 8 - leaving the platform.

According to B. Schetina, lack of data on the phase duration and the tasks solved in the process reduces the effectiveness of these learning materials in practice for training and technical development of competitive exercises.

As a result of the biomechanical analysis of the squat exercise conducted at the Department of weightlifting, Bashkir Institute of Physical Culture, B. Sheiko, selecting a group of athletes as an example, offers a variation of the kinematic structure of the squat technique. He divided it into 3 periods with 6 phases and 10 elements. (Fig. 2.3 shows the kinematic structure of the squat technique with a barbell on the back).

First period - starting position (consists of two phases and five elements)

Phase 1. Preparatory position

Elements: grip the bar, step under the bar, place the bar on the shoulders and remove the bar from the racks

Phase 2. Starting Position

Elements: Step back from the rack, feet placement, adjustment of the barbell

Second period - squat (2 phases and 2 elements)

Phase 3. Lowering to the squat

Elements: Lowering into a squat, achieving depth

Phase 4. Rising back up in the squat

Elements: rising back up in the squat, overcome the «sticking points»

Third period - the end of the exercise (2 phases and 3 elements)

Phase 5. Recovering to the final position

Elements: Complete straightening of the legs in the knee joints, recovering to the final position

Phase 6. Return of the bar to the rack

Elements: the return of the bar

It should be noted that the phases have certain time frames, their duration depends on an athlete's skills, his height and the weight of the bar. The elements of the technique in most cases depend on the individual, morphological and functional characteristics, the proper use of which forms an individual technique for an athlete, making the technique (subject to all the general rules of sound movement) ideal for this particular person. Omission or violation of at least one phase or component makes a motor task completion impossible. The analysis has shown that the application of the above kinematic structure of the squats technique in the training process increases its effectiveness and mastery.

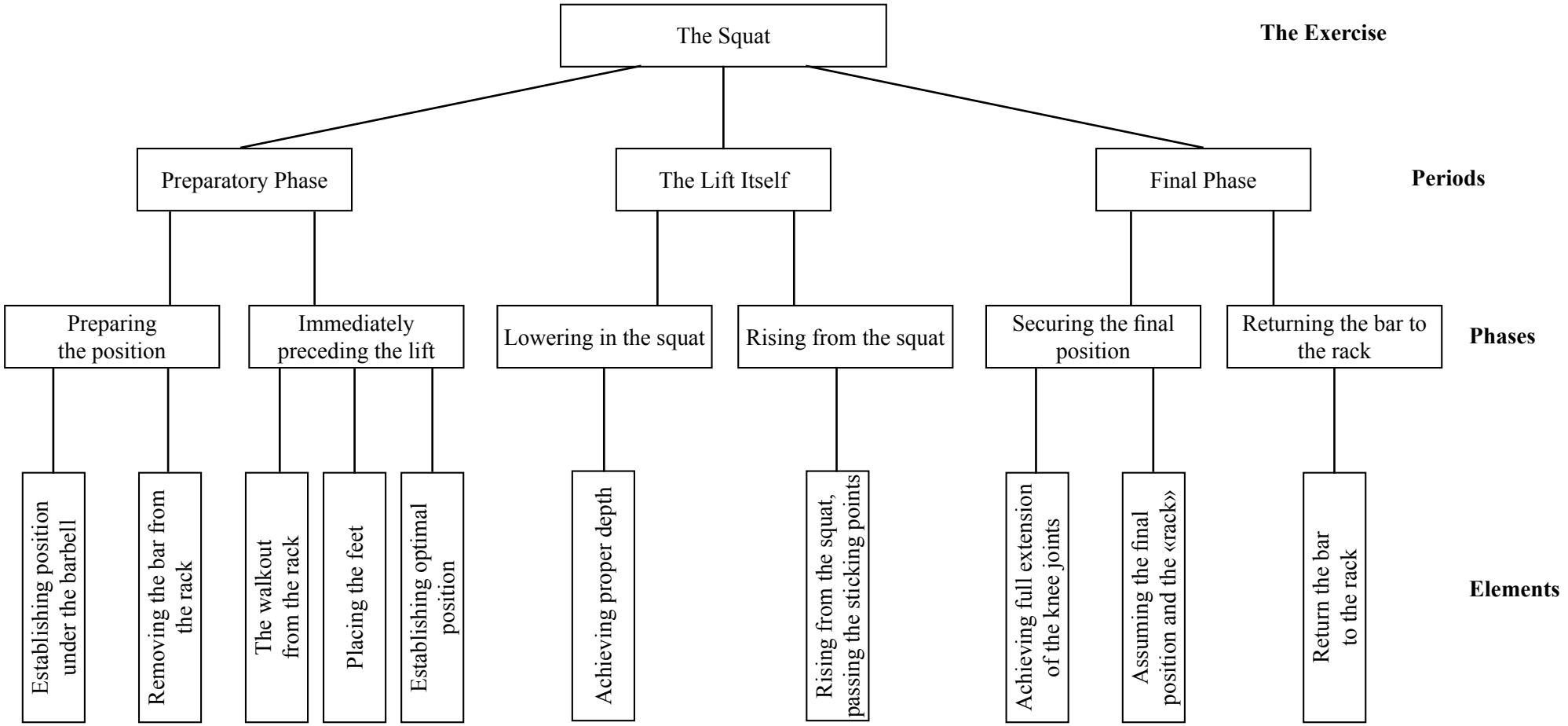


Fig. 2.1. Kinematic structure of squats (B. Sheiko 2005, 2010)

2.3.3 Technique and structure of squats[2]

The ability to perform a particular motor action is based not only on the general physical fitness but also on the knowledge of the technique. The formation of motor action is subject to certain laws, the knowledge of which makes it possible to effectively build the training process. For example, in order to perform the motor structure of squats, one needs to precisely reproduce the phase techniques of the studied action.

In order to teach the basics of squats, it is necessary to teach an athlete to perform the squat in the correct order, taking into account the periods, phases and elements, including all the single tasks on the way to the goal.

Therefore, the squat consists of three periods, six phases and ten elements. Squats consist of three periods of six phases and ten elements. In this section, each period will be described in more details together with visual examples.

First period - starting position (2 phases and 5 elements)

Phase 1. Preparatory position

The first phase lasts from the beginning of the grip to the removal of the bar from the rack. Its motor task is to create the optimal conditions for the starting position.

Preparatory actions:

1. Grip the barbell

At present, the technical rules of the competition allow two grips of the barbell:

with all the fingers facing one direction (false grip)

with four fingers on one side of the barbell and the thumb on the other (photo 2.5-2.6).

Grip width of each athlete is individual and depends on his anatomy, the flexibility in the elbow and shoulder joints as well as muscle mass. Some experts believe that the bar should be kept wider than shoulder width by about 7-10 cm on each side.

Photo 2.5. False grip

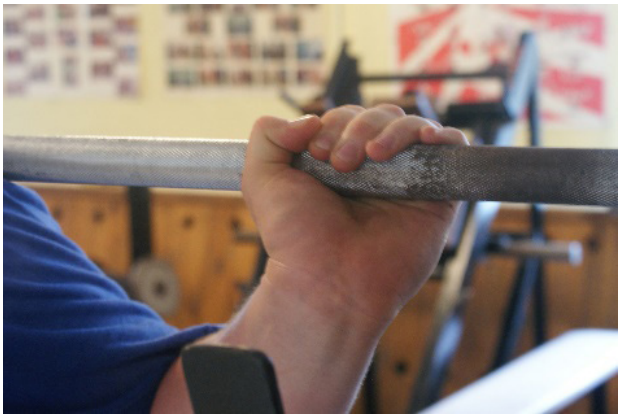


Photo 2.6. Simple grip

[2] When writing this section materials were used from: Mark Rippetoe, A. Samson, N. Kichaykinoy, G. Samsonov, B. Shtetina et al.

Others believe in holding the barbell at a distance of approximately 10-12 cm wider than shoulder width. While carrying out this element, the elbows should be slightly raised to prevent slipping of the barbell. The upper part of the back and both wrists while holding the bar should create a precise «triangle», which allows the bar to safely lie on an athlete's back during all phases of the movement.

Photo 2.7. R. Gadiev

Photo 2.8. A. Tarasenko

Photo 2.9. A. Malanichev

2. Step under the bar

Stepping under the bar should be done in a fairly narrow stance, close to the standard position (standing), so that later, when backing up, the athlete will not lose his balance (photos 2.10-2.13).

Photo 2.10

Photo 2.11

Photo 2.12

Photo 2.13

3. Placement of the barbell on the shoulders

In powerlifting, the barbell is placed on the back of the rear deltoids and the middle part of the trapezius muscles, while slightly tilting forward. In order to retain the bar on this part of the back, it is rigidly fixed by the hands, increasing the burden on the wrist joints. This technique obviously shortens the vertebral lever arm and reduces the load on the back muscles. The greater the forward tilt of the back, the less strain on the legs, and more on the back. Low bar placement on the back provokes an increase in the forward tilt. With this back position, the judge will not give the command to perform the exercise.

4. Removing the bar from the rack

Removing the bar from the rack is performed by tensing the legs and back muscles. When placing the bar on the shoulders, the trunk muscles are tense, the shoulder blades are retracted. (photos 2.14, 2.15)

Photo 2.14

Photo 2.15

Phase 2.

The phase begins with the athlete's walkout from the racks and securing of the position. The motor task of the second phase is to adopt a good position for the firm and effective interaction between the components of the kinematic chain of the athlete's body.

Elements: the athlete's walkout from the rack, foot placement, fixing the barbell from moving from the racks and assuming the starting position.

1. The walkout

This element consists of one or two small steps depending on the design of the racks. Then the athlete assumes a stable position so that the overall center of mass is evenly distributed between the heels and toes of both feet, i.e. in the middle of the foot (photo 2.16-2.19).

Photo 2.16

Photo 2.17

Photo 2.18

Photo 2.19

2. Foot placement

There are many options in terms of the optimal foot width for the squat (see. photos 2.20 - four-time world champion, Nikolay Suslov); to the widest possible (photo 2.22 – two-time world champion, Maxim Podtynny). Most athletes use an average foot width slightly wider than shoulder width (photo 2.21 - six-time world champion, Victor Furazhkin).

Photo 2.20. N. Suslov

Photo 2.21. V. Furazhkin

Photo 2.22. M. Podtynnyy

The wider the position of the feet, the larger the angle of separation in the feet, which will keep the thighs more parallel to the feet. In the narrow position with toes pointing more forward, the knees will move further forward, compared with having the knees wide apart. This is due to the fact that in the narrow position there will be a greater distance from the front of the knee to the back of the thigh. And the greater the distance, the more the knees have to move forward to adapt (Mark Rippetoe, 2007).

A narrow stance with feet turned out shows the same forward position of the knee as in the middle positions with the same rotation angle of the feet. (Fig. 2.4).

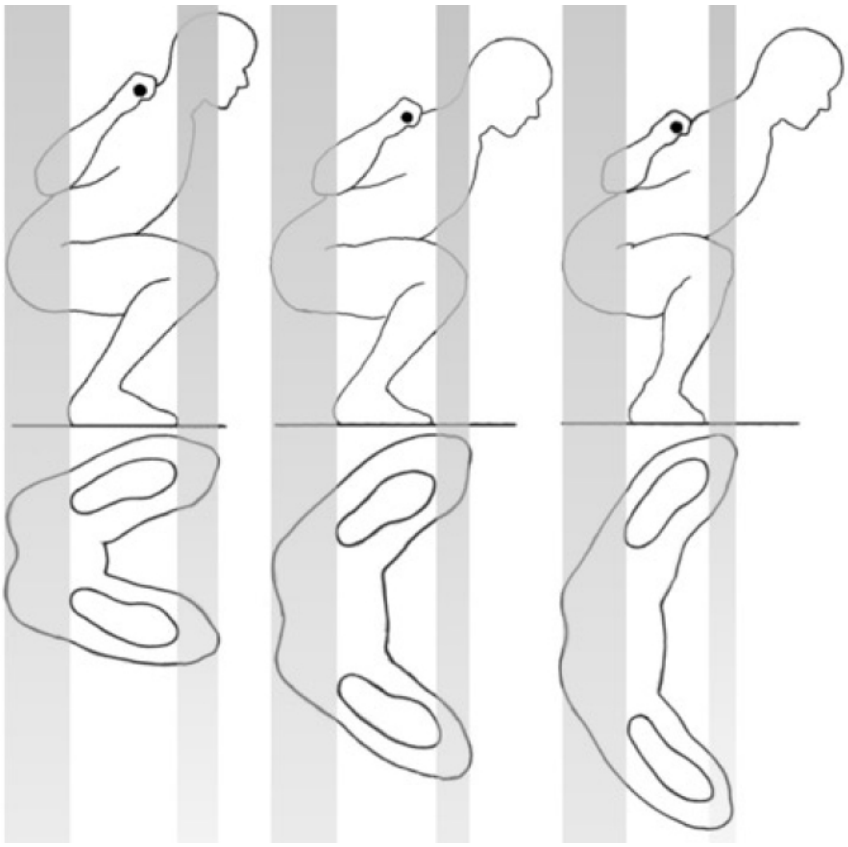


Fig. 2.2. The figure originates from the article by Mark Rippetoe «Popular biomechanics»

With a very wide stance in a powerlifting squat one can observe that the knees go forward slightly and the shins usually remain almost vertical. But the wide stance of the knees will not work if the toes are pointing forward, because it causes the knee joint to be inverted, which is anatomically incorrect.

For each athlete's squat, there is only one optimal stance width, allowing the most efficient use of muscle strength. The knee joint should move in the direction in which the foot is pointed. The head should be slightly raised.

Many coaches believe that the change in the stance width affects leg muscles differently. In order to test this view, Paoli, Marcolin, and Petrone (2009) conducted an experiment. They measured the electrical activity of the leg muscles in a squat with three variants of feet placement with loads of different weights - 3 sets of 10 reps: 1) without weight; 2) 30% of maximum weight; 3) 70% of maximum weight. The rest between sets lasted for 6 minutes. Electrodes were placed on 8 muscles: medial broad, vastus lateralis, rectus femoris, semitendinosus, hamstrings, gluteus maximus, and the gluteus medius. Differences were found only for the gluteus maximus, which was more activated with increasing load in the wide stance. No significant differences in the activity of other muscles were observed (Paoli, Marcolin, Petrone, 2009).

3. Adjustment of the barbell at the start command

Preparing for the squat is performed by tensing the arm muscles, «activating» the knee joints, a few quick breaths to lift the chest, as well as the final breath for about three-quarters of the maximum lungs capacity, and tensing the chest muscles. In order to create a hard muscular frame around the spine, one should also slightly activate the rectus abdominis.

Second period - squat (2 phases and 2 elements)

Phase 3. Lowering into the squat

This phase lasts from the beginning of bending of the knees and ends at the moment of achieving the proper depth ("breaking the angle").

The motor task of the phase is breaking a 90-degree angle, according to the rules of the competition (photos 2.23-2.25).

After the judge's command «start» an athlete takes a small, energetic breath and performs the exercise holding his breath. Research by I. Seregina (1965) proved that the best effect in power movements is observed in a short breath-hold (Valsalva maneuver). When holding your breath, a kind of «support» to the vertebral column is formed that holds a heavy load on the shoulders. The rib cage should be in a slightly lifted position, with the elbows retracted.

Photo 2.23

Photo 2.24

Photo 2.25

Mark Rippetoe suggests that the squat should be performed in balance, i.e. when the bar is right over the middle third of the foot. It does not matter where the barbell is located (on the back or shoulders), if the bar is moving along a straight path up or down, when performing squats with heavy weights, the barbell barely deviates from this position. If this happens, the bar is out of balance and slows down to the point where it will either return to the balance position or the balance position will be lost.

Rippetoe also encourages the athlete to pay attention to the fact that at the beginning of the squat:

• all skeletal components that support the bar - knees, hips, and spine should be locked in extension so that the muscles have sufficient strength to maintain this position of rest;

• The bar should be directly over the middle third of the foot;

Most foreign and local coaches strongly recommend that the center of mass of the «athlete-barbell» system be over the middle of the foot. For example, Frederick C. Hatfield (1982) states: «The widespread problem is improper skeletal-muscle coordination during squats. Centering weight over the balls of the feet in the squat very often leads to a deviation of the bar too far forward. This, in turn, causes rounding of the back, making it impossible to hold the hips through the critical point in the almost complete transfer of weight onto the back muscles. Accordingly, centering the weight through the heel leads to an almost complete transfer of effort on the muscles of the hips. The back muscles are involved in this case only partially. All kinds of squats - wide, narrow or intermediate require centering the weight over the instep, strictly in the middle'.

There is a contrary opinion held by L. Ostapenko (1994), S. Gliydy, M. Starov, Y. Batygin (1998), who propose to shift the center of mass mainly on the heel. Some Russian experts believe that the total displacement of the center of mass on the heel is only possible in the wide stance. The wider the legs, the closer the mass center will shift to the heel area of the foot, in the feet stance wider than shoulder width the center of mass is in the middle of the foot with some offset to the side of the heel. In a narrow stance, the center of mass is in the middle of the foot with a slight shift in the direction of the toes.

When lowering, the knees move in a vertical plane passing through the center line of each foot. The knee position will vary depending on the torso angle and position of the barbell.

Squat position with the bar on the chest and squat with a barbell on your back in the high and low position of the neck are performed with different angles of trunk deviation from the vertical line. In each case, the motion of the musculoskeletal system is different between the top and bottom point of the movement. This occurs because the position of the barbell on the body varies in relation to the hip and knee joints.

The mechanical analysis performed by Mark Rippetoe (2007) implies that the torso angle determines the position of the bar's neck. When the bar is located on the back, its position should be exactly above the middle of the foot, and your feet should be flat on the floor. If this is not the case, the mechanics of your body will be less efficient.

When the bar is located on the upper back (weightlifting style) or just below the shoulder blades, the back is bent forward at an angle that will keep the vertical projection of the barbell above the middle of the feet. The higher the barbell on the back, the smaller the angle between the torso and the vertical to fulfill this condition (Fig. 2.5).

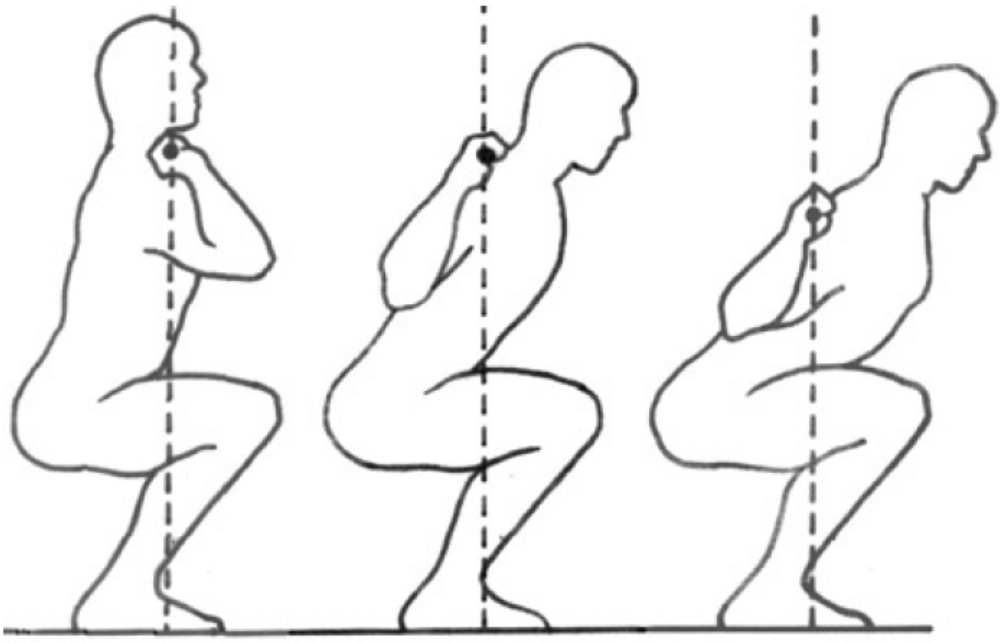


Fig. 2.3. The picture originates from an article by Mark Rippetoe «Popular biomechanics»

This means that in a high position of the barbell on the back, the torso angle relative to the vertical is smaller as compared with the low bar squat. When performing the front squat, the position is the same as in the upper position on the back. However, the angle of inclination of the trunk relative to the vertical is even smaller compared with the position of the bar on the back.

A large tilt forward will increase the load on the spine, and a backward tilt when leaning on the heel will increase the load on the muscles of the lower extremities.

The hips turn should match the feet turn. If the position of the feet is wide enough then it helps to mitigate any non-linear load (twisting) to his knees, and ensures the use of the adductors. The turn of the feet, and hence, the turn of the knee joint (if the feet are turned at the same angle as the knees), is largely determined by the width of the stance. A narrow stance can be made either with the feet turned outwards or with the toes directed forward with an outward turn of five or ten degrees. But the knees have to move straight ahead to keep the thighs parallel to the foot, and knees should always travel in the same plane to keep the linear relationship between the patella, patellar tendon and tibial plateau. These three objects should be in a straight line so that the knee joint can function without excessive wear and tear, especially under load.

Some foreign experts have expressed the view that when lowering into a squat, the tibia must remain in an upright position. With a vertical position of the tibia, the load on the knee will be lighter.

Movement of the knees forward in the vertical position of the back leads to more acute angles between the tibia and femur at the bottom of the squat, to full contraction of the hamstrings and more open hips. Therefore, when performing the squat position with the barbell on the chest, the quadriceps and gluteal muscles are loaded more compared to the back squat with low bar position, because these muscles are still able to contribute to the lift.

Mark Rippetoe (2007) argues that an athlete may slightly deviate from the desired path between the top and bottom of the squat, but if he does not begin and end as described, he makes mistakes in the mechanics, and the execution of the squat will be more difficult than if the athlete did everything correctly. The athlete's musculoskeletal system seeks a solution to the problem of how to most effectively use the muscles to perform squats. The athlete does this within the constraints imposed by the physics of mass of the "barbell-body" system.

At the lowest point in the squat, the mechanical "athlete-barbell" system is in the state of balance

the spine should be rigidly fixed in the thoracic and lumbar sections;

the bar should be positioned vertically above the middle third of the foot;

the feet are flat on the ground;

the thighs are parallel to the vertical plane of the foot (the support);

acetabulum (hip joint center) will be lower than the top of the patella.

For multi-faceted, mobile biomechanical system, which is the human body, the state of balance means that the external torque (moment of gravity of the bar and the forces of mass of individual body parts) with respect to the center of rotation in a particular joint must be balanced by articular muscles overlying this biomechanical chain.

Phase 4. Rising back up in the squat

The phase starts from the beginning of the leg extension in the knee and lasts until the passage of the «sticking point». The motor task of the fourth phase is successful passing of «sticking points».

The rise from the bottom of the squat starts with the extension of the knee joint with an average angular velocity relative to the thigh and tibia of 2.0 - 2.2 rad\s. With the beginning of the knee joint extension, one can observe the reduction of the angle between the thigh and torso for 0,2-0,24s. N. Kichaykina and G. Samsonov (2010) suggest that this occurs as a result of the changes in the dual angles of lower extremities as well as the torso bent in order to maintain the necessary degree of stability in the balance.

All athletes, regardless of their skill and experience, reach the same point («sticking point») when lifting with a barbell from the bottom of the squat. Usually, it occurs at a 30-degree angle between the thigh and the line parallel to the floor, passing through the center of the knee joint. At this point, the muscles of the lower limbs have to perform work in extremely unfavorable mechanical conditions, because, for instance, the activation of the gluteal muscles is reduced to a minimum, and the extensor muscles of the legs experience the greatest load.

The experiment conducted by N. Kichaykina and G. Samsonov (2010) showed that hip extension begins in 0,16-1,20s when rising from the squat. When one squats with a load of 60%, the extension is practically inhibited, there is a plateau-like stage in the change of the mechanical angle for 0.04 s. With a load of 80%, the extension lasts for 0.02 seconds, approximately at the same angular velocity, followed by a plateau at the angle change lasting 0,06 s. The onset of a plateau in the change of the angle at the hip joints is the very passage of the «sticking point», which corresponds to the spatial position of parts of the body, as noted by powerlifting specialists.

Following the laws of mechanics, it is impossible to eliminate the «sticking point». It can only be minimized (meaning minimization of vertical dynamic overloads of the musculoskeletal system of an athlete). The organization and display of the «sticking point» occur within tenths of a second, i.e. for the period of time during which the instant corrections of the motion parameters of the «athlete-barbell» system are practically impossible.

The only way to pass the «sticking point» is by going through it with the effort of the thigh muscles and moving the head back, which increases the impact of leverage to aid the hips. The shoulders and hips have to rise with the same speed. Exhalation takes place only in the upper third part of the squat lift (photo 2.26-2.28).

Photo 2.26

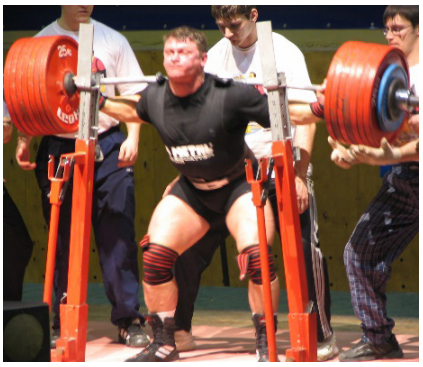


Photo 2.27

Photo 2.28

The third period is the end of the exercise (the stabilization of the target position). It consists of two phases and three elements.

Phase 5. Assuming the final position

This phase lasts from the moment of the maximum leg extension in the knee and hip joints and securing the barbell from moving until the “rack” command.

The motor task of this phase is the maximum leg extension in the knee joints and stabilization of the final position until the command of the chief judge “rack” (photo 2.30 and 2.31).

Elements: straightening the legs in the knee joints, stabilizing the final position.

Maximum leg extension in the knee joints: in the upper third phase of the rise from the squat, the body is straightened at the hip joints simultaneously with the maximum leg extension in the knee joint. If the leg extension is faster than the extension of the back, the overall center of mass shifts from the middle of the foot forward, which increases the load on the muscles of the back, especially the lumbar section and inhibits the success of completing the lift.

Assume the final position and the command of the Head judge «rack”.

After the leg extension in the knee and hip joints has been completed, the athlete, without releasing his back muscles or losing concentration, waits for the «rack» signal of the head judge on the platform (Photo 2.28).

Photo 2.29

Photo 2.30

Photo 2.31

Phase 6. Return of the bar to the rack

The sixth phase starts with the beginning of the movement of the lifter to the rack and ends with the return of the bar on the rack.

The motor task is to return the bar to the rack.

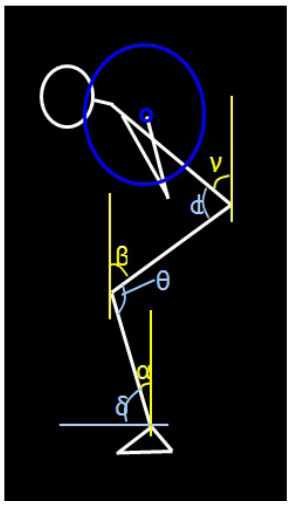
After the “rack” command of the head judge, the returning of the barbell is done by a careful step forward, by preserving muscle tension in the back and a raised rib cage. The technical rules of the competition allow spotters to assist the athlete in returning the bar to the rack. Photos photo 2.29 and 2.30 show spotters supporting barbell who are helping the athlete to return it to the rack.

2.3.4. Squat techniques of advanced athletes

All exercises for learning and retention of skills require practice, as in parts (method of split exercise), and as a whole (holistic method of exercise). In these cases, it seems very effective when the technique of a competitive exercise is shown by leaders, that is, advanced athletes. This method helps in cases of proper technique distortion due to excessive muscle tensions.

2.3.4.1 K. Pavlov’s squat technique

Konstantin Pavlov is the Honored Master of Sports of Russia, ten-time world champion and silver medalist in VI-X of the World Games, world benchpress champion, eleven-time European champion, two-time Junior European champion, three-time European benchpress champion, twelve-time champion of Russia.



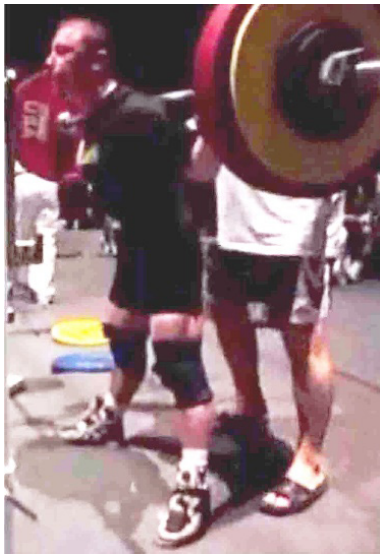
The frame reflects the principal elements of squats:
1 - starting position
2 – bar 12-15 cm below the starting position
3 - bar in the middle of the exercise
4 – bar 5-7 cm from the end point
5 - end point of the squat
6 – bar 8-10 cm above the endpoint
7 – bar approximately in the middle of the exercise
8 - bar 12 cm from the final position
9 - final position

Fig. 2.4. Angular schematic

Frame 1. Starting position

He is taking a conventional grip (in which four fingers are on one side of the bar and the thumb on the other). The hands are at a distance of 15 cm wider than shoulder width. The barbell is located on the back of the deltoid and mid-trapezius, with the back straight, slightly tilted forward. The angle of inclination of the back (γ) - 10 degrees. Due to the fact that it is difficult to keep the barbell on this part of the back, an athlete must firmly hold it with his hand. The chin is slightly raised. The common center of mass is approximately the middle of the foot.

The legs are apart at shoulder width, the distance between the centers of the heels is 40 cm. The angle of feet rotation is about 30-35 degrees. The knees are directed towards the toes. The angle of the ankle (δ) is equal to 89 degrees, the angle of the knee (θ) is equal to 177 degrees at the hip (φ) is equal to 168 degrees



Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Frame 9

Frames 1-9: squats, K. Pavlov

Frame 2. The barbell 12-15 cm below the starting position

At the initial stage of the lowering phase the athlete's speed reaches its maximum value - 30 cm / s. The angle of the ankle (δ) is equal to 77 degrees. The angle of the knee (θ) decreases to 137 degrees, at the hip (φ) it equals to 120 degrees. The back is straight. Squatting, the athlete leans forward a bit, so the angle of the back (γ) increases to 20 degrees.

Frame 3. The middle of the lift

In this segment, the bar is approximately in the middle of the lift. The angle of the ankle (δ) increases to 85 degrees, the angle of the knee (θ) is equal to 120 degrees, at the hip (φ) is equal to 100 degrees. The angle of the inclination of the back (γ) increases up to 25 degrees. While lowering, the speed decreases to 18 cm / s.

Frame 4. The bar at the point of 5-7 cm from the bottom of the squat.

8-10 cm from the end point of the squat, while lowering, the speed again increases to 25 cm /s. The angle of the ankle (δ) decreases to 75 degrees, the angle of the knee (θ) decreases to 85 degrees and at the hip (φ) decreased to 70 degrees. The angle of the inclination of the back (γ) continues to increase up to 30 degrees.

Frame 5. The bar at the bottom of the squat

The athlete has reached the bottom position. The angle of the ankle (δ) decreases by 15 degrees, reaching 60 degrees. The knee angle (θ) reduces to 58 degrees, in the hip joint (φ) the angle decreases to 53 degrees, the back angle (γ) continues to increase up to 35 degrees. The lowering speed has reached zero. The common center of mass has moved from the middle of the foot slightly forward in the direction of the toes.

Frame 6. The barbell 8-10 cm above the bottom of the squat

The barbell is 8-10 cm above the endpoint of the squat. At this stage the lifting speed increases to 48 cm / s. The angle of the ankle (δ) increases to 75 degrees, the angle of the knee (θ) is 85 degrees, the angle of the hip joint (φ) increases to 70 degrees. The angle of the inclination of the back (γ) decreases to 30 degrees. An increase in the knee and hip angles means that the athlete has begun his ascent from the endpoint by activating his leg and back muscles.

Frame 7. The barbell half-way through the ascent

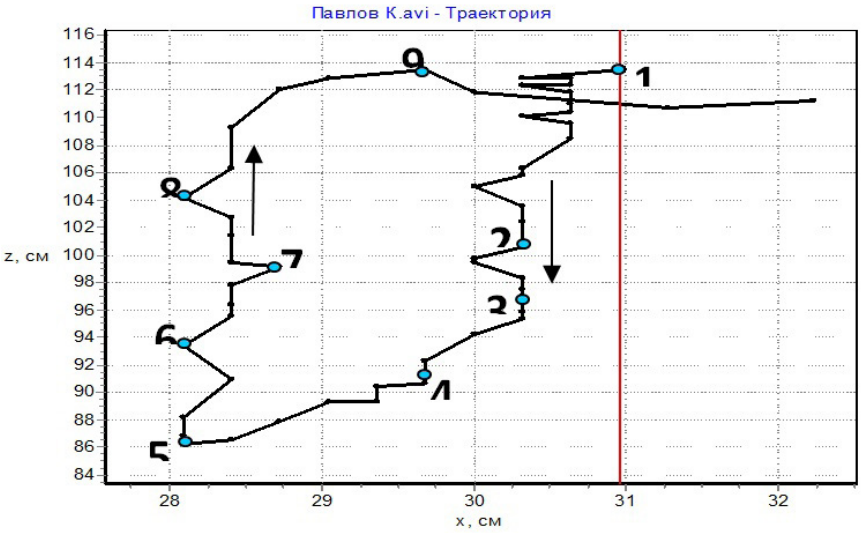
After Phase 6 of the lift, the athlete begins to slow down, and his speed 7th phase decreases to 30 cm / s by Phase 7 (see. Figure 5.7). This decrease in speed indicates that the barbell has reached the «sticking point» zone. The angle of the ankle (δ) is equal to 85 degrees, the angle of the knee (θ) approaches 125 degrees, at the hip (φ) the angle increases to 105 degrees. The angle of the inclination of the back (γ) is 25 degrees. The athlete continues to activate his legs and back muscles.

Frame 8. The barbell 12 cm from the final position

After a short pause, the speed has increased and reached its maximum of 50 cm / s. The angle of the ankle (δ) is equal to 87 degrees, the angle of the knee (θ) has increased to 140 degrees, and the angle at the hip (φ) has increased to 125 degrees. In this portion, the legs and back muscles are still being activated, so the angle of the back (γ) decreased to 20 degrees.

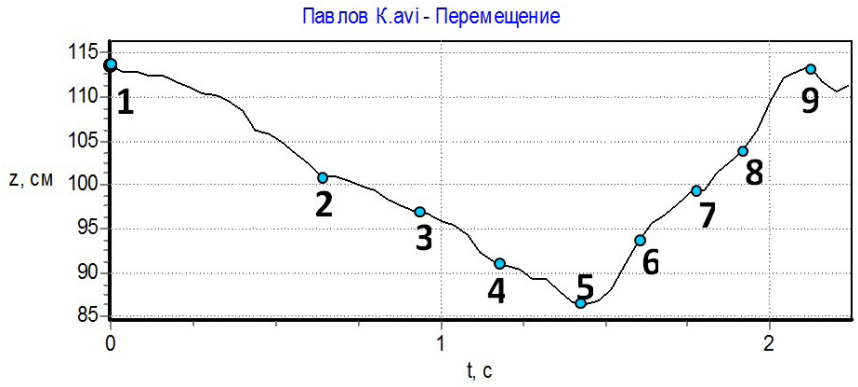
Frame 9. The final position

The athlete comes into the final position, the angle of the ankle (δ) is equal to 89 degrees, the angle of the knee (θ) approaches 177 degrees, the final position of the angles in the ankle and knee came to the initial value of the starting position. Due to the fact that the muscles of the back have gone under a great load, the athlete cannot straighten his back to the original, starting position. In the hip joint (φ) the angle is 166 degrees, the back angle (γ) equals to 12 degrees, in both cases the angle is 2 degrees wider than at the starting position.



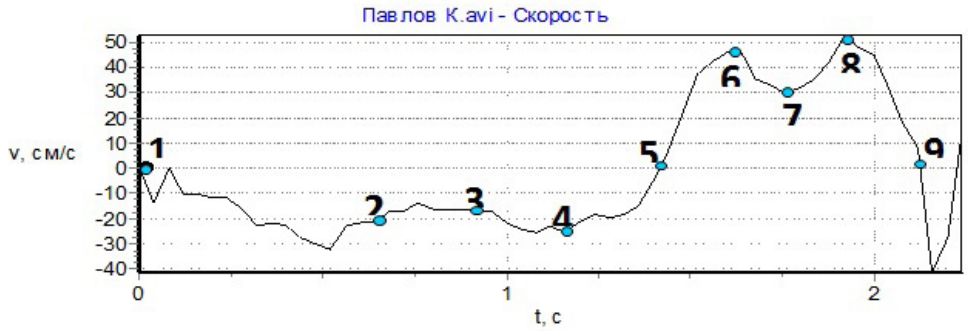
1 - starting position, 2 - bar 10 cm below the starting position,
3 - bar in the middle of the lift,
4 - bar 8-10 cm from the bottom,
5 - bottom of the squat,
6 - bar 10 cm above the bottom
7 - bar approximately the middle,
8 - bar 10 cm from the final position
9 - final position

Fig.2.5. Trajectory to squats performance, K. Pavlov



1 - starting position,
2 - bar 10 cm below the starting position,
3 - bar in the middle of the lift,
4 - bar 8-10 cm from the bottom
5 - bottom of the squat,
6 - bar 10 cm above the bottom,
7 - approximately half-way through the ascent,
8 - 10 cm from the final position,
9 - final position

Fig. 2.6. Vertical movement of the center of mass



1 - starting position, 2 – bar is 10 cm below the starting position,
3 - bar is in the middle of the lift, 4 – bar 8-10 cm from the bottom,
5 - at the bottom of the squat, 6 – bar is 10 cm above the bottom
7 - approximately half-way through the ascent,
8 - bar is 10 cm from the final position, 9 –final position

Fig. 2.7. Speed of the bar's center of mass

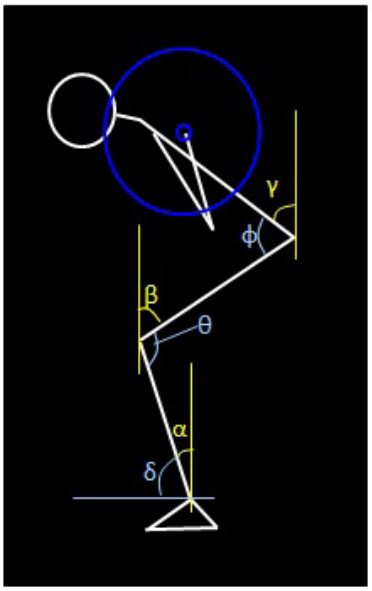
When analyzing the starting position of the lift, we cannot observe any deviations from the proper technique: a simple grip of the barbell, the hands are at a distance of 12 cm wider than shoulder width. The barbell is positioned on the back of the deltoid and mid-trapezius, with the back straight, slightly tilted forward. The angle the back inclination is small. The chin is slightly raised. The common center of mass is approximately the middle of the foot. The feet are at shoulder width apart, they are pointed out up to 40-45 degrees. The knees are pointed in the same direction as the toes.

In Phase 2 and 3, the athlete is not able to hold the angle of the back like it was in the initial state, the inclination angle is increased. Besides that, in Phase 3 and 4 his lowering speed to the bottom of the squat does not increase, which may cause difficulty in passing the parallel due to the wraps on the knees being too tight. The fact that his knees move 3 cm further than the toes and the back tilt increases during the descent, leads to a shift of the center of mass from the middle of the foot to the ball of the foot. The continued increase in the bent of the back during lifting from the endpoint of the squat in Phase 6 and 7 as well as the movement of the knees beyond the toes level, makes the athlete shift most of the load to the back.

2.3.4.2. E. Kovalkov's squat technique

Frames 1-9 show the basic elements of squat technique performed by master of sports of Russia, silver medalist of the Cup of Russia (2008,2009) Evgeny Kovalkov (Ufa, Bashkortostan).

The screenshot series (frames 1 to 9) shows the specific features of the basic elements of Evgeny Kovalkov's squats technique.



The main elements of the technique:
Frame 1 - starting position
Frame 2 - bar 10 cm below the starting position
Frame 3 - bar in the middle of the position
Frame 4 - bar at 6-8 cm from the end point
Frame 5 – end point of the squat
Frame 6 - bar at 6-8 cm above the endpoint
Frame 7 - bar approximately in the middle
Frame 8 - bar 10 cm from the final position
Frame 9 - final position

Fig. 2.8. Angular schematic

Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Frame 9

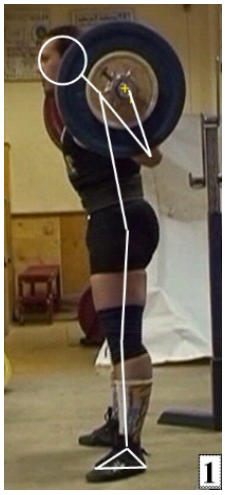
Selected frames of the squat performed by Evgeny Kolakov, MSIC, VK 110 kg, the weight of the bar 275.0 kg (V. Fetisov, 2009)

Frame 1. Starting position

Simple grip of the bar. Hands are at a distance of 12 cm wider than shoulder width. The barbell is located on the back of the middle deltoids and trapezius muscles, the back is straight, slightly tilted forward. The angle of inclination of the back (γ) is 10 degrees. Due to the fact that it is difficult to keep this barbell on back, the athlete must firmly grasp it by his hands. The chin is slightly raised. The common center of mass is approximately over the middle of the foot. The legs are apart at shoulder width, the distance between the centers of the heel is 55 cm. The angle of rotation is about 40-45 degrees. The knees are in the same direction as the toes. The angle of the ankle (δ) is 90 degrees, the angle between the femur and tibia (θ) is equal to 180 degrees, and the hip (φ) angle is equal to 170 degrees.

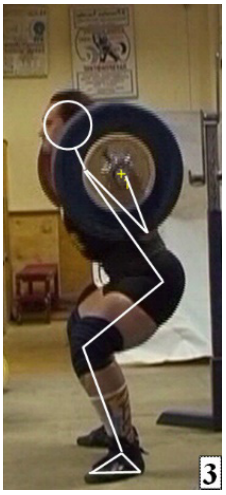
Frame 2. Barbell 10 cm below the starting position

At the beginning of the descent, the speed reaches its maximum (45cm / s). The angle of the ankle (δ) is equal to 80 degrees. The angle between the femur and tibia (θ) continues to decrease to 140 degrees, and at the hip (φ) is equal to 130 degrees. The back is straight and taut. Squatting, the athlete leans forward a bit, so the angle of the back (γ) increases to 25 degrees.



Frame 1

Frame 2



Frame 3

Frame 3. The midpoint of the lift

In this segment, the bar is approximately in the middle of the lift. The angle of the ankle (δ) decreases to 70 degrees, the angle between the femur and tibia (θ) is equal to 110 degrees, and at the hip (φ) is equal to 100 degrees. The angle of inclination of the back (γ) increases up to 35 degrees. When lowering, the speed remains at the same level: 45cm / sec.

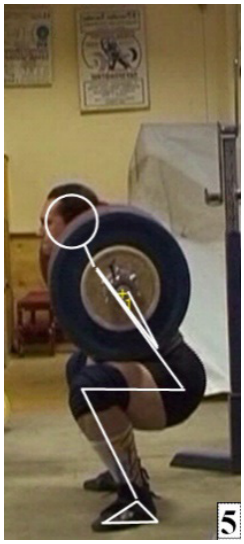
Frame 4. The bar in the 8-10 cm from end point

8-10 cm from the end point of the squat when lowering the speed has slowed down to 40 cm / s. The angle of the ankle (δ) continues to decrease to 70 degrees, the angle between the femur and tibia (θ) has decreased to 90 degrees and at the hip (φ) the angle decreases to 70 degrees. The angle of inclination of the back (γ) is maintained at 35 degrees.

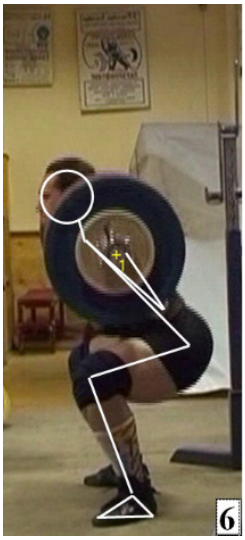
Frame 5. The end point of the squat

The athlete has reached the bottom of the squat. The angle of the ankle (δ) has decreased by 5 degrees, reaching 65 degrees, with a maximum extension of the knees the athlete has gone over the edge of the toes by 3 cm. The angle between the femur and tibia (θ) continues to decrease to 65 degrees, at the hip (φ) the angle decreases to 55 degrees; the back inclination angle (γ) continues to remain at 35 degrees. Lowering, his speed has reached zero. The center of mass has moved from the middle of the foot slightly forward in the direction of the toes.

Frame 4



Frame 5



Frame 6

Frame 6. Barbell 10 cm above the end point

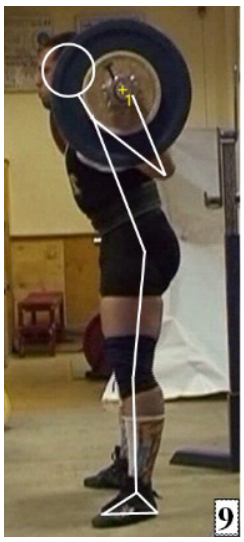
The barbell is located 10cm above the end point. At this stage, the lifting speed increases to 42 cm / s. The angle of the ankle (δ) has increased to 70 degrees, and the angle between the femur and tibia (θ) is 90 degrees, the angle of the hip joint has (φ) increased to 65 degrees. The angle of inclination of the back (γ) increases to 45 degrees. The increase in the knee and hip angle indicates that the athlete has started moving from the bottom of the squat by activating his legs muscles. In this case, the back muscles stay behind the leg muscles in terms of activation.

Frame 7. Barbell approximately mid lift

After Phase 6, the athlete begins to slow down and in Phase 7 his speed decreases to 38 cm / s (see. Figure 2.7). This decrease in speed indicates that the barbell has reached the «sticking point». The angle of the ankle (δ) is equal to 75 degrees, the angle between the femur and tibia (θ) reaches 110 degrees, and at the hip (φ) the angle increases to 80 degrees. The angle of inclination of the back (γ) is 45 degrees. The athlete keeps his leg muscles activated while his back muscles are less so. In this position, most of the load is shifted to the back muscles.

Frame 7

Frame 8



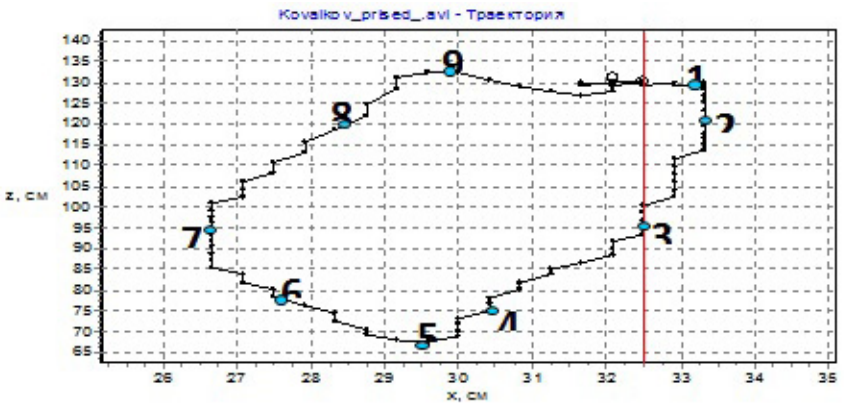
Frame 9

Frame 8. Barbell 10 cm from the final position

After a short pause, the speed continues to increase and reaches its maximum value of 66 cm/s. The angle of the ankle (δ) is equal to 80 degrees, the angle between the femur and tibia (θ) has increased to 150 degrees, and at the hip (φ) the angle has increased to 140 degrees. In this segment, along with the activation of the athlete's legs, he also activates his back muscles, so the angle of the back (γ) decreases to 30 degrees.

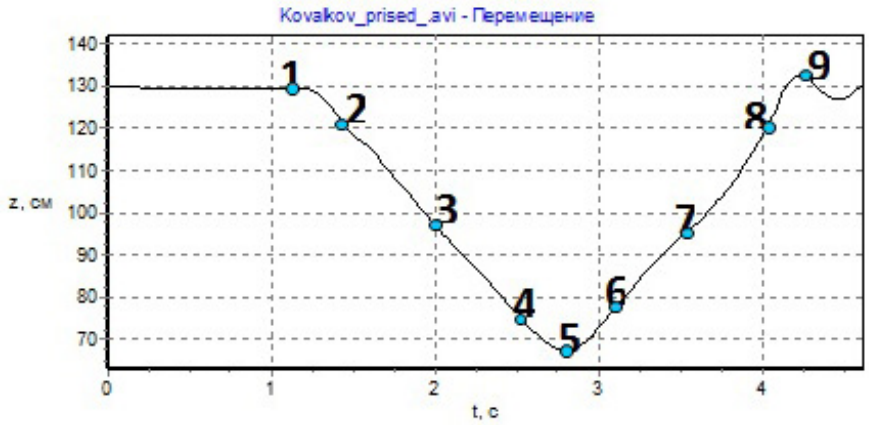
Frame 9. The final position

E. Kovalkov has reached the final position. The angle of the ankle (δ) is equal to 90 degrees, the angle of the knee (θ) is close to 180 degrees. The final position of the angles in the ankle and between the femur and tibia have reached the initial value of the starting position. Due to the fact that the muscles of the back have come under a large load, the athlete cannot straighten his back to the original starting position. The hip joint (φ) angle is 160 degrees, the back angle (γ) equals 20 degrees, in both cases the angle is more than 10 degrees wider than at the starting position.



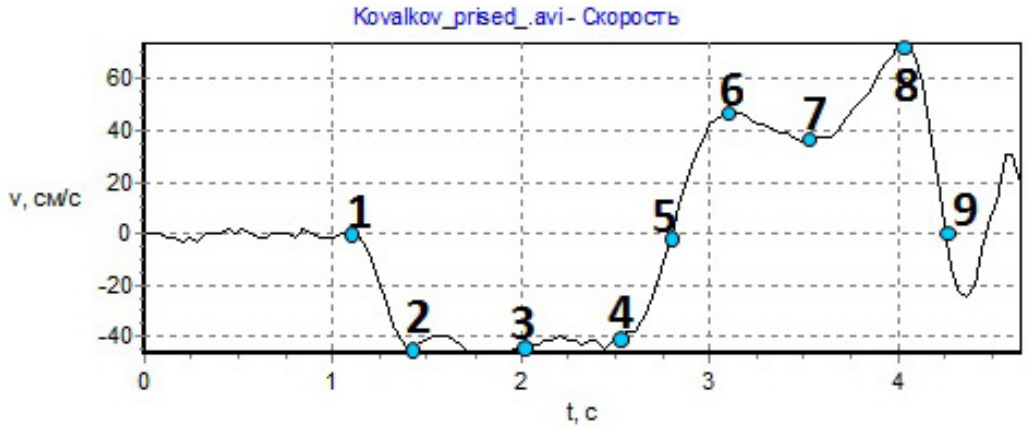
1 - starting position,
2 - barbell 10 cm below the starting position
3 - bar in the middle of the exercise,
4 - bar at 6-8 cm from the end point,
5 – end point of the squat,
6 - bar at 6-8 cm above the end point,
7 – bar approximately in the middle of the lift,
8 - bar 10 cm from the final position,
9 - final position

Fig. 2.9. Trajectory of center of mass of the "athlete-barbell" system while performing squats by E. Kovalkov (V.Fetisov, 2009)



1 - starting position, 2 - bar 10 cm below the starting position
3 - bar in the middle of the exercise, 4 - bar at 6-8 cm from the end point
5 – end point of the squat, 6 - bar at 6-8 cm above the end point of the squat,
7 - bar approximately in the middle, 8 - bar 10 cm from the final position, 9 - final position

Fig. 2.10. The trajectory of the bar's center of mass movement in the sagittal plane (V. Fetisov, 2009)



1 - starting position, 2 - bar 10 cm below the starting position
3 - bar in the middle of the exercise, 4 - bar at 6-8 cm from the end point,
5 – end point of the squat, 6 - bar at 6-8 cm above the end point of the squat,
7 – bar approximately in the middle, 8 - bar 10 cm from the final position
9 - final position

Fig. 2.11. The velocity of the bar's center of mass in the sagittal plane (V. Fetisov, 2009)

Table 2.3

Weight, kg	Maximum vertical displacement during descent, cm.	Duration of descent, s.	Maximum speed of descent, cm/s.	Maximum acceleration of descent, cm/s²
275	66	1,5	40	250

Table 2.4

Maximum distance knee moves past side of foot, cm.	Maximum vertical displacement during ascent, cm.	Duration of ascent, s.	Maximum speed during ascent, cm/s	Maximum acceleration during ascent, cm/s²
3	68	1,5	66	260

During the squat, the athlete cannot sustain the original back angle, the angle of inclination of the back has increased from 10 degrees (starting position) to 35 degrees (bottom of the squat). Besides that, the squatting speed in Phase 3 and 4 has not increased, which may cause difficulty in passing parallel with the wraps on the knees being too tight. Moving the knees 3 cm beyond the toes and increasing the bent of the back while lowering leads to a shift in the direction of the weight. The continued increase in the back inclination during lifting from the bottom upwards (up to 45 degrees) makes the athlete move most of the load onto the back muscles.

The maximum change in the vertical coordinate is 66 cm, the maximum change in the vertical position when lifting is 68 cm, the maximum distance of the

knees over the toes - 3 cm.

Lowering time - 1.5 seconds, top speed of descent - 40 cm / s, the maximum acceleration while lowering - 250 cm/s2.

Concentric time - 1.5 seconds, top speed when rising - 66 cm/s, the maximum acceleration when rising - 260 cm/s2

Duration of the squat - 3 seconds.

2.3.4.3 Alexey Serov's squat technique

Alexey Serov - Master of Sports of Russia, Champion and record holder of the Republic of Bashkortostan in the Junior category.

Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Frame 9

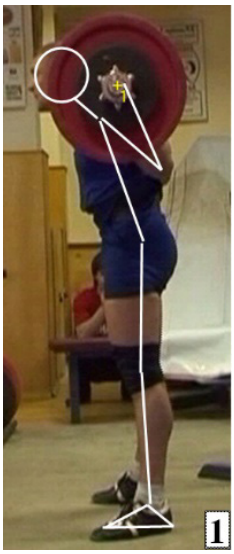
Frames of the squats Alexey Serov, bodyweight - 99.6 kg; height 185 cm, squat 235 kg

Frame 1. Starting position

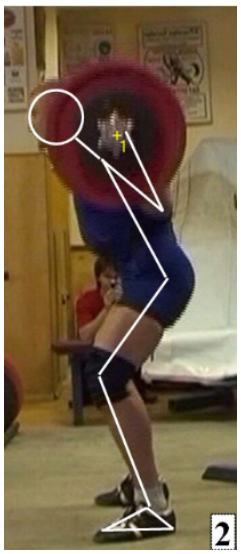
Simple grip (in which four fingers are on one side of the bar and the thumb on the other). The athlete's hands are at a distance of 15 cm wider than shoulder width. The barbell is located on the back of deltoid and mid-trapezius, with the back straight, slightly tilted forward. The angle of inclination of the back (γ) is 20 degrees. Due to the fact that it is difficult to keep the barbell on this part of the back, the athlete must firmly hold it with his hands. The common center of mass is approximately the middle of the foot. The legs are apart at shoulder width, the distance between the centers of the heels is 40 cm. The angle of feet rotation is 15-20 degrees. The knees are pointed in the same direction as the toes. The angle of the ankle (δ) is equal to 88 degrees, the angle of the knee (θ) is equal to 178 degrees, and at the hip (φ) it is equal to 160 degrees.

Frame 2. Barbell 10 cm below the starting position

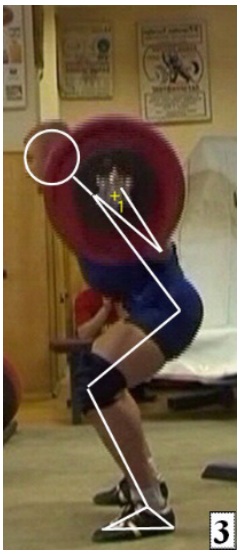
At the stage of the initial phase of lowering into a squat, the speed reaches 70 cm / s. The ankle angle (δ) is equal to 70 degrees. The angle of the knee (θ) decreases to 125 degrees, at the hip (φ) the angle is equal to 116 degrees. The back is straight, tight. Squatting, the athlete leans forward a bit, so the angle of the back (γ) increases to 29 degrees.



Frame 1



Frame 2



Frame 3

Frame 3. Barbell in the middle of the exercise

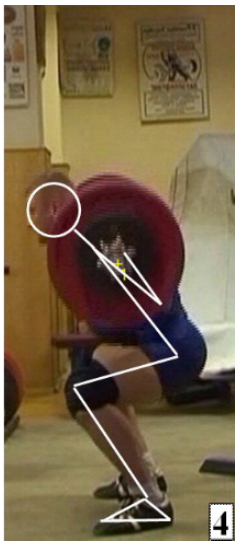
In this segment, the bar is approximately in the middle of the lift. The angle ofthe ankle (δ) has decreased to 65 degrees, the angle of the knee (θ) is equal to 105 degrees, and at the hip (φ) the angle is equal to 94 degrees. The angle of the inclination of the back (γ) is increasing up to 36 degrees. When lowering, the speed reaches its maximum value of 80 cm / s. Maximum acceleration when lowering is 300 cm / s2

Frame 4. Barbell in 8-10cm from the end point

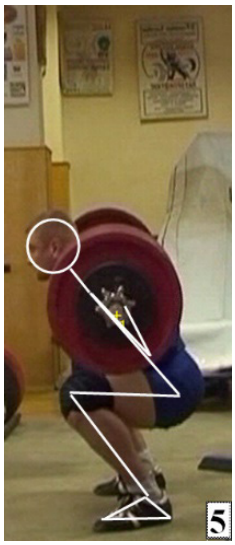
8-10 cm from the end point of the squat when lowering, the speed has slowed down to 60 cm / s. The angle of the ankle (δ) continues to decrease to 56 degrees, the angle of the knee (θ) decreases to 102 degrees, the knees go beyond the vertical projection of the toes. At the hip joint (φ) the angle decreases to 68 degrees. The angle of inclination of the back (γ) increases by 2 degrees and reaches 38 degrees.

Frame 5. The end point of the squat

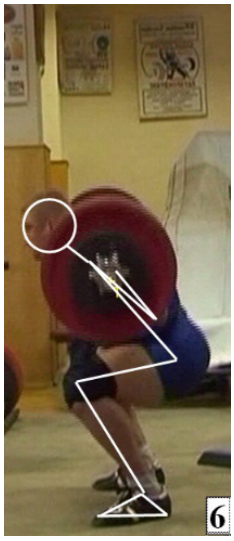
The athlete has reached the end point of the squat. The angle of the ankle (δ) decreases to 54 degrees, with the athlete's knees being 7cm over the toes line. The angle of the knee (θ) has reached 53 degrees, at the hip (φ) the angle decreases to 49 degrees, the angle of the back (γ) increases to 40 degrees. The lowering speed has reached zero. The center of mass has shifted from the middle of the foot slightly forward in the direction of toes.



Frame 4



Frame 5



Frame 6

Frame 6. Barbell 10 cm above the end point

The barbell is 10 cm above the end point of the squat. At this stage the lifting speed increases to 60 cm / s. The angle of the ankle (δ) has increased to 59 degrees,

the angle of the knee (θ) is equal to 81 degrees, the angle of the hip joint (φ) increases to 62 degrees. The angle of inclination of the back (γ) continues to remain at 40 degrees. Increasing the angle of the knee and hip joints means that the athlete has begun his ascent from the bottom of the squat by activating his leg muscles, with his back muscles being less active than the leg muscles.

Frame 7. Barbell approximately in the middle of the exercise

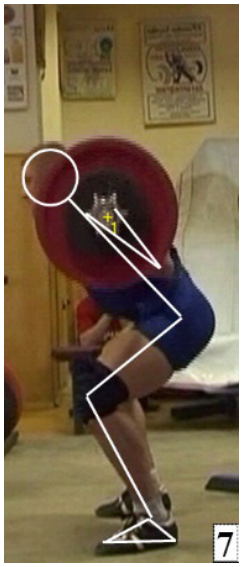
After Phase 6, the athlete's speed is decreasing and by Phase 7 the speed decreases to 45 cm / s (see. Figure 3). This decrease in speed indicates that the barbell has reached the «sticking point». The angle of the ankle (δ) is equal to 62 degrees, the angle of the knee (θ) reaches 100 degrees, and at the hip (φ) the angle increases to 84 degrees. The angle of inclination of the back (γ) is equal to 44 degrees. The athlete keeps his leg muscles activated while his back muscles are less so. In this position, most of the load is shifted to the back muscles.

Frame 8. Barbell 10 cm from the final position

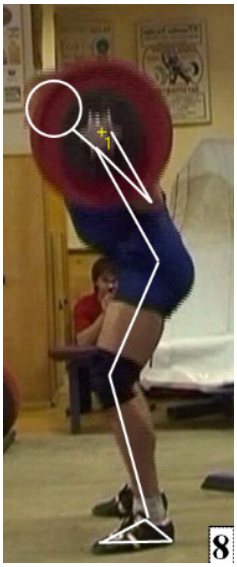
After a slight decrease, the speed continues to increase and reaches its maximum value of 90 cm / s. The angle of the ankle (δ) is equal to 75 degrees, the angle of the knee (θ) increases to 140 degrees, and at the hip (φ) the angle increases to 125 degrees. In this segment, along with the activation of the athlete's legs, he also activates his back muscles, so the angle of the back (γ) decreases to 30 degrees.

Frame 9. The final position

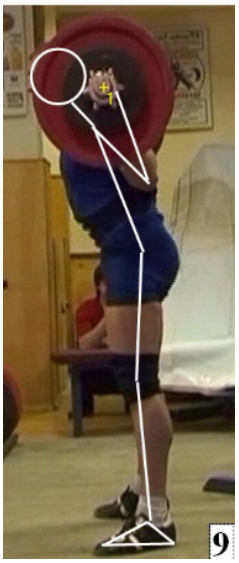
The athlete has reached the final position, the angle of the ankle joints (δ) is equal to 86 degrees, the angle of the knee (θ) approaches 175 degrees, the final position of the angles in the ankle and knee joints have come to the initial value of the starting position. In the hip joint (φ) the angle is 159 degrees, the back angle (γ) is 20 degrees.



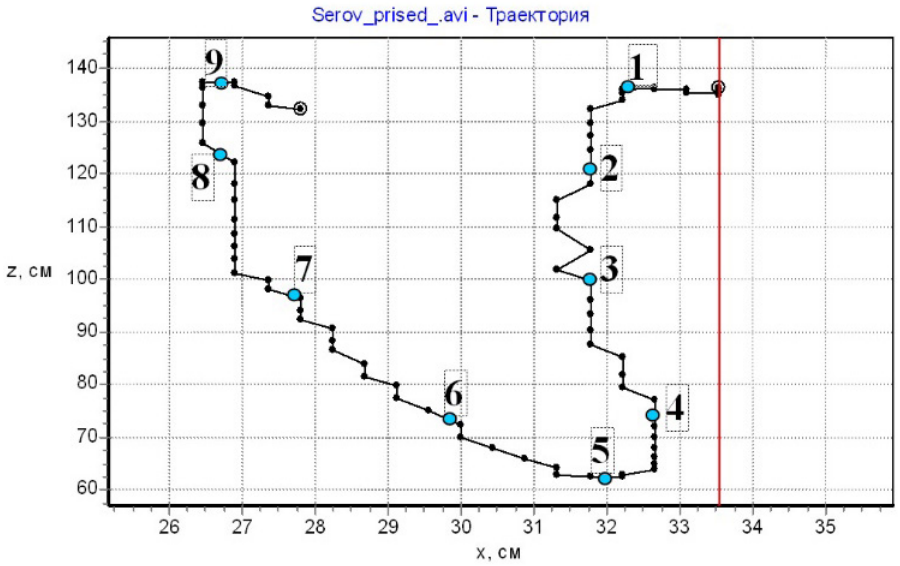
Frame 7



Frame 8

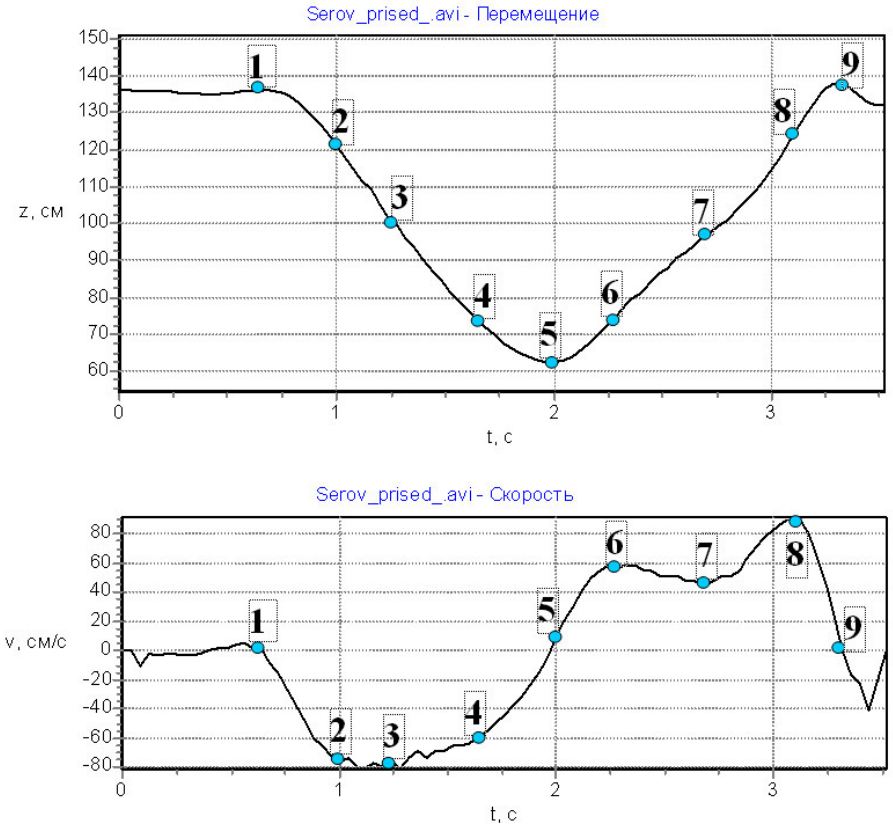


Frame 9



1 - starting position, 2 - bar 10 cm below the starting position
3 - bar in the middle of the exercise, 4 - bar at 6-8 cm from the end point,
5 - end point of the squat, 6 - bar at 6-8 cm above the end point of the squat,
7 - bar approximately in the middle, 8 - bar 10 cm from the final position
9 - final position

Fig. 2.12. Trajectory of the squat performed by A. Serov



1 - starting position, 2 - bar 10 cm below the starting position,
3 - bar in the middle of the exercise, 4 - bar at 6-8 cm from the end point,
5 – end point of the squat, 6 - bar at 6-8 cm above the end point of the squat,
7 – bar approximately in the middle, 8 - barbell 10 cm from the final position,
9 - final position.

Fig. 2.13 Graphs of the vertical displacement and velocity of the barbell

Table 2.5

Distance between center of heels, cm.	Angle of foot rotation,°	Maximum vertical displacement during descent, cm.	Duration of descent, s.	Maximum speed of descent, cm/s.	Maximum acceleration of descent, cm/s²
40	15-20	74	1,3	70	300

Table 2.6

Maximum distance knee moves past side of foot, cm.	Maximum vertical displacement during ascent, cm.	Duration of ascent, s.	Maximum speed during ascent, cm/s	Maximum acceleration during ascent, cm/s²
7	76	1,3	90	300

2.4 Bench Press Technique

One of the main conditions to fulfill in order to improve achievements in the bench press, as in other competition exercises, is to master the technique. This is associated with the improvement of an athlete's ability not only to coordinate, but also to form precise movements. An exercise technique must be regarded as the unity of form and content. The powerlifting bench press (the second competitive exercise) is performed lying on a flat bench with a barbell in the hands. The barbell is first held with locked arms. Then it is lowered until it touches the chest and pressed up until full extension of the arms is reached and the final position is stabilized.

When performing the bench press, the pectoral muscles are developed, as well as arm and upper body muscles. This is why it is used in the training of not only powerlifters, but also bodybuilders, weightlifters, and representatives of many other sports.

2.4.1 Bench press rules and regulations

The technical rules and the order of the bench press are approved by the Presidium of the Russian Federation, which comply with the rules of the International Powerlifting Federation:

The lifter must lie on his back with shoulders and glutes in contact with the surface of the bench. The movement of the feet is allowed, but they must be fully in contact with the surface of the platform or blocks (as far as the shape of the shoes allows it), but in any case not to be lifted from their placement during the exercise. The bar should be taken so that the thumbs are wrapped around (not a false grip). This position is maintained throughout the exercise.

The distance of the hands hall does not exceed 81 cm measured between the forefingers. Both forefingers must be within the 81 cm marks and the whole of the forefingers must be in contact with the 81 cm marks if maximum grip is used. Reverse grip is prohibited.

To ensure firm footing an athlete may use flat plates or blocks not exceeding 30 cm from the surface of the platform. For all international competitions, blocks of different heights: 5, 10, 20 and 30 cm, must be provided for leg support (Photo 2.32).

Photo 2.32. The athlete performs the bench with foot blocks of 10 cm

During the lift, there may be not more than five and not less than two spotters on the platform. Once an athlete takes the correct position to start, he can ask the spotters to help him remove the bar from the racks. In this case, the bar is given directly to him while his arms are locked (Photo 2.33).

Photo 2.33 - three spotters, central spotter gives the hand-off

After removing the bar from the racks (with or without spotters' help), the athlete with fully straightened arms at the elbows awaits for a signal of the judge. The signal is given after the bar is stabilized, and is in the correct position.

The signal for the start of the exercise is the movement of the hand down along with the audible command «start».

After receiving the signal, the athlete lowers the bar to the chest (the chest, under these rules, ends at the base of the sternum) and pauses it in a stationary position. Then the Chief referee gives an audible command «press», after which the athlete presses the arms up to a fully straightened position at the elbows. After the barbell has been fixed in this position, the Chief referee gives an audible command «rack» with the simultaneous movement of his arms to the back.

2.4.2. The phase structure of the bench press

One of the common problems of any competitive exercise training that needs to be addressed is the increase of technical mastery. In recent years, as already mentioned in previous chapters, the kinematic structure of a techniques has received a broader popularity. The structural approach, when analyzing competitive exercises technique, allows to identify its phases in a new way, to have a fresh look at methods and technique mastery. In the structure of the bench press, specialists distinguish between preparatory, main and final periods.

In the bench press, the preparatory period is the starting position. The main period is the actual bench press. The final period is the stabilization of the final position and the return of the bar to the rack.

Table 2.7 shows the phase structure of the bench press in powerlifting as seen by foreign experts, the analysis of which shows that they do not attach much importance to the starting position and completely ignore the phase of stabilizing the barbell in the final position and returning it to the rack. An exception is the phase structure proposed by the honored coach of Ukraine, Professor A. Stetsenko (2008, Ukraine). Unfortunately, none of the foreign experts, including A. Stetsenko, pay sufficient attention to the phase in which the barbell is paused on the chest. Ignoring the phase during the competition is against competition rules, and the attempt will not be counted.

Table 2.7

The phase structure of performing the bench press, as suggested by foreign experts
(L. Ostapenko, B. Sheiko, 2003.2010)

Author / Phase	Bill Jamison (USA)	Rick Well (USA)	John Lear (Canada)	Theodor Arcidi (USA)	Charles Poliquin (USA)	A.Stestenko (Ukraine)
Phase 1	Removing the bar from the rack and obtaining the start position	Preperatory phase	Placement on the bench	Placement on the bench	Placement on the bench	Placement on the bench, removing the bar from the rack
Phase 2	Lowering the bar to the chest	Lowering the bar to the chest and pausing	Removing the bar from the rack	Removing the bar from the rack	Removing the bar from the rack	Beginning position
Phase 3	Pressing the bar	Acceleration	Preparing for the “start” command	Preparing for the “start” command	Lowering the bar to the chest	Lowering the bar to the chest
Phase 4		Pressing upward	Lowering the bar to the chest	Lowering the bar to the chest	Proper upward pressing	Correct upward pressing
Phase 5			Correct upward pressing	Correct upward pressing	Securing the final position	Securing the bar in the final position
Phase 6						Placing the bar in the rack

The names and number of the structural units, presented by Russian specialists, range from three (V. Muravaiev) to seven (B. Sheiko, R. Tsedov, L. Ostapenko) phases, with 12 elements (tab. 2.8).
The table below shows the phase structure of the bench press, proposed by Russian specialists.

Table 2.8

The phase structure of the bench press, proposed by Russian specialists
(B. Sheiko, 2005.2010)

Author / Phase	B.Muraviev	I. Belsky	A.Grachev	Y.Nazarenko S. Tye	A.Zabolotniy	B. Schetina	L. Ostapenko	R.Tsedov	B. Sheiko
Phase 1	Lowering the bar to the chest and pause	Start	Lowering the bar to the chest	Start	Achieving the optimal starting position	Achieving the Starting position	Establishing position on the bench and grabbing the bar	Removing the bar from the rack	Positioning before the lift
Phase 2	The bar comes off the chest	Lowering the bar to the chest	Pause of the bar on the chest	Lowering the bar	Lowering the bar	Lowering and stopping the bar	Removing the bar from the rack	Receiving the “start” command	Start the lift
Phase 3	Moving the bar upward	Pressing the bar	Lifting the bar off the chest	Changing direction of the bar	Pausing the bar on the chest	Temporary pausing the bar	Preparing for the lift	Lowering	Lowering the bar to the chest
Phase 4	Securing the final position	Lockout	Pressing the bar	Pressing The bar	Pressing the bar	Lowering the bar to the chest	The pause	Pausing the bar on the chest	Pausing the bar on the chest
Phase 5			Securing the final position	Securing the final position with straight arms	Securing the final position	Securing the pause	Raising the bar, “squeezing”	Pressing teh bar	Pressing teh bar
Phase 6					Returning the bar to the rack	Pressing supwards	Receiving the “rack” command	Securing the final position	Securing the final position
Phase 7						Returning the bar to the rack	Returning the bar to the rack	Racking the bar	Racking the bar

First period - starting position (2 phases and 3 elements)

Phase1. Preparatory position

Elements: positioning of an athlete on the bench; grip of the barbell, lifting the bar off the rack.

Phase 2. Starting position (1 element)

Elements: fixing of the barbell with straight arms.

Second period – press (3 phases and 7 elements)

Phase 3. Lowering the bar to the chest

Elements: lowering the bar until maximum speed point; from the point of maximum speed until it touches the chest

Phase 4. Fixation of the barbell on the chest

Elements: stabilizing the barbell in a fixed position

Phase 5. Bench press of the bar

Elements: the rise to the point of maximum speed, from the maximum speed point up to the sticking point, passage of the «sticking point», and press from the «sticking points» to the full extension of the arms.

Third period- the final part of the exercise (2 phases and 2 elements)

Phase 6. Bar stabilization

Elements: stabilizing of the bar in the final position

Phase 7. Return of the bar to the rack

Elements: returning the bar on the rack

B. Schtetina (2008) added two additional phases to the phase structure of the bench press suggested by the local experts. According to him, additional phases such as Phase 1- approach to the bar and Phase 8 - leaving the platform, should be added to the kinematic structure pe/rforming of bench press.

2.4.3 Technique and motor structure of the bench press

Motor structure can be described as interaction patterns of movements in space and time (kinematic structure), as well as power and energy interactions (dynamic structure) in the system of movement (D. Don, V. Zatsiorsky, 1979). Every athlete's move should be consciously directed towards achieving a specific goal.

The competitive exercise - the bench press on a flat bench - consists of three periods, seven phases and twelve elements. Each phase has its beginning and end, and includes elements with specific motor task.

First period - Starting position (2 phases and 3 elements)

Elements: positioning of an athlete on the bench; gripping the barbell, lifting the bar from the racks.

Phase 1. Preparatory position

The phase starts from the positioning of the athlete on the bench, gripping the barbell and removing the bar from the racks.

The motor task of this phase is to assume the optimal body position on the horizontal bench before the starting position.

Elements:

1. Positioning of the athlete on the bench: foot placement, barbell grip, arching of the back:

a) Foot placement: an athlete can place his feet at any width, from the widest to the narrowest, under the condition that your feet do not touch the bench (Photo 2.34 and 2.35);

Photo 2.34. Bench press with wide stance of feet

Photo 2.35. Bench press with narrow stance of feet

b) Gripping of the bar: it is necessary to grab at the knurling. In the bench press the athletes can use two grips: wide (81 cm), i.e. the maximum under competition rules, under which the main burden falls on the pectoral muscles, and the medium grip (60-65 cm), which actively engages the triceps muscles and the deltoid muscles. The choice of the grip depends on the length of the arms and the power of the chest muscles of an athlete. The width of the grip is specific for each athlete. But there is a common axiom: the wider the grip, the shorter the path of the barbell from the chest and less work and vice versa, the narrower, the longer the path of the bar and more work (photo 2.36 and 2.37);

Photo 2.36. Bench press with the max legal grip (81 cm)

Photo 2.37. Bench press with medium grip (60-65 cm)

The starting position is taken with the shoulder blades pulled back (this greatly reduces the range of motion) and straight elbows. The athlete should try to arch the back as much as possible, keeping the shoulders and glutes in contact with the bench. The more arched the back, the shorter the path of the bar and less physical effort is required. The height of the arch is not limited under competition rules and depends largely on the flexibility of the athlete in the lumbar and thoracic spine sections.

2. Removing the bar from the racks with a spotter's assistance

At present, athletes remove the bar from the rack with the central spotter's assistance (photo 2.33.). The athlete lies down on the bench further away from the rack and the spotter hands the bar to the athlete who has extended arms. In order to execute this element, the bar should be on the racks at the height at which the athlete will be able to take the hand- off from the spotter without compromising the arch. If the bar is located too high, the athlete will have to reach for it, which will decrease the arch, increasing the length of the torso. When the bar is too low, the athlete will not be able to make a good arch, and will experience difficulties removing the bar from the racks. Taking the hand-off from a spotter, the athlete, slightly lifting the hips, moves the bar forward and takes it down as low as possible on the extended hands, pulling his shoulders blades and shoulders down. Then he lowers the hips on the bench. The front deltoids should be lower than the pectoral muscles.

Phase 2. Starting position (1 element)

The second phase begins after receiving the barbell from the spotter and ends with the athlete in a stationary position.

The motor task of this phase is to gain an optimal position for the effective realization of the motor potential during the exercise.

Elements: Fixation of the barbell with extended arms

The head, shoulders and hips (glutes) are in contact with the surface of the bench. The athlete's soles and heels of the shoes touch the surface of the platform or blocks. The first point of support in this position is the neck and trapezius muscles. The athlete has retracted his shoulder blades and they do not touch the bench, the shoulders are pulled down as much as possible. The leg and back muscles are activated, the hips lay on the bench (Photo 2.38, 2.39). The second pivot point is the athlete's feet. Next, the athlete, with fully extended arms at the elbows and stabilized barbell, awaits the Chief referee's «start» command.

Photo 2.38

Photo 2.39

Starting position demonstrated by the two-time Junior bench press world champion, Chistyakova Yulia

The command is given by the Chief referee immediately after the athlete is motionless and the bar is in the correct position.

Second period - the bench press.

This period is divided into three phases and seven elements.

Phase 3. Lowering the bar to the chest

This phase begins with the bending at the elbows and ends when the barbell touches an athlete's chest. In this phase, the athlete must make a counter movement of the chest to the bar without lifting his support points off the bench. The angle between the arm and the body should be approximately 45 degrees in the downward movement of the barbell as well as the press. Both moves are performed while holding one's breath. The speed of lowering the bar on the chest depends on the athlete's anthropometric features (his arm length), the width of the grip, the height of the arch, the lowering speed and weight of the bar, as well as the quality of the bench shirt. After analyzing over 250 charts of vertical movement of the barbell, we have observed that the lowering phase duration varies from 0.6 to 3.0 seconds, which averages at 1.28 seconds. The motor task of this phase is to select the most favorable biomechanical trajectory of the barbell movement towards the chest.

Elements:

1. Start lowering the bar until it reaches the point of maximum lowering speed of the bar.
2. From the point of maximum lowering speed until the bat touches the chest.

The lowering speed decreases to zero in this segment. The athlete meets the barbell with his body muscles, his legs and shoulder blades provide the biggest support. Many athletes press the barbell deep into their chest. At the same time they relax the arms, their wrists only hold the barbell tightly. Under the rules of IPF and FPR the athlete must lower the bar to the bottom of the pectoralis major muscle.

Phase 4. Pausing the barbell in a fixated position

This phase starts when the barbell touches the chest and the athlete stabilizes it on his chest and ends with the moment of separation of the barbell from the chest.

The motor task of the phase is to keep the bar on the chest in accordance with the rules of the competition and to switch the muscle work from the eccentric to the concentric type.

Elements:

1. Keep a pause with a barbell on the chest

After lowering the bar to the chest, the athlete must hold it in a fixed position on the chest, which means full stop. When analyzing the vertical movement graphs, we have observed that athletes' pauses are different not only in their manner but also duration. The minimum fixed pause duration was 0.3 seconds and the maximum was 1.5 seconds. The average duration was 0.70 seconds. Along with the athletes, who begin the lift of the barbell from the same point where it was lowered, there are those who shift the barbell during the pause on the chest. The barbell can be moved forward (from the head) or backward (to the head).

Phase 5. Press

The phase begins with the separation of the barbell from the chest (since the extension of the arms at the elbows) and ends with a maximum straightening of the arms at the elbows.

The motor task of the phase is to be constantly engaged without losing power from the lowest point to the top of the bar's movement.

Elements:

1. Raising the barbell to the point of maximum velocity

After the “Press” command, the athlete makes an explosive move, developing maximum speed of movement of the barbell upward.

2. From the point at which the barbell reaches the maximum speed of motion to the sticking point. The athlete’s task is to keep the maximum speed of the bar for as long as possible

3. Passing of the «sticking point»

Correlation analysis has shown that the height and the time of the «sticking point» for each individual athlete depends on the qualifications, gender, the weight of the bar and individual maximum results in the bench press. It has been observed that with increasing skill, athletes in competitive conditions show a more pronounced «sticking point.»

Probably, this phenomenon is connected with the fact that the weight of the barbell is close to the record result of a certain athlete. The «sticking point» is manifested through 0.4 - 1.2 seconds after passing the boundary position (BP) between the fourth and fifth phases. Depending on the anthropometric features of the athlete, the distance from the chest at this time is 11-18 cm. The task of the athlete is to keep the movement of the bar with the greatest speed. Successful passing of the sticking point depends on the ability of the athlete to maintain the speed of the press: the higher the speed, the easier it will be to pass the «sticking point» and vice versa.

4. Raising the bar after passing the «sticking point» to the full extension of the arms at the elbows: the locked position.

Lifting the bar after passing the «sticking point» to the full extension of the arms: the locked position. In the final part of the lift (approximately in the upper third) we recommended to make an abrupt exhale. A number of high-class athletes reduce the arch at the end of the movement, so to say, «leave» the bar, it helps to activate your elbows at the end of the movement. The arms extension at the elbows must occur simultaneously.

The third period of the powerlifting bench press is called the final part. It consists of two phases and two elements:

Phase 6. Stabilizing the barbell at the end position

The bar stabilization begins after the complete and simultaneous extension of the arms at the elbows and lasts in a fixed position until the Chief Referee’s signal “rack”. The motor task of the phase is keep the fixed posture in accordance with the rules of the competition.

Elements:

1. Stabilizing the barbell in the final position: fully straightening the arms at the elbows, the athlete stabilizes the barbell in a fixed position and waits for the signal of the head judge «rack.»

Phase 7. Return of the barbell to the rack

The motor task of this phase is to return the bar to the rack.

Elements:

1. Return of the bar on the rack

After the command “rack” the athlete returns the barbell to the rack by himself or with help spotters’ assistance, by shifting the center of mass of the barbell to the back behind the vertical projection of the athlete’s eyes.

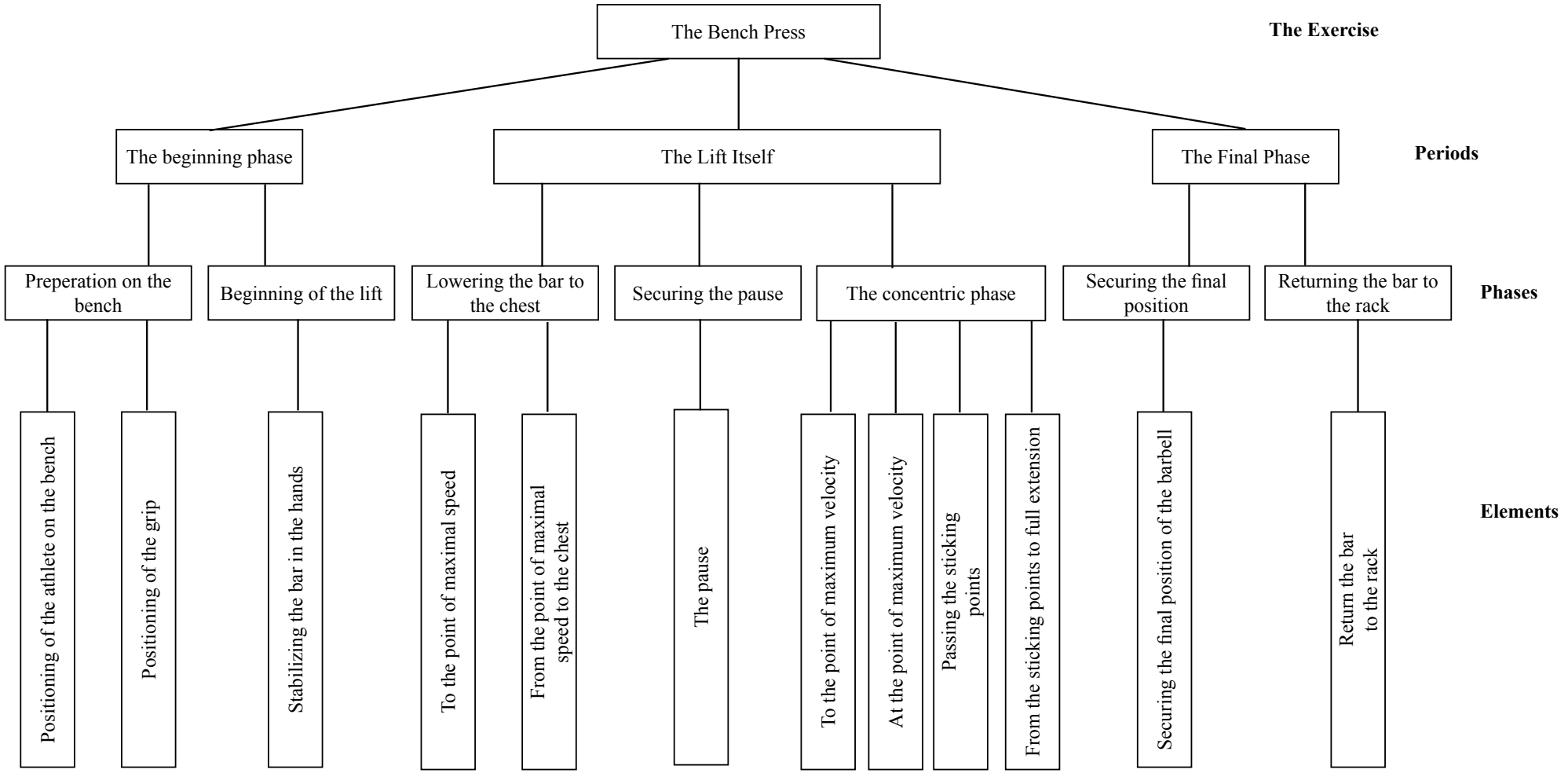


Fig. 2.14. Kinematic structure of the bench press (B. Sheiko 2005)

2.4.4 Biomechanical analysis of the bench press

As a result of the biomechanical analysis of 250 bench press attempts, all the trajectories can be divided, based on their shape, into several groups that can be described by certain features. Pictures 2.15-2.23 show typical examples of the trajectories from each distinct group. In all the pictures the athlete's head is located on the left side (or near the origin of the coordinate plane). The criteria used to form the groups are:

- Presence or absence of the intersection with the vertical line through the trajectory start point (this line is shown in red color in the pictures)
- Presence or absence of a loop
- Presence of a vertical part of the trajectory
- Presence or absence or of a flat area at the bottom (bar movement backwards or forwards before the start of pressing it off the chest)

Let's look at the distinct trajectory groups:

1. Trajectory Type 1 (Fig. 2.17) is a trajectory where the ascent (the press itself) is practically the same as descent. There is no intersection with the vertical line at the start. The trajectory is often almost linear. A stopping point too far from the vertical line means that the athlete let the bar go down too far towards the belly, which is against the competition rules.

2. Trajectory Type 2 (Fig. 2.18) is a trajectory where the ascent intersects the vertical line at the start once and noticeably diverts from the descent in the direction of the athlete's head.

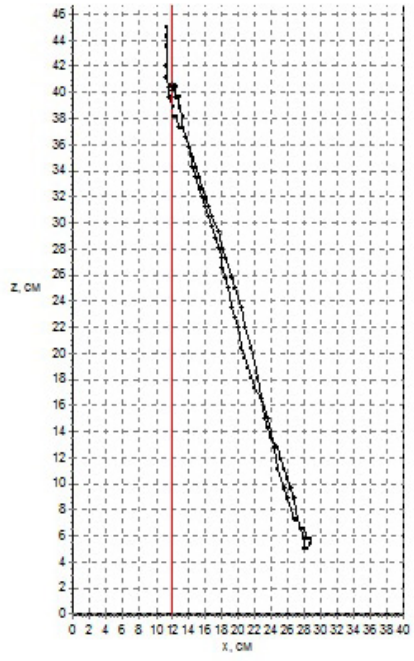


Fig. 2.15.
Trajectory Type 1

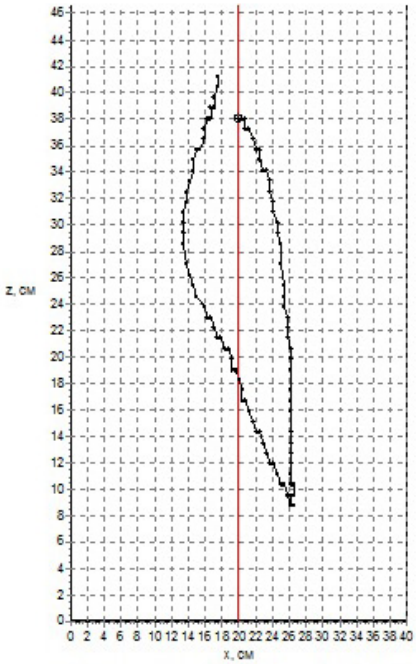


Fig. 2.16.
Trajectory Type 2

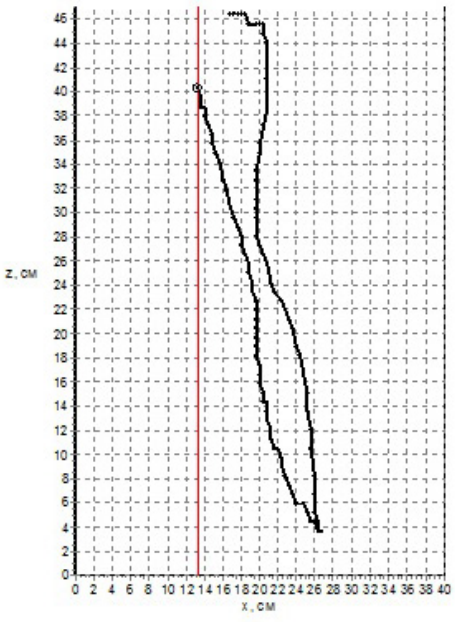


Fig. 2.17.
Trajectory Type 3

3. Trajectory Type 3 (Fig. 2.19) is a trajectory where ascent and descent do not intersect each other or the vertical line at the start, but slightly diverge. The whole trajectory is located to the right from the vertical line at the start.

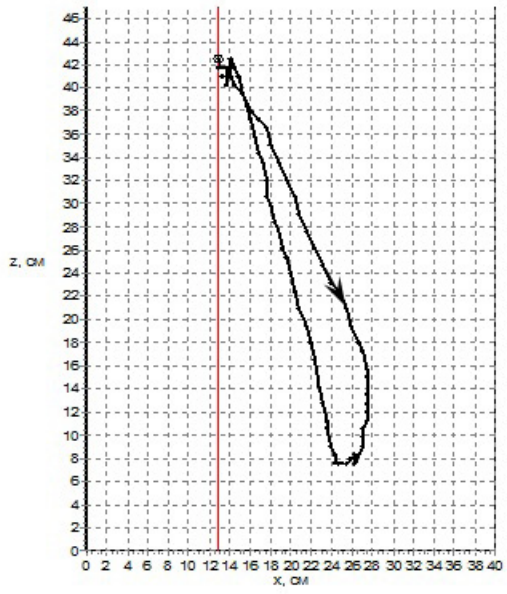


Fig. 2.18.
Trajectory Type 4

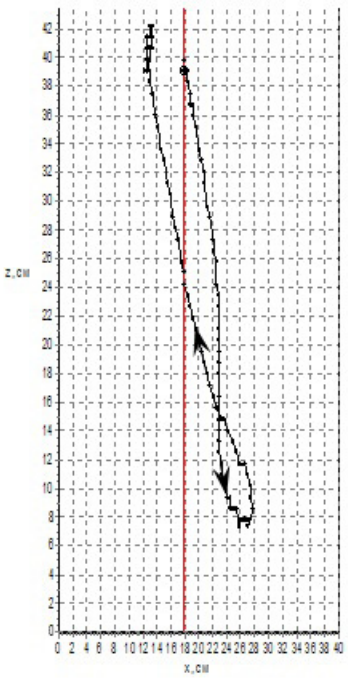


Fig. 2.19.
Trajectory Type 5

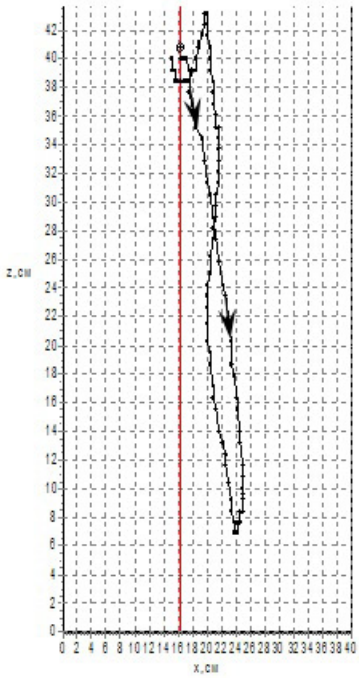


Fig. 2.20.
Trajectory Type 6

4. Trajectory Type 4 (Fig. 2.20) is a trajectory where the ascent is slightly displaced forward relative to the descent and finishes almost at the starting point. There is no intersection with the vertical line at the start.

5. Trajectory Type 5 (Fig. 2.21). The bar path has a loop at the bottom of the trajectory and the start of the ascent is slightly displaced to the right (to the athlete's feet) relative to the finish of the descent. The ascent may or may not intersect the vertical line at the start.

6. Trajectory Type 6 (Fig. 2.22) also has a loop, but the loop is bigger than the one in the previous trajectory type. The ascent start is displaced to the left (to the athlete's head) relative to the descent finish. The upward motion trajectory does not intersect the vertical line at the start.

7. Trajectory Type 7 (Fig. 2.23) is a trajectory where the ascent intersects the vertical line at the start once or twice. In the picture 5.14 it is easy to notice that, after stopping the barbell motion, the athlete pushed it into the chest, which is prohibited by the competition rules.

8. Trajectory Type 8 (Fig. 2.24) is defined by the presence of a sticking point during the ascent phase. This point is the most uncomfortable for the athlete's muscles to display their strength. This area is one of the main factors that limit the result. Correlation analysis showed that both height and time when the sticking point occurs are individual for each athlete. Their values depend on qualification, gender, the weight used relative to the maximum, and the personal best bench press. There may or may not be an intersection of the ascent trajectory with the vertical line at start.

9. Trajectory Type 9 (Fig. 2.25) is characterized by almost vertical downward motion with possible (single or double) intersection with the vertical line at start. The upward motion is possible either with a loop, or along a path that lies completely on the left side of the vertical line at the start.

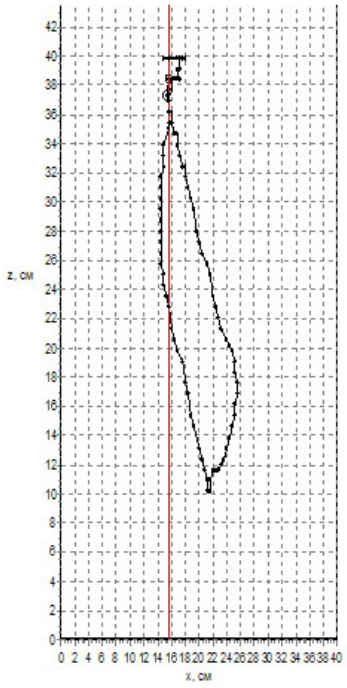


Fig. 2.21.
Trajectory Type 7

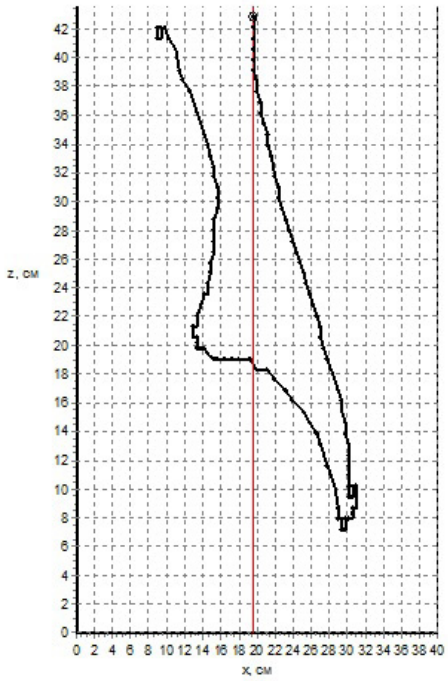


Fig. 2.22.
Trajectory Type 8

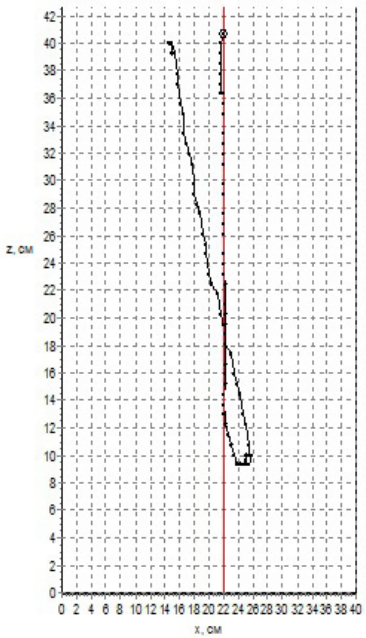


Fig. 2.23.
Trajectory Type 9

A comparison of trajectory configurations has shown that athletes with high and stable results in competitions have trajectories depicted on figures 2.17-2.19 most of the time.

After the statistical analysis of the vertical barbell motion graph in bench press, it was found that:

a) the length of the downward motion varied between 0.6 and 3.0 seconds, with 1.28 seconds average and standard deviation of 0.50;

b) the length of the pause varied between 0.3 and 1.8 seconds, with 0.70 seconds average and standard deviation of 0.26;

c) the length of the upward motion varied between 0.5 and 5.0 seconds, with 1.40 seconds average and standard deviation of 0.68

d) quantitative analysis of the press phase length showed that this phase has a large distribution of time values: from 0.5 to 5.0 seconds with the average of 1.4 seconds.

Figures 2.26 and 2.27 show the time base plots of the BEC (bar end center) vertical path and its speed for two different athletes. On the left side of each picture there are graphs showing attempts with 80-85% of maximum weight. The right side corresponds to attempts with submaximal weight. Black dots mark positions corresponding to the peak bar speed. The pictures graphically demonstrate each athlete's technique.

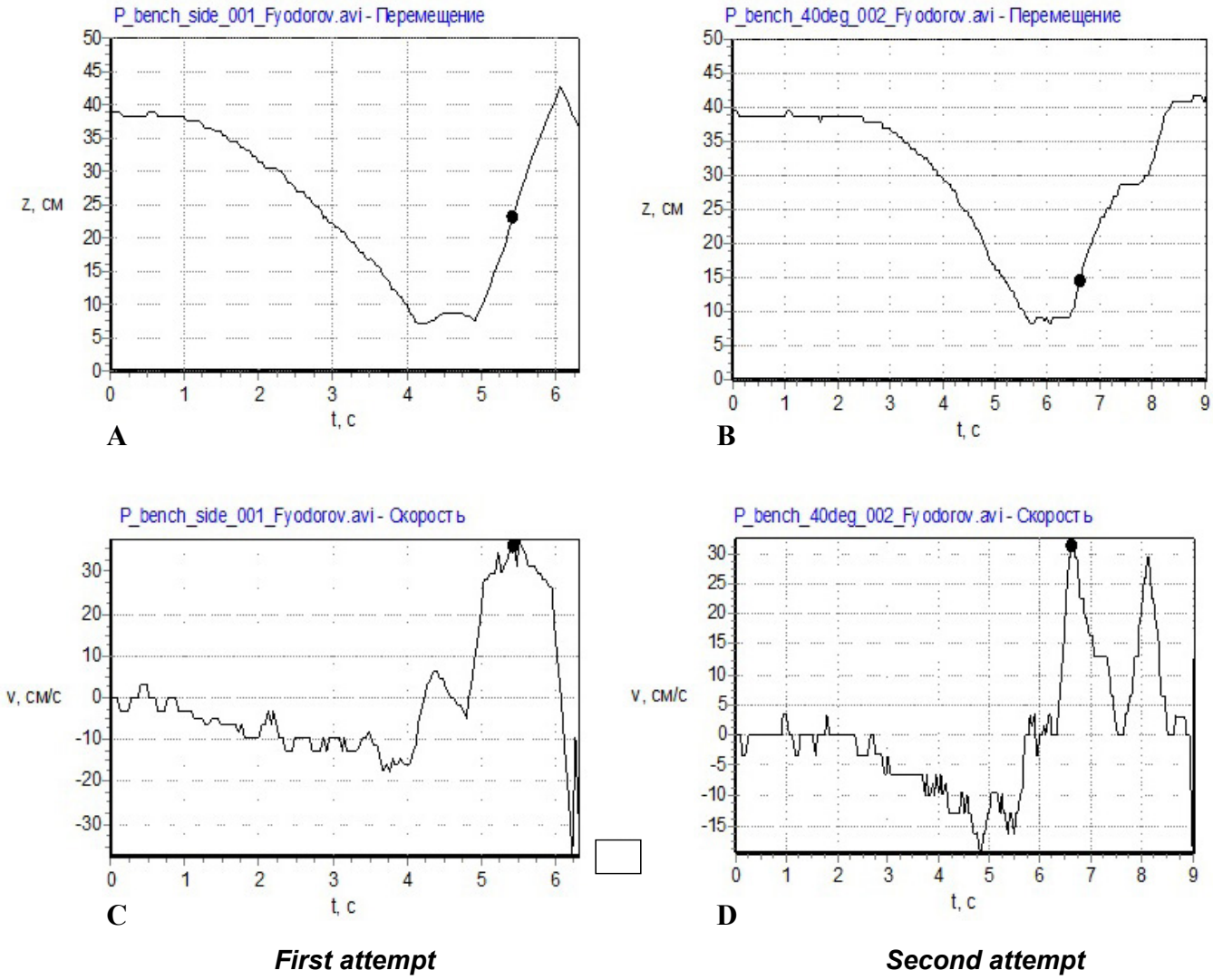
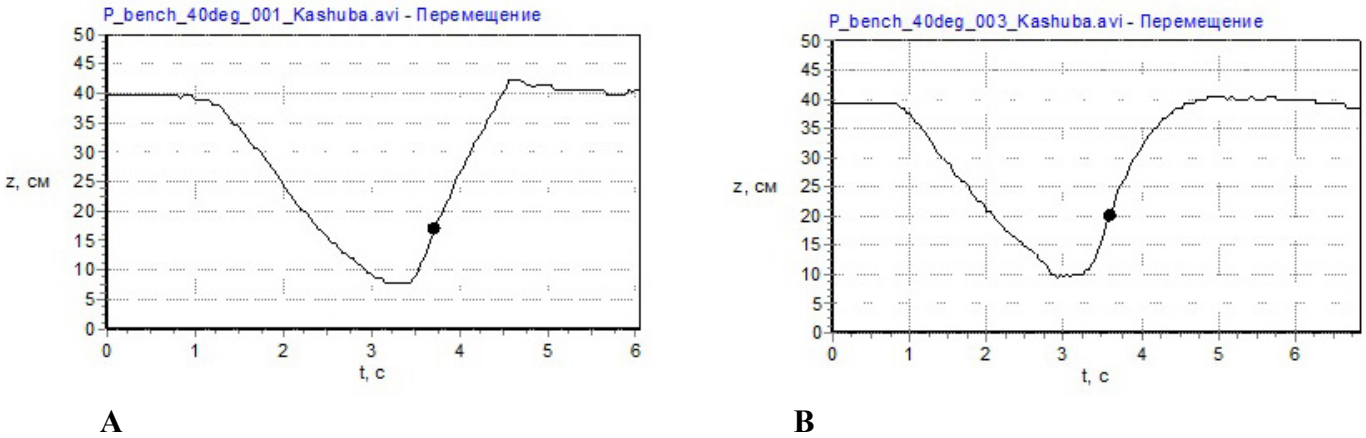


Fig. 2.24. Time base plots of the BEC vertical motion for two attempts of the athlete F.

So, athlete F. (Fig. 2.26) tends to start the descent (about 3s) smoothly, pauses at the bottom (for about 0.6s), and then proceeds to a uniform fast ascent (about 1s). With the submaximal weight of the barbell the ascent stops being uniform: on the time base plot a clear sticking point can be seen between 7-8 seconds, and the plot has two peaks. Such a trend happens for most athletes at submaximal loads. The corresponding areas are often observed on the trajectory graphs as well (see Fig. 2.27).



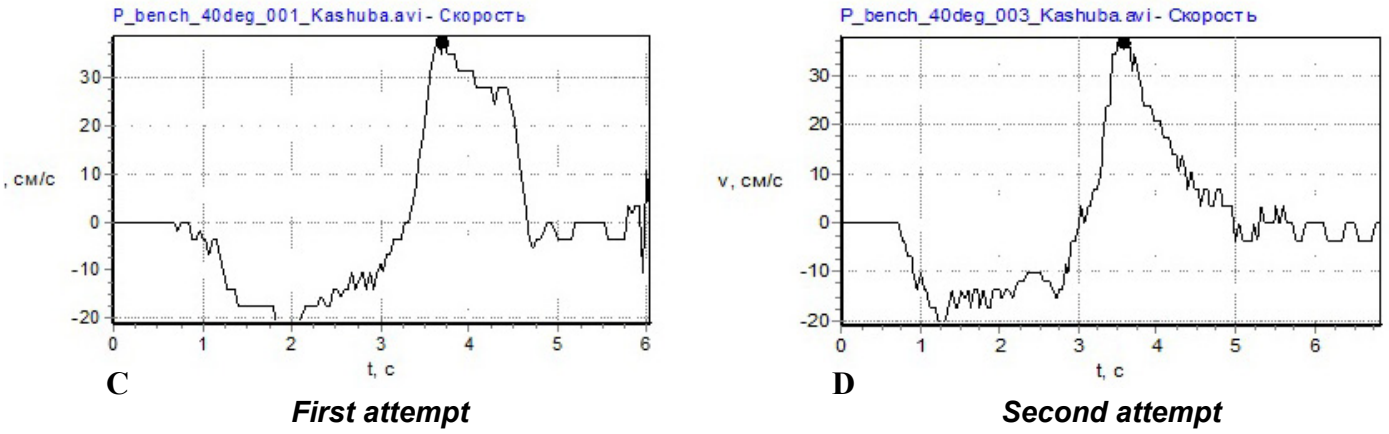


Fig. 2.25. Trajectories and time base plots of the BEC vertical motion for two attempts of the athlete K. (MS, born in 1987, 135 kg bodyweight)

The technique of athlete K. (Fig. 2.27) tends to have a fairly stable trajectory that does not deform when the load increases, a uniform ascent and descent that both take about the same amount of time (1,5 - 2,0s), and a short (≈ 0,4s) pause at the bottom. The load increase also slightly slows down the second half of the ascent phase and shrinks the fast area in the speed graph, although not very much.

Time base plots of the vertical motion also let us control the assessment quality in competitions because they clearly show successful and unsuccessful lifts. As an example, picture 2.28a shows a plot where the press out delay is critical, but such an attempt is usually considered to be a good lift since the negative motion is almost nonexistent. Fig. 2.28b shows an area with the negative motion (downward motion in 4 second) that made the attempt unsuccessful despite the athlete pressing out the barbell successful.

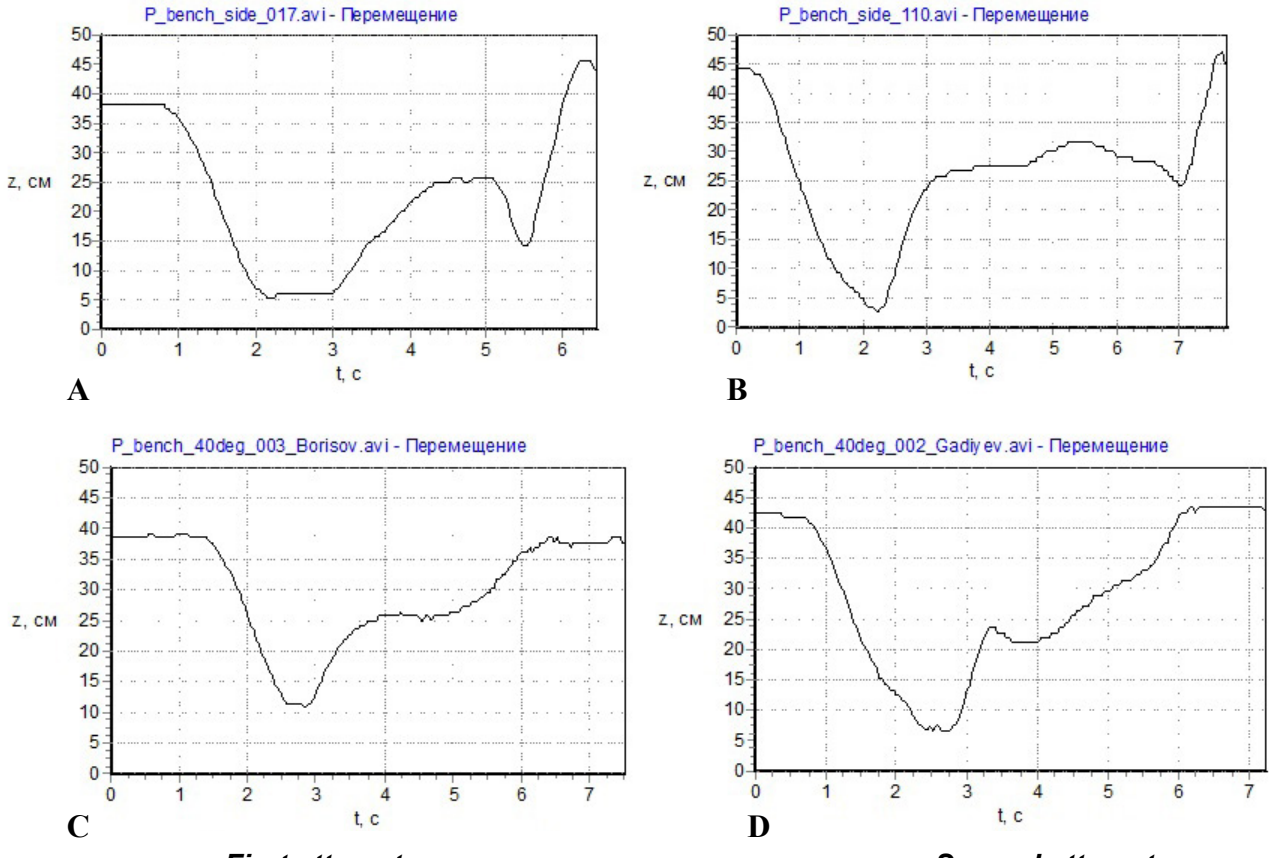


Fig. 2.26. Time base plots of the BEC vertical motion for unsuccessful bench press attempts.

Analysis of typical cases of slowing down during the ascent shows that if either the speed at the start of this phase is not greater than 10 cm/s or the length of the flat area in the middle of the phase is greater than 1 s, the attempt is usually unsuccessful with submaximal weight on the bar. Figures 2.28b, 2.28d show plots of such cases where the barbell was returned to the rack with the help of the spotters.

During the conducted research a biomechanical analysis of bench press was performed, giving us factors that define differences in technique and let us objectively evaluate them. High-class athletes with developed technique tend to have peculiar and persistent type of trajectories, as well as time base plots of the vertical motion and speed.

Trajectory analysis lets us make conclusions about the rationality of motions and presence of clear mistakes in the exercise execution. The detected kinematic features of exercise execution can be used to develop models of rational (efficient) technique in powerlifting exercises.

The obtained results let us perform qualitative express analysis of motion during training or competition without having an accumulated database of videos related to the given athlete.

Thus, by comparing space-time and dynamic properties of the motions performed by athletes with different levels of athletic development during exercise, we can get an idea about the mistakes of a lower level athlete, as well as proper technique parameters that are required for high results.

We have come to the following conclusions:

1. The basic bench press technique has three parts, seven phases and twelve elements (see Fig. 2.16).

2. Dropping or violating even one given element or relation makes it impossible to solve the movement problem. Phases have certain time features. Their lengths depend on athlete's level, height, and weight on the bar. Technique elements may differ from athlete to athlete and mostly depend on their individual morphological and functional features. Correct usage of individual features (along with all the general rules of rational technique) defines the proper technique for the given individual.

3. The speed of competition bench press is a fundamental factor.

4. The lower the speed is, the earlier the sticking point happens.

5. The lower the sticking point is, the harder it is for the athlete to overcome it.

The obtained results let us perform qualitative express analysis of motion during training or competition without having an accumulated database of videos related to the given athlete.

On the basis of the above, the authors recommend athletes who have difficulties at 10-15 cm from the chest include bench press with 40%, 50%, and 55% weight with maximum possible speed. This will help develop speed and ability to perform explosive motions.

2.4.5. The influence of the arch on bench press

In the present time in order to get maximum results in the bench press many athletes arch their lower back (Fig. 2.40) (B. Sheiko, 2013). Back arch helps get higher results because of the following:

1. **The range of motion decreases.**
2. Stronger middle (pars sternocostalis) and lower (pars abdominalis) chest muscles are more actively involved (S. Glyadya, M. Starov,

Y. Batygin, 1999; M. Rippetoe, S. Bradford, 2011; P. Evangelista, 2011).

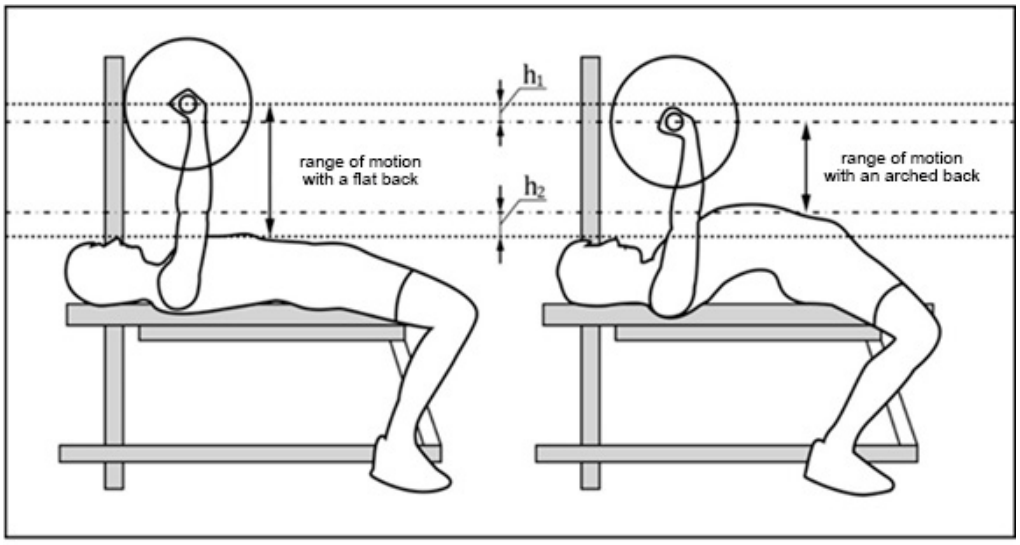
3. Chest muscles stretch significantly more and allow more active usage of their elastic deformation strength according to the length-strength dependence in the muscles.

Let's look at the first two factors.

1. Decreased range of motion.

The range of motion decreases while arching happens because of three reasons:

1) Scapula retraction during the arch significantly decreases the barbell height relative to the rib cage (h1 on Fig. 2.29). One of the main muscles – lats (m. latissimus dorsi) – contributes to that. Data obtained by R. Roczniok at al. (2013) shows that the activity of lats during the descent phase significantly influences the bench press result. Our data also shows that this muscle shows a fairly high activity during the descent phase.



Meaning: h_1 - range of motion decrease from scapula retraction
h_2 - range of motion decrease from lower back arch

Fig. 2.27. Influence of the arch on the range of motion

Significant arch in lower back (h2 on Fig. 2.29).

Lower back can be arched because of the intervertebral mobility that is possible due to the existence of intervertebral disks. C.I. Medrano, D.A. Cantalejo (2008) showed that arching your back causes a significant load on the lower back. The following muscles allow maintaining the back arch: erector spinae (spinal erectors), transversospinalis, and interspinales. It can be noted that elite powerlifters have very hypertrophied erectors (S. McGill, 2009).

Performing the bench press on an inhale widens the chest, increases intra-abdominal pressure, and lifts the ribs

2. More active functioning of the middle (pars sternocostalis) and lower (pars abdominalis) chest, which are stronger muscles. H. Krol et al., (2006) showed that during bench press with back arch some athletes use different fibers of their pectorals (m. pectoralis major). Because of that, the authors hypothesized that involvement of various fibers of pectorals

(m. pectoralis major) depends on the degree of the back arch. Later P. Evangelista, (2011) showed that the arch creates favorable mechanical conditions for the pectorals, just like in decline bench press. An athlete can show a better result in this exercise compared to the competition bench press. This happens because the position of the shoulder bone related to the torso is favorable for higher torque between shoulders and lower (stronger) pectoral (m. pectoralis major) muscle fibers.

Thus, it is very important for the athlete to maintain the arch height since it helps perform less work moving the bar upward.

There are factors that can decrease the arch height. The spine and intervertebral disks can be seen as a spring with a relatively high stiffness that is compressed in the shape of an arch. The generated force tries to “decompress” the arch and “straighten” athlete’s spine. However, during the arch the spinal erectors oppose this force, together with the lower extremities that provide a reliable supporting frame. Providing this reliable supporting frame that prevents decrease of the arch during the bench press is the main function of the lower extremities. If such frame does not exist, during the upward motion of the bar the hips will move towards the legs and decrease the arch height, which is highly undesirable.

2.4.6. The transfer of the mechanical impulse from the legs to the bar

Most works dedicated to the bench press have studied the movement technique of the upper body, while the lower body function has not been taken into consideration. M. Rippetoe and S. Bradford (2011) state that, although the legs are not a part of the kinematic chain in the bench press, the correct position and usage of the legs and back muscles will help stabilize the torso position on the bench (main function), as well as create the necessary horizontal force to increase the arch and keep the chest as high as possible.

During the descent, the back arch tries to decrease if it is not sufficiently supported by muscle strength. However, isometric work by the thigh (both quadriceps and hamstrings) and gluteal muscles prevents the back from losing the arch.

Let’s look at one more technical element that makes an athlete transfer the mechanical impulse from legs to the bar.

B. Sheiko et al. (2007) have proposed a hypothesis that when performing the bench press, the lower limbs can also transmit a mechanical impulse from the legs to the bar. The authors note that during the beginning of the ascent phase some athletes push themselves into the platform with their legs moving in the direction of the head.

The use of this technical element was proven by the further study of the competitive benchpress technique by two athletes of completely opposite qualification level. The low-qualification level athlete performs the benchpress exclusively by using his upper back and torso muscles, his lower limbs muscles are not fully activated. The other athlete, by the simultaneous activation of his lower limbs muscles, creates a firm support for himself, which, first of all, allows him to stabilize his body on the bench, and secondly, to keep the arch in the lowering phase as well as the press itself, and thirdly, to pass a mechanical impulse from his legs to the bar at the beginning of the lift.

A. Samsonova et. al conducted a research aimed at the explanation of the mechanism responsible for the impulse movement from the legs to the bar. They analyzed over 30 videos of the athletes of different qualifications who apply the impulse method in order to examine their benchpress technique.

Fig. 2.30 shows the impulse transfer from the lower extremities to the bar. Due to the activity in calves and quadriceps, the knee joint shifts towards the hips together with a sharp flexion of the gluteal muscles. This leads to a small shift of the hips up and towards the head while the gluteus (m. gluteus maximus) is touching the bench.

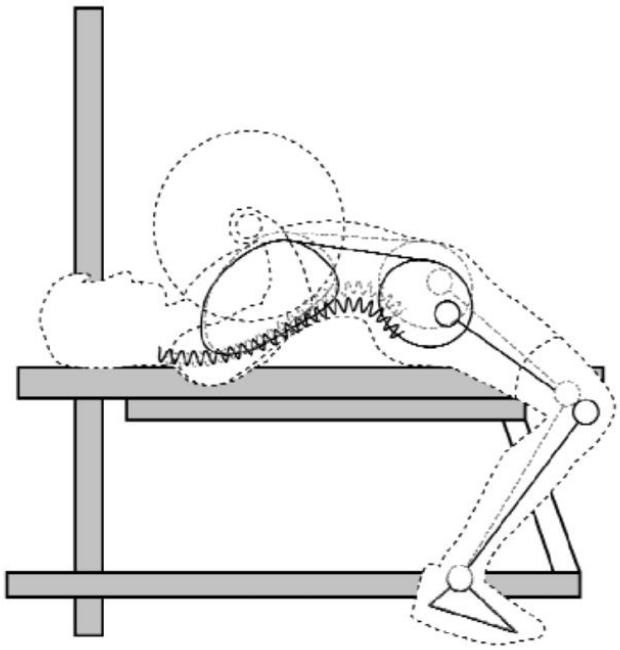


Fig. 2.28. The impulse transfer from the lower extremities to the barbell

It is worth noting that the athletes who apply the impulse method do a slight upward chest movement. This movement happens because their hips, spine, chest, and scapulae are parts of a rigid frame that is held together by torso muscles. The movement of the hips up and towards the head leads to the rotation of this frame along the scapulae and slight chest elevation. As a result, the mechanical impulse is transferred to the bar (that is located on the athlete's chest at the moment). If the athlete can use the arms (powerfully activate upper torso and arm muscles) on this small impulse (impulse is equal to the bar's mass times its speed), it will help to give the bar higher speed at start. However, not all athletes properly use the arms during the jerk with the legs and chest movement. If they don't do it in time, the impulse is lost. Moreover, if the hips move upward too much, the gluteal muscles will lose contact with the bench and the attempt will not count.

2.4.7 Bench press techniques of advanced athletes

2.4.7.1 K. Pavlov's bench press technique

Konstantin Pavlov is an Honored Master of Sport of Russia, ten-time world champion, VI World Games champion, and a world champion in bench press, eleven-time European champion and three-time European champion in bench press. He is also twelve-time Russian champion.

We use a traditional point for such analysis – bar end center (BEC) in order to track Konstantin Pavlov's bench press technique features. In frames 1-9 that came from the videotape, the order of the technique is clearly seen.

Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Frame 9

Selected frames of the bench press (V.Fetisov, 2008)
(K. Pavlov, 56 kg weight class, 185 kg on the bar)

Frame 1. Starting position: fingers are around the bar, thumbs are on the opposite side of the bar “locking” the bar. The grip width is 81 cm – the maximum allowed by the competition rules. With such grip width the main load is on the pectorals. Head, shoulders and glutes are in contact with the bench. Leg and back muscles are contracted. The sole and heels of the shoes are completely on the platform. Waiting for the “start” signal by the Chief referee, the athlete has stabilized the bar in place with his elbows locked.

Frame 2. The bar is 8 cm below the starting position. The descent speed is maximum – 35cm/s.

Frame 3. The bar is halfway to the athlete’s chest. At point 3 (see the vertical movement and speed graphs) the descent speed slows down to 5cm/s.

Frame 4. The bar is 5 cm above the chest. The descent speed goes up again to 25cm/s.

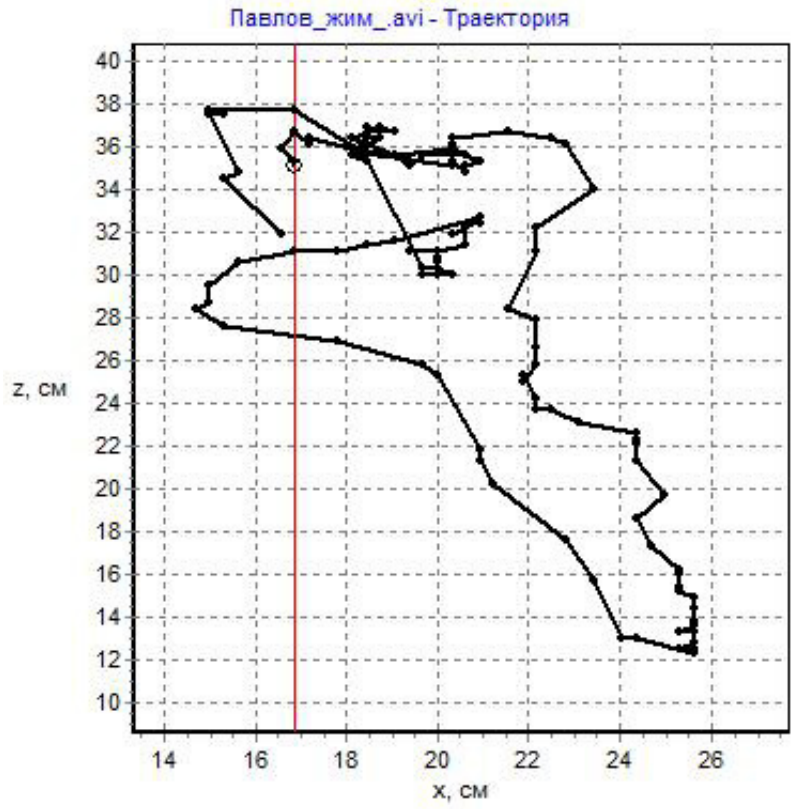
Frame 5. The athlete touches the chest. The descent trajectory length is 24 cm. The pause length is 0.75s. After the “press” command by the Chief referee, the athlete makes a maximally explosive push using his upper torso muscles. The starting impulse goes from the legs to the head. The following muscles are involved in the push of the bar: pectorals, deltoids, triceps, and lats.

Frame 6. The bar is 5 cm above the chest. The bar speed is maximum – 50cm/s.

Frame 7. The bar is about 8 cm from the final position and approaches the sticking point, which explains the sudden speed drop to 5cm/s.

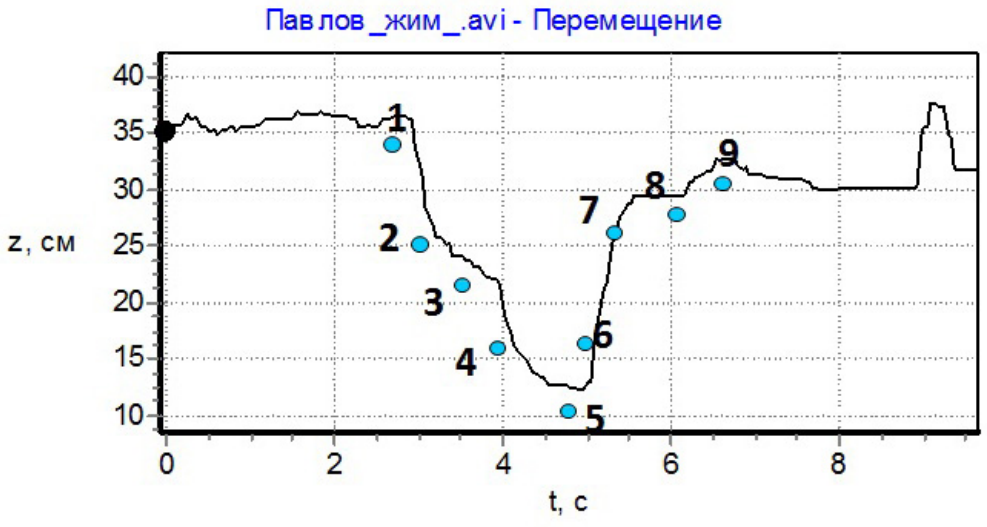
Frame 8. The bar speed slows down to 0, the movement stops, and then the speed increases to 10cm/s.

Frame 9. The final press position of the bar. After the full and simultaneous lockout of both arms, the athlete fixates the bar in place waiting for the “rack” command from the Chief referee. The bar ascent trajectory is 20 cm long, the ascent time is 2.7s, the maximum acceleration during the ascent is 260cm/s^2.



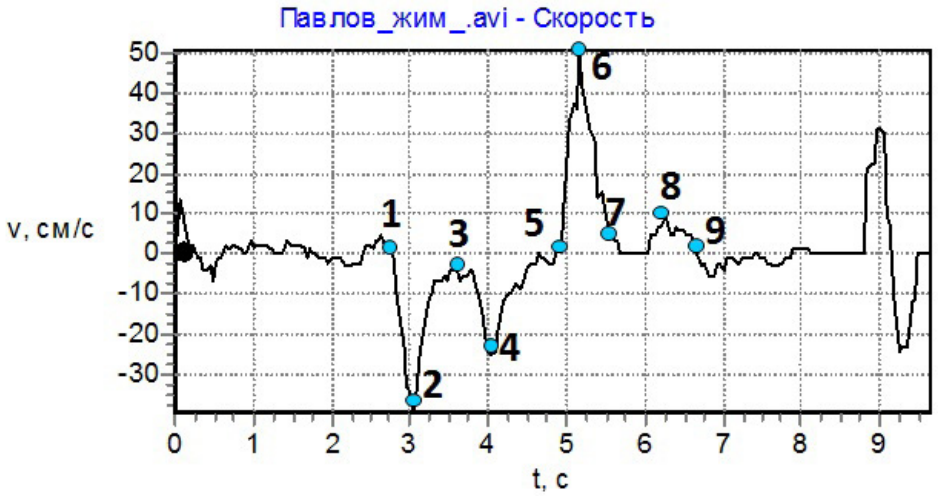
1 – start of the descent,
2 – bar is 8 cm below the start,
3 – middle of the descent
4 – bar is 5cm above the chest,
5 – bar is on the chest,
6 – bar is 5cm above the chest
7 – middle of the ascent,
8 – sticking point, 9 – finish of the ascent

Fig. 2.29. Trajectory of the bar’s center of mass during the bench press. (V.Fetisov,2007)



1 – start of the descent, 2 – bar is 8 cm below the start, 3 – middle of the descent
4 – bar is 5cm above the chest, 5 – bar is on the chest, 6 – bar is 5cm above the chest
7 – middle of the ascent, 8 – sticking point, 9 – finish of the ascent

Fig. 2.30. Trajectory of the bar's center of mass in the sagittal plane (V.Fetisov,2008)



1 – 0cm/s, 2 – 35cm/s, 3 – 5cm/s, 4 – 22cm/s, 5 – 0cm/s, 6 – 50cm/s,
7 – 5cm/s, 8 – 10cm/s, 9 – 0cm/s.

1 – starting position, 2 – maximum descent speed,
3 – descent speed starts to decrease,
4 – descent speed is 22cm/s, 5 – the bar is on the chest
6 – maximum ascent speed
7 – speed "drop", 8 – sticking point, 9 – ascent finish

Fig. 2.31. Speed of the bar's center of mass (V. Fetisov, 2008)

2.4.7.2 Y. Chistyakova's bench press technique

Yulia Chistyakova is an International Master of Sport of Russia, two-time Junior World champion in bench press (2010, 2011), and a silver medalist at Junior World Championship in bench press (2012).

Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Frame 9

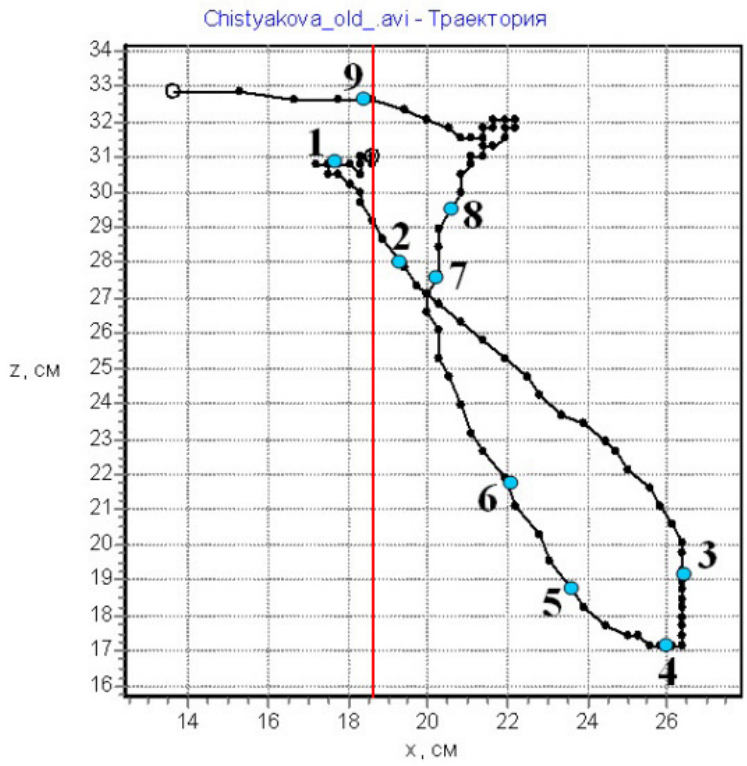
Selected frames of the bench press (V. Fetisov, 2008)
(Yulia Chistyakova, International Master of Sport, 66 kg weight class)

Frame 1. Starting position: fingers grip the bar, thumbs are on the opposite side of the bar "locking" the bar. The grip width is 81 cm – the maximum allowed by the competition rules. With such grip width the main load is on the pectorals. Head, shoulders, and glutes are in contact with the bench. There is a high back arch. Her leg and back muscles are contracted. The sole and heels of the shoes are completely on the platform. Waiting for the "start" signal by the Chief referee, the athlete has stabilized the bar in place with her elbows locked.

Frame 2. The bar is halfway to the athlete's chest. At point 2 (see the vertical movement and speed graphs) the descent speed is maximum – 30cm/s.

Frame 3. The bar is 5 cm above the chest (point 3 on the graphs). The descent speed goes down to 25cm/s.

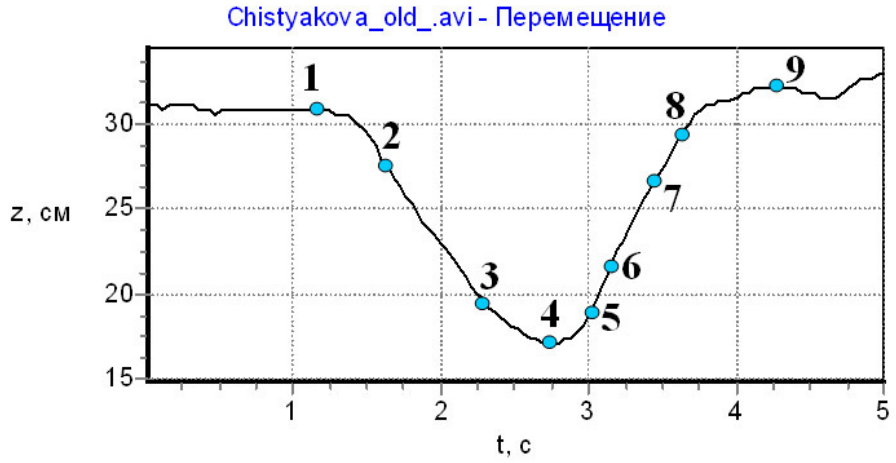
Frame 4. The bar is at the chest. The descent speed is 1.25s. The trajectory length is 32cm. The pause length is 0.25s (the pause is so small that it is barely noticeable, but it can be easily seen on the graph).



1 –starting position,
2 – start of the descent,
3 – bar is 3 cm above the chest
4 – bar is on the chest,
5 – bar is 2cm above the chest,
6 – bar is 5cm above the chest,
7 – bar is 10 cm above the chest,
8 – bar is 12 cm above the chest,
9 –finish of the ascent

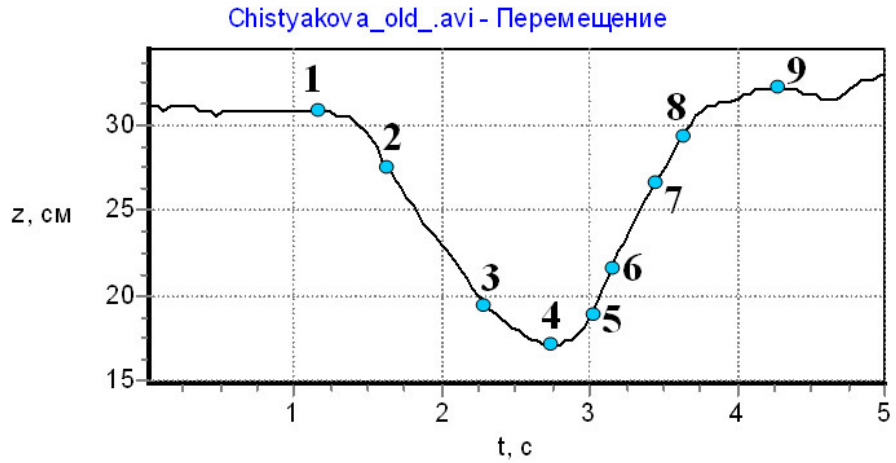
Fig. 2.32. Trajectory of the bar's center of mass during the bench press. (V. Fetisov, 2008)

Frame 5. After the "press" command by the Chief referee, the athlete makes a maximally explosive push using her upper torso muscles. The starting impulse goes from the legs towards the head. The push is made by the following muscles: pectorals, deltoids, triceps, and lats (see point 5).



1 –starting position, 2 – start of the descent, 3 – bar is 3 cm above the chest,
4 – bar is on the chest, 5 – bar is 2cm above the chest, 6 – bar is 5cm above the chest,
7 – bar is 10 cm above the chest, 8 – bar is 12 cm above the chest,
9 –finish of the ascent

Fig. 2.33. Trajectory of the bar's center of mass in the sagittal plane



1 – 0cm/s, 2 – 20cm/s, 3 – 10cm/s, 4 – 0cm/s, 5 – 13cm/s,
6 – 20cm/s, 7 – 17cm/s, 8 – 13cm/s, 9 – 0cm/s.
1 – starting position, 2 – maximum descent speed,
3 – descent speed starts to decrease, 4 – the bar is on the chest,
5 – the speed is 13cm/s, 6 – maximum ascent speed, 7 – speed "drop",
8 – sticking point, 9 – ascent finish

Fig. 2.34. Speed of the bar's center of mass (V. Fetisov, 2008)

Frame 6. The bar is about halfway above the chest. The ascent speed is maximum – 42cm/s (see point 6).

Frame 7. Since the athlete cannot maintain the maximum speed for a long time, it goes down to 32cm/s (see point 7).

Frame 8. The bar's center of mass speed goes down to 8 cm/s fast (point 8). The hardest moment of the bench press, the sticking point, occurs. The athlete goes through the sticking point by concentrating her will power and muscle strength and presses the bar until the full arm lockout (see frames 7 and 8).

Frame 9. The final bar position. After full and simultaneous lockout of both arms, the athlete fixates the bar in place waiting for the “rack” command from the Chief referee. The bar ascent trajectory is 36 cm long. The ascent length is higher than descent length because of the curves in the trajectory. The height is almost the same.

The descent length is 1.3s, the pause at the bottom is 0.2s, the ascent length is 1.3s, the starting phase is 0.6s, the sticking point takes 1.0s, the press out takes 0.5s.

The maximum descent speed is 15cm/s.

The maximum descent acceleration is 80cm/s^2.

The maximum ascent speed is 18cm/s.

The maximum ascent acceleration is 80cm/s^2.

2.4.7.3 A. Babin’s bench press technique

Alexander Babin is an International Master of Sport of Russia, multiple-time champion and record holder of the Bashkortostan Republic.

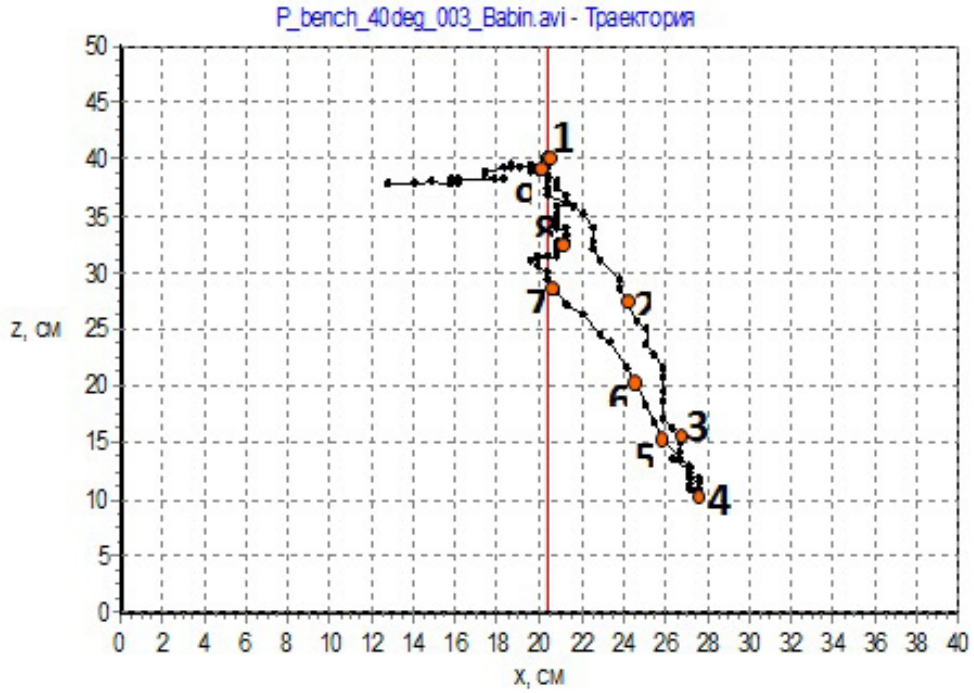
Selected frames of the bench press (A. Babin, 90kg weight class, 245kg on the bar)

Frame 1. The starting position: fingers grip the bar, thumbs are on the opposite side "locking" the bar. The grip width is 81 cm – the maximum allowed by the competition rules. With such grip width the main load is on the pectorals. Head, shoulders, and glutes are in contact with the bench. There is a relatively small back arch. His leg and back muscles are contracted. The sole and heels of the shoes fully touch the platform. Waiting for the "start" signal by the Chief referee, the athlete has fixed the bar in place with his elbows locked.

Frame 2. The bar is halfway to the athlete's chest. At point 2 (see the vertical displacement and speed graphs) the descent speed is maximum – 30cm/s.

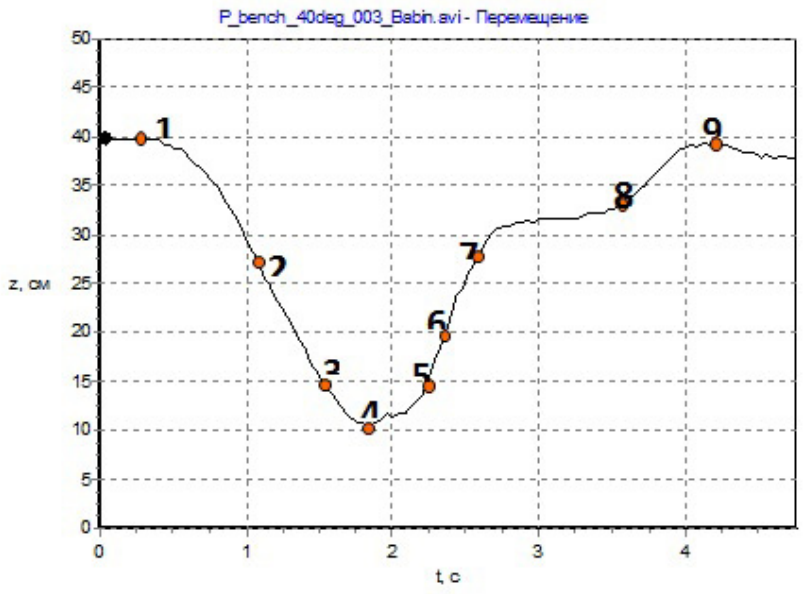
Frame 3. The bar is 5 cm above the chest (point 3 on the graphs). The descent speed goes down to 25cm/s.

Frame 4. The bar is at the chest. The descent speed is 1.25s. The trajectory length is 32cm. The pause length is 0.25s (the pause is so small that it is barely noticeable, but it can be easily seen on the graph).



1 –starting position,
2 – middle of the descent,
3 – end of the descent (5 cm above the chest)
4 – bar is on the chest,
5 – start of the ascent (5cm above the chest),
6 – middle of the ascent,
7-8 – sticking point,
9 – end of the ascent

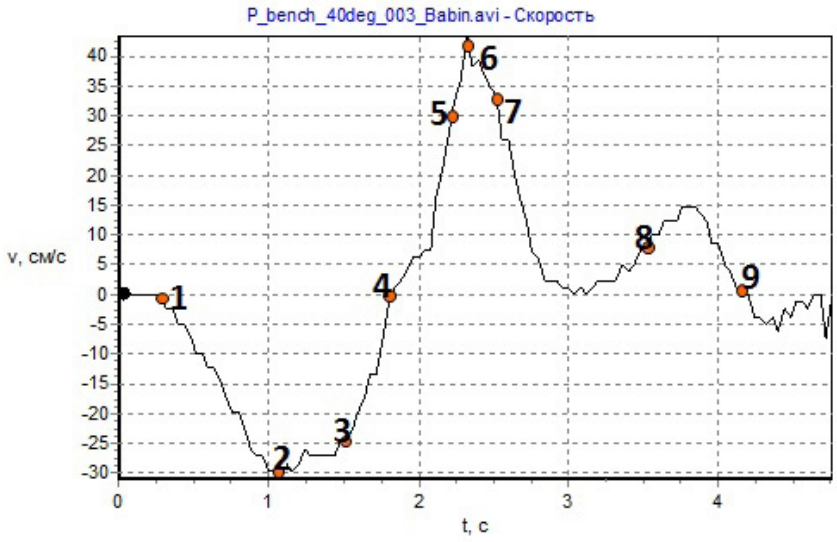
Fig. 2.35. Trajectory of the bench press.



1 –start of the descent, 2 – maximum descent speed,
3 start of the descent slowdown, 4 – bar is on the chest,
5 – bar is 5cm above the chest, 6 – maximum ascent speed
7-8 – speed drop, 9 – end of the ascent

Fig. 2.36. Vertical bar motion graph.

Frame 5. After the “press” command by the Chief referee, the athlete makes an explosive push using his upper torso muscles. The starting impulse goes from the legs towards the head. The following are involved in the push: pectorals, deltoids, triceps, and lats. At point 5, the bar is located 5 cm from the chest and its speed is 30cm/s (see point 5).



1 – 0cm/s, 2 – 30cm/s, 3 – 25cm/s, 4 – 0cm/s, 5 – 30cm/s,
6 – 42cm/s, 7 – 32cm/s, 8 – 8cm/s, 9 – 0cm/s.

1 –start of the descent, 2 – maximum descent speed,
3 start of the descent slowdown, 4 – bar is on the chest,
5 – bar is 5cm above the chest, 6 – maximum ascent speed
7-8 – speed drop, 9 – end of the ascent

Fig. 2.37. Speed of the bar

Frame 6. The bar is about halfway above the chest. The ascent speed is maximum – 42cm/s (see point 6).

Frame 7. Since the athlete cannot maintain the maximum speed for a long time, it goes down to 32cm/s (see point 7).

Frame 8. The speed goes down to 8 cm/s fast (point 8). The hardest moment of the bench press, the sticking point, occurs. The athlete goes through the sticking point by concentrating his will power and muscle strength and presses the bar until the full arm lockout (see frames 7 and 8).

Frame 9. The final bar position. After full and simultaneous lockout of both arms, the athlete fixates the barbell in place waiting for the "rack" command from the Chief referee. The bar ascent trajectory is 36 cm long. The ascent length is higher than descent length because of the curves in the trajectory. The height is almost the same. The overall ascent length is 2.1s where 0.6s account for the starting phase.

2.5. Deadlift technique

The deadlift is the third and the final competition exercise in powerlifting. According to the majority of specialists, it is the main exercise because often the deadlift result plays the decisive role in the athlete's success or failure.

2.5.1 Deadlift rules and regulations

Technical rules and order of execution in deadlift are approved by the Russian Powerlifting Federation presidium and comply with the International Powerlifting Federation rules:

1. The athlete must face the front side of the platform. The bar, which is located in front of the athlete, can be held with both hands using any grip and is lifted up until the athletes is standing straight.
2. At the end of the lift, the knees must be fully locked out and the shoulders must be pulled back.
3. The Chief referee's signal consists of the hand down gesture and a clear "down" command. The signal is not given unless the bar is fixed and the athlete is in the clear final position.
4. Any deliberate attempt to lift the bar is considered an attempt. After starting to lift the bar, it cannot move down until the athlete is standing erect with fully extended legs. If the bar goes down slightly during the backward shoulder motion (a slight downward motion at the end of the lift), there is no reason to count the lift as unsuccessful.

2.5.2 Deadlift phases structure

The segmentation of the deadlift moves, just like in the other two competition exercises, is based on identifying parts of the lift with a definite motor tasks. Let's look at the tables 2.9 and 2.10 and analyze the opinions on the deadlift phases of foreign and local specialists.

Table 2.9

Deadlift phases suggested by foreign specialists (L. Ostapenko, B. Sheiko, 2003, 2010)

Author / Phases	Phase 1	Phase 2	Phase 3	Phase 4
Bill Jamison (USA)	Gripping the bar and assuming the starting position	Lifting the bar	Holding the final position	Putting the bar down
John Lear (Canada)	Starting position	Lifting the bar	Holding the final position	Putting the bar down
Ladislav Philip (CSSR)	Starting position	Lifting the bar	Holding the final position	Putting the bar down
David Pasanella (USA)	Starting position	Lifting the bar	Holding the final position	Putting the bar down
A.Stetsenko (Ukraine)	Assuming the starting position	Lifting the bar	Holding the final position and putting the bar down	

The analysis of table 2.9 shows that foreign specialists differentiate between four deadlift phases, which doesn't give us the full representation of the technique.

Table 2.10

Deadlift phases suggested by Russian specialists (B. Sheiko, 2005, 2011)

Author / Phases	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
I. Belsky	Start	Lift	Lockout			
Y. Nazarenko S. Tyo	Start	Start of the pull	Pull	Lockout		
E. Kudelin	Preparation phase	Start	Main phase	Final phase		
S. Smolov	Preparation phase	Dynamic start	Start of the pull	Pull	Lockout	
R. Tsedov	Assuming the starting position	Start of the pull	Pull	Assuming the final position	Putting the bar down	
A. Ermachenko	Preparation phase	Starting position	Starting phase	Main phase	Final phase	
B. Sheiko	Assuming the starting position	Starting position	Start of the pull	Pull	Lockout in the final position	Returning the bar on the platform
L. Ostapenko	Assuming the starting position	Gripping the bar	Start of the pull	Getting into the final position	Holding the final position	Returning the bar on the platform
B. Shetina	Starting position	Athlete-bar interaction before it leaves the platform	Starting acceleration of the bar	Final effort	Lockout	Returning the bar on the platform

B. Shetina (2008) offers two more additional phases: coming to and leaving the platform. Russian specialists have a wider range of deadlift phases consisting of three to six phases.

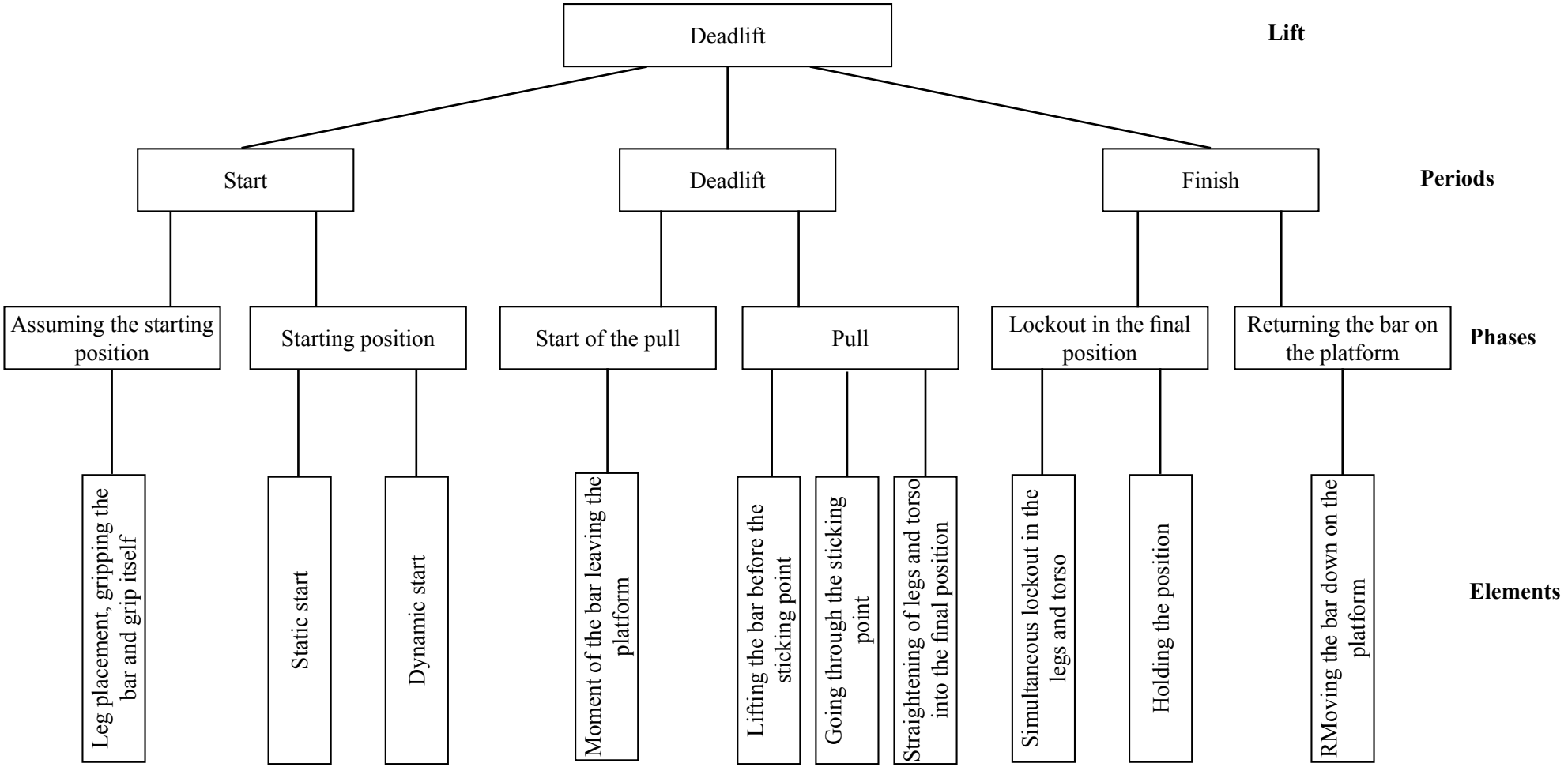


Fig. 2.38. Kinematic structure of the deadlift (B. Sheiko 2005)

2.5.3 Technique and structure of deadlift[3]

The time structure of the deadlift movement consists of three periods, six phases, and ten elements.

The first period is the start. It consists from two phases (assuming the starting position and the starting position itself) and three elements.

The correct starting position in the deadlift is very important. It greatly contributes to the rational usage of the athlete's physical abilities and successful completion of the subsequent movements. According to V. Rodionov (1967), "the starting position influences the bar path, amount of effort required from the athlete, amount of muscle involvement, speed and technique."

In powerlifting, there are three deadlift styles: conventional, sumo, and medium. The conventional style involves foot placement at hip width or a bit narrower, just like in weightlifting (photo 2.42).

Photo 2.40 Fedenko A.

Photo 2.41 Medvedev I.

Photo 2.42 Gadiev R.

Athletes who use the sumo style place their feet as wide as possible (photo 2.40). Those who prefer the medium style, place their feet narrower than in sumo, but wider than in weightlifting style (photo 2.41).

Phase 1. Assuming the starting position

The first phase lasts from approaching the bar to assuming the proper position on the platform.

The motor task of Phase 1 is to create the optimal conditions for assuming the starting position. According to L.N. Sokolov (1972), the main goal of the start is to assume a position that will let the athlete use physical abilities to lift the bar in the most efficient manner.

Elements:

Foot placement - The athlete's base of support is determined by the foot placement in starting position. Foot placement, just like in the squat, varies from classic – feet at hip width, to the widest possible foot placement in the sumo style, with many intermediate possibilities. In any case, it should provide the proper acceleration to the bar and support the "athlete-barbell" system (photos 2.40, 2.41, 2.42). In addition, the wider the foot placement is, the smaller the deadlift height and the range of motion are. The deadlift height depends on the foot placement and length of athlete's legs, torso, and arms. The wider the foot placement is, the closer to vertical the athlete's back must be. The feet are placed in such a way under

[3] This chapter uses works by Mark Rippetoe, A. Samsonova, N. Kichaikina, G. Samsonov, B. Sheiko and others.

the bar that its vertical projection is on the metatarsophalangeal joints. The feet should be turned out at 45-60 degrees. The athlete, maintaining the straight back, goes down, bends the legs, and pushes the knees out. The hips comes closer to the bar. The back is straight. Ankle, knee, and hip angles during the starting position depend on the athlete's morphology (relative lengths of his body parts), so it is impossible to give exact recommendations for everyone. However, there are general principles that are important for everyone. One can recommend to:

a) shorten the distance between the shins and the bar (they should almost touch);
b) place the shoulders joints above the bar. Small deviations are allowed.

The differences are mainly in foot placement relative to the bar, since it depends on an athlete's morphologic features. Most authors think that the center of metatarsophalangeal joints should be placed on the same vertical line as the bar. A. Mulchin (1971) theoretically proved that, depending on the athlete's body type, the projection of the bar center should be between the metatarsophalangeal joints and the middle of the foot.

The shins should slightly lean forward and touch the bar. Several Russian researchers have come to the conclusion that the closer the bar is to the shins, the greater the pressure on the floor is, and the center of the metatarsophalangeal joints is the final point of the support, so it is unreasonable to place the bar beyond this point. Stone and O'Brien (USA, 1987) think that, from the mechanical standpoint, the closest position of the bar to the athlete should help exert the maximum force on the bar. Farley (1995) has a similar opinion. He writes that at the starting position, the bar is usually located quite close to the shins. However, there are several works by Russian and foreign authors stating that the bar being too close to the shins may cause excessive deceleration of the bar and, thus, lower the attempt's efficiency.

In all the types of the starting position the knees should face outward just like the feet. This puts the bar close to the hips horizontally, lowers the torque of the bar relative to their common axis, and, thus, makes it easier to lift it.

Assuming the starting position, the athlete bends the legs, leans forward with the torso, and grips the bar with the hands.

Grip width - The distance between the hands on the bar is called grip width. In the sumo and the medium deadlift styles the hands are inside the knees (photos 2.43, 2.44). In the conventional deadlift the hands are outside the feet (photo 2.45).

Grip type - The way of holding the bar with the hands is called grip type. Athletes mostly use two grip types – mixed grip, with the hands facing different ways, and double overhand grip, with four fingers on the other side of the thumb. Some athletes don't use the full grip, holding the bar in the phalanges of the fingers, just like in "hooks".

Photo 2.43 Mixed grip

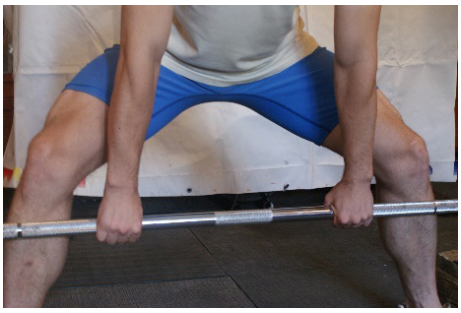


Photo 2.44 Double overhand grip

Photo 2.45 Finger grip

The grip width should be large enough to be outside the hips in the lockout position.

Head position - In the starting position, the athlete's head should be slightly raised. The athlete should look forward. In this particular head position, the tonic neck reflexes improve the tone of torso extension muscles. This makes it easier to maintain tight torso and increases the activation of the mentioned muscles. The studies show that athletes with such head position apply 9% more force at the start compared to the lowered head position. It also lowers the tone of arm flexors, making the "suspension" of the bar more solid, and thus, making it harder to use the arms prematurely.

The head position in the deadlift significantly influences the technique. Any change in the head position leads to redistribution of muscle tone. Its movement relative to the torso in different parts of the lift should contribute to such redistribution of muscle tone that will create good conditions for rational technique.

Torso and shoulders position - Athletes position their torsos differently both in the starting position and in the deadlift itself. Some keep their lower back arched during the lift; others keep it arched at the start, but round it slightly during the exercise; others keep it slightly rounded both at the start and during the lift. Experiments performed by weightlifting specialists have showed that in all cases the arched lower back leads to 8.3% higher force at the start. When the flexion of the erector arches the lower back, it makes the torso act as a rigid lever. This leads to better force transfer from leg and torso extensors to the bar. Regarding the shoulder position, many weightlifting specialists agree that the middle of the shoulders should be exactly above the middle of the bar at the start of the pull.

Mark Rippetoe (2007) expresses a contrary opinion, stating that every correct pull off the platform – regular, clean or snatch (in weightlifting)

deadlifts start with the bar right below the scapulae and in front of the shins. Such position puts the shoulders slightly in front of the bar and the arms behind the bar at a small angle. This is a property of the musculoskeletal system and it works even with bad technique: if the bar is too far from the shins instead of close to them, it will still leave the platform being right below the scapulae.

Phase 2. The starting position

Phase 2 starts with bending the legs and reaching down with the torso, and ends with assuming the starting position.

The motor task of Phase 2 is to assume a rational position for rigid and effective interaction between the elements of the kinematic chain of the athlete's body. The start is divided into consecutive static and kinematic periods.

Static start is the position where an athlete starts interacting with the bar. In this position, the athlete and the bar don't interact yet and have no common center of mass.

In the dynamic start, however, the athlete has actually started lifting the bar, balancing its weight with some force, but it is still on the floor. At this time, the "athlete-barbell" system gets a common center of mass, which is above the athlete's feet. As soon as the athlete starts performing the deadlift, he starts interacting with the bar as well, forming the "athlete-bar" system that tries to gain a common center of mass. This moves the athlete and bar's centers of mass above each other, which explains the change in the athlete's position. Thus, a dynamic start is an instantaneous position of the athlete right before the bar leaves the floor.

At the start, some athletes do preparatory actions in the vertical plane in order to make it easier to lift the bar off the platform. After assuming the grip, they lift their hips high until their

torso is in a horizontal position, and then perform the reverse action with increasing speed. When the shoulders are right above the bar, the legs start extending.

It is important to remember that the transition from the preparatory actions to the pull itself is difficult. Thus, such actions should be practiced only after the athlete has mastered the proper technique without them.

At the start, the athlete places the feet in the most comfortable and natural position; the bar is above or close to being above the metatarsophalangeal joints. The feet are turned outward. The knees are turned in the same directions. The shins are straight, touching or almost touching the bar. The center of mass is above the middle of the feet. The back is straight or slightly rounded. The knee angle is 60-90 degrees – so that the hips can be positioned at knee level or above. The arms should be straight at the start. The shoulders are lowered and located above the bar or slightly forward. The chin is slightly lifted up to help maintain proper back position. Eyes look up and forward.

The weightlifting style of the starting position requires from the athlete to have stronger erectors because the back is more parallel to the floor. The sumo style has more straight and vertical back position and a great load goes to quads (Piper, Waller, 2001).

A. Mulchina (1971) writes that the starting position depends on the athlete's height, limb and torso lengths, torso and leg extensor development, body proportions, and foot placement width. Athletes of different height have different angles between the limbs in the starting position. Shorter athletes have higher knee angles, so in the first deadlift phase they mostly use torso extensors. Average and taller athletes use leg extensors to give the bar the initial speed.

There may be infinite starting positions, but in any of them the athlete slowly increases the force on the bar before it leaves the floor. It is important to never forget the main and required condition: at the very last moment before the bar leaves the floor, the athlete should assume a position where the shoulders are right above or almost above the bar.

The balance of the "athlete-barbell" system is very important in the deadlift. The best condition of such a balance is the position of the common center of mass of the system (further referred to as CCMS) above the middle of the foot. B. Sheiko (2001) and R. Tsedov (2003) agree that during the "assuming the starting position" phase the athlete should assume such a position that CCMS is above the middle of the foot. According to them, this is one of the basic conditions of the optimal deadlift execution.

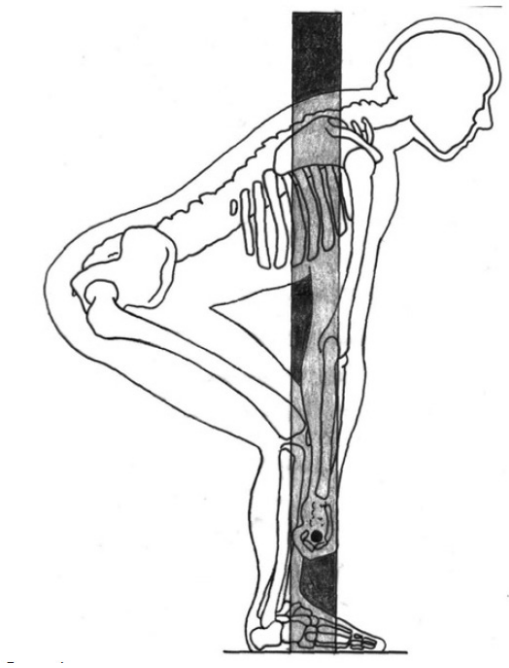


Fig. 2.39.

The bar leaves the floor when it is right above the middle third of the foot (Fig. 2.41).

Why is it the middle third of the foot?

Because the load is distributed across the entire sole of a foot if the weight is balanced and the load is equal to the total weight of the athlete and the barbell acting on the floor right under the "athlete-barbell" system's center of mass.

In the deadlift, the bar is always hanging in the hands and is always under the scapulae when it is below the knees, so the torso angle is mostly predetermined, although it still varies depending on the anthropometry.

If the weight is shifted to the toes or heels, the system's center of mass shifts either forward or backward to achieve the best balance. When the bar is light, bodyweight is the main factor in the balance. When the bar gets heavier as you get stronger, its contribution to the system is greater, and the system's center of mass gets closer to the bar's center of mass.

Thus, the main goal of the athlete assuming the starting position is to optimally align the levers of the body and create a tight interaction between the elements of the kinematic chain, ground, and the bar, which will contribute to the lift of the bar later.

The second period is the deadlift itself. It consists of two phases (start of the pull and the pull itself) and four elements.

Phase 3. Start of the pull

The third phase begins with the start of applying the force to the bar and ends with the bar leaving the floor.

The motor task is to create a tight interaction between the elements of the kinematic chain, ground, and the bar before it leaves the floor, which will contribute to the lift of bar later on. You make a quick inhale and hold your breath before the bar leaves the floor. I. Seropegin (1965) proved that power movements are the most effective when the breath is held for a short time. He discovered that the pull strength is the highest when the lungs contain the amount of air equal to 3/4 of their vital capacity.

The start of the pull (photo 2.46) is on the border between the first and second deadlift phases. At this moment, the athlete and bar become one system. The pull happens due to the initial impulse created by straightening of the legs. Not only the knees, but also the hips extend. The ankles extend slightly as well. Hips and shoulders rise with the same speed, back muscles remain tight. The bar is as close to the shins and quads as possible.

Photo 2.46. Start of the pull – Yury Fedorenko, Honored Master of Sport

L. Sokolov (1967), who studied muscle force in weightlifters, used a dynamographic platform for its measurement in isometric regime. He came to the conclusion that the force increases from the starting position. The data supports the idea that the kinematic chain lever positions influence the athlete's ability to produce force.

Some athletes do grip the bar correctly, they hold it by flexing their arm and upper body muscles. The goal is to have your arms straight and hardly use them during the lift, making them only a connection between the bar and the upper body (shoulder joints).

Phase 4. The pull itself

This phase lasts from the bar leaving the floor to the full leg and back extension.

The motor task of the movement is to give the bar enough speed and an optimal trajectory by constantly increasing the force applied to it.

Elements:

Lifting the bar before the sticking point - The bar smoothly leaves the platform, but further on its motion should have constant acceleration, i.e. the lift should happen with constantly increasing force. Under no conditions should the bar be jerked off the platform. Maximum force should be applied during the entire pull.

The main goal in this phase is to develop enough force for the bar to leave the floor without disturbing the forward-backward balance in the new conditions. This requirement should be met throughout the entire lift. It will provide efficient use of energy and functional abilities of the athlete's body and let him or her fully show the technical mastery. If the center of mass moves forward or backward relatively to the middle of the foot, the athlete will have to compensate for this motion trying to keep the balance.

After the bar is halfway up, when the legs are almost straight, the upper body muscles start working dynamically in order to straighten the torso. Practically, the torso extension and forward movement of the hips happen when the bar is at the knee height.

Going through the sticking point - This is the most difficult part of the deadlift. When the bar is at knee level, it is very hard to maintain exact balance. Fast leg extension makes the knees go backward and leaves the bar hanging in front. As a result, the bar's center of mass moves forward relatively to the athlete's center of mass and many can't continue the motion at this point. Thus, it is important to keep the bar as close to the athlete's center of mass as possible, and keep its speed at least 0.1m/s when it is at knee level.

Straightening of the legs and torso into the final position - In the last third of the pull traps get involved in moving the shoulders back. The head is raised slightly, the upper back moves backwards. The main difficulty is holding the bar.

The third period is the finish. It has two phases (lockout in the final position and returning the bar on the platform) and three elements.

Phase 5. Lockout in the final position

The fifth phase starts with the maximum leg and back extension and ends with holding the bar in place until the "down" command.

The motor task of the phase is to keep the fixed body position until the "down" command according the competition rules.

Elements:

Simultaneous full leg and torso extension - During this element the legs and spine extend at the same time because of the traps and rhomboids. The back is straight. To prevent the back from hyperextension the athlete should contract the glutes. At this point one can exhale a little bit to decrease the intra-abdominal pressure.

Holding the position - Holding the final position starts with the scapulae moving towards each other (shoulders moving behind the bar), hips moving forward, and ends with the athlete holding the position for a short amount of time.

Phase 6. Returning the bar on the platform

The sixth phase lasts from the start of the bar's downward motion until it touches the floor.

Elements:

Putting the bar down on the platform - The descent is performed like a reverse ascent, but a bit faster, mainly due to decreasing leg work. The grip on the bar is maintained until the bar touches the platform, according to the rules.

2.5.4. Comparative analysis of deadlift styles

During the last 30 years there have been constant arguments regarding the best deadlift style. For example, S. Smolov (1990), A. Surovetsky (2000), B. Sheiko (2003) think that the sumo deadlift is more efficient than the conventional deadlift. Fredrick Hatfield (1982), Mike Lambert (1988), Farley K.(1995), Bill Starr (2000) have a contradictory opinion. Other specialists (V. Protasenko, 2003) say that the sumo style is beneficial only for short athletes of light weight classes.

McGuigan and Wilson (1996) looked at biomechanical efficiency of the sumo deadlift. They state that the trajectory is minimized due the involvement of more different muscle fibers. Their research showed that the sumo style decreases the trajectory by 19%. McGuigan and Wilson (1996) also came to the conclusion that in the sumo deadlift the bar is much closer to the body than in conventional deadlift. Since power is defined as work divided by time, so the power needed to perform a deadlift depends on the ascent time. They also found out that it takes about 2 seconds to perform a deadlift using both styles, so they are equally efficient in terms of power.

The biomechanical analysis conducted by V.S. Fetisov and B.I. Sheiko after 2008 Russian nationals showed that the sumo deadlift takes 2.74 seconds on average. The shortest deadlift took 1.5 seconds (A. Bykov, 75 kg). The longest one took 7.5 seconds (S. Omelchenko, 125 kg). The conventional style deadlift took 3.85 seconds on average, the shortest one being 2.4 seconds (D. Reshetov, 110 kg), and the longest one being 7.75 seconds (Y. Vavilova, 52 kg). Moreover, the analysis showed that the knee angle is much larger in athletes who use the sumo style rather than the conventional style. This leads to the conclusion that the sumo deadlift requires mostly quads and knees activation while the conventional deadlift uses spinal erectors to start the pull.

American specialists (Sholevsky et al., 1991) found out that the sumo stance makes athletes not only keep the bar closer compared to the conventional stance, but also reduces the stress of the external force, and relatedly the lower back range of motion. Keeping the bar as close as possible should help to produce more force from the mechanical point of view (Stone and O'Brian, 1987). Grabiner, M. and Garhammer, J (1989) share the same opinion and note that the most important rule in the deadlift is to keep the bar as close as possible throughout the entire lift. Such an action helps shorten the moment of lever arm, which as a result will lead to a significant reduction of the moment of force.

Athletes that use the sumo technique can have either straight or S-shaped bar path, while athletes that use the conventional technique can only have S-shaped bar path.

The deadlift analysis by However, McGuinan, and Wilson (1996) showed that half of the athletes who use the sumo style have a sticking point in the second phase, compared with 15% of the athletes who use the conventional style. However, foreign specialists do not all agree with this opinion. We have found that the athletes who use the sumo style have a sticking point about 5-7 cm above their knees. The athletes who use the conventional style have a sticking point earlier, 5 cm below or at the knee level. McLaughlin et al. (1997) stated that the sticking point happens at the position where the most effective muscles are in the

least effective biomechanical positions. The hardest part of the pull is the moment of the bar passing the knees, since it is very difficult to keep balance at this point. Many athletes cannot continue the motion from this point. One should try to achieve the bar speed of at least 0.1m/s and turn the knees out in order to create the best conditions for a successful pull (S. Smolov, 1990).

The conventional style requires greater involvement of the spinal erectors, since the body is leaning forward more. The sumo style is performed with straighter back and quads do most of the work (Piper T., Waller M., 2001).

It is useful to analyze Russian deadlift records until December 2010, which shows that all of them were set using the sumo style (see table 2.11).

Table 2.11

Russian Records (RPF) on 31.12.2010 (B. Sheiko, 2011)

Name	w/ class, kg	From	Weight, kg	Date	Location	Comp.	Style
Sergey Fedosienko	56.0	Novosibirsk	261,0	16.02.2010	Chelyabinsk	RN	sumo
Mikhail Andryukhin	60.0	Serpukhov	285,0	25.08.2005	Oryol	RC	sumo
Vladimir Pak	67.5	Moscow	300,5	25.08.2005	Oryol	RC	sumo
Vladimir Pak	75.0	Moscow	330,5	02.03.2006	Ufa	RN	sumo
Oleg Chvanov	82.5	Uray	340,0	16.03.2007	Berdsk	RN	sumo
Andrey Belyaev	90.0	Kirov	380,0	21.09.2006	Syktyvkar	RC	sumo
Alexey Zvarykin	100.0	Ruzaevka	380,0	14.03.2008	Ufa	RN	sumo
Yury Fedorenko	110.0	Ufa	405,0	04.03.2006	Ufa	RN	sumo
Andrey Malanichev	125.0	Moscow	403,0	28.0820.05	Oryol	RC	sumo
Andrey Malanichev	+125.0	Moscow	402,5	24.08.2008	Berdsk	RC	sumo

The deadlift style choice depends on an athlete's body proportions. One of the main factors of a successful lift is the deadlift technique.

2.5.5 Deadlift techniques of advanced athletes

The deadlift technique analysis of the highest level athletes is performed using video frames with the technique elements' explanations to let us detect individual features of the optimal technique.

2.5.5.1 A.Belyaev's deadlift technique

Andrey Belyaev is an International Master of Sport of Russia, world champion (2007), two-time junior world champion (2002, 2003), bench press world champion (2005), European champion (2006), junior European champion (2003). He holds Russian, European, and world records.

From the given 8 frames it is possible to analyze Belyaev's deadlift technique using the angles defined on Fig. 2.42.

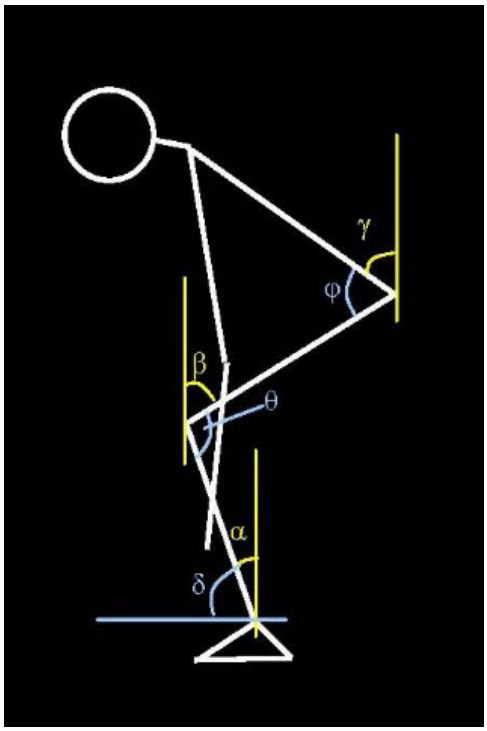
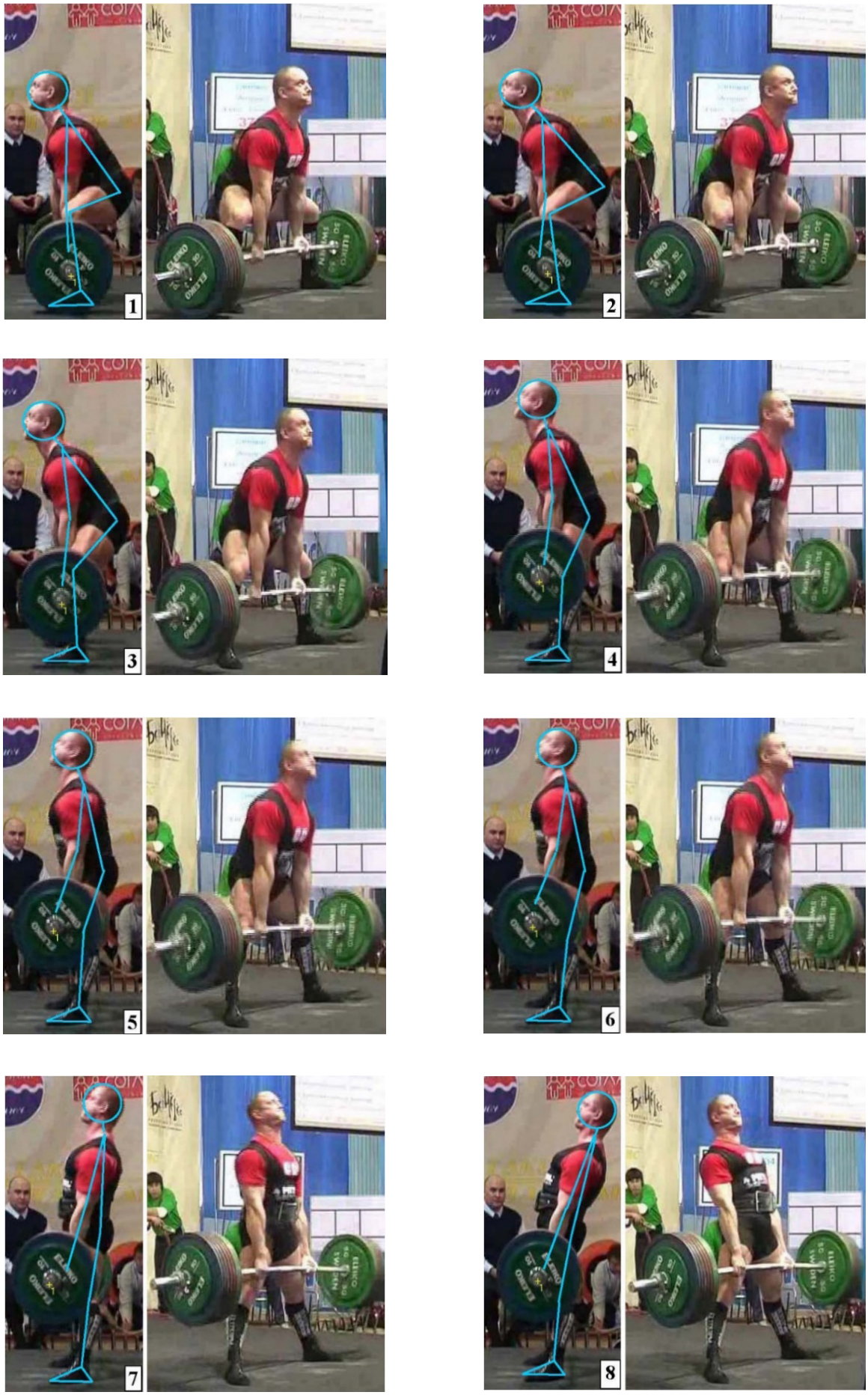


Fig. 2.42:
Frame 1. Starting position
Frame 2. Start of the pull
Frame 3. Bar is at the middle of the shins
Frame 4. Bar is 4-6 cm below the knees
Frame 5. Bar is at knee level
Frame 6. Bar is 3-5 cm above the knees
Frame 7. Highest point
Frame 8. Final position

Fig.2.40. Angle scheme

The athlete's body is not flat, and the movement of the limbs cannot be reduced to the movement only in one plane.

For example, the powerlifter's thigh (especially in the sumo style) moves at an angle relative to the sagittal plane, so the true angles with horizontal or vertical axes are significantly different from the angle between thigh projections on the sagittal plane with the same horizontal or vertical axes. When we look at the athlete from the side, we deal with the projections on the sagittal plane. Thus, all the mentioned angles are between the projections of joints rather than joints themselves.



Selected frames of a deadlift by A. Belyaev, MSIC
(weight on the bar – 375 kg, athlete's bodyweight – 89.6 kg, height – 166 cm)

Frame 1. The starting position

The athlete's legs are in the sumo stance. The distance between the heel centers is 122 cm. The feet are turned out at a 45-degree angle. The knees are turned out in the same direction as the feet. The bar is 1-2 cm in front of the metatarsophalangeal joints. The ankle angle (δ) is 85 degrees. The knee angle (θ) is 105 degrees. The hip angle (φ) is 65 degrees. The shins touch the bar. The grip is mixed. The grip width is slightly narrower than shoulder width. The shoulders are lowered and are in the same vertical plane with the bar. The back is straight. The back angle (γ) is 35 degrees. The hips are above the knees. The common center of mass is above the middle of the foot.

Frame 2. The moment of the bar leaving the platform

The moment of the bar leaving the floor divides the starting position and the pull itself. This is the position that is assumed by the athlete during the moment when the bar leaves the floor. During this and all consecutive moments, the athlete-barbell system has one base of support. Before this moment, the athlete and barbell stability have different bases of support.

When the bar leaves the floor it moves horizontally 1 cm towards the middle of the foot (Fig. 2.43). The ankle angle (δ) is 90 degrees. The knee angle (θ) is 110 degrees. The hip angle (φ) is 80 degrees. The back angle (γ) is 50 degrees. The shoulders are 2 cm away from being above the bar (skeletal frame 2). A. Belyaev starts the pull in a calm manner, without jerking the bar. The starting bar speed is low, about 7 cm/s (Fig.2.44).

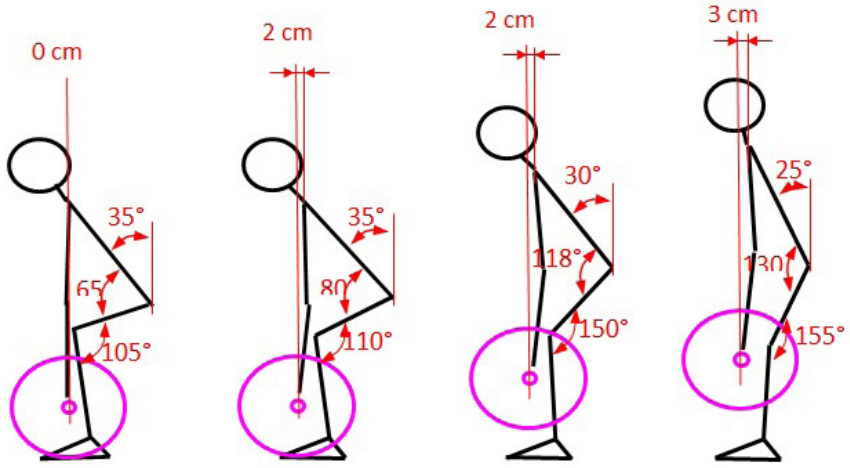


Fig. 2.41. Skeletal model of A.Belyaev's deadlift (V. Fetisov 2009)

Frame 3. The bar is at the middle of the shins

At this point, the bar's center of mass is about 12 cm above the platform. This location corresponds to the middle of the shins. This position has the

following kinematic features: the bar continues getting closer to the athlete by being 3 cm from the vertical line (Fig. 2.44). The bar is being lifted because of active leg extension. In addition to the knees, the hips also extend. The athlete has greatly increased the bar speed, which is 44 cm/s at this point. The ankle angle (δ) is 90 degrees. The knee angle (θ) is 150 degrees. The hip angle (φ) is 120 degrees. Belyaev has involved his back into the lift and the back angle (γ) has decreased to 35 degrees. The athlete's shoulders are 2 cm behind the vertical plane of the bar (skeletal frame 2).

Frame 4. The bar is 4-6 cm below the knees

At this point, the bar is 4-6 below the knees and 19-20 cm above the platform. The speed of the bar is the highest – 47 cm/s. The ankle angle (δ) is still 90 degrees. The knee angle

(θ) is 160 degrees. The hip angle (φ) has increased to 130 degrees. The back angle (γ) has decreased to 25 degrees. Higher hip angle and lower back angle mean that the athlete is actively using his back (skeletal frame 4). The athlete's shoulders are 3 cm behind the vertical plane of the bar.

Frame 5. The bar is at the knee level

The bar is halfway up and at the knee level. It is 26 cm above the platform. The speed of the bar has decreased to 39 cm/s (Fig. 3.30). The ankle angle (δ) is 92 degrees. The knee angle (θ) has increased to 162 degrees. The hip angle (φ) has increased to 145 degrees. The back angle (γ) has decreased to 15 degrees because of the back extension. The athlete's shoulders are 10 cm behind the vertical plane of the bar (skeletal frame 5).

Frame 6. The bar is 3-5 cm above the knees

The bar is 3-5 cm above the knees and 29-30 cm above the platform. The speed of the bar has decreased to 19 cm/s (Fig. 3.36). The athlete has started to experience difficulties. This means that the bar has reached the sticking point. The ankle angle (δ) is still 92 degrees. The knee angle (θ) has increased to 168 degrees. The hip angle (φ) has increased to 155 degrees. The back angle (γ) has decreased to 10 degrees. Throughout the entire pull the bar is as close to the athlete's legs as possible. The shoulders are 11 cm behind the vertical plane of the bar (skeletal frame 6). This can be explained by the back doing more work than the legs at this point.

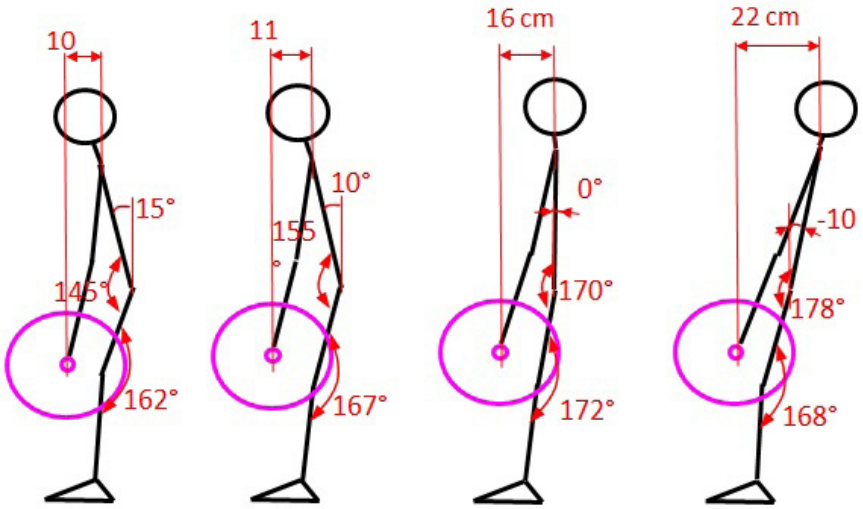


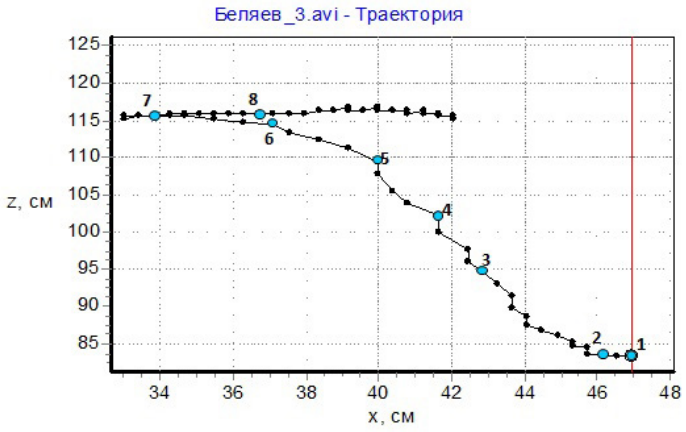
Fig. 2.42. Skeletal model of A. Belyaev's deadlift (V. Fetisov 2009)

Frame 7. The bar is 2-3 cm below the final position

Between frames 6 and 7 the bar speed is minimal and around 2 cm/s (Fig. 2.45). The bar is at its highest point, 33 cm above the platform. The ankle angle (δ) is 92 degrees. The knee angle (θ) has increased to 172 degrees. The hip angle (φ) has increased to 170 degrees. The athlete has fully extended the legs and back. The back angle (γ) is zero.

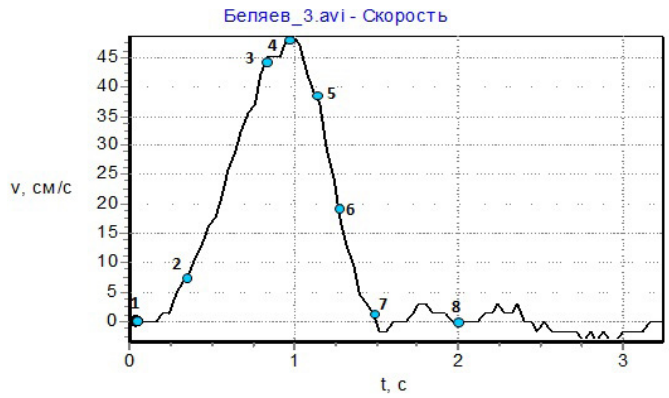
Frame 8. The final position

The lockout is accomplished by extending the back backward while the shoulders are at the same level. The chin is raised. The bar is 33 cm above the platform. The ankle angle (δ) is 90 degrees. The knee angle (θ) is about 168 degrees. The hip angle (φ) has increased to 178 degrees. The back angle (γ) is -10 degrees. The athlete's shoulders are 22 cm behind the vertical plane of the bar (skeletal frame 8). The deadlift time is 1.25 seconds. The maximum bar speed is 47 cm/s. The maximum acceleration is 180 cm/s^2.



1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 4-6 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees
7 – maximum height, 8 – final position

Fig. 2.43. Bar trajectory in the sagittal plane during A.Belyaev's deadlift (V. Fetisov, 2009)



1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 4-6 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees,7 – maximum height,
8 – final position

Fig. 2.44. Bar speed in the sagittal plane. (V. Fetisov, 2009)

As a result of the biomechanical analysis of Andrey Belyaev's deadlift, it can be noted that he shows a great sumo deadlift technique. The athlete assumes the most rational starting position: the leg stance is wide with 122 cm between the heels. The feet are turned out at 45 degree angle. The knees are facing the same directions as the feet. The bar is above the metatarsophalangeal joints. The hips are way above the parallel. The shins touch the bar.

The grip is mixed. The grip width is slightly narrower than shoulder width. The shoulders are lowered and are in the same vertical plane with the bar. The back is straight. The back angle is 35 degrees. The chin is slightly raised, he is looking up and forward. The common center of mass is above the middle of the foot (frame 1). The pull starts with the active engagement of both leg and back muscles (skeletal frames 2-3).

The most important moment in Andrey Belyaev's deadlift is the simultaneous engagement of his back and legs, unlike in many other athletes. The knee angle increases together with the hip angle and the back angle decreases. The coordinated work of the largest (leg and

back) muscles lets Belyaev produce high enough acceleration of the bar (from 7 cm/s^2 to 44 cm/s^2). The maximum bar speed (47 cm/s^2) lets the athlete successfully move past the sticking point. Although at points 5, 6, and 7 the bar speed decreases, it is high enough to finish the pull. Belyaev's deadlift time is 1.25 s. This is the most powerful and fastest deadlift among men competing in 2008 Russian Nationals.

2.5.5.2 M. Baruzdin's deadlift technique

Maxim Baruzdin is a Master of Sport of Russia. This athlete uses the conventional style of deadlift.

Frame 1

Frame 2

Frame 3

Frame 4

Frame 5

Frame 6

Frame 7

Frame 8

Selected frames of a deadlift by M. Baruzdin, MS (V. Fetisov, 2009)
(weight on the bar – 350 kg, athlete's bodyweight – 124.7 kg, height – 175 cm)

Frame 1. The starting position

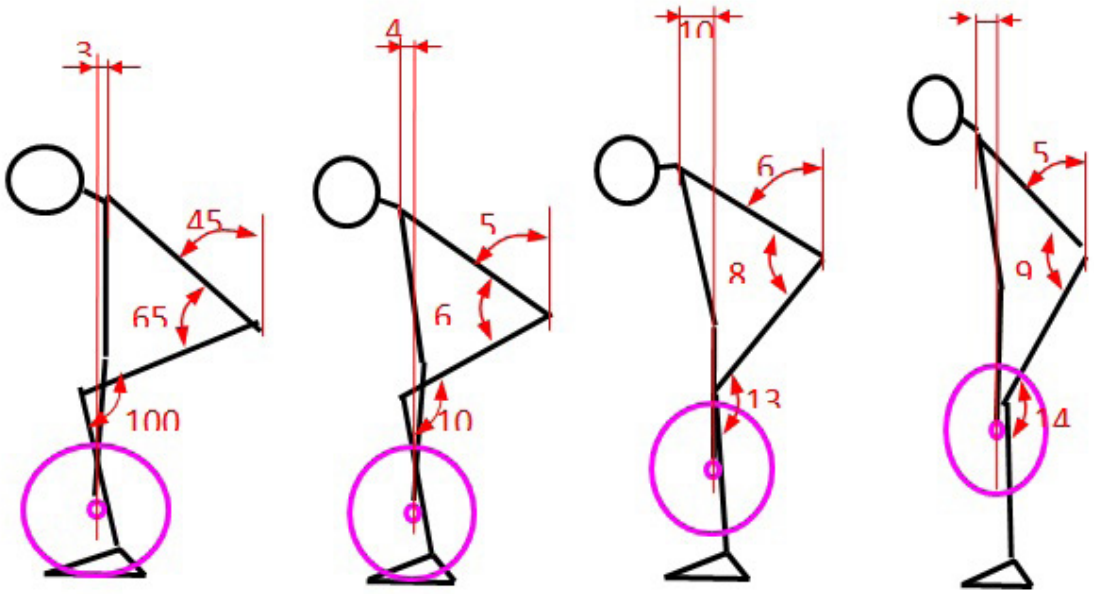
The athlete's legs are in the classic stance. The distance between the heel centers is 35 cm. The feet are turned out at a 5 degree angle. The knees are turned out in the same direction as the feet. The head is looking down and forward. The bar is above the metatarsophalangeal joints. The ankle angle (δ) is 80 degrees. The knee angle (θ) is 100 degrees. The hip angle (φ) is 65 degrees. The shins touch the bar. The grip is mixed. The grip width is slightly wider than shoulder width. The shoulders are lowered and are 3 cm in front of the vertical plane with the bar. The back is rounded. The back angle (γ) is 50 degrees. The common center of mass is above the middle of the foot (see skeletal frame 1).

Frame 2. The moment of the bar leaving the platform

At the start of the pull the athlete actively engages his leg muscles, so that when the bar leaves the floor it moves horizontally 1 cm towards the middle of the foot (Fig. 2.39). The ankle angle (δ) is 85 degrees. The knee angle (θ) increased to 110 degrees. The hip angle (φ) is 65 degrees. The back angle (γ) has increased by 10 degrees and is now 60 degrees. Because of the leg extension and increased back angle the shoulders are now 4 cm behind the bar (see skeletal frame 2). The starting bar speed is insignificant.

Frame 3. The bar is at the middle of the shins

At this point the bar is at the middle of the shins and about 6-7 cm above the platform. This position has the following kinematic features: the bar continues getting closer to the athlete by being 6 cm from the vertical line (Fig. 2.47). The bar is being lifted because of the active leg extension. In addition to the knees, the hips are also extend. The bar speed is at the highest value of 36 cm/s. The ankle angle (δ) is decreased by 1 degree. The knee angle (θ) quickly increases to 140 degrees. The hip angle (φ) also increases to 80 degrees. The back angle (γ) has increased to 60 degrees. Because of the active leg extension the



Frame 1 Frame 2 Frame 3 Frame 4

Fig. 2.45. Skeletal model of M. Baruzdin's deadlift (V. Fetisov, 2009)

following angles have increased: knee (θ), hip (φ), and back (γ) angles. Skeletal frame 3 shows that during the active involvement of the leg muscles the back stays passive and just holds the bar. The shoulders are 10 cm behind the bar (skeletal frame 3). The main load is on the legs.

Frame 4. The bar is 4-6 cm below the knees

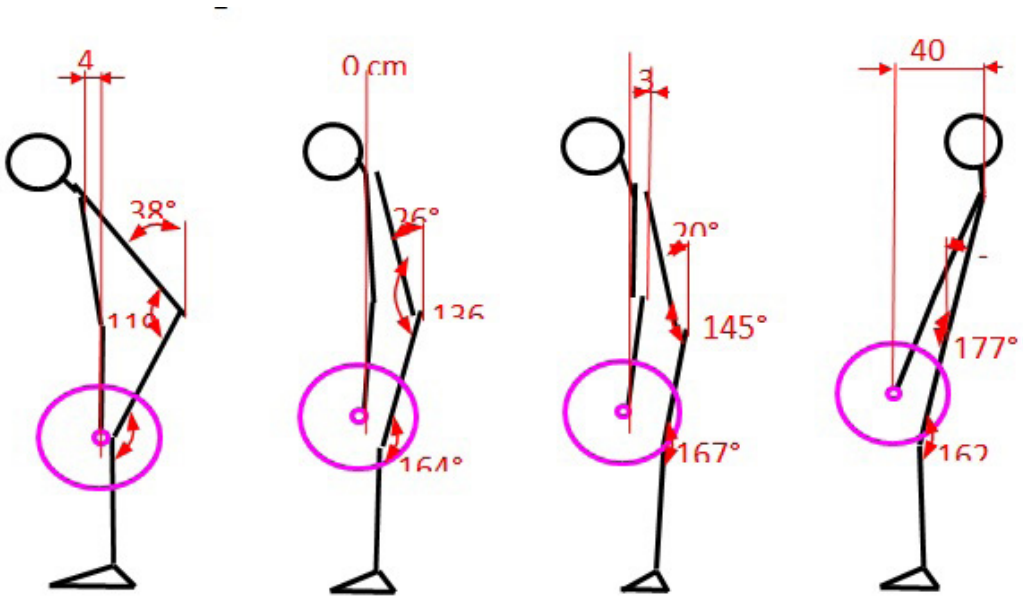
At this point the bar is 4-6 below the knees and 15 cm above the platform. The speed of the bar goes down to 33 cm/s. The ankle angle (δ) has increased to 90 degrees. The knee angle (θ) is 150 degrees. The hip angle (φ) increases to 100 degrees. The back angle (γ) decreases to 50 degrees. While the legs are still actively engaged, the athlete has started to use his back more. The shoulders are now 7 cm behind the bar (see skeletal frame 4).

Frame 5. The bar is at the knee level

The bar is halfway up and at the knee level. It is 30 cm above the platform. The bar has reached the sticking point. The speed of the bar quickly decreases to 11 cm/s. The athlete has started to experience difficulties. The ankle angle (δ) is 90 degrees. The knee angle (θ) has increased to 160 degrees. The hip angle (φ) has increased to 120 degrees. The back angle (γ) decreases to 40 degrees because of the back extension. The athlete's shoulders are 4 cm behind the vertical plane of the bar (skeletal frame 5).

Frame 6. The bar is 3-5 cm above the knees

The bar is 3-5 cm above the knees and 36 cm above the platform. The speed of the bar has decreased to 9 cm/s (Fig. 2.48). The angle (δ) is about the same, 90 degrees. The (θ) angle has increased to 170 degrees because of leg extension at the knees. The hip angle (φ) has increased to 140 degrees. The back angle (γ) has decreased to 25 degrees because of active back work. For the first time during the pull the shoulders are right above the bar (see skeletal frame 6).



Frame 5 **Frame 6** **Frame 7** **Frame 8**

Fig. 2.46. Skeletal model of M. Baruzdin's deadlift (V. Fetisov, 2009)

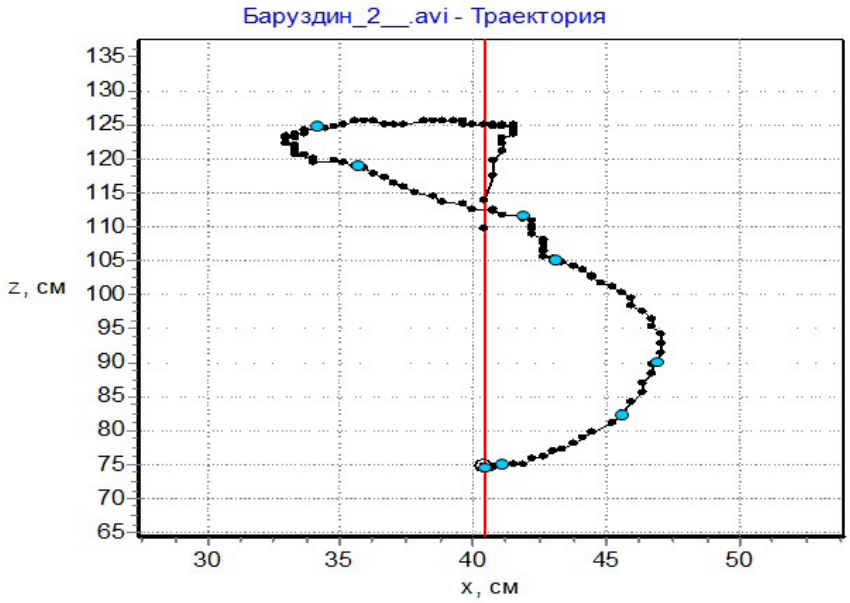
Frame 7. The bar is 2-3 cm below the final position

The athlete has moved through the sticking point and at point 7 the bar speed increases to 19 cm/s. The bar is 45 cm above the platform. The ankle angle (δ) is 90 degrees. The knee angle (θ) is closer to the highest value of 170 degrees. The hip angle (φ) has increased to 170 degrees. The back is still inclined. The back angle (γ) is 20 degrees. The athlete's shoulders are 3 cm behind the vertical plane of the bar (see skeletal frame 7).

Frame 8. The final position

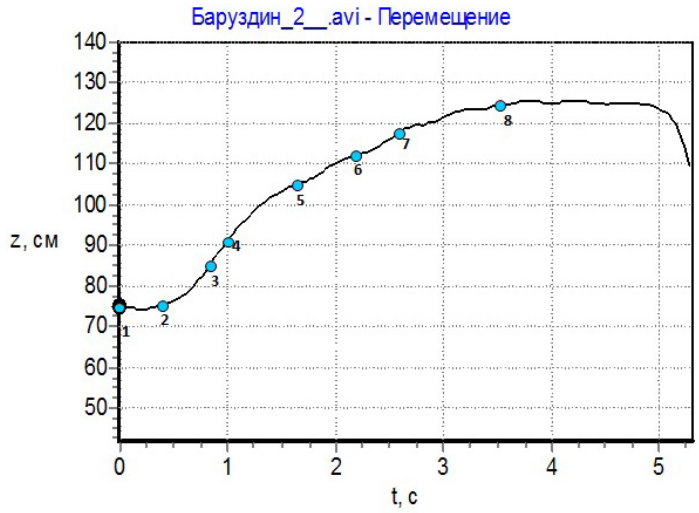
The lockout is accomplished by extending the back. The chin is raised. The bar is 50 cm above the platform. The ankle angle (δ) is 90 degrees. The knee angle (θ) is about 160 degrees. The hip angle (φ) has increased to 180 degrees. The back angle (γ) is 15 degrees. The deadlift time is 3.75 seconds. The maximum bar speed is 36 cm/s. The maximum acceleration is 110 cm/s^2.

The highest bar distance from the vertical bar path is at most 7 cm on each side. This means that M.Baruzdin's pull is S-shaped.



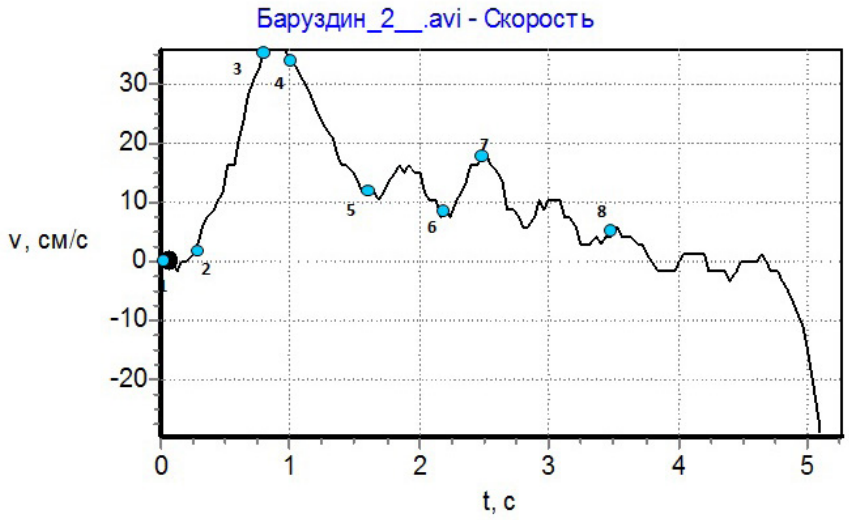
1 – starting position,
2 – moment of the bar leaving the floor,
3 – 12 cm above the platform,
4 – 4-6 cm below the knees,
5 – at the knee level,
6 – 3-5 cm above the knees,
7 – maximum height,
8 – final position.

Fig. 2.47. Bar trajectory in the sagittal plane during M. Baruzdin's deadlift (V. Fetisov, 2009)



1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 4-6 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees,
7 – maximum height, 8 – final position

Fig. 2.48. Bar displacement in the sagittal plane (V. Fetisov, 2009)



1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 4-6 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees,
7 – maximum height, 8 – final position

Fig. 2.49. Bar speed in the sagittal plane. (V. Fetisov, 2009)

The biomechanical analysis of Maxim Baruzdin's deadlift shows that he uses the conventional style (based on his foot placement). The centers of the heels are 35 cm apart. The feet are turned out at 5-degree angle. The knees are facing the same directions as the feet. He is looking down and forward. The bar is above the metatarsophalangeal joints.

The bar is touching the shins. The grip is mixed. The grip width is slightly wider than shoulder width. The shoulders are lowered and are 3 cm in front of the vertical plane with the bar. The back is rounded. The common center of mass is above the middle of the foot. The bar is lifted by means of active leg engagement. When the legs are doing most of the work, the back is passive and is just holding the weight. The athlete's shoulders go 10 cm behind

the bar. The load is mostly on the legs. Maxim Baruzdin's sticking point happens at the knee level, much earlier than Belyaev's one.

The bar path height is 50 cm.
The deadlift time is 3.75 seconds.
The highest bar speed is 36 cm/s.
The highest bar acceleration is 110 cm/s^2.
M. Baruzdin's bar path has a clear S-shaped form.

2.5.5.3 V. Sheglova's deadlift technique

Valeriya Sheglova is an International Master of Sport of Russia, three-time junior world champion, three-time junior European champion, and world champion.

Frame 1

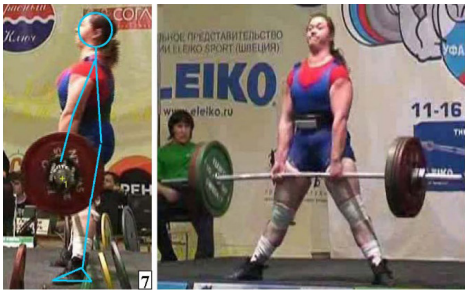
Frame 2

Frame 3

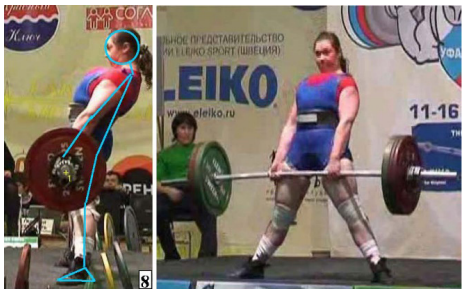
Frame 4

Frame 5

Frame 6



Frame 7



Frame 8

Selected frames of a deadlift by V. Sheglova, MSIC (weight on the bar – 230 kg, athlete's bodyweight – 89.6kg, height – 165 cm)

Frame 1. The starting position

The athlete's legs are in the sumo stance. The distance between the heel centers is 122 cm. The feet are turned out at a 45-degree angle. The knees are turned out in the same direction as the feet. The bar is above the metatarsophalangeal joints. The ankle angle (δ) is 80 degrees. The knee angle (θ) is 120 degrees. The hip angle (φ) is 75 degrees. The shins touch the bar. The grip is mixed. The grip width is at shoulder width. The shoulders are lowered and are in the same vertical plane with the bar. The back is straight. The back angle (γ) is 40 degrees. The athlete is looking up. The common center of mass is above the middle of the foot.

Frame 2. The moment of the bar leaving the platform

The deadlift is a complex interaction system between the athlete and the barbell. It starts when the bar leaves the floor.

At the start of the pull the bar moves horizontally 0.5 cm forward (Fig. 2.52). The ankle angle (δ) is 75 degrees. The knee angle (θ) has decreased to 95 degrees. The hip angle (φ) is 70 degrees. The back angle (γ) is still 40 degrees. Because of a smaller knee angle the hips are now closer to the bar.

Frame 3. The bar is at the middle of the shins

At this point the bar is at the middle of the shins and about 10 cm above the platform. This position has the following kinematic features: the bar continues to get closer to the vertical line. The bar is being lifted because of active leg extension. In addition to the knees, hips also extend. The bar speed is close to the highest value of 35 cm/s. The ankle angle (δ) increases to 80 degrees. The knee angle (θ) is 115 degrees. The hip angle (φ) is 85 degrees. The back angle (γ) is still 40 degrees. Thigh muscles are doing most of the work.

Frame 4. The bar is 3-5 cm below the knees

At this point the bar is 3-5 below the knees and 17-18 cm above the platform. The speed of the bar continues to increase. It is 37 cm/s. The ankle angle (δ) has increased by 5 degrees to 85 degrees. The knee angle (θ) has increased to 130 degrees. The hip angle (φ) has increased to 95 degrees. The back angle (γ) is still about 40 degrees.

Frame 5. The bar is at the knee level

The bar is halfway up and at the knee level. It is 20-21 cm above the platform. The speed of the bar is about the same. Maintaining the same speed of the bar made the athlete pass the sticking point successfully. The ankle angle (δ) is still about 85 degrees. The knee angle (θ) increases to 135 degrees. The hip angle (φ) has increased to 105 degrees. These numbers mean that the legs are still doing a lot of the work. The athlete has started to use her back too. That is why the back angle (γ) decreased to 35 degrees. The bar is 1 cm from the vertical line.

Frame 6. The bar is 3-5 cm above the knees

The bar is 3-5 cm above the knees and 27-28 cm above the platform. The speed of the bar increases to its highest value of 40 cm/s (Fig.2.52). The ankle angle (δ) has increased to 90 degrees. The knee angle (θ) is about 160 degrees. The hip angle (φ) has increased to 145 degrees. These angles have increased because of active leg work. The back angle (γ) has decreased to 15 degrees. The bar got closer to the vertical line by about 0.5 cm. Along the entire pull the bar is as close to the athlete as possible.

Frame 7. The bar is at the highest point

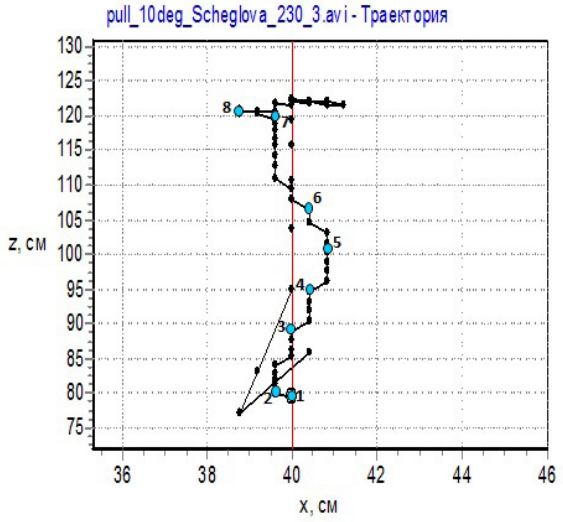
Between frames 6 and 7 the bar speed starts to decrease from its highest value to the average value of the lift (Fig. 2.52). This means that the bar is at the sticking point, which it has passed successfully because of the high speed throughout the pull. The bar is at its

highest point, 40 cm above the platform. The ankle angle (δ) is 90 degrees. The knee angle (θ) is 170 degrees. The hip angle (φ) has increased to 165 degrees. The athlete has fully extended the legs and back. The back angle (γ) is 5 degrees. The bar is 0.5 cm in front of the vertical line, just like at the start of the pull (Fig.2.53).

Frame 8. The final position

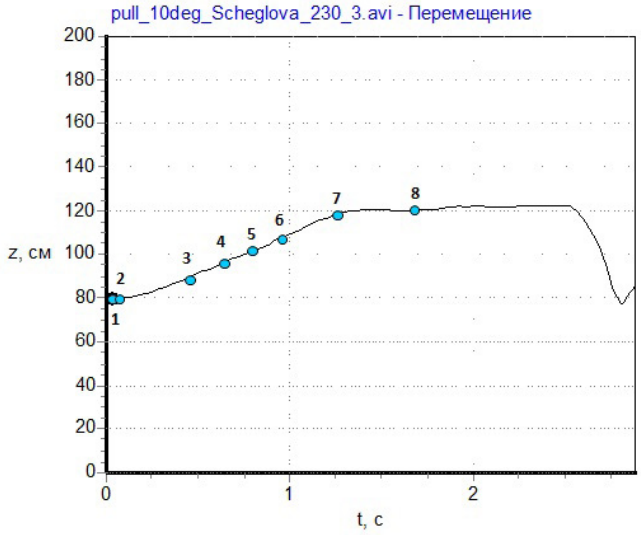
The lockout is accomplished by extending the back and shoulders backward using traps and rhomboid muscles. The chin is touching the chest. The bar is 40 cm above the platform. The ankle angle (δ) is 90 degrees. The knee angle (θ) is about 165 degrees. The hip angle (φ) has increased to 195 degrees. The back angle (γ) is -20 degrees. Leaning backwards makes the hips move slightly forward, so that the bar is 1.5 cm from the vertical line. The deadlift time is 1.25s.

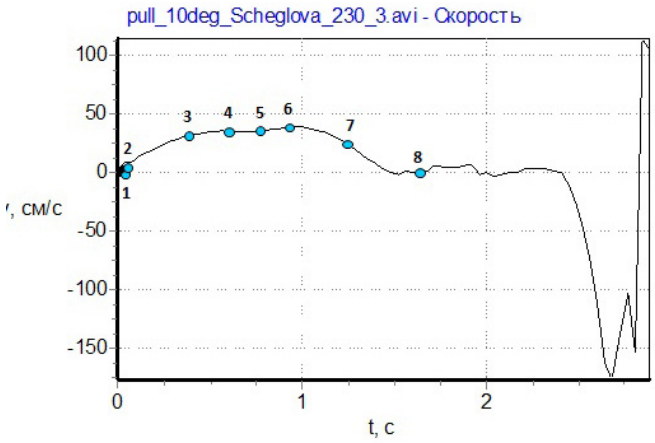
The highest bar distance from the vertical line during the pull is 1 cm. This means that V. Sheglova's bar path is straight.



1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 3-5 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees,
7 – maximum height, 8 – final position

Fig. 2.50. Bar trajectory during V. Sheglova's deadlift





1 – starting position, 2 – moment of the bar leaving the floor,
3 – 12 cm above the platform, 4 – 3-5 cm below the knees,
5 – at the knee level, 6 – 3-5 cm above the knees,
7 – maximum height, 8 – final position

Fig. 2.51. and 2.52. Bar displacement and speed diagrams

Valeriya Sheglova demonstrates an excellent sumo deadlift technique. Valeriya assumes the most rational starting position: the feet are in the sumo stance. The feet are turned out at 45-degree angle. The knees are facing the same directions as the feet. The bar is above the metatarsophalangeal joints. The hips are way above parallel. The bar is touching the shins. The grip is mixed. The grip width is about the same as shoulder width. The shoulders are lowered and are in the same vertical plane with the bar. The back is straight. The back angle is 40 degrees. She is looking slightly up. The center of mass is above the middle of the foot (see skeletal frame 1). The deadlift starts with active leg extension while maintaining the same back angle (see skeletal frames 2-3). The most important aspect of Valeriya's technique is the back engagement at the right time. This lets her not only maintain the speed of the bar, but continue increasing it. This is how she passes the sticking point. The deadlift time is 1.25s. Among 63 participants of 2008 Russian Nationals, only Anna Komlaeva (52kg weight class, 151 cm tall) and Yuliya Medvedeva (67.5kg weight class, 157 cm tall) could finish complete the deadlift in such amount of time. It is important to understand that V.Sheglova is heavier (90kg weight class) and taller (165cm). That's why Valeriya Sheglova has the most powerful deadlift among women. She has an S-shaped pull, but the bar is at most 1 cm away from the vertical line, so we can call the pull vertical.

2.5.5.4 Deadlift mistakes of Stephen Prichard (Australia)

Frame 1. The starting position.

The hips are low and the head is down. This is a bad starting position. It is very hard to start the pull from this position.

Frame 2. The moment of the bar leaving the floor.

To move the bar, Stephen raises the hips and starts the pull. At the start (frame 1) the body position should be the same height as in frame 2.

Frame 3. The bar is 3-5 cm above the platform.

The legs continue the extension. The back muscles are static, they are just holding the bar. Because of the heavy weight the back starts rounding.

Frame 4. The back is still in the static position while the legs continue their extension.

The hips move higher and higher. The back is more rounded.

Frames 5-6 show that the legs are almost fully extended, but the back is still not actively engaged.

Stephen looks like a crane with a long jib.

Frame 7. The bar is at the knee level.

The legs have almost completed their work. The back muscles start to get engaged. They are under colossal load.

Frame 8. The final part of the deadlift.

Stephen's knees are slightly unlocked. The hips are slightly behind. Stephen did not fully extend, so the judge cannot give the "down" command in such a position. These frames (1-8) show that back and leg muscles work separately and not together. This inefficient technique must be taken care of right after the competition in order to learn the proper technique. It will take 2-3 months to correct it. At this very moment the athlete should perform the deadlift in the way he is used to.

Chapter 3. Methods of Teaching Competitive Exercises Techniques

For beginning athletes, as well as those developing towards sports mastership, training mainly consists of learning new skills and mastering the already ones. In this stage the term "educational-training exercise" is applied. That is why learning and training are two sides of the training process. The athletes who have achieved the highest level of sports mastership and take part in Russian and international competitions complete a training process at the level of professional training.

The weakest point of a powerlifters' technical preparation is the process of acquiring and mastering sport techniques. This is caused by the lack of fundamental and advanced methods of this process. This problem explains the existence of even more blank spots in the understanding of the sports technique. In other words, while advanced powerlifters may be strong and may even have a good knowledge of programming, their deep understanding of programming principles and powerlifting techniques is often very limited.

3.1. General Training Principles

The main purpose of training is gaining skills and transforming them into automated and well-executed abilities. A motor skill is characterized by awareness, stability, flexibility, and variability. The stability of a motor skill is based on the conditioned reflex connections solidified during thousands of repetitions. Variability allows for an "on the go" change of the form of common movements based on the purpose of an activity.

Teaching competitive training techniques is connected with solving the problems characteristic to learning any motor activities, whereas training consists of the main principles of sports activity, and is based on common biological regularities of a motor skill formation. Powerlifting training has a few specific features in comparison with other sports techniques, but the general principles that apply to all sports apply to powerlifting as well.

The improvement of sport technique continues throughout an athlete's lifelong training process, which is an integral part of sportsmanship's growth.

The didactic principles, which are the pedagogical basis of teaching and training, express methodological regularities of a pedagogical process; thereby they are obligatory for the implementation of educational and training tasks. These are the principles of awareness, activity, visibility, accessibility and personalization, regularity, and gradual increase of the requirements for maximum physical activity.

1. Awareness and Activity Principle. The effectiveness of an educational process is largely determined by how consciously and actively participants treat an activity. The understanding and creative approach towards tasks speed up the learning process and contribute to the effectiveness of the performed activities. In order to implement this principle it is extremely important to be able to interest an athlete in the subject, explain the studied material clearly, teach the goals and objectives and activate the process of self-study and self-discipline. The ability to communicate clearly (especially verbally) is critical to the ability to create awareness in those learning new techniques. Proper communication is essential to a proper education in sport.

2. Visibility Principle. **Visibility** is directly related to learning, it speeds up the formation of a particular idea about something, and facilitates the material acquisition process; it also involves the widespread use of not only visual but also audio analyzers. Visual perception

plays an important role in learning. It provides an objective presentation of a sport technique in students' minds. The presentation of movements by a teacher or the best trainee (e.g. the bar grip, wrist position on the bar, leg positioning, etc.), is the most important method of the visibility principle. At first, the presentation is carried out as a whole, then step by step, and then finally, back to the whole movement again to integrate the individual steps. Often the elements of a technique are better perceived through the use of posters, pictures, drawings, and models. The deeper the trainee looks into the essence of the proposed technique, the more precise his idea will be of it. Through analysis of a sequence of frames via video analysis, an athlete will be able to better understand the correct motion pattern during exercise.

In the process of teaching competitive exercises in powerlifting, visibility plays a particularly important role, as a trainees' activity is mainly practical. One of visibility's special aims is comprehensive development of senses, or rather, letting students develop an eye for good and bad technique. Demonstration of the studied exercises with a brief explanation, sequence of frames, diagrams, drawings, photographs, posters, or watching a video with the performance of the European and World Powerlifting Champions, are the basic means to ensure compliance with this principle.

3. Accessibility and Personalization Principle. This principle represents correct distribution of the studied material for the entire period of training, improvement of new techniques and the amount of physical activity.

Taking this principle into account, it is necessary to structure the training and teach in accordance with the trainees' abilities, age , gender specifics, the level of their previous fitness and skill level. The principle is highly important due to the fact that in powerlifting, vital body functions are subjected to intensive exposure. It is necessary to distribute the studied material so that it can be manageable (not too much) for trainees and will not cause any health risks. Accessibility of physical exercises directly depends on the coordination difficulty, intensity, duration of physical efforts, etc. The correspondence between abilities and difficulties means that the accessibility is optimal.

Personalization, in this case, means structuring the whole process of physical education and using particular means, methods, and forms of exercises to achieve a personalized approach towards a trainee. This creates the proper conditions for their skill development to be created. Individual characteristics of each trainee do not exclude the presence of general characteristics common to all athletes. Therefore, a personalized approach is compatible with the general requirements for a pedagogical process and can be provided during group trainings. In other words, while general principles of exercise instruction apply to ALL lifters (keeping a straight back during squats, hitting proper depth), all individuals require some personalized technique input to be at their safest and best performance (foot stance width, hand spacing in the squat, for example).

4. Regularity and Consistency Principle. The essence of this principle is in regularity, consistency, and connection between trainings and the system of alternating exercising with rest. It is obvious that regular training sessions give incomparably greater effects than episodic, or sporadic ones. An insufficient break is enough to cause a decline of the emerged conditioned motor connections, reduction of the achieved functionality level, and even regression of some morphological parameters (such as losses in strength and coordination from a lack of recovery). Breaks in training can also be too long, leading to decrements in abilities. According to the data available, certain regressive changes can be detected already on the 5-7th day of a break (Zimkin N. and Co., Ferdman D., Feinschmidt

O., et. al). According to many specialists, unsystematic and chaotic presentation of the material will hinder and slow down the learning progress. It is necessary to structure the training sessions in the way that the basics of competitive exercises techniques will be learned in the first phase, and only then other elements can be taught, moving from greater to less important tasks. Such system will allow for proper preparation of the athletes to competitions and will improve their technical skills. To emphasize, techniques needs to be with important and major components first, and details after. Learning minute details before learning the big picture (learning grip positions in the deadlift before learning how to hinge at the hips, for example) is a much less efficient form of technique acquisition.

5. Gradual Increase of Requirements for Maximum Physical Activity (Dynamics Principle). This is the main principle in an athletic training system. It is based on the accumulation of quantitative changes prior to qualitative changes. Following this principle of learning, the use of maximum physical activity is much more painless if it is gradually introduced rather than slammed down at the front end. This ultimately gives positive results. Such qualities like physical strength, speed and endurance are intensively developed. The experience of the best Russian coaches and masters of sports shows that it is impossible to achieve great results without the use of large and maximum physical activity load in modern sport. If the amount of physical activity does not exceed the point, beyond which fatigue begins, then the larger the load, the greater and stronger adaptive adjustment will be. More intensive exercises contribute to the processes of recovery and «hyperrecovery» (supercompensation, the process of not just recovering but getting better). It has been shown with experiments, in particular, on the example of muscle energy recovery (Yampolskaya L., Yakovlev N. et al.).

Thus, the acquisition of these learning principles takes place in the process of mastering theoretical knowledge as well as practical skills, and comprises a unified educational process. None of these principles can be fully realized if the other ones are ignored. Only on the basis of the unification of these principles can the highest effectiveness of each of them be achieved.

3.1.1. Methods of Teaching

In the process of teaching practical skills, as well as sport techniques, various methods are applied in powerlifting, the main purpose of which is to create a proper and clearly memorable motor presentation of how to do an exercise. None of the methods presented below, all used in powerlifting, can ensure fast and effective learning of a sport technique if used separately. They must all be used to accomplish best results in coaching.

Verbal Method. In sport technique training, speaking is used to provide help during exercises as a complementary description and explanation, instructions and commands, various types of tasks, verbal assessment and enquiries, discussions, as well as dialogue and reports. Very often speaking is necessary to show the correct connection between separate parts of an exercise, to compare a trainee's impression of a task with its actual execution. Analysis and detailed examination of the completed assignments in the form of a conversation allows the coach to evaluate learning outcomes, and identify objectives for next sessions. The wording should be precise and clear. It is impossible to conduct a training session without explaining an exercise by a coach or teacher, without making a verbal remark regarding the technique or mistakes in the exercise.

Reminding and giving tips during exercises helps make the moves better not only when it comes to direction, but also range of motion, display of muscle strength, relaxation or straining. Along with an explanation, the teacher can also apply figurative explanation (such as “pretend your hands are hooks” for the deadlift and “shrug your shoulders like you’re not sure” for the bench press setup), for this purpose using previously acquired skills.

Visual Aids Implementation Method. Visual perception and visual-motor sensations give the most objective mapping of sport techniques in a trainee’s mind. This method involves showing sequences of frames, demonstration of exercises performed by higher class athletes, and use of visual aids such as drawings, diagrams, and photographs that show certain movement characteristics and the conditions of their implementation and the individual phases of motor activities.

Work-out Method can mainly influence the motor areas of the brains of the trainees very directly. In this way, gradual motor representation of an exercise is formed, followed by motor skills, and then competence skills. This method is always used in addition to the methods described above. The correct combination of these methods allows a teacher to teach an exercise faster.

The work-out method is implemented as a whole routine, or in parts.

The Whole Routine Method. The method involves learning an exercise as a whole with singling out some of its parts. This method can be applied when teaching easy exercises or exercises similar to the already learned ones. It is also applied when teaching sportsmen with good coordination.

Routine Fragmentation Method. This method is used in the process of learning fragmented motor activities followed by connecting the fragmented parts into one whole routine. For example, an exercise is separated into parts or phases. Then, an athlete masters each of them individually, and after that the whole exercise. The method is used not only in the learning process, but also in training, when it is necessary to correct or improve the details of an exercise.

There are two types of the fragmentation method:

a) When bigger fragmented sections are used covering one or more phases of the exercise which is being learned (proportioned physical activities);

b) When smaller motor tasks are solved. The method is used to clarify certain moves and phases of an exercise, as well as to correct mistakes in the moves.

The choice of either method depends on trainees’ skills and preparation level, teacher’s qualifications and his creativity.

Combined Method. This method allows the coach to be more flexible about adjusting training and rest periods, as well as to manage the development of the required qualities and skills more effectively. It is used to maintain sufficient physical performance and positive emotions, as well as to create the necessary psychological conditions for improvement of a technique. This is the most effective method that combines both fragmentation and whole routine methods.

Schedules Training Method. The training is based on an optimal schedule of monthly or weekly workouts, with precisely planned content and amount of physical activity, including its value and direction. The method allows the coach to impose the highest requirements toward the functional abilities of the body, and contribute to their greatest development. This method plays a significant role in developing a sense of purpose, determination, perseverance, self-control, an ability to overcome difficulties.

Technical Aids Method. This method provides a methodically organized presentation of exercises, terms and conditions of their implementation. Three types of information are mainly used: visual, auditory, and tactile. In this case, recording devices are the most valuable, such as a video camera, used to demonstrate a technique performed by an athlete. One of the main advantages of this method is the ability to recreate the dynamics of a movement, including slow motion, highlighting the analysis of individual phases (the so-called «freeze frame»).

Joined Method is characterized by simultaneous solution of training tasks in different types of training (for example, the development of physical skills and motor skills). Variations of the joined method depend on the combination of different types of training, such as:

- Physical + technical training;
- Physical + tactical training;
- Physical + technical + tactical training;
- Technical + tactical training, etc.

Variations of the joined method are numerous, and depend on the objectives of the training. Depending on the used variation of the method, it can affect physical qualities of various kinds and types of training. The main feature of the method lies in the complex (joined) solution of training tasks. An example of this method is to have an athlete perform multiple sets of 5 reps of the deadlift, in which each rep is re-started from the ground and technique is the big focus for each rep. Not only is the technical ability to perform proper deadlifts being trained, but the work capacity and muscle size is also being improved through the resulting high volumes of multiple sets of 5 reps. In this case, the weight on the bar should be enough to be challenging, but not so much as to cause a technical breakdown, as that would defeat the purpose of the technique training component.

Game Method is characterized by the presence of a game plot, with simplified or without any rules. It is used not only in the process of physical education for initial training movements, but also for the development of individual skills and comprehensive improvement of motor activity in difficult conditions. It allows for improvement of various qualities and abilities, such as agility, orientation speed, independence, initiative, and also increases the emotionality of lessons. The main feature of the game is the ability to switch between motor and motivational settings. The game method is especially useful for teen athletes, who are much more easily motivated to execute proper techniques when in the context of a fun and competitive setting.

Competitive Method means performing exercises in conditions similar to the ones during competitions. It is used to train moral standing, physical qualities of an athlete, tactical training for the upcoming competitions. It is characterized by complete or partial simulation of competitive activities.

The variations of this method are the following:

- Training games or competitions (with a weaker or stronger opponent; with an unusual opponent, for example, a left-handed person; in abnormal conditions, such as shorter area, etc.)
- Competition with handicap
- Tests

This method improves various aspect of training preparation [1].

Testing the Knowledge, Skills and Abilities Method is used in the process of preparing estimates, competitions, tests, examinations, and routine observations.

3.1.2. Formation of motor skills

Motor skills play an important role in everyone's life. Ushinskiy (1867), the founder of Russian pedagogy, regarding this subject said that if a person did not have the ability to develop a skill, they would not be able to advance even one step in their development. Skills and abilities differ from each other mainly by the degree of their mastery, i.e., control methods on the part of human consciousness. What this means is that some skills are very easy and require minimal thought when being , while other skills are quite complex and require considerable thought and high level brain activity during their learning and execution.

The role of motor skills in physical education may be different. In some cases, abilities lead to acquired skills, if one wants to master a technique of motor actions. In other cases, motor skills are developed without subsequent transition into skills. In this case they are complementary. As a result of the formation of a motor action, an initial motor skill appears. The concepts of «motor abilities», as well as «motor skills», primarily refer to the technical side of the ability to perform physical actions. Acquired skills are the basis for new abilities. Motor activity can be successful if it is based on a variety of solidly formed skills. The greater the skills supply is, the more versatile and efficient activities will be. In sports, an ability to execute precise techniques largely determines an athlete's mastership. Therefore, motor abilities and acquired motor skills primarily depend on the degree of movement expertise.

An acquired motor skill is at the level of expertise when movement control is automatic, i.e. it does not require additional attention.

In teaching sport exercises, the final goal is to form acquired skills, based on habit, to be able to perform activities automatically, without paying additional attention to them, but with awareness. Knowledge of the natural laws by which motor skills are formed makes it possible to build a learning process effectively. According to Matveyev L. (1976), «Motor skills are one of the most common forms of realization of motor abilities of a man, which are expressed in the ability to perform motor activities on the basis of non-automated (or not brought to a significant degree of automation) targeted operations». The range of motor skills movement is limited and varies in different exercises. If you forget a little detail then the accuracy of the movement is affected, within a few weeks' break an acquired skill may disappear altogether. That is why the quality of acquired skills depends on trainees' abilities and on the conditions in which the above have been formed. At the beginning of a motor skill formation, it is necessary to limit new stimuli, i.e. to create constant stereotypical conditions (drilling in practice). Stability in the formation of a motor skill is achieved through repetition and support - evaluation of how precise a movement is being performed. Sequence is a necessary condition in the formation of a new motor skill, as well as increasing the level of activity of the autonomic functions of the body that provide the ability to perform the necessary activities. As new skills are drilled over and over, their execution becomes more and more automatic. As it becomes more automatic, the drilling time spent on these skills can be reduced to make room for drilling other new skills until they become automatic, and so on.

Motor abilities and acquired motor skills in a sense transform into each other. Therefore, it would be wrong to completely separate them. However, we must bear in mind their specific characteristics, their irreducibility, and unequal importance, both in everyday life and in physical education. Acquired motor skills, as the highest stage of motor activity mastery, is of the utmost importance in sports and sports practice. You can only get so far with raw talent, and practice is a must for all trainees.

3.1.3. The Structure of Teaching Motor Activities

The main tasks of teaching motor activities depend on their type and characteristics, as well as the specifics of physical features and their level. Motor abilities and acquired motor skills appear in the process and as a result of motor activities acquisition. When a motor ability, automatized and formed through numerous repetitions, turns into an acquired motor skill, three stages of neural processes (composed of both neural excitation and inhibition) are observed in the central nervous system.

In the first stage, separate movement elements are combined. During the first attempt to perform a new motor activity, the nerve centers responsible for performance, and neighboring centers that do not participate in the process, are activated in the cerebral cortex.

The second stage is characterized by an intensive activation, coordination improvement, and elimination of unnecessary movements. In this case, true “acquisition” of motor skills begins.

The third stage is characterized by stabilization, a high degree of automation and coordination of movements. In fact, all the signs of an acquired motor skill are fully manifested.

Interaction and transition of previously acquired or simultaneously formed motor skills are essential in teaching a motor activity. In sports education, two types of skills transition are observed, i.e. positive and negative transition.

Positive transition means cooperation between skills when a previously acquired skill assists, facilitates and speeds up the process of a new skill formation. The main condition of this type of transition is structural similarity of the main movement stages (separate parts).

Negative transition means cooperation between skills when an acquired skill complicates the process of a new skill formation. This happens when there is similarity of the motions in the preparatory stage but none in the main stage.

There are various types of skills transition:

Unilateral transition. It occurs when the formation of one skill contributes to the formation of the other, while the inverse effect is not detected. Learning to surf will give you better balance for judo, but learning judo takedowns won’t make you any better at surfing.

Mutual transition. It means transition of skills from one motor skill to the other and back. Learning to hinge at the hips in the squat will make you better at the deadlift and learning to hinge at the hips in the deadlift will make you better at the squat.

Direct transition. It is characterized by the fact that forcing a particular skill immediately affects the creation of another skill in any exercise. Learning to retract your shoulder blades for the squat makes you more easily learn to do so during a bench press.

Indirect transition. It is this type of transition a previously formed skill creates favorable conditions for the acquisition of a new one. Already knowing how to create the Valsalva maneuver for the squat will make learning the deadlift easier.

Learning motor skills and their mastery is an important part of an athlete’s training, and is held in a fixed methodical sequence and in several stages. Work at all stages is characterized by certain distinctive features, which are reflected in the features of the acquisition objectives, as well as aids and methods.

Limited (partial) Transition. It occurs when the structure of the learned activities shows great similarities. In this case, the skills transition is carried out in an extremely narrow range. The sequence and continuity of the tasks, as well as the whole system of so called supplementary exercises, are planned on the principle of structural similarities of the main stages of an exercise.

The learning process can be divided into relatively independent and simultaneously interconnected and interdependent links. Platonov V. (2004) identifies three stages of technical preparation of an athlete:

The first stage - the initial practice. In this process a general overview of a motor activity is created, and the conditions for its mastery are set, the main mechanism of the movement is studied, rhythmic structure is formed, mistakes are prevented and eliminated.

The second stage - comprehensive learning and training. The understanding of a motor activity pattern is crystalized, coordination structure of single elements and rhythmic structure are improved, as well as dynamic and kinematic characteristics. The abovementioned elements are adjusted to meet a trainee's individual characteristics. This is done mostly through drilling and repetitive practice.

The third stage - consolidation and further improvement. A skill is stabilized, a variation of an exercise is improved taking into account a trainee's individual characteristics, a variety of conditions, including the highest manifestations of motor characteristics. This is the stage most seminar attendees will be focusing on. When a famous lifter gives a seminar in your area, most of the time he/she will be teaching you how to slightly modify your already solid technique, not just teach you "how to squat" from nothing.

Shtark G (1971), and Schnabel (1982) divide the process of technical training into five stages.

1. Creation of the First Conception of Motor Activity and Setting the Conditions for Learning.

Verbal and visual aids are used to set the conditions and the main option to master a technique. The athlete's attention is concentrated on the main motor activities and methods of their performance. This gives the athlete an idea of what the whole technique is going to look like once it's all put together and well-executed.

2. Initial Skill Formation Stage Corresponding to the First Phase of an Activity Acquisition.

At this stage, the ability to perform the basic structure of a movement is formed, learning the basics of a technique and the overall rhythm of an activity are acquired. The main method of movement acquisition is the method of dividing an activity into separate parts (phases) and isolating the learning process of these parts, followed by their combination. It is necessary to pay attention to the elimination of adverse movements, and excessive muscle tension.

3. Perfect Execution of a Motor Activity Formation. Teaching process is aimed at studying the details of a motor activity. Attention is drawn to the methods based on the use of motor perceptions. A rational kinematic and dynamic structure of movements is formed.

4. Stabilization of a Skill Stage corresponds to the consolidation of motor activity stage. As a rational system of movements consolidates, characteristics of a skill are determined - automation and stabilization of an activity. The educational aim is to stabilize a motor activity and improve its individual parts. At this stage, technical improvement is linked to the process of development of motor qualities, tactical and mental preparation. Notice that the "perfect execution" stage comes before the "stabilization" stage. You do not want to ingrain sub-standard techniques into automation, as they are very hard to unlearn later.

5. Variation Technique Mastery Stage and its Implementation cover the entire period. Through the use of this stage in a variety of conditions and functional stages of the body, a variation skill develops.

Since in most cases, it is impossible to perform a complex motor activity at the first attempt, the exercise separation method should be used. However, this method can be

used only if relatively independent parts of a movement can be highlighted on the basis of a qualified analysis of its technique. Exercise separation method could help overcome the psychological barrier in the early stages of training, reduce physical effort, it will also allow the coach to avoid consolidation of serious motor mistakes. These mistakes, according to the classification by Belinovich V (1991), include:

1. Mistakes in specific motor activities and common mistakes (bad posture, etc.).

2. Individual and complex mistakes (the emergence of a single mistake entails the emergence of a whole range of others, such as when a poor back position leads to poor knee position and ankle position, etc.)

3. Automated and non-automated mistakes

4. Significant (serious) and minor (small) mistakes that do not violate the structure of the movement

5. Typical (common to all) and atypical (individual) mistakes

As a result, the earlier the possible errors are explained, the easier it will be to identify the causes that generate them. The most common reasons are as follows:

a) Misunderstanding of the task;

b) Lack of physical fitness;

c) Lack of confidence about strength, indecisiveness;

d) Negative impact of previously studied activities (bad movement habits such as rounding the back to pick something up applied to the deadlift);

e) Inappropriate environment for lessons, inadequate tools and equipment;

g) Violation of the requirements and the organization of classes that in particular lead to premature fatigue.

Certain rules should be followed in order to correct the mistakes:

1. Mistake must be properly explained.

2. Mistakes should be corrected sequentially, according to their importance. Big mistakes should take precedence over small mistakes.

3. An indication of how to correct a mistake should be appropriate to a trainee's current abilities. This includes using cues that the trainee knows and understands. No matter how good a cue is, if a trainee doesn't know it, it's useless.

4. When mistakes occur that have already been discussed earlier, supplementary exercises where these mistakes have been eliminated should be used, such as using light high-bar good-mornings to teach proper arched back position.

5. It is unacceptable to present a trainee's mistake. To create a clearer idea about his mistake, it is useful for the mistake to be presented by the trainee himself while the proper movements are described. In other words, don't illustrate common mistakes when demonstrating a lift until they come up in the coaching process (as your trainees make them in training). Only when they occur should you address them… otherwise your role is to teach proper technique, not mistakes.

Only after mastering parts of exercises and preventing mistakes, is it possible to proceed to the method of teaching the whole exercise.

3.1.4. Typical Mistakes in Exercises

Significant deviation or distortion of the actual structure of a movement or its parts from the set example is termed here "mistakes in exercises." This deviation has a negative impact on the effectiveness of an exercise due to emerging difficulties, interfering with an athlete's ability

to show off their motor and physical abilities. Distortion of a motor structure can be noticed by a coach, judge and athlete based on the change in spatial and temporal characteristics. Spatial-temporal movement structure of an athlete's body, its parts and a bar during the performance of a routine, is characterized by kinematic components: amplitude, velocity, acceleration, rhythm, etc. In other words, the kinematics of the movement is observed, which is a result of the interaction of an athlete's strength with the forces of gravity, inertia, resistance, strain, and friction. Performance of competitive powerlifting exercises is a vivid example of the fact that dynamics is the cause of kinematics. Since the reasons are hidden in the dynamics, a coach and an athlete should present, above all, a good understanding of the power structure of an exercise. Rational technique of lifting weights, as you know, demands precision when performing an exercise.

As a rule, it is impossible to perform an exercise flawlessly in normal conditions straight away. This fact complicates the process of movements' development. Some errors can be explained by the laws of motor skill formation, while others are related to the lack of understanding, and the rest by non-compliance with certain conditions, etc. Success in the development of movements is largely dependent on how precisely the reasons of motor mistakes origin are determined, and how correction methods correspond to the real reasons of the above. The most typical errors are the following:

- Presence of unnecessary movements in a motor activity
- Movements restraint, inappropriate muscular effort
- Unnecessary involvement of additional muscle groups
- Fluctuations of the amplitude and direction of motion;
- Distortion of the overall rhythm of motor activities;
- Insufficient speed of movement performance

To improve the efficiency of motor activities acquisition and prevention of errors, it is highly important to regulate their performance correctly. The main parameters of this regulation are the number of repetitions and rest intervals between them. Their specific characteristics can be very different, as they are determined by many factors (complexity of movements, stage of development, trainee's individual abilities, etc.). In all cases, you should remember and follow these rules:

- The number of repetitions of a new exercise is determined by a trainee's abilities;
- The movement should be improved at each attempt;
- Reoccurring mistakes in an exercise means that a trainee should take a break and reflect on the movements;
- Rest intervals should ensure optimum preparation to perform another attempt - both physical and mental;
- It is inappropriate and even harmful to continue performing an exercise when a trainee is tired;
- Breaks between lessons should be as short as possible so as not to lose the already acquired skills.

3.1.5. Motor Error Factors

The most typical mistakes encountered when performing a motor task, are, first of all, natural, i.e. they are caused by the natural laws of motor activity initial formation. However, flawed methods of teaching can be another reason for mistakes. The success of teaching

largely depends on how well the causes of errors are defined and what methods for their prevention and elimination are provided.

The main causes of fluctuation and distortion of movements usually are the following:

a) Improper or insufficient understanding of the structure and motor composition of an activity being learned;

b) Improper or insufficient understanding of motor tasks;

c) Lack of motor experience of a trainee;

g) Lack of physical fitness;

d) Lack of confidence, fear, fatigue, etc.;

e) Improper organization of a motor activity acquisition process.

Thus, a variety of physical activities is formed during a person's life, under the influence of many factors, and the process of their formation can take on a different character. The optimization of this process is achieved in a rational educational training structure. The internal logic of the process of motor activities formation and improvement under these conditions is schematically represented as a gradual transition from knowledge and understanding of an activity to the ability to execute it, afterwards, to an acquired skill and finally, a mastered skill.

In order to organize the correct initial learning of an activity performed by a professional athlete, it is necessary to create the correct and complete basis that guarantees flawless performance of all activities.

3.2. Fundamentals of teaching competitive exercises techniques

The basic exercises included in teaching competitive exercises techniques are the exercises that occur in certain conditions and are characterized by rational and efficient motor activities (skills). In power sports, a motor skill means an automated way of performing an exercise. An athlete's motor skills, as well as performance and technique, depend on the degree of automation. The important features of a motor skill are its solidity and stability allowing for continuous repetition of the activity. An athlete must be equally confident when learning to lift different and ever-increasing weight. Only under this condition during sports training can the level of physical qualities increase and the technique become automatic. Technique automation needs to come, in a large respect, before maximal weights are handled in training.

Performing competitive exercises in powerlifting requires appropriate skills. During the teaching process, a coach should not only be guided by the pedagogical principles, taking into account the physiological laws responsible for the formation and development of motor skills, but also to solve some individual training issues. These include, for example, the sequence of development of classical exercises and their components, learning style of squats and deadlift, the choice of weight and application of teaching methods at different stages.

The process of teaching motor activities is divided into three stages:

1. Mastery of a competitive training technique by lifting a barbell with a fixed weight.
2. Mastery of a competitive training technique by lifting a barbell with variable weight.
3. Rational performance of competitive routines in any conditions, including competitions.

Depending on the specific objectives of each stage of the studied motor activity, the learning process has its own characteristics: the first two stages can last from one to two or three months, and the third stage lasts for many years, as long as the athlete improves his technique, bringing it to the ideal model characteristics.

In the process of teaching young athletes, there are five stages of sport technique formation (Ratov I., Nariddinov F., 1991):

1) Creation of a motor activity overview and training guidelines;

2) Mastery of basic techniques and activity rhythm. This stage is characterized by consistent emphasis on some of the most important coordination elements;

3) Formation of skills for perfect motor activity execution;

4) Complete skill acquisition is characterized by consolidation of motor activities, their automation and stabilization;

5) Achievement of variable skill and its implementation.

When organizing a training process, it must be taken into account that the process of teaching a motor activity or sport technique elements should resemble the conditions of competitive practices (Golovin B., 1991). To facilitate the development of the acquisition of competitive exercises in the early stages of preparation, considering the individual characteristics of young athletes, one should use the method of separating a workout routine into a few simple moves, gradually connecting them together into the whole routine. Ivoylov A. and Laputin A. (1986), recommend the use of special exercises for the development of the muscle groups that are involved in a specific movement. Most of these exercises should be selected on the basis of structural similarity with specialized and competitive movements, but each of them must be localized.

The combined method links the advantages of the first two. It allows for an activity to be taught as a whole, while identifying errors in different phases and eliminating them.

In powerlifting, the weight being lifted is of great importance in teaching exercise routines, as it affects motor skills. The weight should not be large. Athletes lifting heavy weights during learning, often show flaws in techniques. On the other hand, the correct dynamic structure of a movement must not be altered because of insufficient weight. Therefore, depending on the complexity of the studied exercises, technical and physical fitness of athletes (gender, age, height, and weight category), the optimum bar weight should be defined for each of them. It has been established that insufficient weight gives insufficient training effect, and larger than the optimum weight leads to fatigue, which ruins the technique of movements performance, and does not provide adequate training effect.

At the initial stage of learning a new technique, for several trainings the bar weight should be constant, and then to consolidate the acquired skill and improve the technique, a routine can be performed with different weight. To improve the efficiency of learning, it is important to include simulation exercises from the main part of the lesson, which should be done with small weights (a metal stick, or a bar).

Duration of a sport technique acquisition largely depends on the individual qualities of an athlete, as well as the knowledge and experience of a coach who should prompt the way toward rapid skill mastery.

3.2.1. Squat Technique Training

In powerlifting, the process of teaching squat technique to beginner athletes is divided into five stages.

During the first stage, it is recommended to begin with the correct approach to the barbell on the racks, hand position on the bar and grip width, the setup, and lastly bar placement on the shoulders (fig. 3.1).

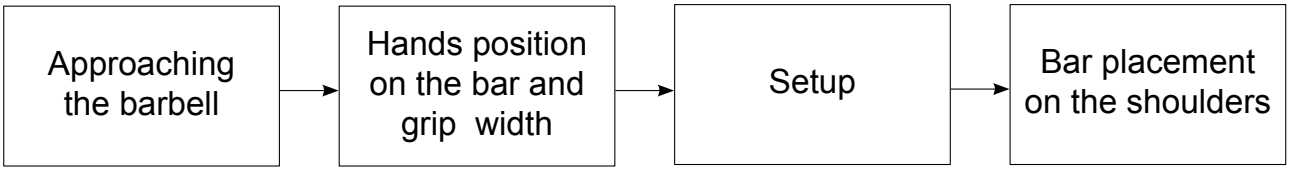


Fig.3.1. Sequence of teaching the setup position

Photo. 3.1

Photo. 3.2

It should be noted that, unlike lifters who, when performing a squat, put the bar on top of their trapezius muscles, powerlifters place the bar on the middle part of their trapezius muscles and rear deltoids. Since the bar cannot be held without support of the arms in this position, it is important to teach beginners to position and hold the bar with their hands.

Photos 3.3-3.5 demonstrate the bar placement during the first phase of training.

Photo 3.3.
High-bar placement

Photo 3.4.
Proper bar placement

Photo 3.5.
Incorrect bar placement

The second teaching stage starts with proper setup, unracking the bar and stabilizing it on the shoulders, followed by racking the bar (fig. 3.2).

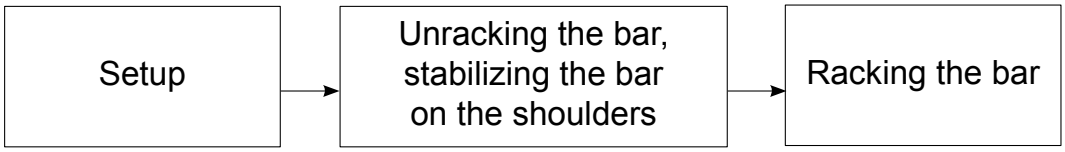


Fig. 3.2. Sequence of teaching the setup position

Photos 3.6-3.8 demonstrate the bar unracking technique during the second teaching stage. The bar is located on a “pyramid” power rack. The athlete sets up and places the bar on his shoulders (photo 3.6.), after which he takes a shallow breath and unracks the bar while straightening his legs at both his knees and hips. After his legs are fully straightened, he stabilizes the bar for 5 seconds (photo 3.7). Then, he returns the bar to its initial position (photo 3.8).

Photo 3.6

Photo 3.7

Photo 3.8

During the third stage, it is recommended to teach beginning athletes to walk out with the bar on their shoulders, stabilizing the bar on their back in the starting position, and racking.

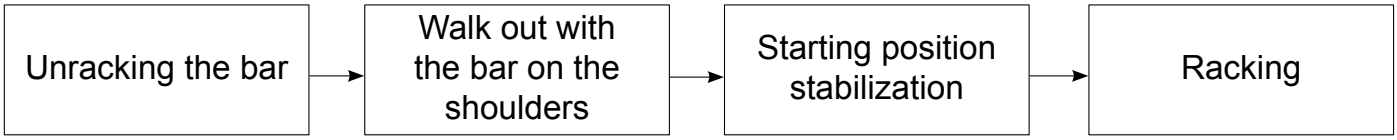


Fig. 3.3. Sequence of teaching the setup position

Photos 3.9 – 3.13 demonstrate the starting position while executing a squat with the bar on the shoulders during the third teaching stage.

Photo 3.9

Photo 3.10

Photo 3.11

Photo 3.12

Photo 3.13

After unracking a light bar (photos 3.9-3.10), the athlete should take one or two steps to the back and take a stance slightly leaning forward (photos 3.10-3.13). The feet should be placed a little wider than shoulder width to stabilize the stance. This position should be held for five to seven seconds, after which the bar should be racked (photo 3.13). During the third stage, a beginning athlete should be supervised by a more experienced one. One spotter is enough if the bar weight is not big (as it shouldn’t be in this stage). The coach’s task is to

observe the execution of the technique, as well as to identify and correct mistakes made by the athlete.

While performing squats, most beginning athletes lean forward too much. This reduces forces passed down into the feet, especially in the early stage during straightening of the legs and the body, which hinders squats execution in accordance with competition rules. Therefore, it is very important to teach beginners to perform squats keeping their knees in line with their feet. Additional exercises can help to achieve this, i.e. box squat (fig. 3.4).

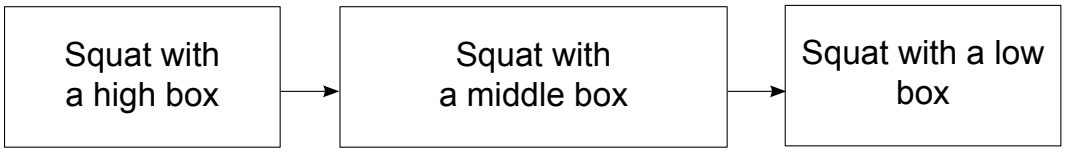


Fig. 3.4. Supporting exercises used in teaching a squat technique with boxes of different height.

At first, a box (block) is put at knee level. The athlete, with a barbell on his back (light weight), stands with his back to the box about half a step away from it. When squatting, he moves his hips back to reach the box without relaxing his back muscles. After touching the box, the athlete stands up and assumes the starting position. After mastering a squat technique at this height, the height of the box should be reduced to create the right angle with the athlete's hips. Only after acquiring squat technique on a stand of this height should the height of the box be reduced, which will allow the athlete to perform squats according to competition rules. The coach should make sure that the athlete does not sit down on the box with his back muscles relaxed. The athlete should touch the box with his hips and then stand up without relaxing his back muscles.

Photos 3.14 -3.17 demonstrate box squat technique.

Photo 3.14

Photo 3.15

Photo 3.16

Photo 3.17

Photos 3.13-3.17 Supporting exercises used in teaching a squat technique with boxes of different height.

Only when box squat technique (at the level below of an athlete's knees) has been acquired we can move on to teaching the following squat stage – half squat, when an athlete's hips are at the parallel level or slightly higher) (fig. 3.5).

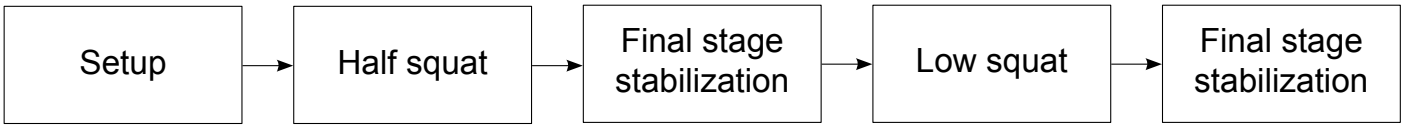


Fig. 3.5 Sequence of teaching squats with a bar on the shoulders (stage 5)

Photos 3.18-3.21 demonstrate a half squat at the fifth stage of learning a squat technique

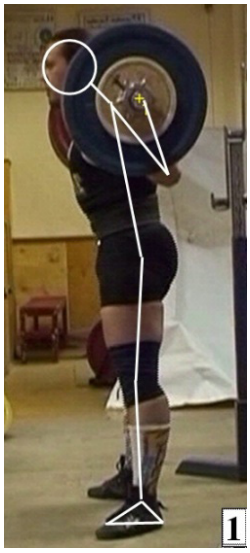


Photo 3.18

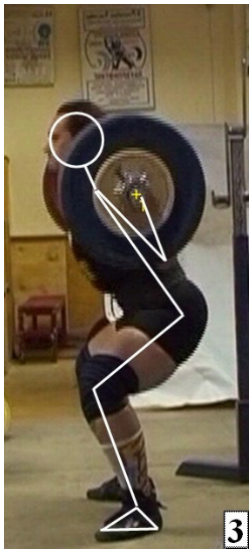


Photo 3.19

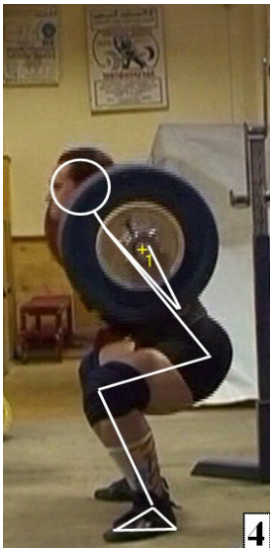


Photo 3.20

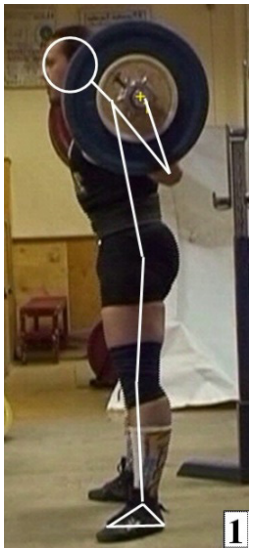


Photo 3.21

Half squat at hip level parallel to the platform or slightly higher.

After an athlete masters a half squat, it is necessary to proceed to teaching a full squat position (Photos 4.22-4.24).

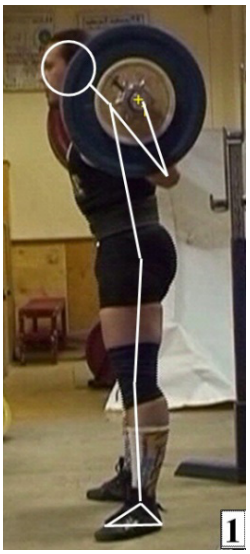


Photo 3.22

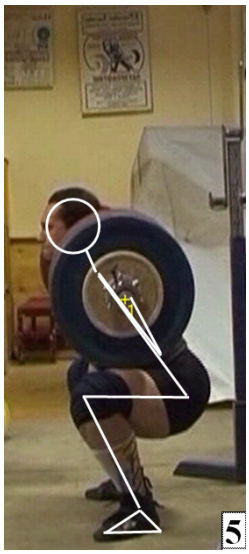


Photo 3.23

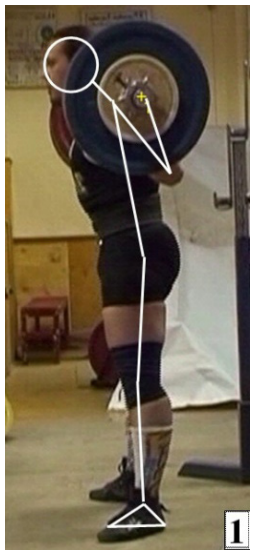


Photo 3.24

Full-depth Squat

Once in two weeks it is recommended to perform a barbell front squat. This exercise will help to teach an upright (not rounded) back position.

It is commonly observed that athlete's knees go past the toes line, which can lead to difficulty squatting below the parallel.

Photo 3.25

Photo 3.26

Photo 3.27

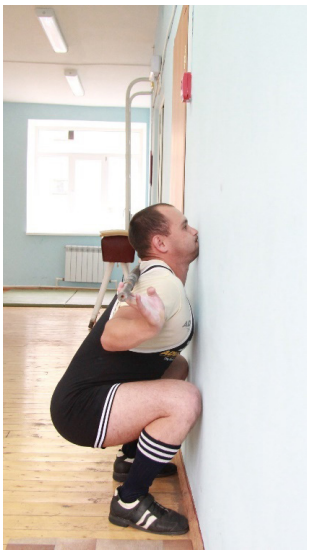


Photo 3.28

Photo 3.29

Photo 3.30

In order to correct this mistake we suggest a special preparatory exercise “wall squat”. This exercise also helps an athlete to keep his back straight (photo 3.25-3.30).

Additional exercises while teaching squats include front barbell squats, squats with “pyramid” power racks from the sticking point, and half squats. In addition to the above, the following developmental exercises are used: leg press in a sitting position, squats in a «hack machine», deep squats while standing on boxes with weights in your hands, etc.

3.2.2. Common mistakes in competitive bar squat technique

1. As soon as the athlete assumes a vertical body position (a slight deviation from the vertical position is acceptable) with his knees straight, the head referee is supposed to give a sign to start the squat. The signal is a downward arm motion with a clear verbal command “squat”. Before the signal is heard, the athlete is allowed to make any movements to assume the starting position that do not involve breaking the competition rules. For safety reasons, the head referee may ask the athlete to rack the bar by making a verbal command «replace» by making an arm motion to the back (as if motioning “come here”) if after five seconds after unracking the bar the athlete was not able to setup properly. After that, the head referee

explains the reason for not giving the signal to start the exercise.

2. After receiving the signal to begin the lift, the athlete must bend his knees and lean so that the upper parts of his legs at the hip crease are lower than the top of his knees. Only one attempt to make a downward movement is allowed. An attempt is considered commenced if the athlete's legs were bent at his knees.

3. The athlete should independently return to an upright position with his knees locked. Double ambulation (double bouncing) from a low squat or any downward movement are forbidden. When the athlete assumes a motionless position (i.e. the move has been completed), the head referee should give the signal to rack the bar.

4. The signal to rack the bar consists of a backward arm motion and a distinctive command «rack». After this signal, an athlete must return the bar to the racks. Any foot movements after the "rack" command should not be the reason to disqualify the attempt. For safety reasons, the athlete can ask spotters for assistance in returning the bar to the racks. In this case, the bar must remain on the athlete's shoulders.

5. During squatting, a minimum two and maximum five spotters/ assistants must be present on the platform. The referees shall decide how many spotters (two, three, four, or five) should be present during squat performance at any time.

Causes for Disqualification of a Squat

1. Mistakes when following the head referee's signals during the initial or final stages of the routine.
2. Double bouncing at the low squat position, or any downward movement while recovering to the upright position from the squat.
3. Mistakes when assuming the upright position with knees locked at the initial and/ or final lift stage.
4. Backward, forward, or horizontal feet movement. Rocking the feet between the ball and heel is permitted.
5. Mistakes when bending the knees and lower the body until the surface of the legs at the hip joint is lower than the tops of the knees, as shown in one of the pictures below.
6. Contact with the bar or the athlete by the spotters/ assistants to facilitate the squat between the head referee's commands during the execution of the lift.
7. Contact of elbows or upper arms with the legs, which is supposed to help the athlete. A light contact is allowed if it does not help the squat performance.
8. Dropping or dumping the bar on the platform after completing the routine.
9. Incompliance with any of the regulations included in the Rules of Squat Performance.

1. Uneven bar position after racking.

Photo 3.31

This mistake can lead to the bar misalignment, which results in failure to perform the exercise.

2. High-bar position on the trapezius muscles

Photo 3.32

This particular bar position makes the athlete's body lean forward from the optimal movement trajectory. It is recommended to place the barbell on the bottom part of the shoulders so that the barbell's top is not lower than 2cm of the rear delts.

3. Grip is too wide or too narrow

Photo. 3.33

Photo. 3.34

The exercises are demonstrated by Vadim Prodanov, Master of Sport of Russia, multiple champion and record holder of Bashkiriya.

A grip that is too wide prevents the athlete from controlling the bar, if it is too narrow it creates a significant and harmful pressure in the elbows. Some coaches recommend a grip five to ten cm wider than shoulder-width, others eight to fifteen cm. At the same time, one must take into account that grip width depends on the flexibility of elbow and shoulder joints. It is obvious that a heavy weight lifter's grip is wider than light weight lifter's. The elbows should be slightly lifted, which will prevent the bar from sliding down during the exercise. The pressure in the elbows will rise if they are lifted too high.

4. Unracking the bar with relaxed arm and back muscles

Unracking the bar with relaxed back muscles can cause the athlete to lose control over the bar. The bar can start "shaking" the athlete and it takes a lot of time to "calm it down". The attempt ends in failure in most cases mostly because that whole process drains too much energy from the lifter for there to be enough to complete a maximal attempt after.

5. Walkout with the bar on one's shoulders is too long

Unnecessary steps when walking out with the bar mean an additional waste of energy and strength. There is a risk for back muscles to relax, which can lead to failure. Advice: the walk out should be just enough not to touch the racks with the bar or one's foot.

6. Foot placement is too wide or too narrow

Despite the fact that foot positioning should be individualized for each athlete, during competitions, one can meet athletes with powerful thighs and poor mobility in ankles who squat in narrow stance. This usually doesn't lead to the most weight possible being lifted. The stance that is too wide can cause injury to the knees and groin area.

7. Excessive leaning forward in the starting position

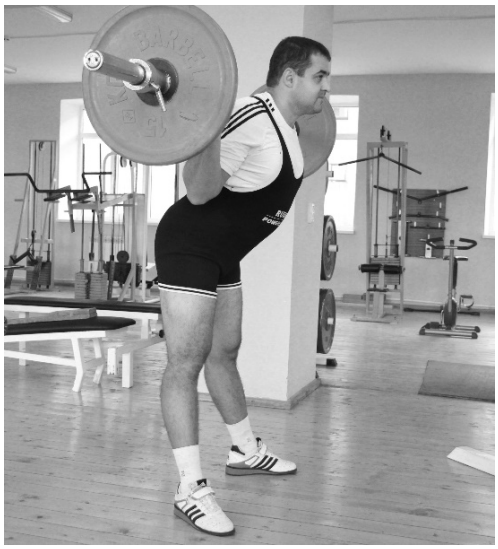


Photo 3.35

In the starting position, an excessive leaning forward can be caused by a number of reasons: lack of flexibility at the ankle joint, insufficient strength of lower back muscles, or improper technique.

8. Premature lift of the hips in relation to the shoulder girdle (hasty straightening of the knees)

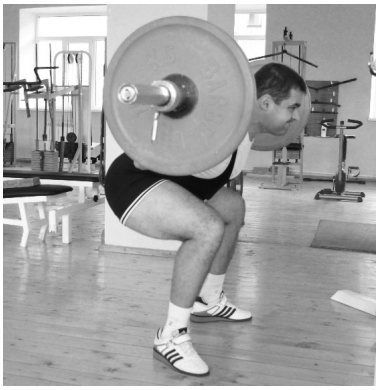


Photo. 3.36

Photo. 3.37

Fast straightening of the legs in the knee joints leads to excessive leaning forward, the back muscles take most of the load. The upward movement of the hips and torso should take place at the same time so that the back does not «collapse» while moving to the upright position.

9. Tilting the head down in any squat phase

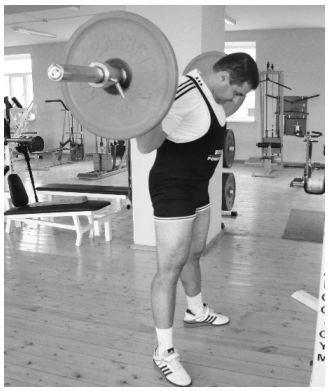


Photo. 3.38

Photo. 3.39

Tilting the head down can lead to rounding one's back and leaning forward with the torso. The more one leans forward, the more pressure he puts on his back muscles and less on his legs.

10. Rounding the back in any squat phase

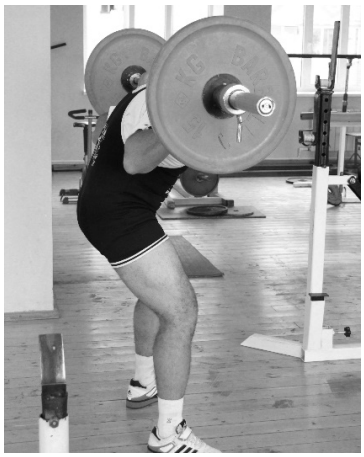


Photo 3.40

Rounding the back can lead to a shift in the center of gravity toward the toes and increase the load in lower back muscles.

11. Valgus drift in the knees when squatting up

Photo 3.41

Dropping the knees inwards is caused by the imbalance between the adductors and abductors of the thigh, which can lead to relaxation in the lower back muscles. In this case, it is necessary to work on weaker muscle groups.

12. Mistakes when assuming the upright position with knees not fully locked at the initial and/ or final lift stage

Photo. 3.42

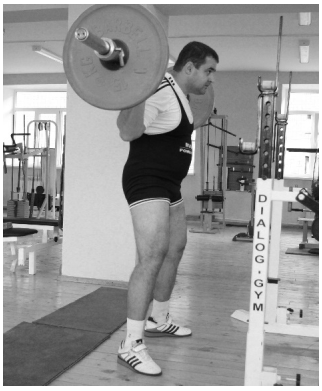


Photo. 3.43

Incomplete locking of the knees at the initial position and after the lift is considered to be a mistake according to the technical rules. If the coach notices this mistake and works on its improvement for two or three weeks by controlling the athlete's knee position at all times, it will be reflected on the athlete's technique.

13. Mistakes while bending the knees

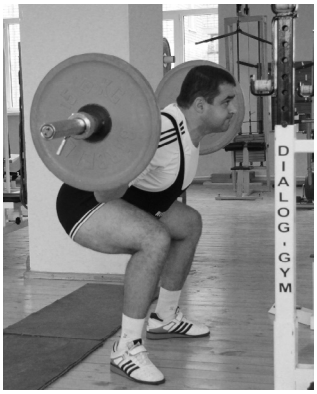


Photo. 3.44

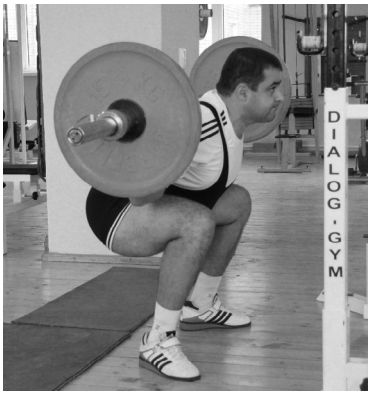


Photo. 3.45

After the head referee's signal to start the squat, the athlete should bend his knees and torso so that so that the upper parts of his legs at the hip crease are lower than the top of his knees. The incomplete range of the downward motion that leads to failure to pass through the proper angles is considered an error according to the technical rules, and the attempt cannot be considered successful.

14. Backward, forward, or horizontal feet movement

Photo. 3.46

Photo. 3.47

According to the international rules, this is an error. Only rocking the feet between the ball and heel is permitted.

15. Physical contact of a spotter/ assistant with the bar or athlete

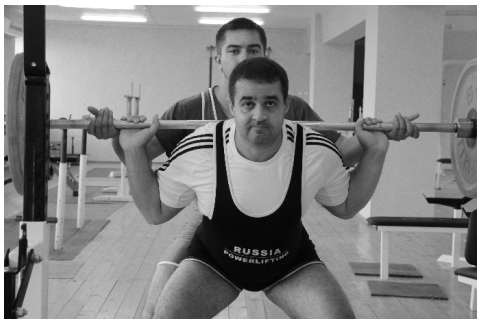


Photo. 3.48

Photo. 3.49

After the referee's command "squat" until the "rack" command, physical contact to facilitate the squat is considered an error.

16. Dropping or dumping the bar on the platform after completing the lift

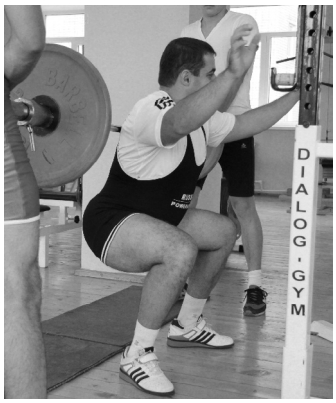


Photo. 3.50

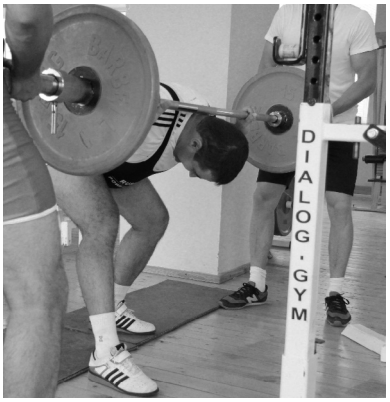


Photo. 3.51

Dropping or dumping the bar on the platform during or after completing the lift is an error. The athlete who dropped the bar gets a card and is disqualified after the second card.

17. Mistakes when following the head referee's signals during the initial, or final stage of the routine.

3.2.3. Bench Press Technique Training

The teaching process of bench press technique on a horizontal bench has several stages.

The first stage starts with approaching the bench, setting up and assuming the starting position.

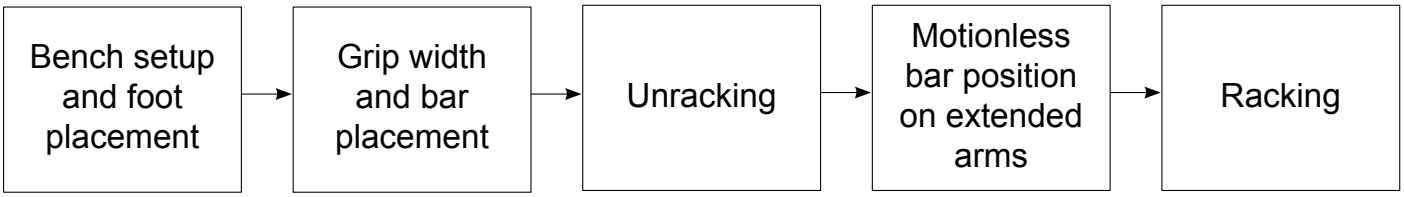


Fig. 3.6 Sequence of teaching the setup

Bench Setup

After approaching the bench, the lifter lays down on his back while his head, shoulders and glutes come in contact with the bench surface. The head should be positioned so that the eyes come even with the bar.

Grip Width and Bar Placement

The athlete holds the bar in the way that his four fingers are on one side of the bar, while his thumb is on the other. (The thumbless grip is not allowed by IPF).

Two grip types are the most common among the athletes: wide (81 cm), i.e. the maximum width allowed by the competition rules, during which the chest muscles are strained the most, and medium width grip (65-70 cm), which activates the triceps and deltoid muscles. The choice depends on the athlete's strength potential and his arms' length. Considering the fact that at this stage, the athletes' chest muscles have not been fully developed, it is recommended to begin with triceps bench press technique. Therefore, the athletes' grip width is going to be around 65-70 cm. At this stage, arching one's back is out of the question. It is crucial to teach the proper technique to lay on the bench pressing his feet to the platform.

Unracking the Bar

At this stage, it is essential to teach the athlete to unrack the bar. After he grips the bar, by slowly extending his arms, he takes the bar off the racks and stabilizes it above the middle part of the chest with fully extended arms. At this teaching stage, it is necessary to make sure that the athlete's arms are fully extended in his elbow joints and he is able to keep the bar motionless for up to five seconds with his arms fully extended.

Racking the Bar

After stabilizing the bar with fully extended arms, the athlete returns it to the racks.

Spotters, and even initial assistance is crucial when teaching the athlete to unrack the bar and returning it to the racks by himself.

Lowering the Bar to the Chest

After mastering the starting position, the next teaching stage is lowering the bar to the chest.

As a rule, during the initial phases of this stage, in order to avoid lowering the bar to different points, it is necessary to divide it into three teaching stages.

During the first stage, the athlete lowers the bar on a ten-centimeter thick board, which is placed on his chest (photo 3.52–3.54). Without making a pause, he presses the bar off two boards and stabilizes it with his arms fully extended.

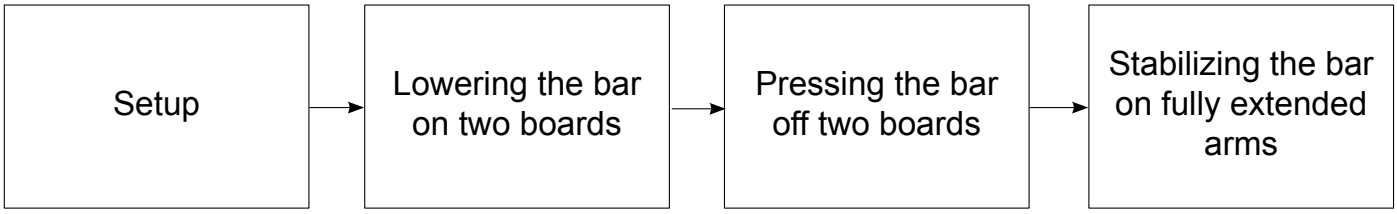


Fig. 3.7. Sequence of teaching the bar lowering technique

Photo 3.52

Photo 3.53

Photo 3.54

After the athlete has learned to lower the bar on two boards with confidence, followed by pressing the bar, it is possible to proceed to the second stage during which the press is executed with one 5cm thick board (Photo 3.55-3.57).

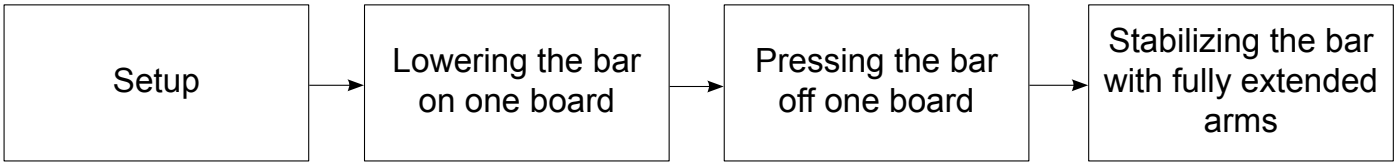


Fig. 3.8. Sequence of teaching the bar lowering technique

Photo 3.55

Photo 3.56

Photo 3.57

It is possible to proceed to teaching the third, and the main, stage only after the initial two stages of the bench press on a horizontal bench have been mastered.

Lowering the Bar to the Chest

Lowering the bar and press are done on the inhale. During the first stages of teaching this particular technique, it is observed that many beginning athletes lower the bar to different points on their chest. It is crucial to teach the athletes to lower the bar to the same point every time, i.e. to the lower part of their chest muscles.

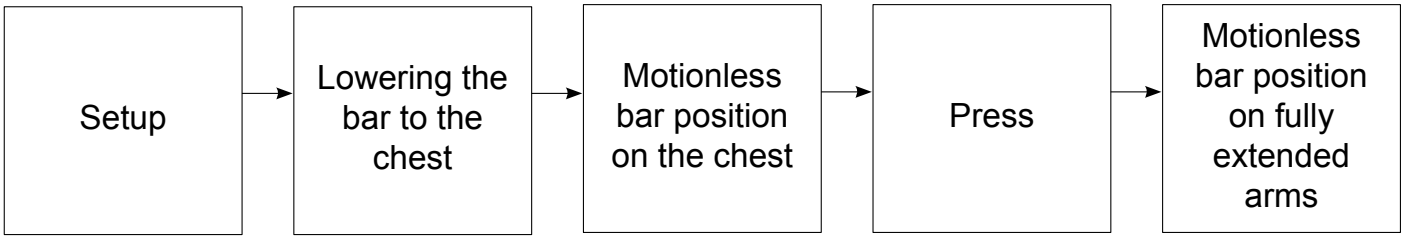


Fig. 3.9. Sequence of teaching the bar lowering technique

Lowering the bar to the lower part of the athlete's chest muscles should be done slowly, with full control of all the muscles groups activated in the exercise. The coach stands next to the athlete's head and by holding the bar helps the athlete to lower it to the same point on the chest.

It is important to remember that if the elbows are pressed to the torso, the deltoid and triceps muscles will be highly strained, if the elbows are spread, the chest muscles will get the maximum load. That is why, in order to perform this routine the optimal elbows position should be chosen, so that the load is equally spread between different muscles groups – chest, front deltoid and triceps muscles.

Motionless Bar Position on the Chest

After the bar has been lowered, the athlete should keep it motionless for a certain and obvious time period. The motionless position means a stop.

Photo 3.58

Photo 3.59

Photo 3.60

Photo 3.61

Photo 3.62

In order for the stabilization of the pause with the bar on the chest not to present any difficulties, during training the bench press technique, it is recommended to make the pause longer than the one during competitions.

Press

During teaching this phase, the attention should be drawn to simultaneous and complete arm extension as well as to the visible stop of the bar in the final stage. Due to the fact that at this stage the athletes' muscles are not fully developed, it is recommended to start with triceps bench press technique.

During the teaching process, the coach should determine what benching style will be used by the athlete. If the athlete prefers to lower the bar to the middle of his chest muscles, it signifies that his chest muscles are strong, and it is more comfortable for him to perform the "chest" press. If the athlete lowers the bar further down to the chest muscles and moves the elbows toward the torso, he should be taught the "triceps" bench press style. In the "chest" bench press style, the bar moves almost vertically and is slightly directed towards the racks, while in the "triceps" style, the bar moves forward from the athlete's head.

Bar Stabilization in the Final Stage of the Bench Press

While teaching the athletes to stabilize the bar during the final stage of the bench press, special attention should be paid to the complete elbow extension as well as the setup for the final stage of the bench press. Very often, the athlete rush through the set (especially when doing 5-6 reps), therefore, they don't lock their arms. If this mistake is not pointed out straight away, it can become automatic. The coach should require from the athletes to lock their arms fully during other exercises such as push-ups, dumbbell press seated, lying, standing.

3.2.4. Back Arching in the Bench Press Technique

Teaching the athlete to press with an arch in his back can start after the press technique without an arch has been acquired. In order to make the back muscles and spine more accustomed to the arch, it is recommended to use a roller that is placed under the back. After the back muscles and spine get used to the arching with a 8-10cm thick roller, the training should be continued with a 13-15 cm thick roller.

Photo 3.63

Photo 3.64

Photo 3.65

Bench Press with a Roller

The bigger the arch, the shorter the bar's path, which leads to muscle strain reduction and the opportunity to lift a heavier weight. Currently, high-class athletes use a number of bench press setup techniques with an arch. The most common are three main types that allow for the maximum arch in the starting position.

The sequence of assuming the starting position in type 1 and 2 are demonstrated by Yuliya Chistyakova, the two-time winner of IPF Junior World Bench Press Championships.

During Type 1, after placing the hands on the bar and establishing the grip width, the athlete places his feet (photo 4.45) and, while lifting his hips (photo 3.65), he tucks in the shoulders and pushing off the bar he places his shoulders as close as possible to the hips, without changing his foot placement (photo 3.66).

The feet should be placed as close as possible to the vertical projection of the hips. The athlete's first support points in this position are his neck and trapezius muscles. His shoulder blades should be tucked in and must not touch the bench, the shoulders should be lowered. The athlete's legs and back muscles are activated. The shoulders and hips should touch the bench. The legs are the second supporting point.

Photo 3.65

Photo 3.66

Photo 3.67

Photo 3.68

Photo 3.69

Photo 3.70

Photos 3.65-3.70. Sequence of assuming the starting (Type 1)

The athlete takes the unracked bar from the spotter, and with his help moves the bar forward while lifting his hips (photo 3.67). Further on, without the spotter's assistance, he holds the bar with his fully extended arms and lowers it down to the maximum, tucking in his shoulder blades and lowering his shoulders. Then, he lowers the hips to touch the bench and assumes the starting position (photos 3.68-3.70).

Type 2. After placing the hands on the bar and establishing the grip width (photo 3.71), the athlete places his feet on the bench (photos 3.72-3.73) and after lifting his hips off the bench (photo 3.74), he tries to move it as close to his shoulders as possible. Then, the athlete lowers his feet to be in line with the vertical projection of his hips (photo 3.75-3.76).

Photo 3.71

Photo 3.72

Photo 3.73

Photo 3.74

Photo 3.75

Photo 3.76

Photo 3.77

Photo 3.78

Photos 3.73 - 4.78. Sequence of assuming the starting position (Type 2)

With his hips lifted, the athlete takes the unracked bar from the spotter (photo 3.77), and with his help moves the bar forward. The following steps of the technique are the same as in Type 1.

Type 3 of assuming the starting position with an arch is demonstrated by Aleksey Nikulin, Master of Sports of Russia (RPF), holder of the „Elita" award (WPC).

After placing his hands on the bar and establishing the grip width, the athlete lifts his body above the bar (photos 3.79-3.80). He spreads his legs wide and firmly presses them onto the platform. Pushing off the bar with his arms, he tries to move the shoulders as close to the hips as possible, depending on the flexibility in his waist (photos 3.81-3.82).

Photo 3.79

Photo 3.80

Photo 3.81

Photo 3.82

Photo 3.83

Photo 3.84

Photos 3.79-3.84. Sequence of assuming the starting position (Type 3)

Unlike in the other two types, the athlete takes the bar from the spotter while his hips is laying on the bench (photo 3.83). After receiving the bar from the spotter, the athlete slightly moves his feet forward, so that the support moves to the mid-foot section. Photo 3.84 demonstrates that the athlete has stabilized the bar in the starting position.

Type 4 is an original way of assuming the starting position demonstrated by the Australian athlete, Ben Polke.

Sitting on the bench, the athlete puts his feet on his toes far behind his torso (photo 3.85). After turning his torso to the right, Ben grips the bar with his right hand (photo 3.86), then he turns to the left and grips the bar with his left hand (photos 3.87-3.88).

After gripping the bar, Ben arches his back and lowers his torso on the bench (photos 3.89-3.90).

Photo 3.85

Photo 3.86

Photo 3.87

Photo 3.88

Photo 3.89

Photo 3.90

Photo 3.91

Photo 3.92

Photos 3.85-3.92. Sequence of assuming the starting position (Type 4)

After assuming the starting position (photo 3.95), the spotter helps the lifter unrack the bar (photo 3.91).

When setting up, it is important to take the height of the bar into account. It should be on the racks at such a height so that the athlete can take the bar from the spotter without changing the arch position (unlike in Type 1 and Type 2 where the athlete takes the bar from the spotter with his hips laying on the bench).

In case the bar is too high on the racks, the athlete would have to reach for it, which implies a smaller arch in the back and bigger length of the press. If the bar is too low on the rack, the athlete would not be able to fully arch his back and it would be difficult for him to unrack the bar.

3.2.5. Common Competitive Bench Press Techniques Mistakes

The ability to assess how correct the techniques is, to notice and correct mistakes, is an important condition of successful teaching. In the process of teaching this technique, a number of common mistakes have been recorded. Here are some of their specifics:

1. The athlete must lie on his back with head, shoulders and glutes in contact with the bench surface. The feet must touch the surface of the platform or blocks (they should be as flat as the shape of the shoe will allow). His hands and fingers must grip the bar positioned in the rack stands with his fingers locking the bar. This position shall be maintained throughout the lift. Foot movement is permissible but must remain flat on the platform or blocks. The hair must not hide the back of the head when lying down on the bench. The preferred hairstyle is ponytail or bun.

2. To achieve firm footing the athlete may use flat surfaced plates, or blocks not exceeding 30 cm in total to build up the surface of the platform. Blocks in the range of 5 cm, 10 cm, 20 cm, and 30 cm, should be made available for foot placement at all international competitions.

3. The spacing of the hands shall not exceed 81 cm measured between the forefingers (both forefingers must be within the 81 cm marks and the whole of the forefingers must be in contact with the 81 cm marks if maximum grip is used). The use of the reverse grip is forbidden.

4. After receiving the signal, the athlete must lower the bar to the chest or abdominal area, hold it motionless, after which the head referee will signal the audible command “Press”. After receiving the command, the athlete must then return the bar to straight arm’s length elbows locked. When held motionless in this position the audible command “Rack” shall be given together with a backward motion of the arm. If the bar is lowered to the belt and remains there for five seconds, the head referee will give a “Replace” command rack.

5. Heaving, or sinking the bar into the chest or abdominal area after it is motionless in such a way as to make the lift easier.

6. Bar is not lowered to chest or abdominal area, or the bar is touching the belt.

7. Failure to press the bar to straight arm’s length elbows locked at the completion of the lift.

8. Contact with the bar or the lifter by the spotter / assistant between the head referee’s signals, in order to make the lift easier.

9. Any contact of the lifter’s feet with the bench or its supports.

10. Deliberate contact between the bar and the bar rests support.

1. The bar is not aligned

Photo 3.92

Not aligning the bar properly can lead to uneven press.

2. Reversed, thumbless grip

Photo 3.93. Reverse-grip Bench Press demonstrated by the World Champion, Anthony Clark

Due to the possibility of injury, this grip is prohibited by the technical rules in many international federations.

3. The head is too close to the racks

In case the bar's projection is lower than the athlete's eyes level when he is setting up, he could touch the racks while pressing, which will create unnecessary contact of the bar with the racks, or the bar will stop entirely.

4. Improper breathing during the bench press

Bench press should be executed on the inhale.

5. Lifting the head after the "Start" command.

Photo 3.94

Lifting the head from the surface of the bench is forbidden according to the IPF rules. However, it is allowed during many other international competitions.

6. Lifting the hips from the bench surface after the "Start" command

Photo 3.95

Lifting the hips from the bench surface is forbidden by the technical rules in all international competitions.

7. The rules do not allow the bar to touch the belt

Photo 3.96

8. Lifting the heels from the platform or block surface

Photo 3.97

Lifting the heels from the platform or blocks surface after the setup is forbidden by the technical rules of IPF. In other international federations it is allowed.

9. Failure to keep the bar motionless on the chest

Failure to keep the bar motionless on the chest, or heaving the bar is forbidden by the technical rules of competitions.

10. Sinking the bar into the chest after its motionless

Photo 3.98

The lift is disqualified if the referees notice that after keeping the bar motionless on his chest, the athlete sinks it after the “press” command to make the lift easier.

11. The bar does not touch the chest or abdominal area

Photo 3.99

When wearing a multi-layered bench press T-shirt many athletes experience problems with lowering the bar down to their chest.

12. Any downward movements of the bar during the press

13. Failure to extend the arms and locking the elbows in the final stage (photo 3.100)

Photo 3.100

The attempt is considered failed if in the final press stage the bar is motionless with the arms not fully extended and the elbows not locked.

14. Contact with the bar or the lifter by the spotter / assistant during the execution of the routine

Photo 3.101

The lift is considered failed if, in between the head referee's signals, there has been physical contact with the bar or the lifter by the spotter.

15. The bar is not aligned during the press

Photo 3.102

16. Any contact with the bench or the rack stands by the athlete's feet

Photo 3.103

Any contact with the bench or the rack supports by the athlete's feet is considered an error according to the competition rules.

17. Contact with the bar and its supports during the bench press

Photo 3.104

If the contact of the bar and its supports does not facilitate the execution of the routine, it is not considered an error according to the competition rules.

18. Failure to comply with any of the items outlined under the Rules of Performance of the bench press.

3.2.6. Deadlift Technique Training

The teaching process of the deadlift technique to beginning athletes can be divided into five stages.

The first stage includes teaching the setup technique. First, the approach toward the bar is taught, followed by the setup, foot placement in relation to the bar. Then, the athlete bends his knees, leans forward, decides on the grip width, grips the bar with “the mixed grip”, and assumes the starting position.

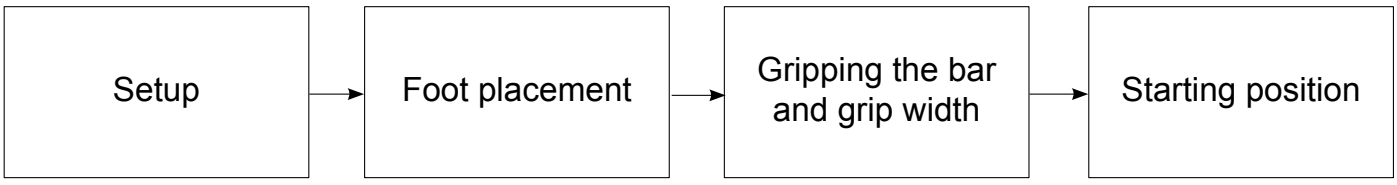


Fig. 3.10. Sequence of teaching the setup and starting position

When teaching the “sumo” technique of foot placement during the setup position, it is recommended to place the feet as wide as possible, which is an individual characteristic of every athlete. At this teaching stage, attention must be drawn to the knees position: they should be in line with the toes.

Photo 3.105

Photo 3.106

Photo 3.107

Photo 3.108

Photo 3.109

Photo 3.110

Photo 3.111

Photo 3.112

Photos 3.105–3.112. Sequence of teaching the setup and starting position

During the acquisition of the technique, special attention should be paid to the position of the athlete's back. Under no circumstances should it be rounded. The back must be straight and arms relaxed. To achieve the correct posture, the coach stands next to the athlete and, by pushing down on the lower back with one hand, and up on top of the chest with the other hand, helps the athlete to assume the correct back position at the setup. The shoulder joints are of special importance in this case, as they should be in line with the bar.

After mastering the starting position technique, one can proceed to the stage in which the athlete interacts with the bar before lifting it from the platform as well as preliminary acceleration. A static start is used for this purpose. During the execution of this routine, attention should be drawn to the athlete's shoulder position when lifting the bar off the platform – the shoulders should be in vertical alignment with the bar.

The second stage is the deadlift itself. At this stage, it is recommended to use instructional techniques of separated exercises, which are:

Reversed order with blocks of various height;

Reversed order of the starting position: hang deadlift

The method of the reversed order with blocks of various heights appears to be the most effective (fig. 3.11). Let's analyze what happens during this stage.

When teaching the stage of moving from the starting position to the complete knees and torso extension, it is advisable to separate the movement into two parts. One should start from the second part of the deadlift starting position when the bar is placed at the mid-thigh level. In order to achieve this, the bar is placed on high blocks. With such a high start, it is easier to assume the proper starting position and finalize the move. The athlete executes the pull until his legs and torso are straight with the obligatory stabilization of the position at

the end of the move. Usually, one training session is sufficient to master the proper deadlift technique using a high blocks, and with training further on, one can proceed to learning the deadlift from a lower position.

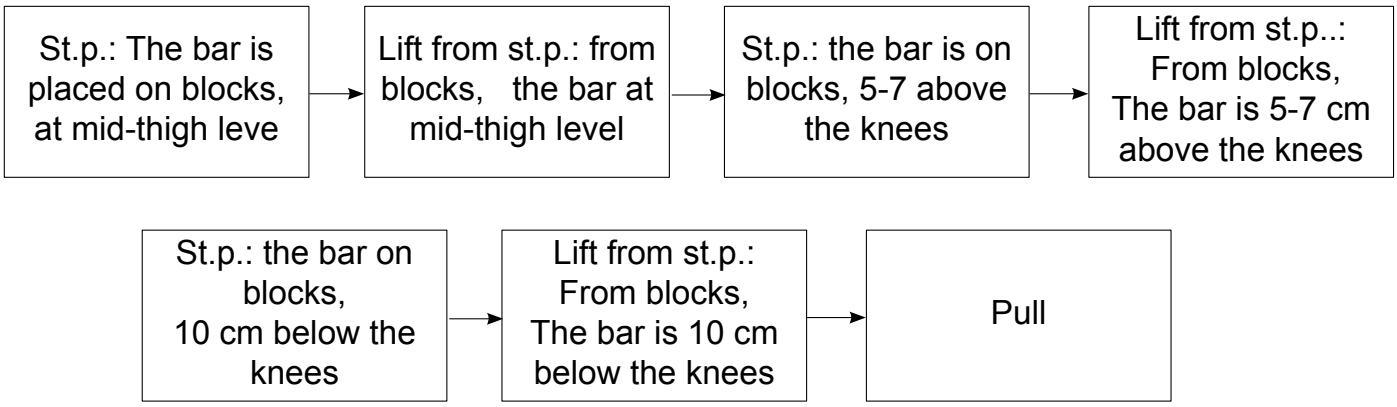


Fig. 3.11. Algorithm of teaching deadlift technique with the reversed order with blocks of various height

Photo 3.113

Photo 3.114

Photo 3.115

Photo 3.116

During the second session, athletes perform the deadlift with the blocks when the bar is placed 5-7 cm above the knees. During the third lesson, the bar is placed 10 cm below the knees.

The most important condition of the reversed method with blocks of different height (fig. 3.12) is the fact that one can move on to performing the deadlift from lower height blocks only after the former technique has been acquired (photos 3.117-3.20).

The third teaching stage, i.e. execution of the complete routine from the previously learned starting point, can begin after the athlete is able perform the deadlift with various boxes correctly and with confidence (fig. 3.12).

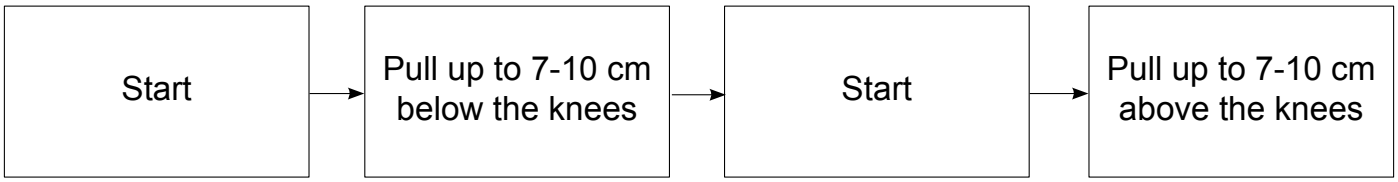


Fig. 3.12. Sequence of teaching the deadlift technique

After assuming the starting position, the athlete pulls the bar to 7-10 cm below his knees, makes a 1-2 seconds pause, and lowers the bar to the platform. Three sessions with 3-4 reps should be done (photos 3.122–3.124).

Photo 3.117

Photo 3.118

Photo 3.119

Photos 3.117 – 3.119. Pull up to 7-10 cm below the knees

After mastering the pull up to 7-10 cm below the knees with 1-2 seconds break, the athlete performs the pull up to 7-10 cm above his knees, and after 1-2 seconds pause, he lowers the bar on the platform (photos 3.120 – 3.122).

Photo 3.120

Photo 3.121

Photo 3.122

Photos 3.120– 3.122. Pull up to 7-10 cm above the knees

Provided that the abovementioned exercises are performed correctly, one can move to the fourth teaching stage.

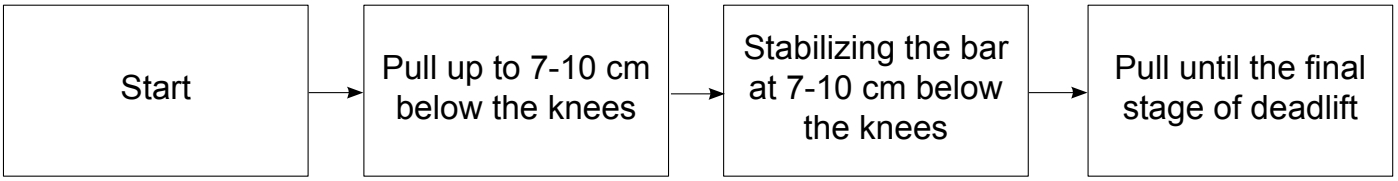


Fig. 3.13. Sequence of teaching the deadlift technique

From the starting position (start), the athlete pulls the bar up to 7-10 cm below his knees, pauses for 2-3 seconds, after which he continues the execution until the final stage (photos 3.123-3.125).

Photo 3.123

Photo 3.124

Photo 3.125

Photos 3.123 – 3.125. Pull up to 7-10 cm below the knees, stabilization and pull until the final stage

The task becomes more complicated at the next stage: from the starting position the athlete pulls the bar up to 7-10 cm above his knees, pauses for 2-3 seconds, after which he continues the execution until the final stage (fig. 3.14).

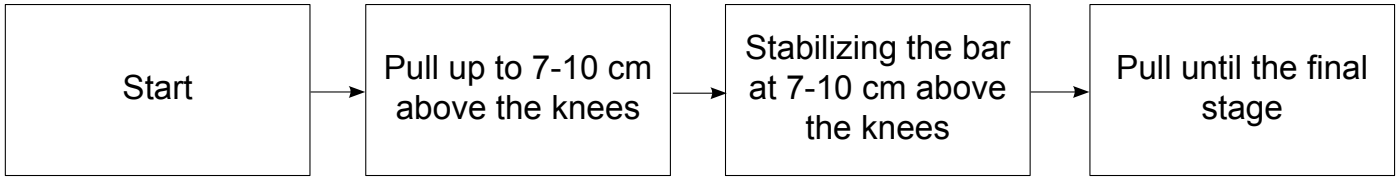


Fig. 3.14. Sequence of teaching the deadlift technique

From the starting position the athlete pulls the bar up to 7-10 cm above his knees, pauses for 2-3 seconds, and continues the execution until the final stage (photos 3.126-3.128).

Photo 3.126

Photo 3.127

Photo 3.128

In order to master all the acquired skills of all the deadlift stages, it is recommended to apply the method of dividing the exercise into parts. The application of this method will help the athlete to master the competitive deadlift technique.

Assuming the starting position (photo 3.129).

Competitive deadlift and bar stabilization in the final stage (photo 3.130).

Photo 3.129

Photo 3.130

Photo 3.131

Photo 3.132

Lowering the bar back to the starting position: hang deadlift, bar at 5-7 cm above the knees, stabilization for 2-3 seconds (photo 3.131).

Pull from the starting position to the final position and bar stabilization (photo 3.132).

Assuming the starting position (photo 3.133).

Competitive deadlift and bar stabilization in the final stage (photo 3.134).

Photo 3.133

Photo 3.134

Photo 3.135

Photo 3.136

Lowering the bar back to the starting position: hang deadlift, bar at 5-7 cm below the knees, stabilization for 2-3 seconds (photo 3.135).

Pull from the starting position to the final position and bar stabilization (photo 3.136).

Photos 3.137 – 3.138. Sequence of teaching the deadlift technique with reversed order from the starting position: hang deadlift

After the athlete has acquired the complete deadlift technique, one can proceed to the deadlift technique when the athlete steps on blocks, which works the legs and back muscles at the initial stage: raising the bar off the platform (photos 3.139-3.140).

The final teaching stage – deficit deadlift - deadlift while stepping onto 7-10 cm high blocks.

Photo 3.137

Photo 3.138

Deficit deadlift, sumo style

The starting position: stepping onto 7-10 cm tall blocks. It is recommended to place the feet in the position used during competitions: Sumo style (photos 3.137-3.138), or conventional style (photos 3.139 – 3.140). During the execution of the deficit deadlift, the bar's path is extended while the load on the extensor muscles of the legs and torso increases.

Photo 3.139

Photo 3.140

Deficit deadlift, conventional style

This particular exercise helps increase the force when raising the bar off the platform. It is not advisable to use this exercise when training novice athletes as the starting position is different from the classic one, which can lead to mastering the improper skill.

Depth squat (or belt squat between blocks) appears to be a good facilitating exercise when learning the deadlift technique. A number of additional exercises are used, e.g. goodmornings - standing or seated. It is better to teach the final stage from the starting position: the bar is on the blocks and above the knees, simultaneous straightening of the legs and back, lifting the bar to the final stage.

3.2.7. Most common deadlift errors

Any downward movement of the bar before it reaches the final position.

Failure to stand erect with the shoulders back.

Failure to lock the knees in a straight position at the completion of the lift.

Supporting the bar on the thighs during the performance of the lift. If the bar touches the thigh but is not supporting any weight directly with his thighs, this is not a reason for disqualification. The referee has the right to decide whether the contact was legal or not.

Stepping backward or forward, or horizontal feet movements, although lateral movement of the sole and rocking the feet between the ball and heel is permitted.

Lowering the bar before receiving the signal from the head referee.

Allowing the bar to return to the platform without maintaining control with both hands, i.e. releasing the bar from the palms of the hand before it has reached the platform).

Failure to comply with any of the items outlined under Rules of Performance.

Asymmetrical grip.

Photo 3.141

This particular grip may lead to improper bar alignment.

- **Too narrow feet placement**

Photo 3.142

If the distance between the feet is reduced, the bar's path increases. Consequently, the wider the foot placement, the shorter the bar's path.

- Transferring the athlete's center of gravity to his toes or heels.

If the bar is closer to the toes, in the initial stage the lever arm length between the footing and the center of gravity will increase. As a result, the load on the back muscles will increase and raising the bar off the platform will become more difficult. The pull, in which the center of gravity is towards the feet, will result in the athlete pulling the bar too close to his body, and the bar will come into close contact with the body, which will lead to absolutely undesired friction between the bar and the legs, or the lifter's arms and his/her legs.

- **Feet turned out too much in the starting position**

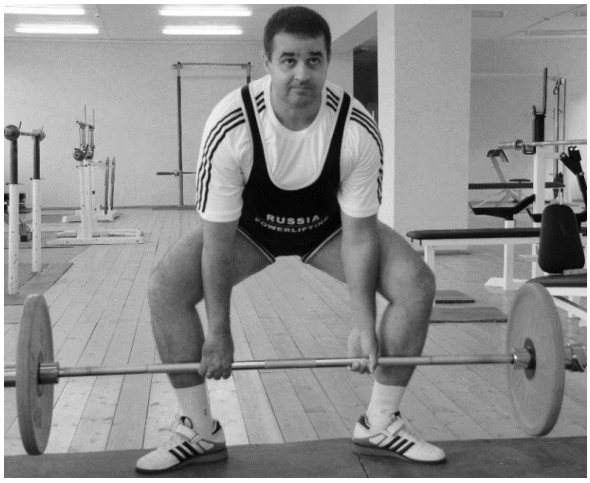


Photo 3.143

The foot placement with the feet turned out too much leads to poor stability in the final phase and therefore requires more precise execution of the movement.

- **Knees not pushed out sufficiently in the starting position**

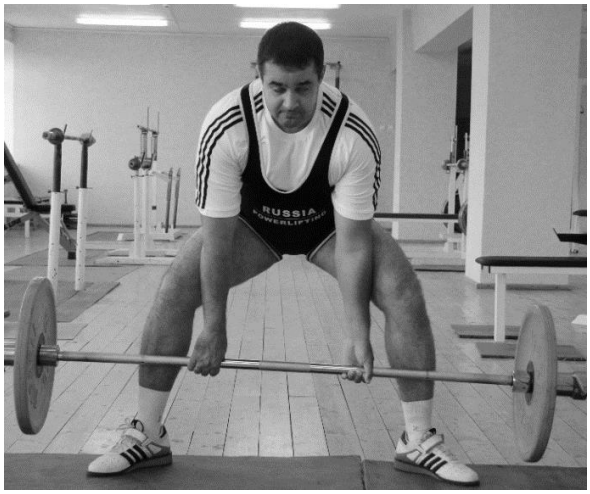


Photo 3.144

If the knees are not pushed out sufficiently, the hips move away from the bar, which increases the load on the lumbar and creates difficulties when raising the bar off the platform. During the deadlift, the knees should be in line with the toes at all times.

- **Improper starting position when the shoulders are not in line with the bar.**

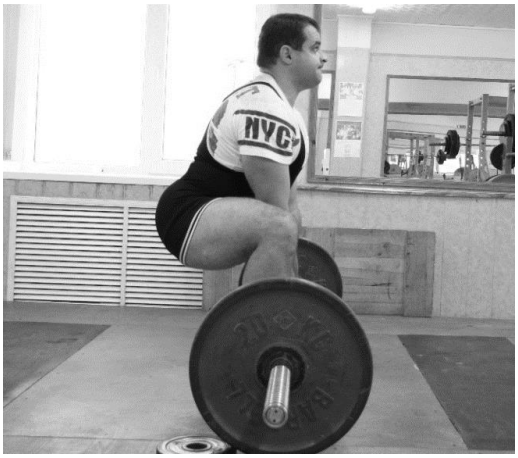


Photo 3.145

Photo 3.146

This error may lead to the bar position being too far from the legs, which will increase the load on the lumbar, especially if the shoulders are too far ahead of the bar. If the shoulders are too far behind the bar, the lifter can end up pulling the bar into his shins, wasting upward force on overcoming the friction between the bar and his shins; force that could have gone to propel the bar upward instead.

- **Jerking the bar**

The athlete may lose control of the bar, which will result in failure to perform a proper deadlift. Jerking the bar removes the athlete's body position from the best position, making the rest of the lift needlessly difficult.

Premature straightening of the legs at the initial stage

It may lead to a shift of the center of gravity of the bar forward, leading to a sharp increase in the load on the back muscles.

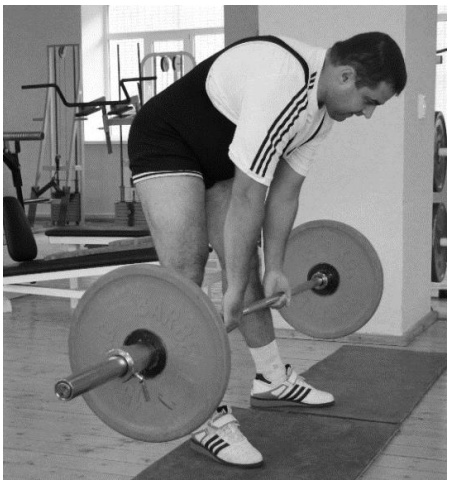


Photo 3.147

This mistake may also lead to bending of the torso and increasing of the load on the lumbar, which will complicate straightening of the torso in the final stage.

- **Head bent forward**

Bending the head forward will relax the back muscles and will round the back.

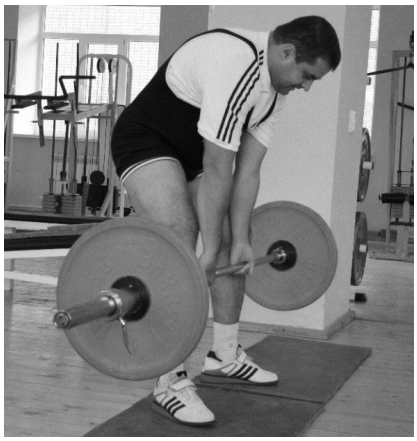


Photo 3.148

- **Rounding the back in any deadlift phase.**

Rounding the back leads to the whole load being placed on the lumbar. The bigger the rounding of the back, the lower the leg muscle load and the bigger on the back muscles. It can lead to injury.

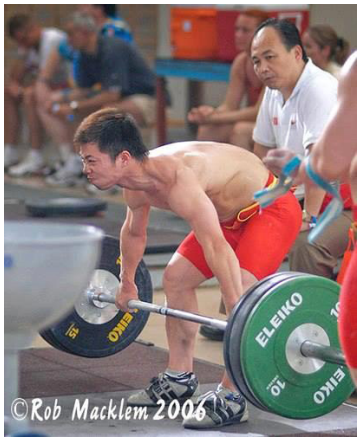


Photo 3.149

13. Failure to lock the knees at the completion of the lift

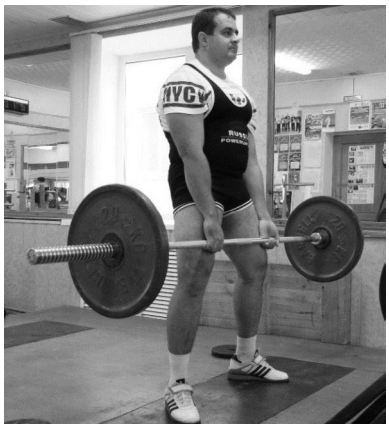


Photo 3.150

14. Failure to stand erect with the shoulders back.

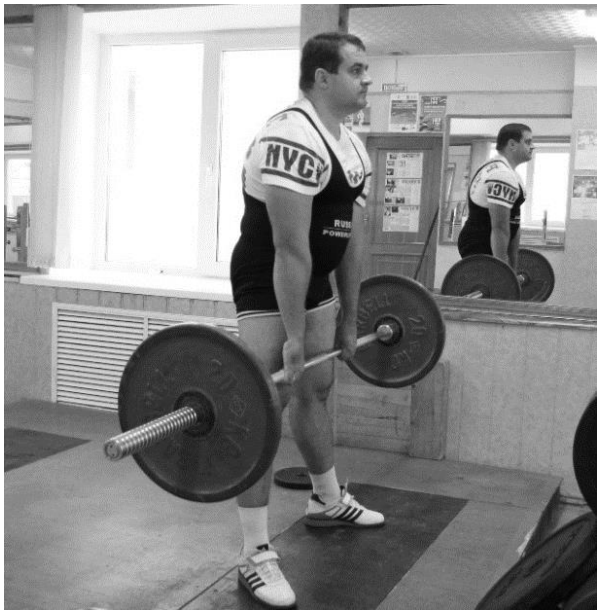


Photo 3.151

15. Any downward movement of the bar before it reaches the final position.
If the bar sinks lower when moving the shoulders back, this is not a reason for disqualification.

16. Supporting the bar on the thighs during the performance of the lift.

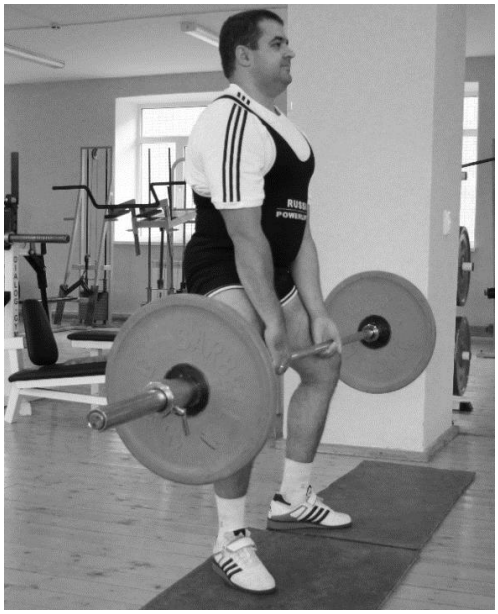


Photo 3.152

If the bar touches the thigh but is not supported, this is not a reason for disqualification.

17. Lowering the bar before receiving the signal from the head referee.

18. Allowing the bar to return to the platform without maintaining control with both hands.

Photo 3.153

According to the competition rules, the athlete should maintain the control of the bar with both hands until it reaches the platform.

19. Stepping backward or forward, or horizontal feet movements

Lateral movement of the sole and rocking the feet between the ball and heel is permitted.

20. Failure to comply with any of the items outlined under Rules of Performance.

The abovementioned mistakes do not allow the athletes to show the competition results that correspond to their potential. Coaches and athletes need to remember that the best way to correct the error is to find and eliminate the reasons that cause them. If there is more than one error, it is very important to highlight the main one. The main errors are the ones that cause the biggest disruptions to the whole movement. When selecting the exercises to correct a particular error, the following methodological guidelines should be applied:

- Lower the bar weight to 50-60%, and after the new skills have been mastered, increase the weight.
- Clarify the motor performance and «muscular sense» of the learned exercises.
- Break down the exercise into parts.
- First of all, use such facilitating exercises that would make the athlete assume the required position, which will lead to the improvement of the technique.
- Work on correcting the mistakes and acquiring new skills with 60-70% of the weight, followed by a gradual increase of the weight.

Chapter 4. Powerlifting Nutrition (written by Leonid Ostapenko)

Nutrition is one of the most important recovery measures in just about every sport, and powerlifting is not an exception. Due to the fact that there are weight classes present in powerlifting, nutrition, apart from providing for energy and recovery, should allow an athlete to keep his body weight close to his current weight class, prepare for weighing in competently, and properly recover the fluid and electrolyte balance before lifting on meet day.

First, a few words about the physiological and biochemical characteristics of powerlifting as they pertain to nutritional demands.

4.1. Physiological basis for the development of muscle strength

Among many muscular (peripheral) factors determining the maximal voluntary force, the diameter (thickness) of the activated muscles is of high importance. Under equal conditions, the bigger the total diameter of the voluntary contracted muscles, the bigger muscle strength manifested. The second important factor is muscle composition, i.e the ratio of fast and slow-twitch muscle fibers in specific muscles responsible for the performance of a competitive exercise move.

Fast twitch, high-threshold motor units of muscles are the major contributors to muscle force development, they contain more myofibrils, they are thicker and contribute more to the increase in the diameter of the working muscle, and therefore also to the developing of strength. As a result of physical training, working muscle hypertrophy occurs – almost or exclusively due to thickening of the existing muscle fibers. With a significant thickening of muscle fibers, their longitudinal mechanical splitting may occur when «daughter» fibers with a common tendon may form. During strength training the number of longitudinal splitting of the fibers increases.

There are two extreme types in the mechanisms of muscle hypertrophy - sarcoplasmic and myofibrillar hypertrophy. Sarcoplasmic hypertrophy means thickening of muscle fibers due to the increase in sarcoplasmic volume, i.e. non-contractile muscle parts. Hypertrophy of this type occurs due to the increasing number of non-contractile (in particular, mitochondrial) proteins and metabolic reserves of muscle fibers: glycogen without nitrogenous substances, phosphocreatine, myoglobin and other cell non-contractile cell contents. Slow (I) and fast (II-A) oxidative muscle fibers are most susceptible to sarcoplasmic hypertrophy. Muscle hypertrophy of these types has little effect on the increase of muscle strength, however, it greatly improves the ability for continuous work, i.e. it increases muscle endurance.

Myofibrillar hypertrophy is associated with an increase in the number and volume of myofibrils, i.e. contractile apparatus of muscle fibers. The density of myofibrils in muscle fibers also increases. This particular hypertrophy of muscle fibers leads to a significant increase in maximum muscle strength. Moreover, the absolute muscle strength significantly increases, and it remains the same or slightly decreases in Type I hypertrophy. Fast (II-B) muscle fibers are most prone to myofibrillar hypertrophy.

Exercises with large muscle force development (more than 70% -from an arbitrary maximum force of the engaged muscle groups), specific to powerlifting, mainly contribute to the development of the second type of hypertrophy. The basis of this particular hypertrophy is an intensive synthesis and decreased disintegration of muscle protein. Creatine, the content of which increases in the contracting muscle, can also stimulate intensified synthesis of myosin and actin, thus contributing to the muscle fiber hypertrophy.

It has been established that strength training does not change the ratio in the two basic types of muscle fibers - fast-twitch and slow-twitch. However, it can change the ratio of the two types of fast-twitch fibers, increasing the percentage of fast glycolytic fibers (FG), thus reducing the percentage of fast oxidative-glycolytic fibers (FOD). As a result of strength training, the level of the fast muscle fibers hypertrophy is considerably greater than that of the slow oxidative fibers (SO).

Powerlifting training is associated with a relatively small number of repetitions of maximum or similar muscle contractions that involve both fast and slow muscle fibers. All this is accompanied by an active physiological deterioration of the protein of the actin-myosin complex, which in the process of recovery is to not only given a chance to come to its original state, but also to get the conditions for supercompensation, as only supercompensatory mechanisms provide the stable growth of sports performance and a successful sports career of an athlete. In this process, nutrition, as one of the most important factors ensuring the supercompensation, plays a pivotal role.

4.1.1. Biochemical basis for the development of muscle strength

Structural factors of speed and power abilities of an athlete (length of sarcomeres in the myofibrils, content of fast and slow-twitch fibers in muscles) are genetically determined, so the main methodological way to improve power qualities is to choose the means and methods that could improve the ATP activity of myosin and increase the synthesis of contractile muscle proteins. In powerlifting, two basic methodological techniques are applied - maximal effort method and the method of repetitive ultimate exercise.

In order to develop the ability to maximize the strength in powerlifting the exercises applied are similar in biodynamic structure to the competitive exercises, or competitive exercises themselves. The maximum amount of exercise with maximum strength, speed or power is determined by the critical concentration of PCr (phosphocreatine) in muscles, below which it is no longer possible to maintain the maximum speed of ATP resynthesis. Due to this amount of PCr, typically up to 5-6 repetitions of exercises are performed continuously. PCr and other energy-rich compounds deplete when performing a large volume of exercises leads to the destruction (physiological wear) of muscle proteins and the accumulation of their decay products (low molecular weight peptides, amino acids, etc.). The byproducts of proteins breakdown, as well as free creatine, are activators of protein synthesis during rest periods after speed and power work when normal oxygen supply and enhanced delivery of nutrients to tissues is restored. The accumulation of lactic acid during extreme workload and the consequent changes in intramuscular osmotic pressure contribute to the storage of rich in nutrients interstitial fluid in muscles. During systematic repetition of such trainings, the content of contractile proteins in the muscles significantly increases as well as total muscle mass. Thus, the primary role of quality nutrition emerges again, being a condition for increasing the volume of contractile proteins and guaranteeing of the steady growth of strength performance.

4.1.2. Anatomical characteristics of powerlifting

The focus of training loads in powerlifting is quite specific, i.e. it aims to create a base for the implementation of maximum dynamic force in three exercises. These exercises, according to anatomical criteria «the topography of strength» affect the largest and the most contributory muscle groups. One part of these muscle groups are prime movers, the second

one - auxiliary movers or synergists and, finally, the third part – stabilizers of the body as a whole or its parts during training or competitive exercise performance. In other words, when a big part of muscle mass is activated during training, this leads to a strong metabolic shift that requires powerful recovery processes occurs in the body.

Since during strength training the main load is received by fast twitch and intermediate muscle fibers, protein metabolism plays a very important role. During strength training, recovery creates conditions for the adaptive growth of actin and myosin myofilaments, which consist mainly of protein. This is what myofibrillar hypertrophy is.

Muscle elasticity plays an important role in the maximum strength indicators during competitive powerlifting movements' performance. It depends on the degree of muscle cells fullness - in other words, hydration. This phenomenon can be qualified as a different type of hypertrophy - sarcoplasmic. Hydration, in turn, depends on the reserves of creatine, glutamine, and glycogen resources of a muscle cell. In this way, the supplementation of the body with the appropriate types of carbohydrates and energizers of muscle cells is one of the most important tasks of powerlifting nutrition.

It should be noted that in the course of an intensive and very hard physical training the cartilage surface of the joints is subject to very heavy loads, while tendons and ligaments are subject to tensile stress. This entails an increase in connective-tissue protein, i.e. collagen as the major protein in the human body that accounts for about 40% of all protein resources of the human body.

Finally, when dealing with extreme weights, phosphorus and creatinine mechanism of cell energy plays a very important role. Apart from phosphorus salts, it requires the presence of other important minerals, which leads to an increased demand for other mineral substances as well. Vitamins and vitamin-like substances must not be neglected, as the powerlifting training increases the need for the above at least 2-2.5 times compared with an average inactive person of the same weight and age.

Keep in mind that the human body is a complex and multi-functional system, and it responds to any exposure as a system. That is, if an important nutrient or an active substance is missing, not an isolated part of the body is affected – the coordinated work of the entire system is somewhat compromised. It is for this reason that an inadequate and unbalanced diet contribute to overtraining or a plateau among a significant number of powerlifters.

Nutrition is one of very, if not the most important means of body recovery after strenuous workouts. The right choice of diet determines the success in strength training (muscle density increase and strength abilities improvement) in 50% and according to some experts - 70-80%. In order to make the right decisions in relation to the diet construction, it is not enough to understand what is included in the diet and how it is built during the day. It is important to understand the main points of an adequately balanced diet, and strictly follow them during powerlifting training.

4.2. Importance of adequacy and balanced diet

Adequacy refers to the correspondence between food consumption and total energy expenditure made up of home and work activities, as well as training loads. The term "balanced diet" refers to the right balance of macronutrients in a daily diet, that is, the energy contribution of proteins, fats and carbohydrates in the total energy.

An adequate balanced diet is crucial for recovery and prevention of overtraining, if it is able to fully provide the compensation processes with energetic, plastic and other biologically important substances, taking into account the immediate or delayed needs and requirements imposed on the body. It has been observed that if an athlete eats poorly, without balancing his diet, his training, in spite of the wisest methodological approach toward the training sessions, becomes a factor that does not improve performance and health, and a circumstance that undermines the body's defense mechanisms and leads to a deterioration in the functional state.

In our approach to the construction of such a diet you will read further on, and here we offer a simple test, based on which an athlete can roughly determine whether his diet is adequate and balanced. Those points that meet an athlete's nutritional situation should be marked, then the total amount should be calculated, and the diet should be evaluated based on the proposed scheme.

«An athlete should get at least 55-60% of the daily caloric intake from foods that contain natural carbohydrates» (Duclos M. (2008).

Table 4.1

Assessment of carbohydrate intake adequacy
(Ostapenko L.A. 2004)

	Never	Sometimes	Often	Always
1. I eat fruit minimum once a day	0	1	2	3
2. I prefer freshly-squeezed vegetable or fruit juice, not soda	0	1	2	3
3. I prefer wholegrain bread and pastry, not white	0	1	2	3
4. I eat cabbage, carrots, radish, squash, greens every day	0	1	2	3
5. I eat whole grain or grain products	0	1	2	3
6. I limit sugar an sweet consumption	0	1	2	3
7. I take carbohydrate supplements (gainers)	0	1	2	3
Total:				

"At least 25-30% of calories per day should be provided due to the protein part of the diet" [1].

Table 4.2

Assessment of protein intake adequacy
(Ostapenko L.A. 2004)

	Never	Sometimes	Often	Always
8. I regularly consume low-fat milk, cheese, yogurt	0	1	2	3
9. Every meal contains meat, fish, or eggs	0	1	2	3
10. I eat legume products in addition to proteins of animal origin	0	1	2	3
11. I have protein powders and amino acids to optimize the protein metabolism	0	1	2	3
Total:				

«Fats should provide no more than 10-15% calorie daily diet». [1]

Table 4.3

Assessment of fats intake adequacy
(Ostapenko L.A. 2004)

	Never	Sometimes	Often	Always
12. I choose only 0% or low-fat dairy products	0	1	2	3
13. I avoid eating lard, margarine, cream, sour cream, cream products	0	1	2	3
14. I do not use fattening sauces, fattening spices and mayonnaise and other cream dressings in salads	0	1	2	3
15. I do not eat canned meat, smoked products (bacon, sausage) and fish canned in oil	0	1	2	3
16. I limit the amount of fats/oils when cooking	0	1	2	3
17. I monitor the fats amount in dishes served in restaurants	0	1	2	3
18. When shopping I choose low-fat products	0	1	2	3
Total:				

After calculating the points in every single parts of the test summarize the result. Nutrition assessment:

0-20 points: An athlete may have difficulties in achieving high sports result. Strength and muscle mass gain will be slow, he will hit plateaus early, and will easily get tired as well as recover slowly. It is more likely that the athlete has reached an overtraining stage which might also be chronic. The approach to the organization of his diet should be drastically changed.

21-30 points: He has a chance of success in the early stages of powerlifting training, but later, when trying to improve efficiency and push trainings loads higher, will experience very incomplete recovery. Strength performance gains will slow down. Periodically bouts of extreme fatigue will occur, which will be impossible to compensate with changes in training. It will be extremely difficult to prepare for competitions, the athlete will lose strength and body weight. It is necessary to adjust the consumption of protein and carbohydrates.

31-40 points: The athlete is definitely watching his diet, and he should not encounter any problems in reaching the goals at powerlifting training. However, this diet may not be fully effective in preparation for competitions, especially if the athlete is trying to remain within the chosen weight class. It is necessary to adjust the diet depending on whether he is gaining excess weight or losing muscle mass.

40 and above: He correctly calculated the diet, and the closer the total number is to the ideal value of 54 points, the greater the chance of success the athlete will have not only in powerlifting training, but also in eliminating many risk factors and diseases, including obesity, low immunity, atherosclerosis, and diabetes. If with this assessment it is not possible to lose excess body fat, without changing the existing balance, he should reduce the total amount of food intake. This will reduce caloric intake without distorting the structure of the diet.

Apart from analyzing the total sum of the test, it is useful to analyze the result of every single table on its own.

If the total number of points in Table 1 is lower than 7, you should be careful: the athlete might experience weakness, sleepiness, lack of motivation to train, muscle mass plateau, and/or inability to move to a heavier weight category. During the competitive period he might not be able to recover when moving to extreme weights. All the above may be accompanied by an unplanned body weight and decrease of strength indicators. It is quite possible that these are the symptoms of vitamin deficiency and an inefficient energy value of the diet, leading to hypoglycemia. The solution is to increase the intake of fruit and vegetables as well as whole grain products about 1,5 times.

If the total number of points in Table 2 is lower than 4-6, the athlete might experience difficulties in gaining muscle mass between competition phases or when he decides to move to a heavier weight class. If he gains body mass, it is possible that his strength indicators will remain the same. The sportsman may not recover fully after training sessions. It happens due to the insufficient amount of protein necessary to build the most muscle, and the exhaustion of natural sources of creatine. The solution would be to increase by 1.5 times the proportion of animal products as well as introducing protein and amino acid food supplements.

If the total number of points in Table 3 is lower than 14-16, the main problem may be excess fat and difficulties when losing it, especially if the total number of Table 1 is low. There is a high risk of experiencing metabolic disorders, diabetes, digestive disorders, and decreased efficiency of the cardiovascular system. The athlete may experience lethargy, sleepiness, reluctance to exercise. The consumption of saturated fats should be lowered.

If the score in Table 3 is high, while the score in Table 1 and 2 are low, the athlete may experience constipation, flatulence, intestinal cramps and other disorders of the digestive system. Greater attention should be paid to the consumption of fresh fruits and vegetables.

The adjustment of the daily diet should be done taking into account the results of this simple test.

4.3. Algorithms for calculating an adequate diet and its components

4.3.1. Short-term and long-term requirements

Regardless of the fact if you are an amateur sportsman or an experienced master of strength sports, your nutrition must obey an indisputable rule – it should be adequate and balanced.

To be more precise, an adequate and balanced diet means such a nutrition that satisfies energetic, plastic, short-term and long-term needs of a human body and provides all the necessary nutrients and biologically active substances taking into account the requirements imposed on the body.

Short term needs stand for the needs for certain substances that are present in every single moment of human life. For example, a certain level of glucose must constantly be maintained in human blood, which is the fuel source for many biochemical reactions and motor acts, and a certain level of amino acids, which are the plastic material for the synthesis of new tissue and replacement of the broken down tissues. In the body, at any given time a certain balance of vitamins and minerals is maintained, which is important for optimal functioning. In addition, it is important to take into account that certain substances in the human body cannot be synthesized, so they must be delivered only through the food that we take in. The substances include vitamins, minerals, fiber, water, and certain types of proteins (or rather, the essential amino acids that construct the proteins) and essential fats.

Long-term needs are the needs for the same substances which cannot be satisfied with the level of short-term needs. Such needs occur as a response to the revitalization of the body, e.g. during exercise, training, and enhanced mental performance. Long-term needs are felt only during the development of adaptive reactions, and they are always above the level of short-term needs. Consequently, the physical demands placed on the body always cause an increase in the level of long-term needs. This conclusion is crucial for the planning of an optimal powerlifting diet, if an athlete wants to do succeed in powerlifting.

Chaotic and random diet content is unacceptable in diet creation. In order to create a diet, one should take into account all the necessary components, to use an elementary mathematical approach, and then a simple mathematical correction based on the observed reactions of the body. The adequacy of any diet, particularly energetic adequacy, must be based on concrete figures. Of course, these will only be approximate figures that do not characterize the specificity of a single athlete's body, and certainly they do not reflect his goals and objectives in a particular stage of the training process. However, these approximate figures can be adjusted by tracking the changes happening in the body and its functional systems under the influence of training loads and diet, i.e. it is possible to adjust the diet of any person based on their specific needs.

4.3.2. Calculation and adjustment of energy requirements

The calculation of adequacy should be based on a person's energy requirements. It is expressed in kilocalories, or abbreviated as kcal. One gets the daily human need for energy from the so-called basal metabolism, and the additional energy expenditure (for household and work activities, plus training), as well as to a large extent - psycho-emotional expenditure (the energy required to attend to stress,

excitement, and other mood states), which together give an idea of what a person's total energy consumption is per twenty four hours.

Basal metabolism reflects the amount of energy that a person needs only to maintain the necessary life-saving functions - breathing, blood circulation, some important enzymatic reactions. It is enough to perform the most basic motor actions such as movement in the house and another not highly demanding activity, but not for intense sports training, which are specific to powerlifting.

The calculation of basal metabolism is made in different ways, but we recommend not to complicate it and instead apply simple shortcuts. We need to take a person's weight in kilograms, and multiply it by 27.77 for men and 24.44 for women. You will get an approximate value of the calories needed for basal metabolism. These coefficients are determined on the basis of long-term practice and are drawn up for diet programs for athletes, which makes them a good working tool.

Adjustments based on household and other activities: if an athlete is engaged in mental sedentary work (such as school or computer work), 10% should be added to the value of the basal metabolic rate; if he is engaged in physical labor, or his work requires constant travel on foot or by car, 20% should be added to the value of basal metabolism.

Adjustments based on the psycho-emotional type: if an athlete is a sanguinic or melancholic, you should not introduce any adjustments; if he is a pronounced choleric, then you should add 10% to the value of basal metabolism. In other words, for athletes that tend to be emotionally calmer and even-keeled, no additions of calories are required. For athletes that are excitable and emotionally high-energy, more calories need to be consumed.

The most difficult part of the calculation of the adjustment factors is to count the training expenditure. This component of the daily ratio can vary widely depending on an athlete's body type, body composition, and individual response to the training program. However, for the initial calculation of daily calorie intake, you can use the following approach:

- Training three times per week, add 10% to the basal metabolic rate;
- Training four times per week, add 15% to the basal metabolic rate;
- Training five or more times a week, add 20-25% to the basal metabolic rate.

Thus, daily caloric intake would be the sum of the basal metabolism, the adjustments based on household and work activities, the adjustments based on the psycho-emotional type, and the ones based on the training expenditure. All other adjustments should be made on the basis of how the body reacts to the diet.

4.3.3. Tools for conclusions about the adjustments

It is not complicated to make the abovementioned adjustments if an athlete has a training journal that shows body weight fluctuations. These changes will show if the daily caloric intake should be increased, decreased, or kept the same.

The most important tool to adjust an athlete's nutrition is caliperometry method (using skinfold calipers) for determining body composition, i.e. the ratio between water, muscle and fat components. One rule should be followed: if the caloric value of the daily diet is not enough for the body, it usually uses lean body mass (muscle) by processing muscle proteins to compensate energy deficit, and stores fat. This paradox cannot be resolved as it results from a powerful survival mechanism, which is tightly «sewn» into the human body. Since fat has a higher energy cost (about nine calories per gram compared to four calories per gram of protein or carbohydrate), the body saves it as a means of survival in limited diet conditions. All the athletes who resort to periodic weight reduction should take into consideration. If we take into account that the lifters of all weight categories, except for the super heavyweight, it is advisable to keep the weight as close to the upper limit of a particular category as possible, and maximize the muscle component as well as minimize the fat component, caliperometry allows for very precise corrections in daily diet to maintain the best body composition. It is recommended to do calipelometric tests once a month.

4.3.4. Balancing the daily diet

Another issue as important as total daily caloric intake is balance, that is, what the main contribution of macronutrients should be (i.e. proteins, carbohydrates and fats) to the formation of the caloric value of a daily diet. Even some athletes have misconceptions on the subject that are caused by myths and ignorance. Unfortunately, among many powerlifters there is a misconception that says that nutrition during training, aimed at strength and maintaining a stable body weight, does not need such a balance. This way of thinking is wrong. In power and speed-power sports, a basic formula is used according to which 25-30% of daily energy is provided by proteins, 10-15% by fat and 55-65% at the expense of carbohydrates. At one time or another during an annual training cycle, these ratios may vary slightly, but the average balance remains roughly the same.

4.3.5. Protein balance

When testing the quality and balance of nutrition of many people who have experienced a plateau in their physical performance which can catch some people by surprise. Despite the fact that caloric intake in their daily diet is quite high and sometimes it even exceeds the real caloric needs, they do not succeed in strength gain, muscle mass gains, or getting rid of fat deposits under their skin. This can be explained by several reasons, the first of which is lack of protein.

Protein is not only muscle building material. It is a component of connective tissues (tendons and ligaments, fascia) and cartilage. Protein is part of the major peptide hormones, such as growth hormone and gonadotropin, without which the growth of tissues and normal functioning of a human body is impossible. Protein performs a transport function, for example, is part of the hemoglobin molecule that carries oxygen to cells and removes carbon dioxide from them. Protein means healthy skin, hair, teeth and nails, ligaments, tendons and cartilage. Of course, these are not all the functions of protein, but they should be enough to let us draw a conclusion about its importance.

For some of the athletes experiencing difficulties with reaching the objectives of their training, the lack of protein ranges from 10 to 15%, and sometimes more. This stems from the fact that protein is an essential nutrient and cannot be synthesized in the body from

fats and carbohydrates. This deficit, caused by an insufficient supply of protein with food, is compensated by the body by breaking down the currently inactive protein tissues or the tissues which are less important to the body at the moment.

In practice, this means that if, let's say, a lifter has been training his thigh muscles, calf muscles and his hips by doing squats, and the torso and arms muscles have been loaded less, the protein deficiency in the diet will cause the body to break down the muscle tissue of the arms and torso, and extract essential amino acids (protein building blocks). Tomorrow, when the athlete will be working on his arms and upper body when performing a bench press, the body will break down the muscle tissue of the legs and lower body. High-quality body recovery, especially adaptive protein synthesis in these conditions are much less attainable.

It is clear that in this approach muscle mass and strength gains are a near-impossible task. It is also significant that if protein level falls below the critical protein intake, the production of sex hormones, particularly testosterone, drops sharply. Since it is the testosterone that the growth and preservation of the muscle tissue integrity depends on in large part, it is clear that if protein deficiency is observed, an athlete will be functionally deficient.

A balanced protein intake can be achieved only by consuming adequate amounts of meat, fish, low-fat dairy products and eggs. And when necessary, it is strongly recommended to use specialized protein supplements (proteins).

4.3.6. Carbohydrate balance

Protein deficiency is not the only reason for plateau in performance, and the most severe overtraining. Sometimes it is accompanied by carbohydrate deficiency. Carbohydrates are the main source of fuel for working muscles, they provide energy for most of the important processes in the human body. All foods containing carbohydrates after a meal are broken down into glucose, then it is transported to muscle cells where it is stored as glycogen (muscle fuel that is activated during strength training). Part of glucose is stored in the liver, also as glycogen, and where it is extracted if there is a sharp decrease in glucose levels in the blood. Finally, the other part of glucose continues to circulate in the blood in the form of reserve that may be needed to replenish the muscles needs when they exhaust their own glycogen.

Carbohydrates deficiency causes the body to break down proteins to provide an energy supply. First, protein is broken down into amino acids, then the body breaks down parts of those certain amino acids into glucose. But this does not solve the problem, as the body diverts part of an important protein to form glucose and when protein compensates for carbohydrates deficiency, the body faces a severe deficiency in protein! A vicious circle has been formed.

Moreover, the body reacts to carbohydrate deficiency by reducing the metabolic rate and testosterone production. We already know what the latter phenomenon leads to, but the former makes you sleepy and weak, an athlete will not be able to be as agile and strong as he used to if in a low carbohydrate state. For competitive powerlifters these phenomena are certainly unacceptable.

Testing the athletes with an unbalanced diet shows that, as a rule, their diets are 10-20% deficient in carbohydrates, and a simple addition to the diet of grains, wholegrain bread, fresh fruits and vegetables allows us to solve this problem successfully. If this does not help you can use carbohydrate-protein supplements, or «mass gainer» products.

4.3.7. Fat balance

Unlike with proteins and carbohydrates, there is a risk of overdose or excessive fat consumption. The overdose is mainly caused by saturated solid fat consumption (fatty sausages, ham, bacon, confectionery fats). Recently, nutritionists have come to a very important conclusion that, in spite of an excessive consumption of fat by a modern man, there are not enough unsaturated fats in our diet. Unsaturated fatty acids are an important part of the cell membrane, and the health of every cell in our body depends on their intake. Unsaturated fatty acids are found only in the unrefined natural vegetable oil, for example, grape, olive, flaxseed, corn, or sunflower. So, without any risk of gaining excess body fat 2-3 tablespoons of oil can be incorporated into the diet, e.g. with salads or grains. But it is important to limit the intake of saturated animal fats (solid fats), which tend to be reserved in the form of subcutaneous fat. The rules are simple: do not fry but boil or grill meat, eat more fish, remove the skin from chicken before cooking it, periodically eat white poultry meat (e.g. chicken breast), avoid excessive amounts of red meat, e.g. pork, choose veal or turkey.

However, inadequate intake of fat can sometimes cause problems too. It is proved that a sudden restriction of fats in the body cause it to decrease the production of cholesterol. Despite all the horrors people tend to attribute to cholesterol, without this substance the body cannot produce certain steroid hormones, particularly testosterone that helps build muscle and maintain high physical performance. In addition, low consumption of fats can sometimes be accompanied by bile build-up in the gall bladder and liver (cholestasis), which in itself is unpleasant and sometimes cause dull pain, requiring special procedures to remove bile clots. Thus, a certain balance in fat consumption should be maintained, and if an athlete reasonably reduces the consumption of animal fats and increases vegetable fat consumption, then this problem will be solved much easier.

4.3.8. Vitamin and mineral balance

Unlike the macronutrients (protein, fats and carbohydrates), the requirements for which are in tens, sometimes hundreds of grams, micronutrients enter the human body in the quantities expressed in grams, milligrams and micrograms. These quantities, however, do not diminish their high importance for the coordinated work an athlete's body performs. It is thought that in comparison with a person leading a sedentary lifestyle, athletes require 2-2.5 times increased amounts of micronutrients. But there is a serious problem here. According to the World Health Organization, modern methods of farming and crop production, storage and industrial processing of agricultural products, as well as unsatisfactory environmental conditions have reduced the content of vitamins and minerals in vegetables, fruit and cereal products to a level that provides in total about of 60% of the daily requirement. And this is a global phenomenon, independent of the economic situation of a country. Attempts to solve this problem by incorporating maximum number of vegetables and fruits into a daily diet face the prospect of such amount of food consumption with which the human digestive system cannot cope. In these circumstances, the only acceptable and optimal solution is to use specialized sports vitamin and mineral supplements.

4.4. Food for a balanced diet

It has been already mentioned that there should be a strict ratio between protein, fats and carbohydrates in a daily diet. This ratio must apply not only to the overall daily food consumption but to every single meal. It is clear that it is a nearly impossible task to keep this balance, however, it can be simplified if an athlete consumes food supplements with high biological density, i.e. without the presence of ballast substances capable of distorting the structure of not only a single meal but also the daily ration.

In healthy eating theory and practice, especially when it comes to sports, it widely accepted to think that the main task is to ensure the widest possible variety of foods in a separate portion of food. All sorts of contemporary nutrition theories, in particular the so-called «separation» diets (eating mostly high protein or just low fat or just high carb for periods at a time), are inappropriate for a healthy lifestyle, especially that of an intensively training athlete.

In connection with the above, we will consider hereinafter only the models of mixed rations and meals, when all macronutrients proteins, fats and carbohydrates are present in roughly even amounts in each meal.

4.4.1. Food groups and their properties

Below, we present a simple and practical classification of food products.

I. Protein products (lean meat, animal organs, fish, and eggs). A powerlifting athlete should go for chicken breast, turkey, lean beef, heart, liver, and kidneys. Low-fat red meat is of particular importance because it is rich in heme iron, essential for ensuring high oxygen transport properties of blood. It is recommended to consume the meat of wild animals and game, as it is richer in protein and creatine. In the “fish” category there is actual fish and all kinds of seafood (crab, squid, shrimp, scallops, octopus). Seafood, which mainly contains «light» fats, with a large number of unsaturated fatty acids, can be any, as their fats are not the same as that of fatty meat, and do not tend to be deposited under a person’s skin. Preference should be given to oceanic fish species as they are richer in minerals and unsaturated fatty acids, as compared to the river or lake species. Predatory fish species contain a slightly higher percentage of protein. Eggs must be eaten whole because egg yolk contains more protein than egg white. Moreover, the presence of cholesterol contained in egg yolks is a very important precursor of testosterone, which is important to ensure the optimal synthesis of sex hormones. Eating exclusively egg whites is appropriate only in the periods when an athlete is attempting to enter into a lower weight class and, therefore, he should limit the intake of dietary fat to a minimum.

II. Protein products (fat-free and low-fat dairy products). In this product type the lowest fat content is important, otherwise you get mostly not a protein product but a fat one. Low-fat dairy products are an indispensable component of athletes’ nutrition because their consumption ensures a constant supply of organically bound calcium - an essential condition for the bone apparatus that undergoes very intensive stress and axial loads during training and competition. Liquid sour-milk products (such as yogurt and kefir) provide a high-quality of bacteria in the gastrointestinal tract and promote effective digestion, especially at night.

III. Carbohydrate products (grains and grain-based products). In essence, we mean bread, cereals, and pastry products. In this case, the rule is quite simple - an athlete should choose products from the least processed grains. We must choose wholegrain bread instead

of bread made from white flower, even better bread with crushed cereal grains; pasta should be from dark flour or hard grains; it is highly recommended to include coarse cereal in the diet (oat, wheat, millet, barley, pearl barley and unrefined brown rice).

IV. Fat products (nuts, seeds, avocado, unrefined vegetable oil). Nuts are very rich in beneficial amino acids and fats, which are light, with a high percentage of unsaturated fatty acids, which are very useful for an athlete's body. However, with a high consumption of nuts there is a risk of exceeding the daily dose of fats, and thus it is recommended to give preference to three nuts types that contain a relatively low percentage of fat and are rich enough in good proteins, the quality of which is not inferior to soy protein. These are hazelnuts, almonds and peanuts. Of course, it is advisable to include raw nuts in the diet, so as not to disturb the composition of fatty acids in nuts (they change their composition when heated and become less beneficial). Among seeds, we should mention pumpkin seeds that contain a unique set of fatty acids that can improve the performance of the liver cells. Vegetable oils should be selected in the following way: use unrefined, unflavored, preferably cold-pressed oil. In any case, you should read labels and verify the presence and the percentage of unsaturated fatty acids.

V. Carbohydrate products (vegetables and legumes). The value of vegetables is in providing low glycemic carbohydrates, as well as supplying the body with fiber, vitamins and minerals. The protein fraction in vegetables is insignificant, but one should pay attention to legumes (peas, beans, lentils) that can be a good source of vegetable protein, in combination with the proteins of the first and second groups they make up an almost ideal amino acid profile of the ingested food. Fiber plays an important role in preventing the absorption of dietary fat (it connects them and helps rid of them from the digestive tract), it transports the food through the gastrointestinal tract (helps intestinal peristalsis). Root vegetables are particularly rich in fiber (carrots, beets, ginger), as well as cruciferous (turnips, radish). Tuberous vegetables (potatoes, sweet potatoes) can be a source of good carbohydrate. Including significant amounts of vegetables in the diet is always a good practice, as in the present conditions, we can somewhat prevent the state of hypovitaminosis, which is problem for a contemporary society. It recommended to include spicy leafy vegetables in the diet (parsley, coriander, dill, celery, lettuce, basil, watercress, and tarragon), because they are rich in essential oils, which improve appetite and digestion, and optimize metabolism. Seaweed is a remarkable food product in this group as it is a source of minerals, in particular iodine, which is involved in the formation of thyroid hormones and regulates the rate of metabolic reactions.

VI. Carbohydrate products (fruits and berries). This group gives us a major share of carbohydrates, the most important of which are fructose and glucose. However, some representatives of this group are rich in other biologically active substances, such as pectin, provitamins and minerals, the nutritional value of which cannot be overestimated in a serious strength athlete's diet. A powerlifter's diet should include bananas, pineapples, apples, pears, plums, cherries, strawberries, blueberries, and all citrus fruits.

VII. Water. Despite the fact that water seems to be a completely non-significant part of the diet, it plays a vital role for any athlete. First of all, we should remember that the human body consists of no less than 60% liquids (blood, lymph, internal contents of cells, extracellular fluid). Optimal hydration (i.e. supplying the body with enough water) is a serious issue, the lack of water in the body causes it to try to maintain the fluid balance and store the water formed in the course of active biochemical reactions that occur during strenuous muscle activity. Moreover, waste and impurities are "filtered" from the used water in the renal tubules, otherwise they would remain in the bladder and kidneys creating serious risks of

sand and stones appearing in an athlete's body. Moreover, water is one of the richest and most essential sources of minerals. Unfortunately, their number is not sufficient in ordinary drinking water, so it is recommended to supplement it with mineral water. It should be chosen accordingly, you should go for alkaline mineral water since it helps alkalize the body and remove acid build-up that occurs while intense strength training is occurring. During intensive training glycogen oxidizes and lactic acid builds up, which shifts the pH balance to the acidic state. Acidification of the body slows down the process of recovery and prevents an effective workout. It is recommended to drink at least one glass of mineral water in addition to the water an athlete drinks throughout the day. The total amount you drink can be roughly determined by the following formula - for every 100 grams of protein in your diet you should drink at least 1 liter of plain water (in addition to soups, teas or other drinks).

Leading nutritionists are right to claim that in order to balance every single meal, it should include at least one product from each product groups mentioned above. It means that, for example, breakfast should consist of meat and cereal, cottage cheese, vegetable salad, and, let's say, an apple or banana, and a glass of mineral water or green tea. Lunch and dinner should be balanced in the same way. In this case you can reach a relative natural balance of protein, fats, carbohydrates, minerals and vitamins. Unfortunately, even if the food selection is ideal there are no guarantees that the balance should be in accordance with the formula 25-30%:10-15%:55-65%. However, it can be adjusted by consuming food supplements, first of all protein ones since an almost balanced macronutrient-wise meal will likely be protein deficient for the strength athlete's needs.

4.4.2. Optimum daily nutrition scheme

A very important methodological and practical question is what should be a single portion size. If an athlete has decided to consume around 3000 calories per day, his breakfast, lunch and dinner should contain such an amount of food that would be equivalent to 1000 calories. Unfortunately, contemporary ideas about physiological, balanced and adequate nutrition do not support this massive number of calories in a single portion. From this amount of food, the body digests and absorbs only what it needs to satisfy the short-term energy needs and plastic (muscle building) processes, and the rest of the food that is present in your system becomes a risk of all the excessive calories to be transferred into fat deposits. This is exactly what happens if a single portion is too big and high in calories. The potential for fat increase is huge in this case. This is not the best option for a powerlifter, especially for the one who is trying to stick to a particular weight category. Nutritional program design is also complicated by the fact that, according to contemporary ideas, the insulin reaction which can increase the lipogenesis, occurs as a reaction to consuming high glycemic index foods as well as to oversized portions.

Therefore, it is necessary to resort to other, more scientific and more rational nutrition schemes. Under these conditions the only possible solution is to eat often but with smaller portions. It is more physiological, and means having a meal not three times a day but and 5-6 times in approximately equal intervals. The energy value of an individual portion becomes lower, i.e. about 500-600 calories. By the time the next meal is consumed, there are no unused reserves remaining in the body and a tendency to accumulate excess fat is greatly reduced, which can even be eliminated completely if all the food an athlete eats is properly calculated. Of course, it requires practice and precise tracking of which body component increases during exercise - muscles or fat.

When eating 5-6 times a day there is a possibility to diversify the food you eat, and also exclude the possible risks associated with the formation of caloric «inverted pyramid». The idea is that by the end of the day it is recommended to reduce the calorie content of each meal, at the same time maintaining its nutritious qualities. To apply this rule we advise the use of the table below to help distribute the desired products in the diet to each of the six meals.

An optimal «food pyramid» requires making right food choices for each meal during the day (Ostapenko, L.A., 2005).

Table 4.4

Food distribution for every meal
(Ostapenko L.A. 2011)

№№	Food products						Meals
	Products	**Breakfast**	**Snack**	**Lunch**	**Snack**	**Dinner**	**30-40 minutes before going to bed**
	1	2	3	4	5	6	7
	I						
1	Beef	+		+	+		
2	Turkey	+	+	+	+	+	
3	Liver	+	+	+	+		
4	Chicken breast	+	+	+	+	+	+
5	Kidneys			+	+	+	+
6	Heart			+	+	+	
7	Fish	+	+	+	+	+	+
8	Squid	+	+	+	+	+	+
9	Shrimps				+	+	+
10	Eggs	+	+	+	+	+	+
	II						
11	Milk	+	+	+			
12	Yougurt, kefir, curdled milk	+	+	+	+	+	+
13	Cottage cheese	+	+	+	+	+	+
	III						
14	Bread	+	+	+	+		
15	Pasta	+		+	+		
16	Cereals	+	+	+	+		

Table 4.4

	1	2	3	4	5	6	7
	IV						
17	Vegetable oil	+	+	+	+	+	+
18	Avocado	+	+	+	+	+	+
19	Nuts	+	+	+			
20	Seads				+	+	+
	V						
21	Beans	+		+		+	
22	Potatoes	+	+	+	+		
23	Root vegetables	+	+	+	+	+	
24	Cruciferous vegetables	+	+	+	+	+	+
25	Spices	+	+	+	+	+	+
26	Seaweed	+	+	+	+	+	+
	VI						
27	Bananas	+	+	+	+		
28	Apples, pears, plums, sour cherries		+		+		+
29	Citrus fruit	+	+	+	+	+	+
30	Berries	+	+	+	+	+	+
	VII						
31	Mineral water				+	+	+

Note: «+» means that this particular product is the most appropriate during a meal [4].

Obviously, this table is just an example. If an athlete applies it he can establish quite a varied and tasty diet, and he can build each meal over, say, a week, with an unrepeated pattern of combinations of various food products. The food preferences of each individual athlete can also be taken into account.

After the daily diet scheme has been built, you should carefully calculate its protein content and add appropriate supplements (high-protein content) if a protein deficit occurs. It is likely that it will be necessary to slightly reduce the daily amount of carbohydrates intake to keep the overall caloric intake intact.

4.5. Methods of objective control over diet quality

The most informative and reliable methods of objective diet quality control are the ones allow us to monitor the following:

- body weight dynamics
- body composition dynamics
- anthropometric dimensions dynamics
- functional state dynamics
- strength ratio stability

A powerlifter's weight is a serious indicator of the adequacy of his diet. If the weight is increasing consistently it is likely that there has been a surplus in the diet. If the weight is decreasing, most probably the diet is inadequate in calorie amount. However, this indicator is not perfect as it doesn't allow us to clarify which component causes weight fluctuations. In other words, from just weight alone we cannot reliably tell if the increases and decreases are the fluctuations observed in muscle weight or fat deposits.

Nonetheless, it is recommended for the athlete to weigh in at least once a month during the preparation period, in the pre-competition period at least once a week, and every single day in the period within 7-10 days before the competition.

The caliperometric control should be as systematic as the body weight control (we believe that the simplest and the most precise method to check the thickness of the skin and fat folds on up to 10 different sites on the body). Thus, skinfolds should be taken and recorded once a month on average. There are many formulas to calculate the fat, water and muscle content, however, it is recommended to use the ones that take into account the maximum quantity of measures taken at different places as it allows to increase the data precision. This method allows us to differentiate between muscle and fat gain and to make subtle adjustments of nutrition by increasing or decreasing the daily macronutrient intake.

The anthropometric (tape) measurements of the most important body parts (chest, waist, pelvis, hips, legs, shoulders and arms) is a technique that complements the two previous ones as it allows, together with caliperometry, for us to determine the localization of the key gains or losses at particular body parts, not only to determine the changes in total body muscle and fat mass.

The functional state indicators important for determining the diet and training adequacy are load adaptability (how quickly you're gaining strength), recovery, blood pressure, results from a functional changes index, and results from a stress indicator. We have established that the overtraining syndrome can occur in response not only to the irrational schemes the training process, but also as a reaction of the body to the chronic low calorie state, and protein deficiency in particular. In any case, during plateaus or lower performance periods, even if they are accompanied by a constant body weight, you should review the composition of food intake for possible protein deficiency.

Finally, if the diet is not adequate to the amount and intensity of training loads in the current training cycle, it can also cause plateaus and lower performance indicators. This inadequacy is especially clear when performance is decreased even if an athlete's weight remains unchanged.

Of course, none of the abovementioned methods is completely indicative and accurate on its own. The more multidirectionally a powerlifter or a coach approach the use of these techniques, the more accurate the diagnosis can be and the best decision about the diet adjustments and its daily content can be made. In other words, by tracking body weights, skinfold caliper measures, tape measures, and performance, stress and recovery abilities

together, the powerlifter or coach can develop the most complete possible picture of the nutritional state of the athlete and make needed adjustments in the most informed and effective way possible under the circumstances.

4.5.1. Nutrition in preparatory period

Based on the main objectives of the pre-contest period (to increase strength, improve speed, flexibility, agility, and versatile general physical preparedness), most often the power's diet should be hypercaloric. This is especially true as amount of training load and weight on the bar increases through this period, stressing recovery demands. During this phase, an athlete's normal caloric intake (to maintain their weight) can be increased about 10-15%. It has been noticed that for most athletes seeking to keep their competition weight it is difficult to improve their strength results. This can be explained by the following: a reasonable increase in body weight allows an athlete to be able to improve his absolute strength indicators (correlated with total body weight) and then by reducing his weight to the upper limits of the weight class before the competition he can achieve improvements in his relative strength (calculated per kg of body weight). The same logic applies to the situation where a powerlifter (or coach) decides to move to a higher weight category. However, during a higher weight class transition, caloric intake can be increased by about 15-20%, while the ratio of proteins, fats and carbohydrates in their daily diet should not undergo major changes (see Table 4.5).

What are the limits of body weight increase when the task is to remain in the same weight class? It is obvious that the increase of body weight should preferably occur through muscle and liquids, and not due to added fat deposits. According to the results of long-term caliperometric monitoring of competing powerlifters and bodybuilders, the optimal body weight gain at the expense of muscle mass not posing any the risk of increasing the percentage of body fat is within 4%. However, if an athlete rapidly increases his body weight fast in the pre-contest period up to 10%, in most cases the increase does not occur as much in the muscles as it does body fat, and in the pre-contest period it is a virtually impossible task to move to a desired weight class keeping the same muscle weight. Put another way, increasing body mass up to 4% above normal levels (within a period of several months) seems conducive to a large proportion of muscle gain. Gaining much more than that will result in higher levels of fat gain if done within several months' time.

Table 4.5

Risk arrangement of exceeding weigh class limits
(Ostapenko L.A. 2015)

№	Excess percentage	Analysis and prognosis
1	1-1.5%	Optimal, allows to keep performance withing weight class
2	1.6-2.5%	Acceptable provided that metabolism is fast and experience in weight loss is present
3	2.6-4%	Risky, may lead to heavier weight class or decrease performance in the previous category after weight loss
4	> 4.1%	Dangerous for keeping the same weight class and performance results
5	Around 10%	Unacceptable, eliminates chances for successful results during competition

The abovementioned risks should be considered while taking into account the individual characteristics of an athlete, i.e. his training and competitive experience, his own experience when entering a particular weight class, and personal metabolic rate.

It is also important how stable an athlete's psycho-emotional state is and how he handles pre-contest psychological pressure. It is not rare to observe the situation when an athlete has a perfect upper limit weight before a competition and during the competition weigh in his weight is one or two, sometimes even more, kilos lower than the top end of the weight class. This should be predicted after careful observation of weight fluctuations, especially in athletes prone to high levels of emotion before competition, as it is thought among physiologists that the central nervous system under psycho-emotional pressure can use up to 70% of glucose circulating in the body. As a result, if its level drops because of stress, the amount of energy available for recovery and performance can be lowered and this can negatively affect competitive results.

The daily macronutrient ratio may also need to be adjusted during the pre-contest period. If the proper ratio of energy contribution of protein, fats and carbohydrates is considered to be 25-30%:10-15%:55-65% of total caloric intake, it can be slightly adjusted depending on what weight category an athlete plans to compete in. If the athlete plans to compete in his/her usual weight category, there is no need to adjust anything unless it is necessary to improve the weight within this weight class. However, if the athlete is new to their weight class and has no prior competitive experience at this weight, some changes can be made.

Table 4.6

Macronutrient ration when changing weight class
(Ostapenko L.A. 2004)

№№	Macronutrient	Moving to heavier class	Moving to lighter class
1	protein	15-20%	20-30%
2	fats	20-25%	25-30%
3	carbohydrates	45-55%	40-50%

The precise ratio can be established for an athlete only by monitoring his body weight fluctuation. His nutrition must be adjusted in the way that his weight increase comes from muscles gain and not fat deposits.

Apart from macro and micronutrients, in pre-contest period it is recommended that athletes consider the use of supplements that help achieve the goals of current mesocycle. In strength sports, proteins, complex amino acids, BCAAs, creatine, and glutamine seem to be quite effective. In varying degrees, they help to increase an athlete's strength performance. In order to meet the challenges of increasing the elasticity and strength of ligaments, tendons and cartilaginous structures such as collagen, glucosamine, chondroitin, and shark cartilage can be used. If the task is to control the body weight by decreasing fat tissue we can use L-carnitine, chromium picolinate, and oxy-citric acid. It is clear that prior to the use of the above supplements and biologically active substances, an athlete should test their effectiveness when taken separately, since during polypharmacy (the use of many substances at the same

time) it is impossible to indicate which substance has positive, neutral or negative effects. The same applies to the dose, i.e. it is strictly individual, and reliable general universal recommendations do not exist for many of the aforementioned supplements.

4.5.2. Nutrition in pre-contest period

When planning a daily diet it should be noted that the pre-contest period goals are to preserve form (levels of fitness established in earlier phases) and ensure the implementation of the acquired abilities in sports achievements, and the main feature is the decline in workload but increase of its intensity in competition exercises. The increased intensity and reduced work load lead to a slight decrease of total energy expenditure. However, this is accompanied by increased catabolism of muscle tissue, and increased requirements for efficient and complete recovery. A powerful metabolic shift requires improving the quality and balance of amino acids of the consumed protein. If in the previous mesocycle an athlete managed to solve the problem of reducing the body weight to acceptable parameters, the macronutrients ratio might look like this: energy contribution of protein - 30%, of fat – 10%, of carbohydrates - 60%. If it remains an urgent task to enter the desired weight, protein intake can be increased about 5%, 10% - fats, but carbohydrates intake should be lowered to approximately 45%. It is recommended that the athlete include additional doses of L-carnitine, chromium picolinate, and oxy-citric acid.

Metabolic acidosis that inevitably arises with increasing intensities of training (as well as lactate accumulation) impose additional requirements on the buffer systems of the body. In this regard, we recommend taking additional amounts of alkaline mineral water, replacing a part of carbohydrates with vegetables (pumpkin, carrots, beets, zucchini, spinach, celery) and fruit (bananas, watermelon, dates, mango) of alkaline nature. By these methods, the bicarbonate buffer system in the body is assisted in its activities of reducing acid levels.

To promote the amino acid buffer, the intake of complex amino acids should be increased, and it is advisable not to take them on an empty stomach but with meals. Besides affecting the buffering system, it has a beneficial effect on the amino acid profile of the consumed food, increasing the bioavailability of the protein fraction.

Finally, to promote the hemoglobin buffer it is advisable to eliminate white meat and replace it with red meat of cattle as well as liver and kidneys. Among plant foods that increase hemoglobin in the blood are beans, buckwheat, parsley, whole grains, beets, carrots, tomatoes, zucchini, spinach, black currants, pomegranates, cranberries, plums, and peaches.

It should be noted that the pre-contest period, due to the increase of intensity, imposes high demands on an athlete's hormonal system. Especially vulnerable are testosterone levels which, according to the studies by Duclos M. (2008), Hayes L.D., Bickerstaff G.F., Baker J.S. (2010) may drop when the volume and intensity of physical activity approach the limits of momentary adaptive capacity of an athlete (4, 5). In this case, nutrition can be of great service to an athlete. Under no circumstances should it be monotonous, in this period it is important to introduce the products that have proven themselves to be testosterone boosters. Above all, these include all kinds of beans (they are rich in fiber, zinc and vegetable proteins, which, combined with animal protein provide an optimal amino acid profile, and zinc is important for spermatogenesis and helping to maintain a normal reproductive system). Further, a powerlifter's diet in the competition period must include nuts - peanuts, almonds, cashews, hazelnuts, pistachios, and walnuts (they are rich in protein and unsaturated fatty acids, the presence of which in the body is important for the synthesis of testosterone). In

this period, nuts can partially or fully replace vegetable oils, which will raise the contribution of fats to the caloric balance. Finally, the third food adjustment important for the synthesis of the optimal testosterone levels may be the introduction of seafood - squid, lobster, shrimp, crab, oysters, mussels (all of these foods are rich in zinc, vitamin D, B3, B5 and B12).

During this period, the balance of macronutrients should provide the daily energy from protein, fat and carbohydrate based on the scheme characteristic for the transition to a lower weight class (see Table 6). The proportion of carbohydrates is reduced, but the proportion of protein is increased (at the expense of high-quality protein) and fats (nuts). Moreover, some acceleration of metabolism, leading to a safe weight loss due to the athlete's body fat, is achieved through the use of the dynamic and specific protein effect. To implement this technique, one should begin any meal with the food products that are mainly of protein nature, and ideally - a protein shake. According to the academician Yaglov V.V., such an adjustment of meal order can improve metabolisc rate by almost 60% (Yaglov V.V., 1986).

Among food supplements, the most useful in the competitive period are ZMA, glutamine, collagen, and CLA (conjugated linoleic acid). In connection with the increase of protein percentage in the daily diet it is important that protein is fully digested, so it is advisable to double the intake of vitamin B6 (pyridoxine), and if necessary to take enzyme supplements to enhance digestion.

A few words about the use of creatine. Some powerlifters report that this supplement, when taken in the pre-contest period, often causes fluid retention in the body, which prevents him/her from reaching the optimal competition weight within the chosen weight class. In connection with this, we suggest experimentation (mainly in the base period of preparation) as follows: do not use creatine supplementation for seven days and follow a standard diet while take other supplements. During this period, accurately record the body weight, and if a more accurate diagnosis is possible, record the water component fluctuations of the body (with something like a DEXA scan). You should try to determine whether creatine leads to a hydration drop when stopped. If so, stop the use of creatine about 7-10 days before the event. In addition, you need to determine if there has been a possible decrease in strength and performance. If this happens, creatine should be kept as a supplement, but additional measures should be taken to ensure that water retention has not led to a higher than desired weight and a difficulty in making the planned weight class weigh in.

First of all, these are the measures of dietary nature that can reduce water retention. Namely, it is necessary to increase the content of products with a mild diuretic effect - apples, pears, peaches, lemons, artichokes, asparagus, borage, cauliflower, celery, cucumber, eggplant, grapes, grapefruit, leek, parsley, watermelon. Sometimes it is useful to eliminate dairy products, which, according to many athletes, lead to some retention of water in the body.

In general, 24 hours before the competition, an athlete's diet must allow him to weigh about 1-1.5% higher than his competitive weight. 12 hours before weighing he should stop taking any liquids, including water. If this does not affect the body weight, it is recommended he/she take herbal diuretics.

Diuretic mix 1 - dry bearberry leaves, blue cornflower flowers and licorice root in proportion 3: 1: 1.

Diuretic mix 2 - dry licorice root, juniper berries, bearberry leaves in proportion 1: 2: 2.

Diuretic mix 3 - dry wheat grass roots, juniper horsetail leaves and fruits in proportion 1: 2: 2.

Diuretic mix 4 - dry horsetail leaves, birch, cumin seeds, chicory root and wild strawberry leaves in proportion 5: 5: 1: 2: 2.

Diuretic mix 5 - Knotweed dry leaves, birch, mint and rose hips in proportion 2: 2: 1: 1.

Put a tablespoon of any mix in 200 g of boiling water (use only distilled water!) and cook in water for thirty minutes. Drink 1 tablespoon every 2.5 hours. Of course, due to the pronounced individual reactions to any herbal infusion, it is recommended to check the effect of every single mix and determine which one works best.

4.5.3. Competition day nutrition

Competition day nutrition essentially begins after the first weigh-in session, which takes place one day prior to the competition. After weighing in and determining the exact weight, an athlete makes a decision on how to build his diet for the remaining time. The last meal on this day should be high-protein, low-sodium, without any spices or anything that can upset his stomach. He should also drink 1 cup of distilled water. If the weight is still slightly higher than expected for this particular weight category, it is recommended to eat a medium-sized lemon with peel before going to bed. The use of herbal diuretics before going to bed is not recommended because it can disrupt the night sleep pattern. They can be used in the morning if the body weight has not normalized for the chosen weight class. The second day nutrition for the formal weigh in session, also depends on how much an athlete's weight is different from the selected weight category. The general rules are as follows: food should be high-protein, low-carbohydrate with a high glycemic index, with a moderate amount of fat (nuts are the best source of fatty acids in this case). Experimenting with new dishes is not allowed. Athletes should also avoid any products that are able to induce increased gas formation in the gastro-intestinal tract. If an athlete balances on the verge of the weight class, the best option is to drink shakes containing whey protein throughout this period.

If between the first and second weigh-in session an athlete had used herbal diuretic teas, he should stock up on hypotonic sports drinks containing essential minerals and electrolytes, and typical for him easily digestible food: usually starchy carbohydrates (baked potatoes, for example), lean meat (preferably minced), tangerines or grapefruits, and alkaline mineral water. After the second session, it is advisable to immediately restore the balance of minerals in the body. The entire volume of mineral water should not be drunk at once, but over 15 minutes in small sips.

The supplements that may be useful in the day of competition are BCAAs (taken immediately before and after the performance of each competitive routine) and glutamine (taken 30-40 minutes prior to the event).

4.6. Fundamentals of body weight regulation technique

When trying to reduce weight one can sometimes utilize rather radical methods that can hardly be called rational or reasonable. Drastic hydration restrictions, starvation diets, strong diuretics and laxatives, grueling endurance training, rubber suits and prolonged exposure in saunas or steam rooms are some of the examples.

The negative consequences of this approach can present themselves quite quickly. Below are the potential resulting phenomena affecting strength performance:

- Reduced levels of muscle strength and aerobic capacity
- Increase in heart rate at rest and during stress
- Irritability
- Depletion of important minerals and electrolytes
- Impairment of motor coordination

We should also present the problems that arise with regard to the health of the athletes who abuse such techniques:

- Arrhythmia
- Hormonal imbalance
- Decrease in the level of the immune system
- Disorders of the pancreas
- Deterioration of the rheological (smooth flow) properties of the blood
- Excessive load on the heart and the cardiovascular system in general

In powerlifting, it is more common to attempt to reduce weight rapidly rather than gain it before competitions. When it comes to gaining weight, it should be said that it is highly unlikely that rapid weight gains will result in high levels of muscle and in the long term this practice is unproductive. The increase in an athlete's body weight due to the consumption of excessive food amounts or large amounts of fluid, of course, will increase the weight, but in terms of performance the athlete will not receive the slightest of advantages, since the increase in absolute strength is minuscule, and the levels of relative strength clearly drop.

The best tactics is to prepare an athlete in advance to compete in the most appropriate weight class. In all preparation periods, like we have mentioned before, it is recommended to maintain the weight close to the upper limits of a particular weight category. Even for super heavy class athletes it is advised to have minimum amount of fat deposits, which do not benefit either absolute or relative strength. In this chapter we are going to look at weight regulation mainly by means of precise food planning.

If an athlete or a coach decide to move to a heavier weight class, this transition should be planned in a way so that the body weight increases due to «working» mass (muscles and not body fat). Experience shows that if an active athlete gains more than one kilo a month it means that the fat levels have risen, which results in very limited increase strength in proportion to weight gain. This means that weight gain needs as many months as the number of kilos an athlete plans to gain. The main dietary mistake is to increase the daily caloric intake drastically, which is made worse by consuming high-glycemic carbohydrates – simple sugars. However, years of practice show that increasing the daily caloric intake by 10-15% above the normal level combined with intensive training mainly leads to the increases in the muscle and insignificant fat gains.

This advice, of course, needs to be checked in every single powerlifter's case, and the best method here is taking caliperometric measurements every other week. The dynamics of muscle and fat changes will suggest the best option to build a diet taking into account the macronutrients as well as daily caloric intake.

We would like to repeat that radical methods of rapid weight reduction cause decreases in performance. Moreover, after a short restrictive diet aimed at weight reduction to move to a lighter weight class, the tendency to gain fat increases even more and body weight regulation becomes more complicated.

We believe that rapid weight gain makes sense only in cases when there are certain tactical intentions present, including the strongest representative of your own team. This can occur when you have a stacked weightclass and the weightclass above has relatively low competition. In this case one team member from the lower class can temporarily gain lots of weight to compete in the higher class and score more points for the team. Moreover, you should be aware of possible losses in performance when the transition to a lighter weight class happens prior to competitions.

We would like to remind you that rapid weight loss happens in 60% due to muscle loss and only in 40% due to fat. It means that weight reduction should be slow. On the other

hand, if one loses 200 grams per week there is no muscle mass loss observed. This is a safe reduction rate which will allow a lifter to maintain his muscles and, as a result, his strength. When weight reduction is connected with losing muscles and muscle glycogen - the main fuel for strength production - strength regression is inevitable.

The inevitable water loss that occurs when the rapid weight reduction method is applied disrupts body fluid balance. The body then tends to regain everything it has lost and more. Active sweating and the use of diuretics are an excessive and unreasonable burden on the circulatory system and the function of the kidneys.

Thus, weight loss is appropriate only when fat tissue is decreased. A crucial role is played by three factors:

- Physical activity (intensive training)
- Energy deficit (mild «malnutrition»)
- The composition of food (mainly a low-fat, high protein diet); highly recommended is the principle of food «fractioning», i.e. when there are not 3-4 but 5-6 meals a day.

There are three stages of the rational body weight regulation system:

1. Reduction in advance (approximately 90 days) includes:

- Restriction of carbohydrate intake, low-calorie food diet, high in protein (cottage cheese, cheese, beef, vegetables, fruits)
- Restriction of liquids, salty and sweet food, tea and coffee
- Increases in motor activity of aerobic character (jogging, brisk walking, swimming or biking are introduced in a weekly cycle); due to the fact that endurance exercise can have a negative impact on strength performance, it is recommended to do short low-intensity exercise and the «explosive» types of exercise (high intensity short sprints, jumping, running up and down the stairs)
- Conducting training in warm clothing to enhance sweating, use of saunas and steam bath
- Increase in the volume of work on the abdominal muscles;
- Reducing the amount of food during dinner, moving cereals to the first half of the day, and vegetables to the second.

2. Intensified reduction (1-2 weeks) requires the following rules to be added to the above:

- Elimination of salt, with the exception of various sauces and gravies, limiting to a minimum liquids intake
- Warming the body and increased sweating (sauna, hot tub, training in a few T-shirts)
- The use of the dynamic specific food effect (eating protein first in each meal).

3. Rapid reduction (3-5 days) involves following all of the above rules, plus the use of diaphoretic and natural diuretic substances (the recommended herbal teas above, lemon and other natural products). Of course, you need to pre-test the impact of these teas and infusions to determine the most effective compounds.

The final body weight adjustments that go beyond the weight category are made after the first weigh in session. Usually an adjustment is possible in the range of 500-800 grams. In other words, up to 800g of water weight can be lost in the 12 hours between the two weigh-in sessions if needed.

To accomplish such a loss, we recommend an intense workout in warm clothes, urination and bowel movement, and spitting saliva into a bottle. If you need to free the bowel (promote pooping), use a micro-enema or drink freshly squeezed red beetroot juice.

If an athlete needs to use sauna to enter a lighter weight class, he should follow these rules: sit or lay in the sauna only until the moment when the body is covered with big drops of sweat. You should leave the sauna after that, wipe the body with a dry towel or wrap a sheet around it, and sit still until sweating subsides. Repeat the routine twice, and do not take a shower or jump into a swimming pool to avoid the possibility of water absorption through the skin.

To enhance sweating it is recommended that the athlete make a mixture of liquid honey with fine ground salt and rub it on the body before entering the sauna. A very strong effect is observed when you rub 75% alcohol on your body in the sauna.

In all cases, it is better when all these procedures are carried out under the supervision of the team's doctor.

Chapter 5. The earliest studies in the field of muscle strength training

Theoretical studies, which were focused on the foundation of training and evaluation of developing muscle strength, have been an interest of both domestic and international scientists for many decades. The earliest studies in the field of muscle strength training of athletes first appeared at the end of the XIX century. As W. Roux (1895) has stated in one of his publications, the methodology of strength training lies in increasing the strength of muscle, not by the length of a contraction, but by increasing the muscle tension higher than a regular daily load. During the same time period, a different scientist P. Morpurgo proved that "The core of change in muscle during training lies in the muscle's hypertrophy", which was later confirmed by; W.Thorner (1930) and B. Hoffman (1938), J. Dubois (1915), I. Solonevich (1925) stated that the three time weightlifting world champion (1903-1905) P. Bonn in order to beat the record, trained with the weights which were only 10 kilograms below the current world record, and once a week, he has attempted a one rep max, giving it all he could. One of the most outstanding weightlifters, a Frenchman C. Rigoulot, also trained with heavy weights at 75-90%1RM.

In 1907 the methods of T. Zibert gave a huge drive in the strength development possibilities. Firstly he has introduced two sets in each exercise, secondly, he proposed a three sessions per week training schedule. In the first week, the exercises performed were focused mainly on developing the muscles of the upper body with five repetitions, and the development of the lower body, with ten repetitions. With every week, the amount of repetitions has increased by one, finally reaching 10-20 repetitions in each exercise. Next, he has increased the weight of the bar, thus dropping the amount of repetitions to the initial amount. This cycle would be repeated over and over, which has indeed contributed to some improvement in the final result, however, the gain of strength was extremely slow and lengthy. This appeared to be a serious drawback of the current method.

An American scientist and philanthropist, B. Hoffman, has enriched the system of strength training by introducing the wavelike change of intensity technique. Peary Rader, attempted to promote a system where an exercise was attempted with multiple sets. Unfortunately, as a result of using this method, the athlete has spent an extensive amount of time performing a high number of sets in the same exercise; causing the time dedicated to other exercises from his planned sessions to decrease.

In the early XX century, J. Weider, while developing methodology for bodybuilding training as an independent sport, was one of the first to come to the conclusion that the achievement of great strength is practically impossible without large amounts of muscle mass, but this can be achieved (as he claimed) «only with the use of significant weight loads».

In Russia, the first studies in the field of methodology of muscular strength training could be found in the book of an organizer of the weightlifting team, Dr V. Kraevsky, "Development of Physical Strength With and Without the Use of Weights". In that book, however, the author only cited a few exercises with dumbbells, and did not say anything about the methodology or the quantity of exercises.

There were other Russian specialists in the field of sports. Then, A. Taushev (1902), F. Olshanik (1905) and A. Stoltz (1908) wrote in their recommendations that one should exercise for 10 reps per set with low weight and through every week add another rep.

B. Skotak (1906) argued that the load of the kettlebells must be between 3 and 12 pounds, and should be lifted 10 times, with the exception or some exercises it should be performed 4-5 times.

G. Hackenschmidt. (1912) In his book «The Way to Power: How to Become Strong and Healthy» offers the following teaching methods: «First, the exercise should be repeated 5 times, then every two weeks add one repetition, and so on until reaching ten reps, then increase the weight by 2 kg «.

In his book «Strength and Health. A Guide to Make a Strong and Healthy Man «(1912) I. Lebedev similarly writes that the weight of the dumbbells for all of exercises is to be increased gradually, and only in the third year of training, the weight can be brought up 8 kg. That same year, for the first time in history a magazine called «Hercules» published a concrete individual training program of I. Lebedev: «120x10, 130x10, 140x10, 150x5, 160x5, 170x3 and in reverse: 160x5, 150x10, 140x10, 130x15, 120x20 «This he has recommended for athletes who can bench press 200 pounds.

Among experts there were also people such as E. Sandov (1900) and B. Leitner (1915), who argued that in the beginning of a training session, it is necessary to exercise with 5-pound dumbbells, lifting them freely, and not to failure.

In 1925, the principal of gradually increasing weight has become recognizable by the world of physical education. Hence G. Birzin proved gradual increase in the training load to be the law in optimal stimulus of strength. Most experts then believed that the load needed to be lifted a several times in a row (Order of Ten), avoiding the maximum load and increasing the load gradually, smoothly, and with no sudden leaps in weight increments. The recommendations «The objective of one set is to increase the number of reps» can be found in the book of M. Yakovlev (1927) and in the manual A. Bukharov (1933).

The Soviet specialists in the field of weightlifting at the time favored training with a medium load. Therefore, M. Yakovlev (1927) pointed out that during a normal workout the main load must be carried out with an average weight of the bar, the average weight was considered by the author to be 2/3 of the 1RM.

In 1928, A. Zass (Samson) published a system of physical development called «Samson» This system consisted of dynamic and isometric exercises. He wrote: «I stand by the rule of gradual increase of the load, not the abrupt start of heavy work». Zass has warned that isometric exercises will give a good effect only in the event where they are combined with dynamic exercises using weights, dumbbells, and expanders.

In 1933, in a guide for weightlifting A. Bukharov recommended, «The objective for one set was to perform a large number of repetitions.

In 1939, A. Bukharov recommended an approach to lift the weight no more than 8 times.

In 1940, N. Luchkin published a study manual in the division of weightlifting. In it he summarized the training experience of the best weightlifters. This guide gives evidence for the methods and the immediate preparation for the competition. At that time this was a step forward in the method of training. Unfortunately, in this manual and in its following edition (1948), as well as in textbooks of N. Luchkin (1956, 1962) it does not mention what weights the athletes should have been training with.

The Weightlifting textbook which was under the editorial of N. Luchkin (1947) described a technique in which the routine proposes the following: «If an athlete can press 50 kg 10 times in a row, and 55 kg only 6 - 7 times, it can be concluded, that the working weight for his training is to be 55 kg. In the end an athlete must lift the weight 10 times in a row, and do so with every following weight «.

K. Grantyn (1939), N. Ozolin (1949) L. Matveev (1964), M. Sholih (1966) and some others considered the gradual increase in the load as the basic principle of sports training. To ensure the growth of athletic performance, they said, it is necessary to continuously

progress the development of an athlete's body. Therefore, in 1947 N. Luchkin in one of his textbooks on weightlifting recommended the following system: «If an athlete can press 50 kg 10 times in a row, and 55 kg only 6 - 7 times, in can be concluded that the working weight for training is to be 55 kg. In the end an athlete must lift the weight 10 times in a row. After 55 kg the athlete should now go to the next weight of 60 kg, which he, of course in the beginning cannot press more than 6-7 times. Bringing the number of repetitions to 10, again add weight, and so on». Years passed, and the same N. Luchkin, having gained experience, in his newer book published in 1956, noted: «We must recognize that a positive role in the development of maximum strength using classical exercises, as well as associations of work with the form of the movement, should be played by a weight close to the limit».

In 1950, A. Bukharov proposed a method of training, similar to the technique of I. Lebedev: «To the initial results of 40 - 50% adding 5 kg, to reach 70-85%, and then by reducing the weight of 5 kg and perform 3 – 4 sets».

V. Romanov (1952) proposes that the optimal working weight for training is a weight that the athlete can lift at least four times in a row.

Mc. Morris and Elkins (1954) conducted an experiment in which two groups trained using the same load volume. The first group starts work with low weight which increases gradually from set to set, the second with the highest weight and continues in descending order. It was found that muscle strength increased more in the second group. Explanation of this fact can be found in the works of B. Chikvaidze: «In an annual training cycle the load must not differ significantly from session to session. However, it should be variable in volume, intensity, number of competitions, and a particular set of means and methods «.

N. Luchkin (1956) and R. Moroz (1957) were in favor of preferential use of limiting and near-limiting weights of the barbell for the development of maximum strength of a weightlifter. However, an examination of training diaries of male athletes of the Russian Powerlifting Team showed that the constant increase in load due to the maximal and submaximal weights in competitive periods is not possible, because it leads to fatigue.

According to A. Yanchevski (1958), sessions with weights of 70% of the maximum are more effective in both the growth of speed and power, than training with weights at 20% of maximum.

V. Monogarov (1958) found that in the initial period of training with a medium load is appropriately effective in the development of strength. R. Roman (1958) reached a similar conclusion.

G. Chikvaidze (1959), studying the power of movement during competitive exercises with a barbell, conducted an experiment, which revealed that working with the near-limit and the limit of the load during training shows a positive effect on the development of muscle strength in advanced athletes.

D. Mateev (1959, 1962) also noted that exercising with a weight of 100% of the maximum, gives the greatest increase in muscle strength.

In his thesis V. Chudinov (1961) convincingly demonstrated the effectiveness of heavy loads in the development of muscle strength. The higher the intensity (the average weight of the bar) of course subject to a certain optimum, the more effective is the strength developing training.

A. Chistyakov (1965) during a study of the strongest skiers determined, that the highest strength gain and improved results are obtained using weights of 70-80% of the maximum load.

Berger compared the effect of the training program, consisting of two sessions per week with weights 2/3 of the maximum load, with the effect of the training program, which includes sessions only once a week with a maximum load. As a result of a comparison of the two programs he discovered that with the weight of 2/3 of the maximum load the total increased results were at average more than 15.76 pounds. According to the author, only the combination of training with weights 2/3 of the maximum load 2 times a week and with a maximum load of 1 times a week can lead to a greater growth in strength.

Berger's study results contradicted with the data of Muller and Romerta (1963), who came to the conclusion that the strength gains are greater when training with a maximum load 6 days a week than when training with weights 2/3 of the maximum load 5 days a week, combined with maximum load 1 time per week. But their conclusion, in turn, contradicted the studies of Hettinger (1966) who found that training with weights 2/3 of the maximum load 5 days a week and 1 times the maximum load increases strength similarly to a daily workout with a maximum load.

Many experts - R. Berger, T. Hettinger, E. Mueller, D. Matveev, G. Chikvaidze, A. Vorobyov, N. Saksonov and others – are all in favor of systematic training sessions with a maximum load.

For example, A. Vorobyov (1989) indicates that only large loads are effective for improving athletic performance of trained athletes. Heavy loads only give an effect when alternated with light and medium loads when the body is recovering from the heavy load workouts. Average loads allow the body to maintain efficiency and athletic performance at a certain level. Small loads after large and medium contribute to the restoration of the body together with a sharp increase in athletic performance «.

Other specialists prefer to use lighter loads, offering an effective stimulus to counteract a less sufficient stimulus of high number of repetitions in the set (V. Kapitonov, 1965; S. Vaytsehovsky, 1965; N. Ozolin, 1967; A. Chernyak, 1973; R. Roman, 1975, et al.).

Deciding based on the previously listed studies, it can be concluded that the most effective method in the development of muscle strength is a load of up to 2/3 of the maximum weight limit.

The research of the head coach of the USSR national weightlifting team A.S. Prilepin (1974) discovered that the most optimal training is the following number of lifts:

70% of the working weight (3-6 reps) - 18 lifts
80% of the working weight (2-4 reps) - 15 lifts
90% of the working weight (1-2 reps) - 7-10 lifts

A training effect is reduced if the number of lifts with the working weight used in one exercise is significantly above or below the goal weight. In this case, A.S. Prilepin recommends the following number of lifts with the working weight:

70% of weight - not less than 12 and not more than 24 reps.
80% of weight - not less than 10 and not more than 20 reps.
90% of weight - of at least 4 and not more than 10 reps.

From his experiment A. Prilepin concluded that the highest growth results in the first 5 weeks of training proved to be while lifting 90% of the load, and the second 5 weeks of only 80%. This suggests that the subject gets adjusted to the constant use of the same method, hence giving it a lesser effect.

In a retrospective analysis of training loads of the barbell A. Pakov et al. (1985) provided data showing a wide range of loads, which positively influenced the increase in muscle strength of a human organism (Table 5.1).

Table 5.1

The training load range of a barbell affecting the quality of strength in weightlifters (A. Pakov et al, 1985)

% 1RM	Developing Attributes	Qualification of Athlete.	Condition of Data Obtained	Author, Year
40 – 50%	Strength	Beginner		R. Moroz (1957), E. Muller, T. Hettinger (1953)
52% (45-60)	Strength	Beginner	Experiment	N. Zimkin (1956)
To 67%	Strength	Intermediate	Experiment	M. Yakovlev (1959)
67% (67-75) 70-80%	Athletic Performance	Intermediate, Advanced	Training, Experiment	A. Medvedev (1968) R. Roman (1970) N. Atanasov (1972) A. Prilepin (1974)
To 80%	Athletic Performance	Advanced	Training	I. Solonovich (1925) F.Bogdanovsky (1962)
82%	Strength	Intermediate	Experiment	N. Zimkin (1961)
Minimum	Athletic Performance	Advanced	Training	A. Bukharov (1939) J. Popplevel (1952)
Average	Athletic Performance	Advanced	Training	A. Bukharov (1939) I. Lebedev (1912) Y. Dushanov (1952) J. Popplevel (1952) T. Lomakin (1953) M. Ravtsov (1956)
Maximum	Strength	Intermediate	Experiment	V. Chudinov (1961)
80 – 85%	Strength, Athletic Performance	Intermediate, Advanced	Training, Experiment	N. Shatov (1958) N. David (1950) G. Luchkin (1952) R. Hilleger (1953) D. Hollidav (1954) J. Harris (1955) D. Weider (1960) M. Sorokin (1963)
70 -95%	Strength, Athletic Performance	Beginner, Intermediate, Advanced	Training, Experiment	I. Murray (1954) B. Konnikh (1954) R. Karpovich (1957) I. Knepst (1958) A. Vorobyov (1956) K. Onuma (1963) A. Falameev (1970)
85-95%	Technique, Athletic Performance	Intermediate, Advanced	Training, Experiment	T. Chikvaidze (1957), (1961) L. Sokolov (1963) R. Vorobyov (1965) R. Roman (1965)
89-93%	Technique	Intermediate	Training	S.Harutyunyan (1965) A. Korobkov (1953)
20-80%	Strength	Beginner	Experiment	V. Gerasimov (1953) I. Vasiliev (1954)

The wide range of the weight load is indeed the most appropriate for the development of strength, which reflects the need for such training, where there is an increase of the strength stimulus. Also, the researchers could not always have the methods, which evaluate the developing of strength and the total amount of load in the experimental groups.

As a result of laboratory experiments more definite conclusions were obtained: if the weight of the bar is approximately 85%, then out of the three sets with the bar, 6 repetitions become ideal (R. Berger, 1963). At the same time, A. Pakov warns that the conclusions drawn by researchers in laboratory experiments should be transferred to practice with great caution, since the experiments were performed with great limitations.

5.1. Modern Methods and Means for Strength Training Athletes

Modern methods and means of strength training have a powerful impact on an athlete's body, especially on his musculoskeletal and nervous systems. When they rationally use and plan their training routines, there is a high effect in the development of various strength qualities. Thus, a number of authors (G. Semenov and V. Chudinov 1963; V. Petrov, V. Chudinov, 1966) believe that to further enhance athletic performance requires the addition of muscle building exercises into the sports training.

Training Session - specialized instructive process, built on the basis of a system of exercises that are aimed at improving the direction and development of certain abilities that allow the athlete to achieve the best results. Training Sessions are an integral part of the training of athletes. The meaning of «training» contains the process of development of athletic perfection.

Guided by the basic theories and techniques of the sport for the development of muscular strength of advanced athletes, experts suggest using a number of planning methods: The Method of Repeated efforts, the Maximal Work Method, the Method To Failure, the Impact Method, Method of Isometric Exertion, etc.

The method of Repeated Efforts - The key to this method is a repeated lifting of the load, the weight of which gradually increases with the increasing muscle strength. The effect of training depends on the lifted weights, as well as on the physical ability and muscle strength of an athlete (H. Kabat, 1947, F. Hellebrant, S. Houtz, 1956).

A standard stimulus has a specific «strength limit», therefore instead of the repeated method; a US Army Physician named T. Delorme applied a **Progressive Overload** in 1945. He began to use this technique as an indicator in the rehabilitation of patients, recovering from various disorders, including fractures and injuries of bones, ligaments and joints.

The key to this method was the development of strength by the repeated method, where the weight was gradually increased not only in separate sessions but also from session to session as strength increased as well. In the beginning of sessions, a weight that can be lifted 10 times was determined.

By the end of 1960 there were no less than 10 methods of progressive overload. Therefore, it is wrong to only follow the method of T. DeLorme's, which consisted of performing three successive sets: the first set 50% with 10 maximum reps; second set 75% with 10 maximum reps; and the third set with 10 maximum reps. According to the data of special studies a weight for 5 maximum reps and 10 maximum reps for any muscle is around 89.8% and 78.9% of the 1RM weight, respectively (R. Berger, 1961). The progressive increase in resistance by DeLorme's method has practical value for the development of strength, endurance, and muscle hypertrophy (E. Faulkner, 1950; A. Lindervold, 1952; A. Montgomery, 1954).

G. Birzin (1925) demonstrated the gradual increase in the training load with the *Optimal Strength of the Stimulus Law.* K. Grantyn (1939), N. Ozolin (1949) L. Matveev (1964), M. Sholih (1966) and others considered the gradual increase in the load to be the basic principle of athletic training. Training load should increase or decrease strictly gradually (V. Gorinevsky, 1922; G. Birzin, 1925; N. Ozolin, 1960, 1970; L. Matveev, 1964, et al.). Such arguments can be found in many training manuals and textbooks. Certainly, in strict obedience of this principle one's organism can achieve gradual adaptation to the changing load. However, as noted above in connection with small load changes, the body relatively quickly adapts to what appears to become like a static stimulus. Thus, since the reaction to this load is almost unchanged, adaptation occurs as a result. As soon as there is adaptation, then the needed reaction doesn't occur and necessary development doesn't happen.

There are known modifications to the method of DeLorme (D. Hoag, 1946; S. Houtza. O. , 1946; A. Zinoviev, 1951; A. Watkins, 1952; R. McGovern, H. Luscombe, 1953; I. McQueen , 1954), which are connected to the decreasing number of repetitions by increases the weight and other several variations during the training cycle. Furthermore, if the method of DeLorme is completed in a reverse order, this is called the Oxford method or a method of Zinoviev (A. Zinoviev, 1951), and a method to increase the weight followed by decreasing it is referred to as the McCloy method. (C.McCloy, 1945). Although these modifications do not add anything new to the method of DeLorme, they do however provide different options of developing strength.

Evidently, with the improvement in the methods of absolute strength development, attention was given to issues such as the relative number of sets to weight and number of reps in one set. For example, an experiment was conducted, which has tested the training effect of nine different dynamic training programs of the bench press, and one motionless program (the powers developed in a position where the bar was on the chest, and the angle of the elbow was 90 degrees). Criteria used to measure the strength of growth were dynamic (bench press). The groups performed various numbers of sets ranging from one to three, and a different number of repetitions in each set. Training Sessions took place three times a week for 12 weeks. Group, that has trained in motionless program, once a week were given the opportunity to attempt one set with 6 or 10 repetitions in the bench press to have the ability of press the bar (R. Berger, 1962). The experiment revealed that the method of training by completing three sets with 6 repetitions (with the weight of a 6 rep max) was the most effective.

In the second half of the 50s and early 60s R. Berger, conducted a series of experiments of the maximum and near-maximum load using a series of repetitions with different repetition maximums.

In his first publication, he studied the effects of training three times a week using three different exercise programs in the bench press. Developed on the foundation of repeated maximums 2 RM, 6 RM and 10 RM, each of these programs was performed in 3 sets of the maximum or near- maximum loads. After five weeks of training the authors found no difference in the effectiveness of these programs. This meant that applied repeated maximums were equivalent or even identical to the maximum load of an achieved training effect.

In 1965, R. Berger and B. Hardage found a more effective program for the development of strength for beginners. The configuration of this protocol was not the traditional maximum series, but simply consisted of 10 repetitions with a maximum load. This program was compared with the traditional protocols where the repeated maximum was performed at 10 RM, offering maximum resistance only in the last rep.

In the late 60s there was the McCloy program, consisting of not one but two first sets of 10 RM, and then in the third set, with a weight of 75% with 10 RM which had to be lifted for a one more repeated maximum, but with a maximum number of reps in a set. (Number of reps/Set).

G. Palmieri in the late 80s studied the squat exercise program using different configurations of repetition maximums, consisting of 2 - 3 sets at 1-10 number of reps/set with a varying pace of the exercise: fast, slow and combined. The resting break between the sets is approximately 3 minutes long. Each of the three experimental groups trained with the appropriate method of muscle training for 10 consecutive weeks with a maximum and near-maximum loads, starting in the first week with 10 RM and with growing intensity from week to week, by the 10th week has been brought up to a maximum of a 1 RM. G. Palmieri (1987) found high and statistically indistinctive results for growth and the effectiveness of the training program.

In agreement with the studies K. Rooney (1994), training to failure with greater efforts in the greatest measure helps with the stimulation of maximum strength development.

In the works of S. Borst and J. Marx (2001) it has been experimentally shown that the use of the three sets in the exercise using RM, caused significantly greater strength gains than training with only one set with a RM. S. Borst and J. Marx conducted a comparative study of two programs with low (1 set) and high (3 sets), with the volume of load in women during a time span of three workouts a week for 24 weeks:

1) 8-12 RM
2) 12 RM (3X6)

Testing the maximum (1 RM) was carried out before the start of training, after 12 weeks and after the completion of the 24-weeks. Both groups improved their power rates after 12 weeks, but the groups used in the high volume load (3 sets) has seen a greater result and furthermore, only in the second group there was an increase of strength at the end of the 24 week period.

The analysis of theoretical resources, which was conducted by M. Rhea (2002,2003), also showed the advantage of the multi-set training methods compared to using only one set using a rep max.

It is known that many experts supported the methods of exercises where the resistance is progressively increasing with multiple sets and lots of reps to failure, is not only for beginners and young competitors, but also for highly skilled athletes. Thus, A. Pakov et al. confirmed «All variations of the T. DeLorme method with certain modifications and improvements have been widely used in the training of Soviet weightlifters «.

Y. Verkhoshansky (1988, 1997) suggested the **Repeated-Series Method**, which is characterized by the volume of work due to the large number of reps and sets. Movements are to be performed slowly, without relaxing the muscles between the reps. A several sets are combined into series, which is repeated a several times. The author identifies three main alternatives for the Repeated-Series method, which differ in the primary focus of training impact with a moderate increase in muscle mass, a significant increase in muscle mass, and aerobic-strength development.

1. Performing a series of weight load of a 70- 80%, in one set, 5 - 6 reps. In the second series 2-3 sets with a rest of4-6 min. 2-3 series with a break of 6-5 min.
2. Performing a series of 3 sets:
   with the weight of 80%, for 10 reps;
   90% for 5 reps;
   93-95% for 2 reps with a break of 4 - 5 minutes;
   in a single training session 2-3 series with breaks of 6 - 8 minutes.

3. Performing a series of 4 sets with a break of 5 -6 min in between.
   70% for 12 reps;
   80% for 10 reps;
   85% for 7 reps;
   90% for 5reps.
4. Two series with a break of 8-10 minutes.

Y. Verkhoshansky states that this method is not for beginners, but for advanced athletes, and it promotes a good maximum strength during slow movements, but is ineffective for the development of explosive power and speed. Therefore, it is suitable in small amounts at the beginning of the annual training cycle.

According to P. Poletaev (1998, 1999, 2005) such variations from the real potentials of athletes, programs can be employed only if the indicated 100% of the weight, is at the time of the lift no longer a 1 RM for the athlete, having become much less than the maximum.

In the cases where a rapid onset of absolute strength is needed, a training method of short-term maximum weight is preferred. The difference of this method from the method of progressively increasing resistance is the predominant use of weights in the range of 85-95% of the maximum (3-5 RM), which is combined with the lifting of smaller weight (in one training session) and more. In this case, the number of sets has to be increased to more than three (R. Berger, 1962). So, a weightlifter in one training session was recommended to perform exercises with 5-6 exercises with 6-10 sets of 1-3 reps (A. Medvedev, A. Vorobiev, 1967).

The method of Short-Term Maximum Pressure ensures the development of neuromuscular concentration, which provides more of an effect than a method of progressively increasing resistance, in the development of absolute strength, which has to do with the need to quickly develop it. It contributes to the increase in strength without a significant increase in muscle mass, which is important for sports, which mainly require relative strength. In weightlifting, there is no doubt about the effectiveness of single and double lift series. It is experimentally and theoretically proven and tested in practice, especially in the competitive period of training.

Thanks to the monitoring of trainings of the strongest athletes in the world, it was discovered that every athlete has a maximum level of contraction, but it is important to strictly limit them using individualized specific to the athlete frames. (A. Vorobiev, 1966).

The following experts: R. Berger, T. Hettinger, E. Muller, D. Matveev, G. Chikvaidze, A. Vorobiev, N. Saksonov and others, favor systematic training with a maximum load. Other professionals prefer to use smaller weights in training, offering enough to compensate for an effective stimulus with a large number of reps (V. Kapitonov, 1965; S. Wojciechowski, 1965; N. Ozolin, 1967; A. Cherniak, 1973; R. Roman, 1975, et al.).

Method of Maximal Efforts is characterized by exercises in which the athlete exerts the most amount of force, of which he is capable of during any given training session. It must be remembered that the best efforts are those in which a demonstrative lift allows the athlete to exceed the limit of training. This method is intended for exercises with submaximal, maximal and supramaximal weights.

On the basis of the power of movement when performing the classic barbell exercises and subsequent experiment involving qualified athletes, G. Chikvaidze (1959) talked about the positive impact of near-maximum and maximum loads during training to increase strength. D. Mateev (1959,1962) also noted that the greatest increase in muscle strength is during an exercise performed with a weight of 100% of the maximum. R. Berger, T. Hettinger,

E. Muller, D. Mateo, G. Chikvaidze, A. Vorobyov, N. Saksonov and others were all in favor of a systematic training with a maximum load.

N. Ozolin (1970) emphasized «borderline weight loads make it difficult to control the technique of action and increases the risk of injury and overtraining, especially in children and beginners. Therefore, this method is not to be done by qualified athletes, it is often not applicable 2-3 times per week. Weight greater than the limit training weight, is to only be used occasionally about once every 7 - 14 days «.

In connection with the study of the above experts, it was observed that the implementation of the method of maximal effort most importantly requires strictly following the rules of gradualism. So, N. Shatov and D. Markov advised to increase the weight of the load very gradually over a long time. This applies to any exercise performed with the maximum intensity. A similar view is found in the work of Y. Kuramshin (2004): «The number of repetitions in one set, overcoming limits and supramaximal resistance may be reached at most - 2 reps. When you exercise with near-limit loads (weight 90-95% of maximum) number of possible repetitions of movements in one set to be 2-3, and 2-4 number of sets. «However, the «maximal effort method ensures that the neuromuscular concentration provides a great boost of strength».

A. Vorobyov (1988, 1989), discussing the application of the method of maximal effort, said: «At present, a number of authors suggest a maximal effort method. It would be better not to talk about the maximal effort, but about the periodic, with a load of 90% of the maximum, and more. Analysis of training of the strongest athletes in the world has shown that the maximum contraction must be included in the preparation of each athlete, but should strictly be limited in accordance of their specific, individual frames».

The following method for developing muscle strength is called the **Method «To Failure»** and is primarily used in bodybuilding to increase muscle mass and strength. It is characterized by a continuous, relatively slow execution of exercises with 50 - 70% of the maximum weight as long as it takes to fatigue the muscle. Exercise should be stopped only when it becomes too difficult to continue. In many cases, it requires a several sets of such method per exercise.

The Impact Method is based on stimulating muscle groups by using the kinetic energy of the falling weight or weight of your own body. The muscle is trained by the energy of the falling mass, which promotes an abrupt contraction of the muscle and the rapid development of the work force, creating additional capacity in muscle contraction, which provides plenty of power and speed of the repulsive motion and a quick transition from overcoming a lower level of work.

Method of Isometric Exertion is a complementary method for the training of development of maximum strength. It is characterized by performing the short-term maximum contractions without changing the length of the muscle.

Isometric movements have been studied by many scientists (Hellsten, 1907; Bethe, Fischer, 1928; Ferrari, 1932; Seits, 1933; Fessard, Langier, Monnin, 1935; V. Shokhrin, 1934; M. Marshak and A. Shlykova, 1934; et al.), but in those years, none of them considered such exercises in relation to development of power.

T. Hettinger and E. Muller (1953) after conducting a series of experiments have suggested isometric exercises in order to increase muscle strength and mass. T. Hettinger (1966) believed that the optimum amount of a load could be considered equal to 40-50% of maximum. In a load of 20-30% of the maximum muscle strength does not change. A session of 5-7 times a day can give greater strength gains than single daily workouts.

Experiments conducted by Ezengas, showed that at two contractions a day there was improving effect of 9%, at three 16% than one daily contraction. However, at 10, 100 and 600 the effect corresponded to an effect of different contractions (for T. Hettinger 1966).

In the 60s in our country, a lot of research on the effect of isometric exercises to develop muscle strength has been conducted D. Donskoy, V. Dyachkov, D. Markov, M. Mirskiy, I. Ratov, 1962; V. Chudinov, 1964; N. Laputin, 1966; Y. Ivanov, 1966 and others.

V. Zatsiorsky (1966) considered isometric exercises as an additional mean to the training of muscle strength. The duration of isometric tension is usually 5-10sec. The magnitude of force should be 40-50% of the maximum, and the static strength schemes must consist of 5-10 exercises aimed at developing strength of different muscle groups. Each exercise is performed 3-5 times at intervals of 30-60 seconds of rest. It is advisable to perform isometric tension in the positions corresponding to the position where the maximum effort during an exercise takes place.

Comparative experiments have shown that athletes who in training use only static exercises, strength growth was slower than those who used the dynamic exercises (Rush, Mohauz, Peterson, B. Monogarov et al.). In this regard, isometric exercises should be used as an additional means to the development of strength. They are executed in the form of repeated maximum contractions with duration of 5-6 seconds each (both shorter and longer duration yields a lesser effect). Prolonged use of the isometric contractions leads to a significant proliferation of intramuscular connective tissue, which increases the strength (rigidity) of muscles, but reduces their elasticity. Therefore, for sports in which a fast dynamic muscle performance is needed, long-term use of isometric exercises is not justified (P. Hutz, 1975).

Thus, in reviewing the methodology of strength training and their variations have their advantages and disadvantages. Rational method involves the skillful combination of all areas in accordance with the characteristics of the subject engaged in the matter of training and level of physical education. Using the proposed methods, we must not forget that each of them has a preferred direction, and improves the quality of strength and equipment of the exercises.

5.2. Training methods of Foreign and Russian Powerlifting Specialists

5.2.1. Training Methods of Foreign Specialists

Since the late 80's to the mid 90's, Russian power lifters and their coaches enthusiastically «grabbed» the works on methodology of foreign experts on bodybuilding and powerlifting, which were published in various bodybuilding magazines. Moreover, the authority of these experts was so high that Russian coaches (who mostly came from weightlifting) completely took advantage of these training methods.

A famous specialist, the American Scientist Dr. Frederick K. Hatfield (USA) in his book «Comprehensive Guide to Strength Development» (1983), says that the maximum output of strength training (on average) is only when training loads exceed 80% of the maximum of an athlete. F.K. Hatfield claims that the average three-lift power lifter is able to perform at least 10 reps with such a load. But most power lifters perform fewer repetitions. In general, in the off-season an athlete should perform 6-8 reps with weights in excess of 85% of their maximum. At the peak of their training cycles an athlete includes sets with three repetitions with loads exceeding 90% of the maximum.

American powerlifting specialist Tom McCullough, in his article «How to Design Your Workout» offers a method of periodization. He describes the method of periodization. Periodization - a gradual cycling of specificity, intensity and volume of training to achieve peak levels of strength. Cycle gradually goes from high volume and low intensity to low volume and high intensity training. Cycle length (usually a several weeks) is selected based on the day of the competition. A typical powerlifting cycle consists of three phases: hypertrophy phase, strength phase, and power phase. Usually a single phase lasts from one to six weeks. During that time, the sets performed are consisting of 8-10 reps with the load of 65% to 79% of the one rep max. During the phase, sets of 5-8 reps are used, and intensity grows to 80-90s of the one rep max. Obviously, this is the phase for increasing muscular strength. Then, we use sets of 1-4 repetitions with intensity, gradually increasing from 90% to 107% of a one-rep max. During this phase the athlete reaches their peak strength for competition «.

One of the strongest athletes in the world, Rick Weil (1986) believes that the key to the training method is the proper cycling of weights during training. Cycling is an approach of reaching the peak of your performance not by leaps, but by small steps. He argues that the number of repetitions from 4 to 7 in a given set is the only thing that really builds power and strength. Rick warns that during this cycle athletes should not attempt a one-rep max in the bench press. In order to achieve the desired strength, sets with 1-2 reps are not appropriate.

L. Simmons (2003) adheres to his own opinion on methodology: «Simple progressive method of training (6, 8, 10 repetitions: a progressive increase in the load) works well in the beginning. Extensive studies have shown that for elite athletes 4-6 reps with 70% of the maximum is the optimal load. According to the method of the progressive weight increase, which is used most frequently, in the beginning exercises are performed at high intensity and low volume. Later in the cycle, and closer to the competition date this changes to the exact opposite. Let us forget the method of the progressive weight increase. This is a dead end».

An accomplished head coach of the Ukrainian men's national team, (2004 – 2006) A. Kotlubey has a different opinion on the development of training routines. He states that «The workouts should take an athlete to a certain level of fatigue (Lack of recovery) with de-loading just before the competition». «This, says A. Kotlubey, is something I constantly demand from all those who I help with in their preparations».

World record holder in the bench press R. Gaugler used a system of «light» and «heavy» days in a weekly micro-cycle. «Light» day consisted of 4 sets of 5 reps with a weight of 75.7 - 77.1% of the maximum. On the «heavy» days R. Gaugler performed 3 sets with 2 reps in each.

T. Arcidi suggests: «If you want to improve the results, you have to train with the highest possible load. If you can perform even one repetition beyond what you have done, then your load is not heavy enough».

L. Simmons (USA) for 24 years was the leader of the «Elite» club in five weight categories for the combined 3 lift total. We also know that Simmons invented the reverse hyperextension machine. Currently, he is the owner and manager of the world-famous "Westside Barbell" club, which produced 17 world champions, and 24 national champions. Louie Simmons has published more than 100 articles devoted to strength training.

Technique of his methods is unlike that of any other professional's. He claims that much of his methods were adopted from trainings of Russian experts: Y. Verkhoshansky,

V. Zatsiorsky, A. Prilepin and A. Medvedev, and he specializes in areas of such techniques as: "Percent training method", «Conjugate method», and «Dynamic Method».

E. Blinov, after analyzing the technique of L. Simmons, concluded: «Training of Simmons divided into two groups: speed developing (dynamic) and strength developing (maximum effort). The prerequisite for the method is that speed and power-training sessions must be 72 hours apart. Squats, bench press and deadlift are not trained in competitive form. Instead Simmons extensive use of the principle of «conjugation», which replaces competitive movements with exercises that have similar mechanics, which on power days are subject to constant change once every 1-2 weeks in order to avoid burning out the Central Nervous System.

Simmons pays particular attention to special preparatory exercises, as they allow the lifters to focus on their weak areas, to achieve the necessary hypertrophy. Neither the squat nor bench press are used as a tool to build muscle mass. The volume of preparatory exercises specifically does not decrease prior to the competition, and often increases, where at the same time the weight of the basic movements, decrease.

The duration of the training session, as well as all the breaks between sets are strictly limited (usually no more than 60-70 minutes, within which the basic work must be completed within 45 minutes). On one hand this allows you to train at a peak of the hormonal level, and on the other, improves general physical preparedness of an athlete.

Simmons widely uses the following principles: The principle of «Accommodating resistance», which uses chains and bands. The principle of «Separation of Eccentric and Concentric Phases», which uses boxes and boards of different heights; the principle of « Maximum Effort and Dynamic Effort Loads»; the principle of «Duration of the Process While Under the Bar» according to which the set takes as much time as needed to perform competitive movement, as well as the method of «Repetition Method». In addition to the 4 basic training sessions in the weekly cycle, included often are a several short additional exercises that can be carried in the same day or on other days. This is done both to further develop lagging muscle groups as well as to accelerate recovery and prevent muscle soreness after exercise.

The use of equipment is limited both in terms of time (just before the competition) and volume (in the squat and deadlift only belt and briefs without straps, and in the bench press, a shirt at the discretion of the athlete). The basic idea of L. Simmons is that full equipment and maximum weight is only to be used on the day of the competition.

He does not recommend that athletes burnout their CNS by preparing themselves mentally as before attempting the lift at the competition.

In his planning of training routines Simmons widely used the method of conjugation, using similar exercises, to basic competitive movements, such as the bench press (but not the bench press itself), changing them every 1-2 weeks.

J. Wendler, who is the chief editor of “Elite Fitness Systems” proposed a method of «5/3/1» reps to fluctuate the number of repetitions throughout the year. The program «5/3/1» as he suggests, is performed three or four days a week, and every workout is designed around one basic exercise: squats, deadlift, bench press, and standing bar press.

Each training cycle lasts four weeks. For each of the basic lifts the routine of sets and reps is as follows:

Week 1 - 3x5
Week 2 - 3x3
Week 3 - 3x5; 3; 1
Week 4 – Re-load

The author suggests using the weight of a certain percentage of a one-rep max. In this case, a maximum weight used is not taken as the maximum weight for one repetition, but 90% of it.

D. Pitts is convinced that his method of «Super Condensed Training» is one of the most advanced in the procedure of planning routines. In the super condensed training only two or at most three exercises, such as squats and bench press, deadlift and standing press, or other exercises like the squat, bench press and deadlift.

Each exercise is performed once a week, or in one single-standing set, or by the classical scheme of 5x5. Pitts claims that he, training bench press on Monday, and squats on Thursday, got great results adding more than 90 kg in the squat and about 40 kg in the bench press during an 18-month period. His training partner, who trained the bench press and deadlift, bench press increased by 27 kg and deadlift by 80 kg during a 12 month period.

He explains this by saying that they trained in these exercises, constantly adding weight and moving forward. When progress was stationary, they took a week off. Then again in a few weeks returned back to heavier weights and continued to train. Pitts says: «This kind of training is incredibly intense. You take one exercise and work on it! Week after week, you add more weight to the bar, and then go home to rest in order to sufficiently add more weight at the next training session. «Pitts offers several training routines (First for beginners and intermediate athletes):

1. Training Session 1- Bench Press 5x5
2. Training Session 2- Squats 5x5.

The author warns that the stronger the athlete becomes, the more rest he requires. Each exercise is performed once every 7 days. Many power lifters during competition season, and in the off-season, use approximately the following system:

1. First Training Session- Squats 1x5
2. Second Training Session - Bench Press 1x5
3. Third Training Session - Deadlift 1x5

The Athlete must select only one program and then work on it for a period of at least one year, preferably two years. While you will have a disproportion in strength, the total strength will grow, so that when you return to the regular program, your loads in other exercises very quickly surpass previous personal records «.

Some strong German power lifters, including junior world champion, R. Gierz, have used Korte's "3x3" training program. The basic concepts of this program have been used by almost every weightlifter, including many of the world champions in the last 40 years.

The 3x3 system is an eight-week training program, which consists of two phases. Phase one is the high volume phase, while phase two is the competition phase. This part is similar to the style used by L. Simmons. The similarity lies in the fact that there are no off seasons. The working weight is 58-64% of the maximum and the main focus is given to the 3x3 during the high volume phase. Another similarity is that there is a decrease in the amount of exercises, when an athlete starts working with weights nearing 80-95% of the one rep max.

S. Korte states that: «One thing that makes the 3x3 system unique, compared with a system of Simmons and other powerlifting systems, is that here there are only those exercises that you perform in the competition, no supplementary movements. Why? The answer is very simple. To gain strength in the squat you need to train the quadriceps, hamstrings, hip flexors and the muscles of the lower back. Other programs use different ways to exercise these muscles. You can perform many different exercises and use it in conjunction with squat. Another option is to train the squat and only the squat. We already know that

this exercise works all the muscles mentioned above. The most important advantage of this option is that the squat develops exactly those muscles, which are needed during competitions».

5.2.2. Training methods of Russian specialists

A famous training specialist and author of books on power sports V. Plekhov (1988) recommended that the bench press should be performed three times a week. «The first training session - competitive style, the second would be to concentrate on the development of major muscle groups, and the third would be supplementary pressing exercises. Each exercise is performed from three to eight sets, with one to six reps «.

S. Cherednichenko (1991) suggested that for competitive exercises, the working weight should be 60-80% of the maximum. The pace of exercise should be moderate, and from 6 to 12 number of reps during a single series, with a constant weight of the bar. Weight of the load must constantly grow and reach 85-95% of the maximum and the number of repetitions in the series should be reduced to 4-6. Once a week, the author recommended using a pyramid to bring the weight of the bar to the near-limit and the limit of an athlete «.

R. Shagapov and O. Sliva (1998) proposed routines with even heavier loads. For example, the authors recommend that during the off-season, an athlete should perform six sets of ten reps with a load of 80% of a 1RM in the squat. In the next training session, they suggest deadlifting eight sets with six reps each with a load of 80% of a 1RM. In the bench press the author suggests an athlete should perform ten sets of eight reps each with a weight of 75% or a 1RM.

P. Rybalskyi (2000) during the competitive month suggests 2 single workouts per week, during which an athlete should train in a competitive mesocycle with weights of 85% for 5-6 reps, 90% for 5x5, and 95% for 3x5.

P. Chernyshev (1999) in his article «Bench and nothing but Bench», wrote: «Avoid at all costs any additional exercises which train the triceps, biceps, and others. Only the bench press is to be performed and that is all. I can explain why; the strength, and energy with which you start the training session, should only go into the exercises that you need, otherwise it is a waste of energy. We must not forget about the recovery process, it is easier to recover from one exercise, than from all of them».

V. Muraviyev (2001) in his program also suggested that for those who are working to develop strength, it is necessary to eliminate virtually all the additional exercises for chest, triceps and deltoids, only bench press (classic), standing press, and the close-grip bench press.

The same opinion is shared by A. Faleev (2006, 2008): «For rapid muscle growth and positive results you should only do three exercises: bench press, squat, and deadlift. Everything else will only cause harm. Performing extra and unnecessary exercises will cause you to waste precious energy reserves, which could potentially give you an amazing spike in the basic exercises. These exercises work and develop all muscles of the body, thus making all the other exercises useless».

With a completely opposite opinion I. Belsky (2003) states, «You can not go to extremes, and even in the competitive period one should not avoid exercising muscle groups which not directly involved in the bench press. By the way, this is a mistake that is made by a lot of athletes, justifying this by conventional wisdom: «If you want to press then press», or, in other words, the result which they focus on is only the bench, they consider doing other

exercises a waste of energy, in addition, they claim that it is easier to recover from one exercise than from many».

A. Gorbov (2004) argues that many successful power lifters successfully use a routine where in one weekly cycle, when the first training session in the microcycle remains almost unchanged throughout the mesocycle; the intensity is increased only in the second and third session.

Thus in such a way an athlete develops a necessary gradual build-up of intensity and speed needed for such development.

In the «Nine Week Competitive Cycle» proposed by the author of the book “System of Effective Training « an unrealistically heavy load is suggested during the 6th, 7th and 8th weeks. On Wednesdays, a heavy burden is put on the leg and back muscles during the deadlift; In week 6, 80% 6x6, week 7, 85% 5x5, week 8, 90%4x4, with an even more unrealistically heavy load on squat days on Fridays” week 6, 90% 4x4, week 7, 95% 3x3, and in week 8 100% 2x2.

It has been proven that if the load (volume or intensity) increases smoothly and gradually, you will inevitably encounter the stress, which exceeds your adaptive capacities, adaptation does not occur at the same time, thus you find yourself in a state of overtraining. Conversely, if the intensity of the load increases gradually but not smoothly but rather sporadically, by the principle of «step back - two steps forward», then this condition usually does not occur, and productivity and training success increases».

I. Belsky (2003), at the same time suggests performing quite the number of repetitions. In his book, the author presents a table of the maximum number of repetitions, in which he recommends performing these with various loads.

Table 5.2

%1RM	100%	95%	90%	85%	80%	70%	60%	50%
Maximum Number of Reps	1	2-3	5-6	7-8	10-12	12-16	15-20	20-25

A. Faleev (2006, 2008), holds quite the opposite opinion, which strictly states: «There should be no lifts with the maximum load! For these purposes we have competitions. Such attempts are harmful in terms of training. Attempting sets with the maximum loads causes you to miss out on a full day of heavy lifting. Secondly, it is very traumatic. It is because of such frequent and reckless actions, athletes have to heal micro muscular tears for six months at a time».

K. Rogozhnikov (2008) recommends benching for 8-10 repetitions in 3-4 sets, with one repetition to spare. The last bench press workout should be 4-7 days before the competition. In this training session, Rogozhnikov suggests performing a 2-3 sets of 6-8 reps not to failure. Do not do any assistance work for the bench. For pure benchers, he recommends 3 to 4 days of complete rest between workouts. You can go to the gym, and do some work, but it is important not to put any load on the groups of muscles that are involved in the bench, which are deltoids, triceps, and pecs. Do not overload yourself before any heavy training sessions».

Since the late XIX and early XX century theoretical investigations of both foreign and domestic sources, showed that the main principle of most methods proposed by national experts of strength sports, was to increase the weight of the bar with the gradual addition of the number of reps (5 to 10 or more) of one set. This led to training sessions with large volumes, at low intensity and thus slowed the development of athletic performance. Even though training volume and intensity increased, the differences of opinions of various specialists still remained.

Based on the data breakdown of the above-mentioned methodological information, it should be noted that professionals working with advanced athletes, did not, and still do not have a universal approach to the methods of training power lifters in the competition period. In order to properly plan the training, it is necessary to know how different loads affects the athlete, the difference in level and path, as well as taking into account the performance of the athlete during the entire training cycle.

Chapter 6. Powerlifting training structure

In order to ensure the improvement of sports performance it is essential to continuously develop the functionality of an athlete's body. This functionality is achieved through systematic increase of training stimuli and thorough full-scale planning. In accordance with the objective of continuous training stimulus increase, a powerlifting training plan should include the following points:

- Improvement of overall, general physical development
- Further development of more specific physical qualities
- Further improvement of technical and tactical training
- Acquisition of knowledge on the theory and methodology of training, hygiene, and self-control
- Acquisition and development of fitness form (actual lifting ability) at a particular training stage to show the planned result (hit PRs).

The integrity of training process is based on a particular structure, which has a relatively constant order of its components, their fixed correlation with each other and their overall sequence.

Given the length of time within which particular training process units are formed we distinguish:

- Microstructure - the structure of a single training sessions and a microcycle (weekly);
- Mesostructure - the structure of training stages, including a relatively complete range of microcycles (total duration, for example, about a month);
- Macrostructure - the structure of long training cycles such as semi-annual (meet to meet), annual and multi-year.

6.1. Single training session and micro cycles' structure

Sports training is an integral part of the preparation of an athlete. It is represented by a pedagogically organized process of sports improvement aimed at the development of certain qualities, abilities and the formation of necessary knowledge and skills that determine if an athlete is ready to achieve the best results in a selected sports activity.

Y. Kuramshin describes the training process in three main points (2004):

1) Sports training should be structured as a year-round and multi-year process, its units must be interconnected and subordinated to the task of achieving maximum results;

2) The impact of each subsequent workout, microcycle, stage, or period should be layered on the results of previous training, consolidating and developing them;

3) The intervals between training sessions must be regulated in such a way as to ensure the development of skills and overall steady increase of athletic performance, with its underlying necessary qualities and abilities.

A single training session has three typical parts, i.e. introductory, main, and final.

In the introductory part, the warm-up promotes the optimal preparation of an athlete for the upcoming work.

In the main part two basic tasks are solved. The routine can be varied and it must ensure the improvement of various sides of physical and psychological fitness, the optimal technique, tactics, etc.

In the final part an athlete's body is brought as close as possible to the conditions that allow for intensive recovery processes, the workload is gradually decreased.

Training is structured in cycles different in length. In 1964, L. Matveyev divided a general multi-year training process into micro (small), meso (middle), macro (big) cycles (part) of a training.

A microcycle is known as a series of workouts carried out over several days that provide a comprehensive solution to the tasks at this preparation stage. Usually the duration of a microcycle is one week. Microcycles consist of the routines whose primary focus varies (mostly regarding training volume) but whose general focus is the same. Each of these routines has a certain effect on an athlete's body, so it is important to know how to build a microcycle to effectively solve all major problems related to a particular stage of preparation. The methodology of constructing a microcycle depends on several factors. First of all, it should take into account the specifics of fatigue and recovery processes as a result of workload of a single training session.

Microcycle structure includes exercises, training tasks, parts of training sessions, training sessions, and several training sessions. The number of training sessions in a microcycle can range from 2 to 12 sessions.

Microcycle types

Microcycle types can be broken down into five categories, i.e. conditioning, shock, preparation, competition, and recovery.

Conditioning cycles are aimed at preparing an athlete's body for intensive training. They are applied during the first preparation stage, and they often come first in mesocycles. These microcycles are characterized by a relatively low total training volume in relation to the volume of the upcoming shock microcycle.

Shock microcycles are characterized by a large total training volume and heavy loads. Their main task is to stimulate the adaptation processes in athletes' bodies, addressing the major tasks of the technical, tactical and integrated training.

Preparation microcycles are directly aimed at preparing an athlete for competitions. The content of these microcycle can be varied and depends on the competition preparation system, an athlete's individual characteristics and the specifics of the final stage.

Preparation microcycles programs preceding the start are highly individualized. Athletes with increased recovery and adaptive abilities, unlike the ones who do not possess such, can handle a more intensive microcycle workload.

Competition microcycles are structured in accordance with competition program. The structure and duration of these microcycles is determined by competition specifications in various sports disciplines, program routines in which an athlete participates, total number of starts and breaks between them.

In competition microcycles, above all, attention must be paid to full recovery and ensuring conditions for peak performance of athletes during major competitions. This requires the establishment of a special regime of training sessions and rest, proper nutrition, psychological preparation, and the application of recovery tools.

Recovery microcycles are usually applied after a series of shock microcycle sessions. They are also planned after intensive competitive activities.

6.2. Distribution of training loads to weekly cycles and training sessions

Training load is a certain value that describes the impact exercising has on an athlete's body, as well as the level of objective and subjective difficulties that are overcome while exercising.

In sports theory and methodology [N. Ozolin, 2002; V. Platonov. 1986, 1997], the amount of training and competitive load can be described from an «external» and «internal» point of view.

The external side of the training load can be generally represented by the indicators of the total training volume. These include: total training volume in hours spent in training, cyclic training volume (jogging, swimming, boating, etc.) in distance covered, number of training sessions, and number of competitions.

To evaluate the «external» load the indicators linking training volume with its intensity are widely used. These indicators include: the pace of movement, speed of execution, time to overcome training distances and lengths, amount of weight, distance length, number of sets and repetitions, and total training volume.

The "inner" aspect describes the training load more precisely, i.e. it is based on the reactions of the body towards the work fulfilled. Along the indicators carrying the information about the immediate exercise effect (changes in the functional systems of the body while training and right after it), one can use the data on the character and duration of the recovery process: time of motor reactions (reaction time), amount and character of the developing force (how quickly the athlete can reach peak force and how much maximal force he can produce), bioelectrical muscle activity, heart rate, respiratory rate, lungs ventilation, cardiac output, oxygen consumption, and the rate of lactate accumulation in the blood. Training value, in addition to these parameters, can be characterized by the recovery of efficiency and glycogen supplies, activity of oxidizing enzymes, speed and mobility of nervous processes, and others such indicators.

The external and internal training load characteristics are closely linked: an increase in the volume and intensity of training work lead to increased changes in the functional state of different organs and systems, as well as to the emergence and deepening of the processes of fatigue, and slowing down of the recovery process.

Training volume at a workout session

The main factor determining the impact of a training session on an athlete's body is the amount of training. Training can be divided into small, medium, large, and maximum categories.

R. Roman (1974), A. Medvedev, A. Chernyak (1980) believe that the amount of training should be considered small when the number of total reps performed in the session is below 50, medium - 51-100, and large - 100 or more.

The experimental data obtained by M. Aksyonov (2006), is similar to the results of the research conducted by weightlifters A. Medvedev A, A. Chernyakov and R. Roman (1980). In his thesis, Aksyonov writes, «To some extent the amount of training sessions load, in our opinion, can be compared with the amount of training done by double event weightlifters, and conclusions can be drawn about the type of the most commonly used training by weightlifters».

R. Roman (1974) recommended that a weekly cycle have not more than two training sessions (at the beginning and end of the week) with large training volume.

A. Vorobyov (1964), for three training sessions per week, suggested distributing the volume in the following way: on Monday, after two days of rest, to train with large and maximum weights, on Wednesday the volume to be reduced by 20-40%, on Friday to increase it to the level of the first training day and sometimes even exceed it.

A. Chernyak (1972) for three training sessions per week proposes to use 24%, 28% and 48% of the weekly training volume .

With three workouts per week, training volume will not be as varied as in 4-6 workouts. But even in these conditions it is possible to achieve diversity. For example:

Table 6.1

Training loads distribution within microcycles
in a preparatory mesocycle, 3 workouts a week
(Boris Sheiko, 2005)

workouts / **weeks**	**week 1**	**week 2**	**week 3**	**week 4**
1st workout	Small	Medium	Medium	Small
2nd workout	Medium	Small	Small	Large
3rd workout	Small	Medium	Large	Medium

Weekly training volume (total reps) gradually increases from week one to week four (see Table 6.1.). Such a structure is possible during the second months when training beginning athletes and during the first month while training experienced athletes at the time when they focus on the elements of competitive exercises technique with light weights.

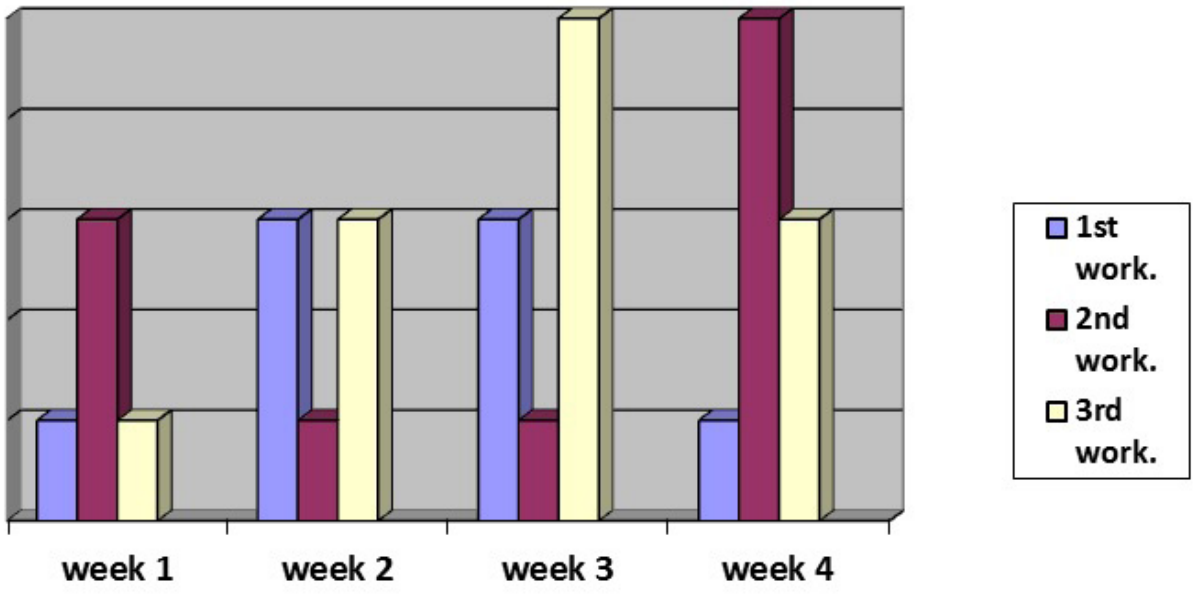


Fig. 6.1. Training volume distribution within microcycles in a preparatory mesocycle, 3 workouts a week

In the table 6.2 weekly training volume is much larger than in the table 6.1, and it changes abruptly. The load reaches its maximum during the second week. During the first and third week it is in the middle, and it increases again during the fourth week.

Table 6.2

Training loads distribution within microcycles
in a preparatory mesocycle,3 workouts a week
(Boris Sheiko, 2005)

workouts / weeks	week 5	week 6	week 7	week 8
1st workout	Large	Large	Small	Medium
2nd workout	Small	Medium	Large	Large
3rd workout	Medium	Large	Medium	Medium

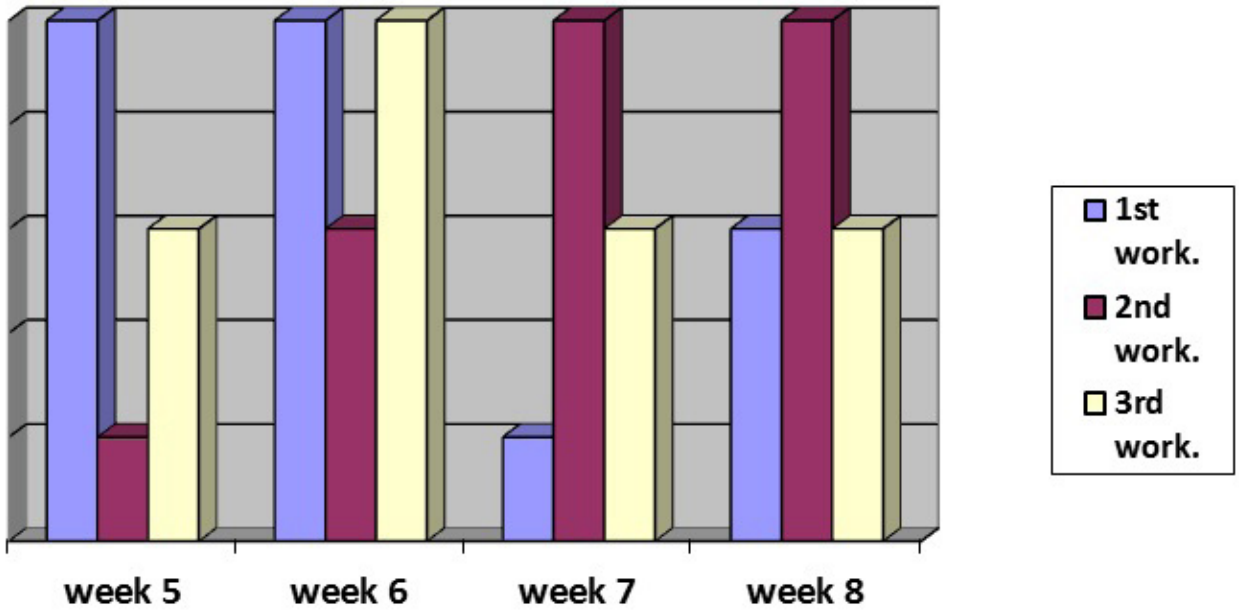


Fig. 6.2. Training loads distribution within microcycles in a preparatory mesocycle, 3 workouts a week (B. Sheiko 2005)

Any well-structured plan for a group of athletes, even if it is written by a highly qualified coach, needs to be adjusted during the process of training sessions. It can be explained by the fact that a group includes athletes with different anatomical and physiological characteristics, which means that their reaction to the training volume will be different, as well as recovery, and errors in the technique of competitive exercises. Thus, coaches must provide additional exercises (not included in the plan) in order to eliminate any technical errors during training.

Table 6.3

Training load distribution within microcycles
in a competition mesocycle,3 workouts a week
(Boris Sheiko, 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Large	Large	Medium	Small
2nd workout	Small	Medium	Small	Small
3rd workout	Medium	Small	Medium	Rest

A large training volume is planned for Monday of the first week after two rest days. The last large volume is planned during the second week 18-20 days prior to competition. During the third week there are two training sessions with average volume and in the fourth week (competition week) two minimum training volume workout sessions. In fig. 6.3. one can see that after the second week training volume gradually decreases.

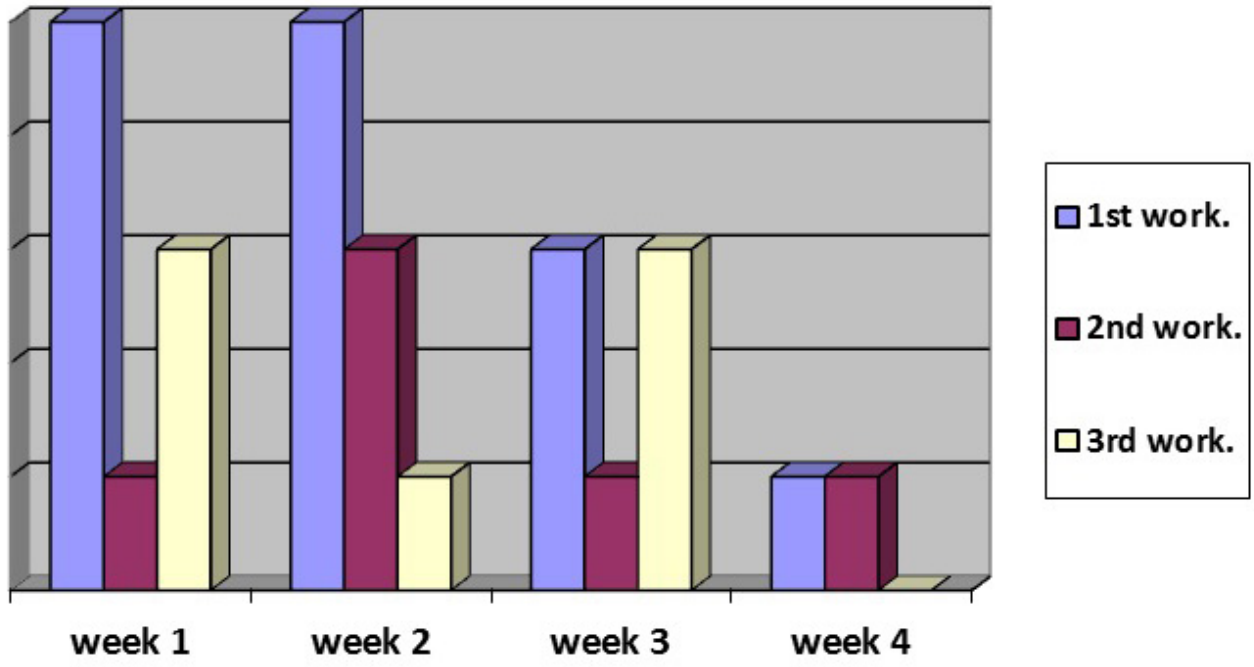


Fig. 6.3. Training load distribution within weekly microcycles in a competition mesocycle

** Note: The athletes who compete on Saturday have a last workout on Wednesday, the athletes who compete on Sunday have a last workout on Thursday.*

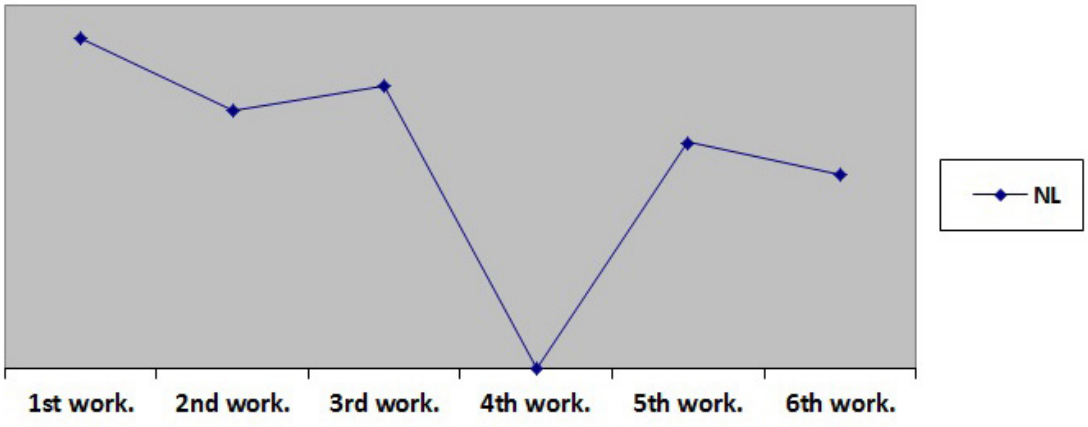


Fig. 6.4 Number of Lifts (aka reps) (NL) distribution in a competition microcycle (2 weeks before the competition)

A. Chernyak (1972), with four trainings per week suggests using 15, 22, 28 and 35% of the weekly training volume. He points out that the numbers sequence (actual training volume structure) is determined by a coach depending on the level of an athlete's physical preparedness and the aim of the training.

An approximate monthly training volume distribution within weekly cycles with 4 trainings per week can be the following (Table 6.4.).

Table 6.4

Training loads distribution within microcycles
in a preparation mesocycle, 4 workouts a week
(Boris Sheiko, 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Medium	Large	Large	Medium
2nd workout	Large	Medium	Small	Small
3rd workout	Medium	Large	Large	Large
4rd workout	Small	Small	Small	Small

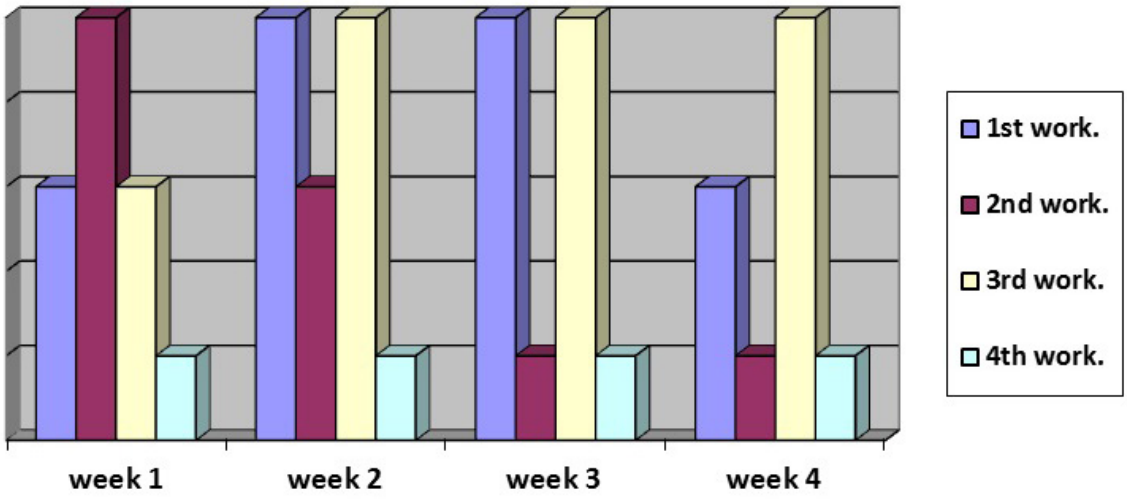


Fig.6.5. T raining load distribution within microcycles, 4 workouts a week

Table 6.5

Training load distribution within weekly microcycles
in a preparation mesocycle, 4 workouts a week (Boris Sheiko, 2005)

workouts / weeks	week 5	week 6	week 7	week 8
1st workout	Medium	Medium	Large	Large
2nd workout	Small	Medium	Small	Large
3rd workout	Medium	Small	Small	Small
4rd workout	Small	Large	Medium	Medium

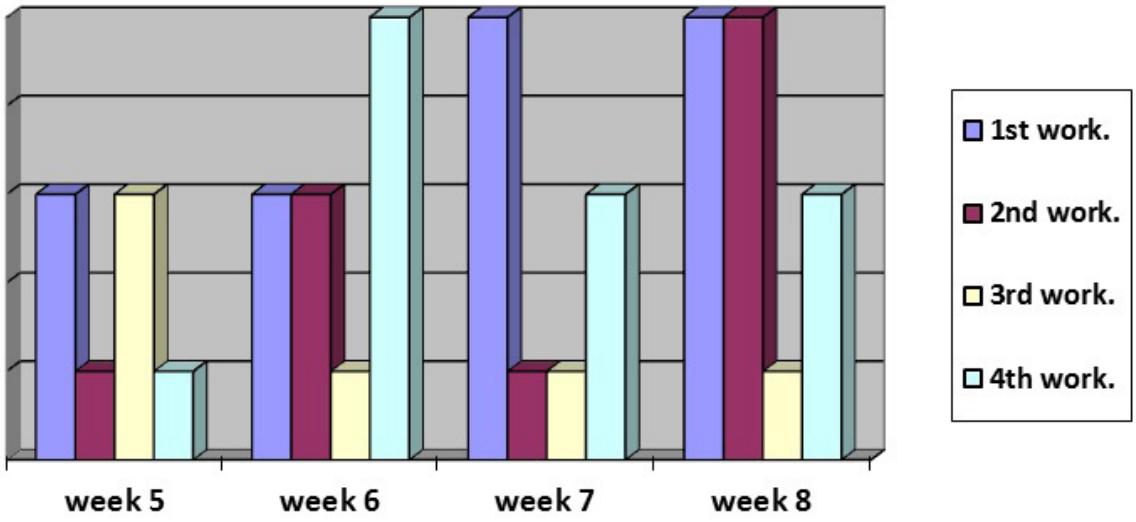


Fig.6.6. Training load distribution within microcycles, 4 workouts a week

Increasing training session number to four times per week is a big step towards increasing training volume in general. However, a number of leading powerlifting coaches claim that four training sessions are not enough. They schedule five or six sessions per week.

During five training sessions a week, training volume variation is slightly larger than during four, and much more than three sessions. However, it should be noted that Albert Fomin (Ulyanovsk), a distinguished Russian coach, achieves good results with world-class athletes during Russian, European and World championships with three training sessions per week.

Table 6.6

Training load distribution within microcycles
in a competition mesocycle, 4 workouts a week (Boris Sheiko, 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Medium	Large	Medium	Small
2nd workout	Large	Medium	Small	Small
3rd workout	Medium	Medium	Medium	Rest
4rd workout	Small	Small	Rest	Competition

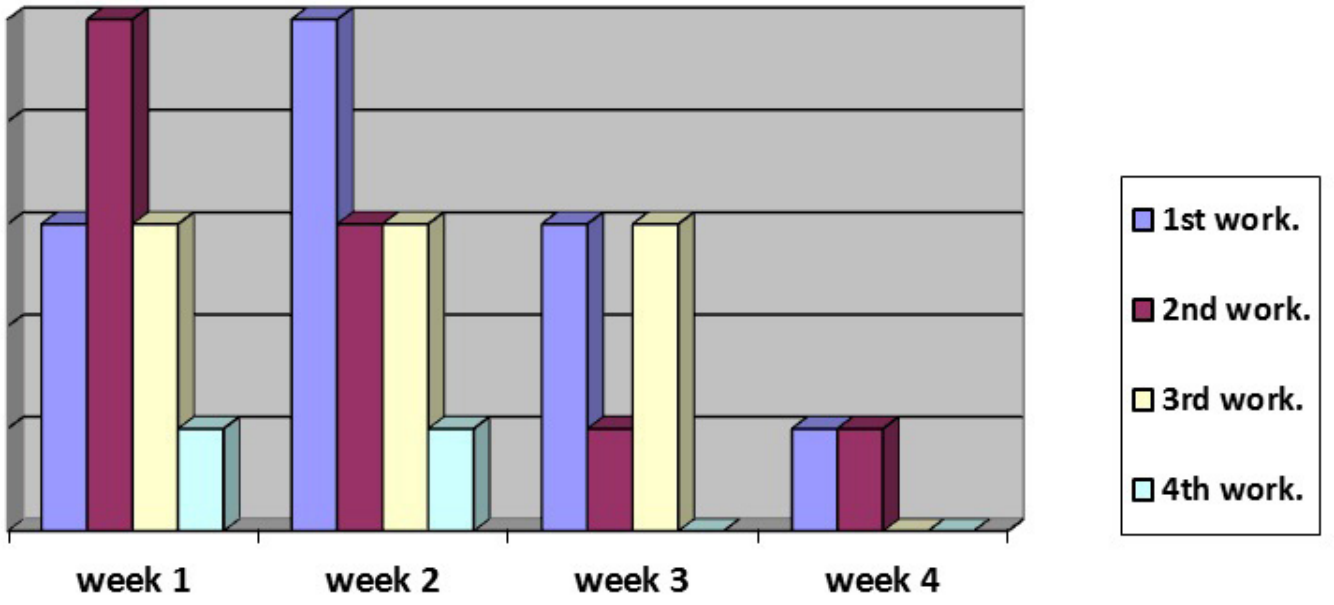


Fig. 6.7. Training load distribution within microcycles in a competition mesocycle, 4 workouts a week

Taking into account the body's resistance to adapt to constant training, the training volume should be distributed variably, i.e. unevenly in weekly cycles.

Depending on the number of training sessions with large volume within a microcycle, the recovery process of an athlete's body may either end in a few hours after the last training session or take up to several days to run their course. The following microcycle can be carried out based on the recovery from the previous microcycle load or the expressed fatigue (Hegedus, 1992).

However, beneficial effects will be achieved only in cases where a few micro-cycles (each of which exacerbates the fatigue caused by the previous one) are followed by a relatively light microcycle, which allows to restore the functionality of an athlete and to ensure the effective flow of adaptation processes. Ignoring this situation will inevitably lead to physical and nervous exhaustion (Ramm, Bude, 1986, Platonov, 1997).

Table 6.7

Training load distribution within microcycles
in a preparatory mesocycle, 5 workouts a week
(Boris Sheiko, 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Large	Medium	Large	Medium
2nd workout	Small	Small	Small	Small
3rd workout	Medium	Large	Large	Medium
4rd workout	Large	Medium	Large	Large
5rd workout	Small	Small	Small	Small

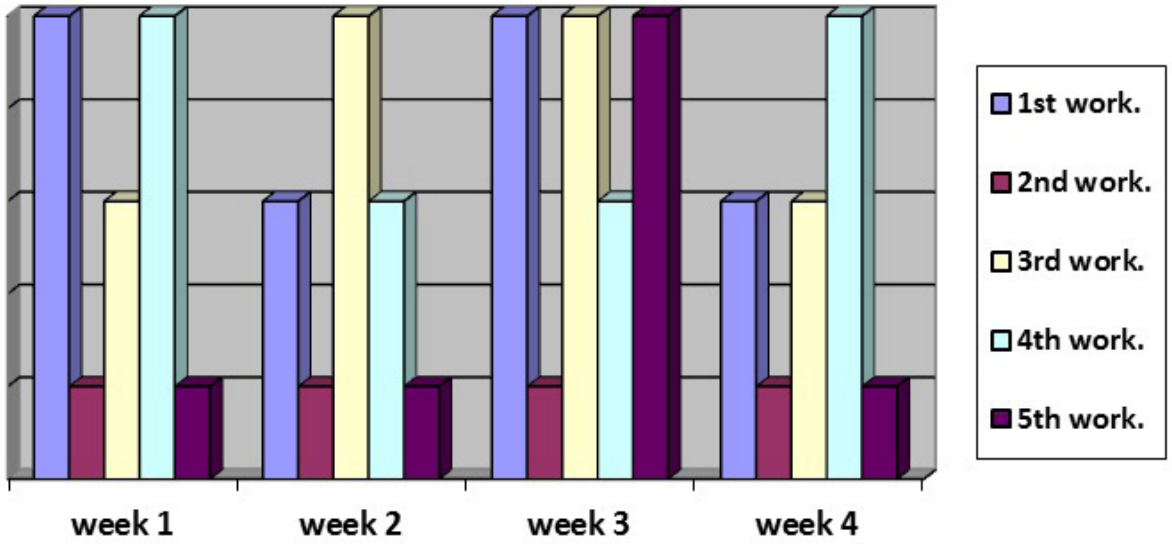


Fig.6.8. Training load distribution within microcycles in a preparatory mesocycle, 5 workouts a week

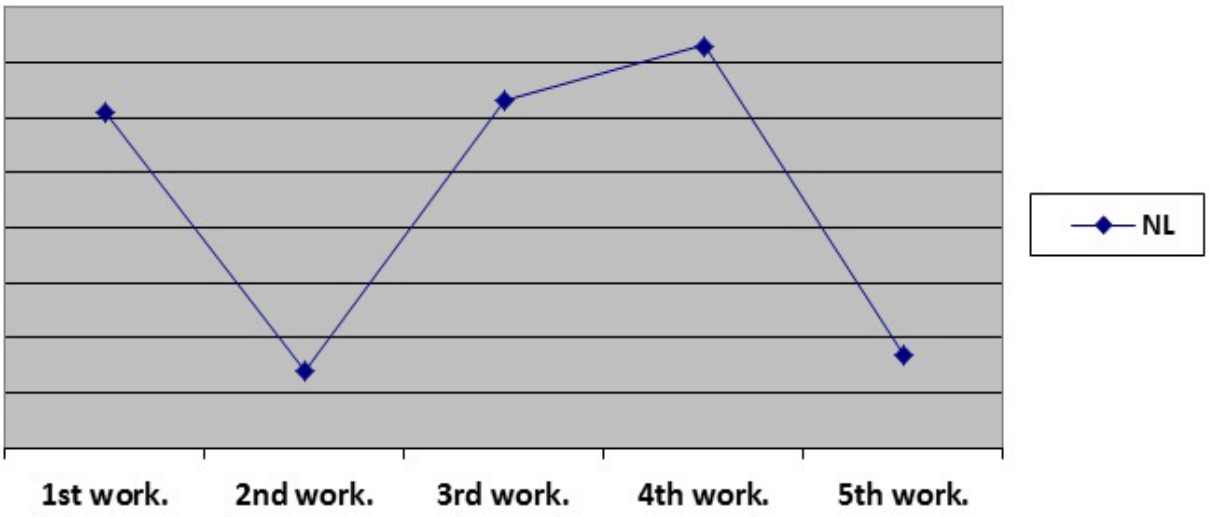


Fig.6.9. Training load distribution within microcycles in a preparatory mesocycle, 5 workouts a week

Table 6.8

Training load distribution within microcycles in a preparatory mesocycle, 5 workouts a week (Boris Sheiko, 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Large	Small	Large	Medium
2nd workout	Small	Large	Medium	Large
3rd workout	Large	Small	Large	Medium
4rd workout	Medium	Large	Small	Small
5rd workout	Small	Small	Medium	Medium

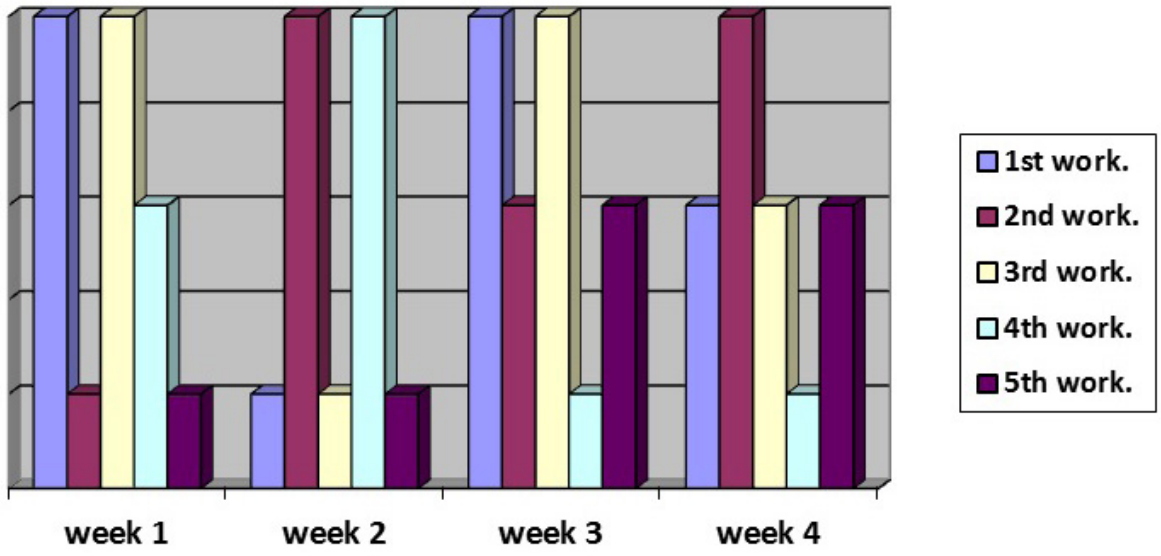


Fig.6.10. Training load distribution within microcycles in a preparatory mesocycle, 5 workouts a week

Table 6.9

Training load distribution within 4 weeks before a competition, 5 workouts a week (Boris Sheiko 2005)

workouts / weeks	week 1	week 2	week 3	week 4
1st workout	Large	Medium	Medium	Small
2nd workout	Small	Small	Rest	Rest
3rd workout	Medium	Large	Small	Small
4rd workout	Medium	Medium	Medium	Rest
5rd workout	Small	Small	Rest	Competition

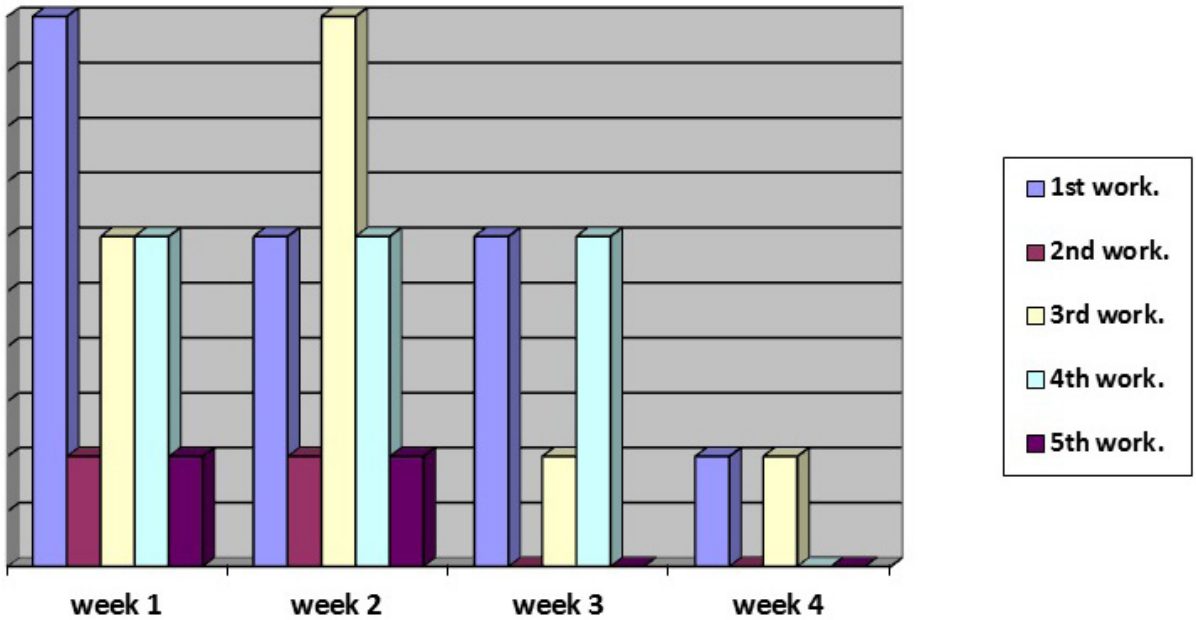


Fig. 6.11. Training load distribution within 4 weeks before a competition, 5 workouts a week

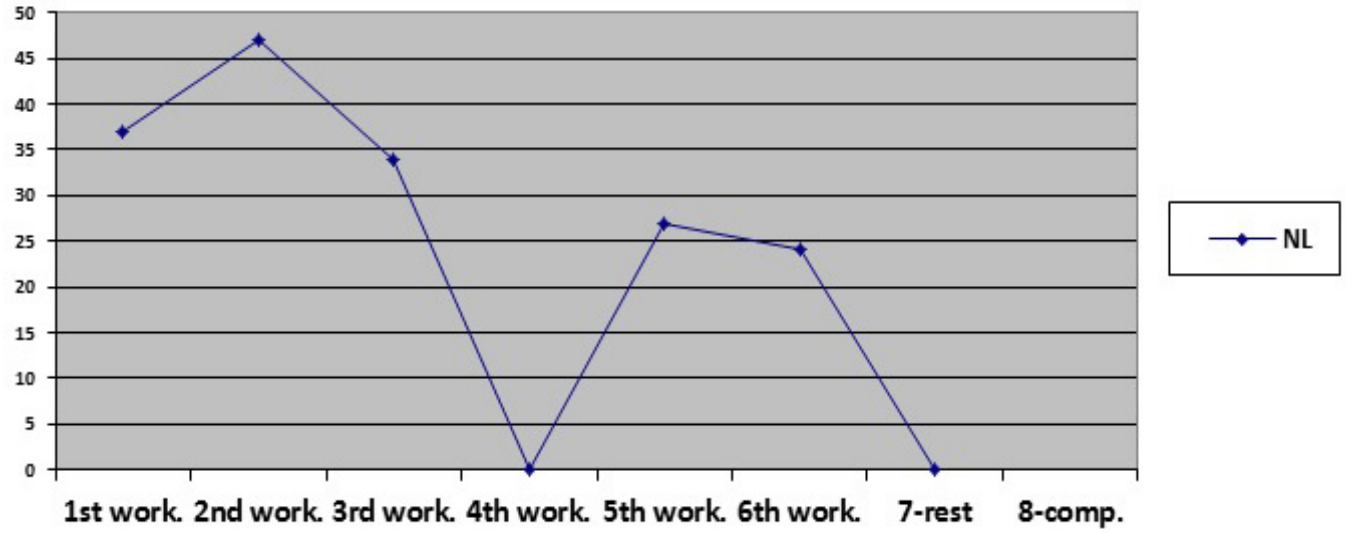


Fig. 6.12. Training load distribution within 2 weeks before a competition, 5 workouts a week

6.3. Mesocycle Training Structure

Mesocycle is a training cycle of intermediate length, lasting from two to six weeks and incudes complete microcycles in proper sequence. Building the training process based on mesocycles allows you to organize it in accordance with the main objective of a period or preparation stage, to ensure the optimum training and competitive loads, as well as a suitable combination of different means and methods of training. It also provides the opportunity to accord pedagogical influence (letting athletes learn proper sport movements and perfect them) and recovery activities, and to achieve continuity in teaching different qualities and abilities (Holodov J. 2007).

L. Matveev (1976) divides mesocycles into the following types: conditioning, basic, test and preparation, precompetition, competition and recovery.

A conditioning mesocycle is characterized by an increase in training volume with a gradual increase in intensity. This mesocycle type usually initiates the preparation period.

During the basic mesocycle the main training focus is aimed at functionality improvement, the development of basic physical abilities and the already developed technical and tactical approaches. It's done with high volumes.

The test and preparation mesocycle is a transition from basic to competition mesocycles. The actual training work is combined with participation in a series of events that mainly serve test purposes with their task being to prepare an athlete for competitions. This particular mesocycle type may consist of two or three training microcycle and one competition microcycle type. This kind of "mock competition" allows the programming for the competition mesocycle to be more precise and reflect the athlete's actual state of preparedness.

Precompetition mesocycles are typical during the preparation for one or a few major events. Mesocycle structure should resemble the entire requirements of upcoming competitions, and provide adaptation techniques to their specific conditions. The optimal conditions must be created so that an athlete can express all of his abilities in decisive competitions.

Competition mesocycles represent a typical training structure during a competition period. The number and structure of competition mesocycles are determined by the sports calendar, programs, competition requirements, athletes' qualifications and their conditions at the moment. Every single competition mesocycle consists of conditioning, competition, and recovery microcycles.

Recovery mesocycles are divided into preparation-and-recovery and recovery-and-maintenance cycles.

Training volume in a weekly cycle, as well as a single training session, often has four categories: small, medium, large, and maximum. A. Chernyak (1978) believes that the relative values of training within weekly cycles are best determined as the percentage of the monthly volume. If a weekly cycle training volume is less than 20%, it can be considered small, if it is 20 to 30% it is medium, over 30 - 40% is large, and above 40% it is maximum.

Training volume is also determined by the number of lifts of the bar (NL) (aka reps) (A. Roman., A. Falameyev, 1969). This parameter (NL) does not depend on the absolute strength of an athlete, and does not include an average bar weight (unlike the tonnage, which does), neither does it depend on the qualifications, age or height of the athlete. This parameter only reflects that the bar has been lifted a certain amount of times. Though not deeply informative, it is simple and convenient when monitoring training volume in weekly, monthly or annual cycles.

When planning training volume in a preparation period for master of sports and international master of sports class athletes, Boris Sheiko suggests doing certain exercises by varying training volume and relative intensity once in 10-14 days. For instance, for squat and benchpress routines with constant weight, Sheiko applies one of the following planning options:

For Squat and Bench press:
1 option: « Pyramid of increases»
50% 5reps x 1set, 60% 5reps x 1set, 70% 3+5+7+9+8+6+4 reps
(70% for 3 reps, rest, 70% for 5 reps, rest, etc)
NL = 52 lifts, Relative Intensity (RI) - 67,1%
2 option: «Ragged method»
50% 5reps x 1set, 60% 5reps x 1set, 70% 3+7+5+8+4+9+6 reps
NL = 52 lifts, RI - 67,1%

The 1st and the 2nd options have the same Number of Lifts and Relavtive Volume (67,1%).

The only difference in the Number of Lifts of 70% barbell: in the first option the NL is sequentially going up (3 + 7 + 9), then downwards (8 + 6 + 4). In the second option, the athlete performs a bench press with a weight of 70%, dramatically changing the number of lifts: 3 times, rest, 7 times, rest, 5 times, rest, etc.

«Ragged method» is easier to adapt to psychologically then the «Pyramid of increase». The athletes of intermediate level perform a bench press with a weight of 65%.

Let's analyze several options for planning of «pyramids», compiled by B. Sheiko, in which the athlete increases the intensity and NL, but when he reaches the top, he starts reducing the intensity and increases NL. (« McCloy Method» 1945).

For Squat and Bench press:

1option: «Mega pyramid» 50% 7reps x 1set, 60% 6reps x 1set, 70% 5reps x 1set, 75% 4reps x 1set, 80% 3reps x 1set, 85% 2reps x 2sets, 80% 3reps x 1set, 75% 4reps x 1set, 70% 5reps x 1set, 65% 6reps x 1set, 60% 7reps x 1set, 55% 8reps x 1set, 50% 9reps x 1set

NL =71; RI = 63,8%

2 option: 50% 6reps x 1set, 60% 5reps x 1set, 70% 4reps x 1set, 75% 3reps x 1set, 80% 2reps x 2sets, 85% 1rep x 2sets, 75% 4reps x 1set, 70% 5reps x 1set, 60% 6reps x 1set, 50% 7reps x 1set

NL = 50; RI = 66,5%

3 option: 50% 6reps x 1set, 60% 5reps x 1set, 70% 4reps x 1set, 80% 3reps x 2sets, 85% 2reps x 3set, 75% 4reps x 1set, 65% 5reps x 1set, 55% 6reps x 1set

NL = 42; RI = 68,6%

4 option: 50% 5reps x 1set, 60% 4reps x 1set, 70% 3reps x 1set, 80% 3reps x 2sets, 85% 2reps x 3sets, 80% 3reps x 2sets

NL = 30; RI = 72,3%

5 option: 50% 3reps x 1set, 60% 3reps x 1set, 70% 3reps x 1set, 80% 3reps x 1set, 85% 2reps x 2sets, 90% 1rep x 2sets, 85% 2reps x 2sets

NL = 22; RI = 74,5%.

The options are arranged in regards to the increase in relative intensity, in the first four options repetitions are planned up to 85% increase of the maximum result (the athlete's 1RM), with 62.8% increase of relative intensity in Option 1 up to 72% in Option 5. In Option 5 an athlete reaches up to 90% of maximum weight.

Option 1, Mega Pyramid, is very difficult. It is mainly aimed at the development of strength endurance. It is recommended to be performed only by well-prepared master of sports and international master of sports class athletes in a preparation period not more than once a month. Out of all athletes who have been training with Sheiko's plans, the biggest Mega Pyramid belongs to the multiple world champion, Aleksey Sivokon. In his case it consisted of 97 repetitions with maximum weight of 90%.

The shortest and the lowest of all the options when it comes to volume, but with the highest intensity is Option 5. It is recommended to be performed no more than once every two weeks. Options 2, 3 and 4 can be carried out each week, with changes in the number of repetitions with the obligatory «top» of the pyramid at 85%. «Pyramid» planning in a competitive mesocycle is not recommended.

The names of all the options are related to the maximum parameter during one of the weeks.

Extensive modification of exercises and conditions for their performance, dynamics of the loads and a variety of methods and their applications, updating of the form and content of the training sessions – all these are known as a variational method. The variability of training is one of the most important principles of training process structuring. Exercises, volume, and intensity are varied. Variability is the basis of a steady progress in strength sports.

The options of weekly training load distribution within a basic mesocycle (Table 5.11) show that relative training volume in weekly cycles vary from low, medium, large and maximum. The abovementioned options are not exclusive. During preparation months other options may be used.

Table 6.10

Options of the weekly load distribution in a preparatory mesocycle (1 month)

Options	Load								Number of Lifts per month
	% from monthly load				Number of Lifts				
	Weeks								
	1st	2nd	3rd	4th	1st	2nd	3rd	4th	
1	46	20	22	12	138	60	66	36	300
1-2	34	30	24	12	119	105	84	42	350
1-3	36	16	27	21	144	64	108	84	400
1-4	35	22	14	29	158	99	63	130	450
2	22	38	25	15	110	190	125	75	500
2-3	20	34	30	16	110	187	165	88	550
2-4	21	35	13	31	126	210	78	186	600
3	15	28	35	22	97	182	228	143	650
3-1	28	15	35	22	196	105	245	154	700
3-2	22	27	33	18	165	203	247	135	750
3-4	17	21	35	27	136	168	280	216	800
4	18	26	12	44	153	221	102	374	850
4-2	15	28	22	35	135	252	198	315	900
4-3	22	15	28	35	220	150	280	350	1000

Table 6.10 shows that relative training volume in weekly cycles range from low, medium, large and maximum. The abovementioned options are not exclusive. Other options exist, especially during preparation months.

A single number in the options stands for the number of the week with maximum training volume. If there are two digits, the first one represents the week with large training volume, whereas the second one stands for the week with the lowest training volume.

Analyzing the training diaries of highly skilled athletes (MS, MSIC, Distinguished Masters of Sport) A. Chernyak has been found that options 1, 2, 1-3 and 3-1 predominate. Last week before the competitions is a deload week.

Table 6.11

Options of the weekly load distribution in a competition mesocycle (1 month)

Options	Load								Number of Lifts per month
	% from monthly load				Number of Lifts				
	Weeks								
	1st	2nd	3rd	4th	1st	2nd	3rd	4th	
1	40	27	20	13	60	41	30	19	150
2	29	38	22	11	58	76	44	22	200
3-1	28	24	34	14	70	60	85	35	250
1-3	38	20	28	14	114	60	84	42	300

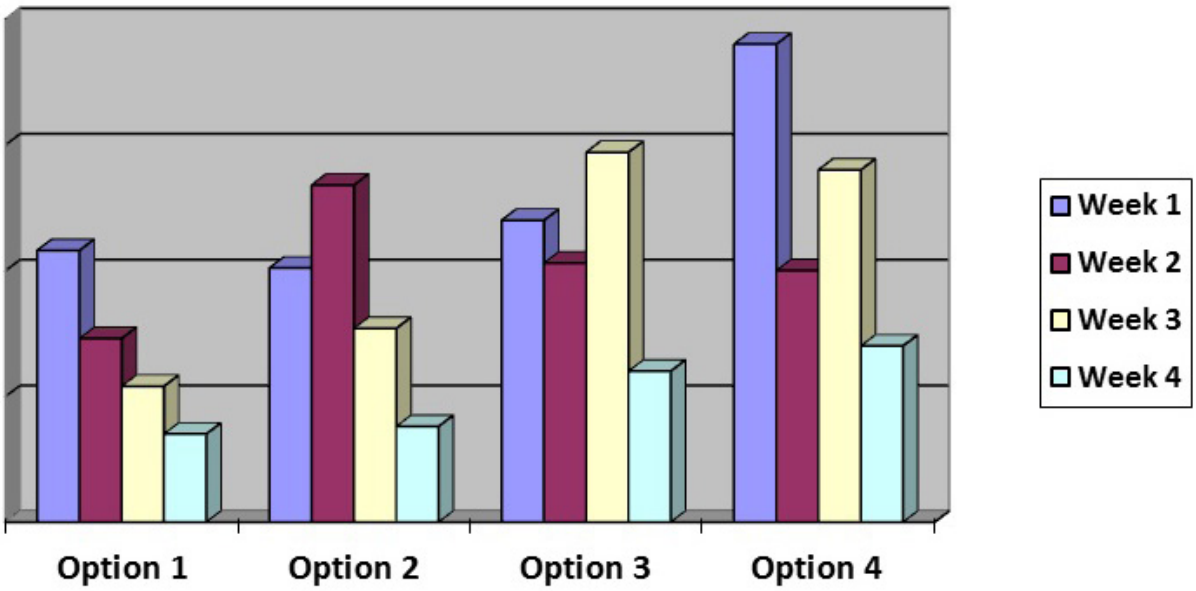


Fig. 6.13. Options of the weekly load distribution in a competition mesocycle (1 month)

The abovementioned schemes of NL distribution in weekly cycles within a month period can be implemented by powerlifters with varied qualifications and weight categories. The option with abrupt NLB changes from session to session is known as alternate training, and the option with a continuous increase or decrease in load volume within three or more trainings is called gradual.

The largest training volume is usually used during one or two weeks before competitions, it is quite rare during the third week, though, which means that few athletes will need a 3 week or longer taper into competition and for most, two weeks or less will be best.

The principle of variability is applicable to athletes of any level in any sport. This particular training structure can be also explained theoretically. Inorganic and organic nature is characterized by the so-called stair-step functions. They are constant in certain ranges, but change abruptly from one interval to the other. In living organisms, the conversion of chemical energy into mechanical, electromagnetic energy as well as heat energy occurs in steps.

Selecting the amplitude of variable weekly training volume also depends on the “training tolerance” (efficiency). An athlete with low efficiency avoids abrupt changes in NLB within weekly cycles, those who are able to withstand larger loads are not discouraged by abrupt changes. However, these athletes do not often implement these changes. Highly efficient athletes, in contrast to the ones with average efficiency, increase their NLB every week by increasing the number of training sessions. In the end, they accomplish a great amount of work within a monthly cycle.

6.4. Long cycles (macrocycles) training structure

The Macrocycle is a long training cycle type lasting six months (in some cases 3-4 months), one year, or many years (e.g. four-year cycle); it is associated with the development, stabilization and temporary loss of physical preparedness; it includes a series of periods,

stages, and mesocycles. An annual cycle is divided into three phases: preparation, competition and transition.

Preparation period is aimed at the establishment of physical preparedness, i.e. creation of a solid foundation (general and specialized), preparation for major competitions and participation, and improvement of various aspects of preparation. This period is characterized by an increase in strength, speed, flexibility, and agility during versatile general physical training. It is defined by the largest training volume and a gradual increase in the intensity of competition exercises. The increase in training volume should happen in waves, i.e. months with large volume should be alternated with months with lower volume. Two phases are distinguished in this period: general preparation and specific preparation.

General preparation period creates basis for physical preparedness. An athlete must achieve the following:

a) to develop general physical abilities (strength, speed, agility, flexibility) using a variety of physical exercises, including barbell routines;

b) to master specific detail of a particular sports technique, and to form new or restore special pre-existing skills;

c) to increase the strength of particular muscle groups (especially the least developed ones) by incorporating additional barbell exercises with a significant weight;

d) to increase the level of special endurance by gradually increasing NLB during training.

During general preparation period, young novices and lower sports class athletes learn and master the technique of classic exercises. High level athletes primarily concentrate on additional exercises, which contain a bigger part of training volume during a session. Competition exercises are used less and mainly with moderate loads. The main role of this stage is repetitive training sessions with constant load.

During competition period, physical preparedness is stabilized through improving various fitness parameters, integral training is ensured, and athletes are prepared for main competitions.

The duration of a competition period depends on the time during which certain physical preparedness can be maintained as well as on the competition calendar. The goal of this stage is to maintain the form and practice the acquired skills in sports. This period is characterized by lower training volume and increased intensity of competition exercises, which create the basis for the aim of the period – to achieve an increase in sports results.

The main goal of a competition period is to maintain and improve the already achieved physical preparedness, and to help an athlete fully use his skills during main competitions. This can be achieved by implementing competition and competition-like routines.

The transition period is aimed at recovery of physical and mental potential after intensive training and competition loads as well as at the preparation for the next macrocycle.

The transition period connects various training cycles. Its goal is to prepare an athlete for the next cycle with vitality and with such level of preparation that would allow for reaching the goals of the following cycles in the best possible way. In order to achieve this, the following points are necessary:

a) to provide active rest for the body;

b) to maintain such physical preparedness that would allow to start the next cycle on a higher level than the previous one;

c) to eliminate any flaws in technique without spending too much time and effort on the task.

In this way, transition period allows us to exclude the possibility of overtraining, and ensures a transition to a different training mode without intensive training load. It also help keeps the fitness level very high. This period lasts 5-7 days after the first training cycle, and 10-14 days after the second one.

During the transition period, in order to provide active rest, various exercises from different sport disciplines are used. However, exercises with a barbell are also used, i.e. specially prepared exercises with low weights. It can be explained by the fact that a break in barbell training lasting over two weeks has a negative impact on the training process. After a month period, even with active rest present, many athletes experience difficulties when coming back to their usual routines.

The transition period of training is characterized by low total training volume and insignificant loads. The amount of training, compared to the preparation period, is 3-4 times lower; the number of training sessions within a weekly cycle as a rule does not exceed 3-4 times; intensive exercises and routines with heavy weight are not planned in this period. The main content of the transition period are various means of active rest and general exercises. In the process of active rest as well as general exercises, it is recommended to use new means (types of training) that have not been widely used in previous workouts.

At the end of the transition period the load is gradually increased as well as the number of general and facilitating exercises, while the amount of active rest exercises is decreased. This particular strategy allows for a gentle transition from the transition period to the first stage of the preparation period within the next macrocycle.

An annual training sessions plan must be extremely detailed and accurate. It is based on a general annual plan and contains the same points, however, training volume is divided into monthly cycles. The plan should specify the training volume in general and specific physical training, its intensity and variation by months, the amount of various exercises, and the number of competitions. The plan also contains the possible results an athlete should achieve at every training stage throughout a year in regular training sessions as well as additional exercises.

When planning an annual training volume one should base it on the number and scale of competitions in which an athlete is going to take part. If an athlete plans to take part in four to six competitions, it is crucial to choose two of them where he is supposed to show the best results. Athletes cannot show equal results at all competitions taking place throughout the year, so the preparation should not be equally intensive. If an athlete is a clear leader, he can take part in various competitions, such as city, districts, or regional events. Athletes must be well prepared for the Russian Cup, when it comes to the load volume as well as the intensity of trainings. An athlete's results are planned ahead for every single competition regardless of its scale.

Let us look at the annual training plan of Alexey Sivokon, Distinguished Master of Sport of Kazakhstan, who took part in competitions in 1994 in the category of up to 67.5 kilos.

At the beginning of 1994 Alexey's goal was to take part in five competitions. In May at the Championship of Kazakhstan he was supposed to achieve 760-770 kilos in the three lifts and take the first place. At the Junior World Championship (June 26, Indonesia) he was supposed to achieve the same results, then in September at the championship in Asia, he was supposed to lower his result to 740-750 kilos and win with this result. His highest result was planned for November at the World Championships (South Africa) at 845 kilos, and also to win at the World Championship (bench press) in Finland.

In order to achieve everything planned for the year, it was necessary to make two peaks with large load. The first peak of the largest monthly volume in tons (520) and NL (3049) was scheduled for March, the second peak, i.e. 452 tons and 2549 NL in August.

The lowest NL (420) was scheduled for September (the championship in Asia) and November (World Championship) with 449 NL. The biggest average barbell weight in April was 181.1 kilos, and 185.5 in October. The training volume was decreased before important competitions and during transition period, which meant between 7 and 10 days of active rest.

Table 6.12

Alexey Sivokon's load in the first half of 1994

	January	February	March	April	May	June
NL	1602	1486	3049	1321	1017	848
Average weight lifted, kg	163,9	174,1	170,5	181,9	176,1	185,0
Relative volume	262612	258660	519885	240355	179109	156858

Table 6.13

Alexey Sivokon's load in the second half of 1994

	July	August	September	October	November	December	Year
NL	2069	2549	420	1149	449	1025	16984
Average weight lifted, kg	175,8	177,5	180,5	185,5	173,6	178,1	175,7
Relative volume	363685	452396	75834	213191	77968	182930	2983483

As a result, the volume of the scheduled training was quite high, and the average relative intensity per year was 69.2%. Besides, the volume fluctuated considerably in wide limits, unlike the average training weight. Thus, the lowest average barbell weight was 170.5 in March, and the biggest was 185.5, while the load volume fluctuated between 179 and 520 tons, i.e. it was three times bigger or smaller at any one point.

Sivokon's annual plan is presented in the graph below. Months are presented horizontally, and the load volume in NLB as well average training weight in kilos vertically.

Fig. 6.14. The dark line (at the bottom) shows the changes in training volume in NLB within months, the light line (at the top) shows the changes in intensity in kilos.

The graph shows how the training intensity goes up, leading the athlete to his best physical preparedness, and then slightly drops, allowing for recovery after important competitions. In order to increase it, NLB in competition exercise is increased with 90% or more weight. Consequently, if the volume needs to be lowered, NLB is decreased. However, it should be noted that decreasing the intensity for a longer period of time leads to a lower level of physical preparedness. It is not recommended to increase the intensity for a longer period of time (especially in competition exercises), as it may lead to the exhaustion of the central nervous system, and consequently, to lower results.

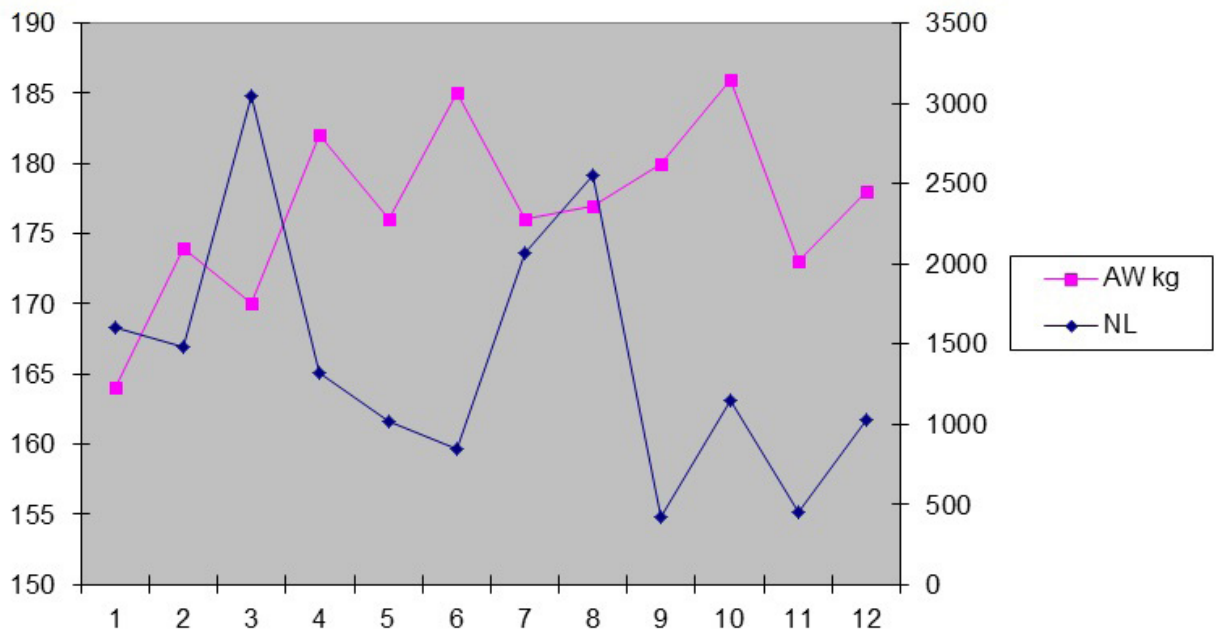


Fig. 6.14. Mesocycle training distribution options

It should be noted that an annual planning is an approximate training program. It is impossible to predict an athlete's exact condition at any given preparation stage, that is why an annual plan should and must be adjusted depending on an athlete's abilities.

6.5. Principles of long-term (multi-year) planning

The main feature of multi-year planning of preparing an athlete is continuous (year after year) increase in training intensity. Weightlifting research shows that training volume per year is not closely linked with sport results. Thus, some athletes show good results with lower volume, while others with large volume, but in every case, the intensity constantly increases. R. Roman suggests the following possibilities of multi-year planning at the stage of high sports mastery:

1. Training volume and intensity increase simultaneously.
2. Volume remains at the same level, but intensity increases.
3. Volume decreases, but intensity increases.
4. Volume remains at the same level, then decreases and increases again, intensity increases all the time.

The greatest increase in the intensity occurs in the first years of training.

Increasing the volume from year to year is as follows:

a) by increasing average NLB (per training);

b) by increasing the number of training sessions (while the average NLB per workout remains average);

c) by increasing the former and the latter parameters simultaneously.

Sports results and average training weight of qualified (advanced) and highly qualified (elite) athletes are highly correlated. The increase in intensity leads to a proportional increase in sports results. However, this is justified only when an optimum ratio between general and facilitating exercises is maintained during training, as well as a certain number of lifts with medium, large, submaximal and maximal weights is performed.

Consequently, the fact that the optimum average training weight has been established does not guarantee that the planned results will be achieved. An athlete and a coach should take into account and keep within limits all the important factors and training parameters, such as exercise selection, NLB of medium, large, submaximal and maximal weight, muscle activity mode, etc. Apart from the aforementioned points, nutrition, daily routine, rest, development of the recovery system after intensive training, as well as psychological state play an important role. If the optimum conditions for training and recovery are ensured, one can count on successful performance during competitions.

In multi-year planning, an athlete's physical preparedness should play an important role at every stage of their physical development, from novice to a high class athlete. However, the growth in mastery inevitably leads to narrow specialization, which will lead to a decrease in the proportion of general physical training in a training session. The emphasis is on the development of the sporting qualities that directly or indirectly influence the increase of the results or improve the recovery conditions and efficiency (the effect of «switching» – active rest).

In the first years of exercising the basic physical qualities, i.e. strength, speed and endurance, develop more quickly. The greater the duration of an athlete's training, the more his body gets used to its effects, and the less these physical qualities develop.

Rapid ability and competition result improvement is observed in general for an average of eight years, but the higher the weight category, the longer it lasts. If an athlete does not hold back the growth of his own weight, his results increase for a longer time, so you should plan ahead for the transition to the next weight category. In a weight class transition year, an athlete can add between 30 and 70 kilos in the three lifts. For instance, the world champion, Yuri Fedoryenko, in the category up to 90 kilos, at the Russian Cup in 2000 showed the result of 852.5 kilos. After that, he moved to 100 kg category at the World Cup in 2001 he accumulated 920 kg. The increase for the year was 67.5 kilos. At the World Championships in 2002, Fedoryenko performed in 110 kg category and showed a total of 985 kilos, adding 65 kg for the previous year. And in 2005, in the same category at the World Championships in Miami (USA), his result was 1,087.5 kilos.

In this way, after paying attention to the main peculiarities of multi-year planning, it is necessary to point out the components that are included in the long-term powerlifting training:

1. Goal (e.g. to gain master of sports or international master of sports class).
2. Duration of training stages (annual, semi-annual cycles), focus and goals of each stage.
3. Main competitions at every stage.
4. Rest intervals.
5. Results an athlete should show in three lifts at different stages.
6. Barbell programs volume and intensity.
7. General physical activity volume
8. Training means of specialized and general physical activity.
9. Mastery of theory and methods of training, self-control, and competition rules.
10. Pedagogical and medical control system.

6.6. Training volume and intensity planning

The subject of planning a powerlifting training cycle has been widely discussed in various articles, books and textbooks. S. Smolov (1990) and A. Surovetsky (2000) describe the most common issues regarding the combined method of strength and muscle mass development,

competitive training methods such as lifting. V. Novokreschyonov's works (1999) are dedicated to the choice of competitive exercise technique in powerlifting, as well as classification of barbell exercises used in training. P. Perov (2004) gives advice on the methodology of increasing training efficiency in bench press. In other articles general questions on training in athletic sports disciplines, for instance in powerlifting and bodybuilding, are discussed (S. Cherednichyenko, 1991).

Some works analyze typical misconceptions about strength training, including powerlifting (B. Hmelnitsky, 2004). L. Dvorkin and A. Slobodyan (2005) in their course book «Weightlifting», discuss issues in methodology of competitive powerlifting training. B. Shchetina (1997) addressed the issue of training load planning in powerlifting. Y. Yakubenko (2006) made a comparative analysis of powerlifting training load volume in training of males depending on the qualifications and body weight.

Through research in this field, many experts have concluded that the main factor determining the impact of a training session on the body of an athlete is its volume, so they plan it according to a certain bar weight in each exercise.

Currently, there are several methods of training volume analysis. One of them is based on the sum of lifted kilograms (N. Luchkin, 1940), another one on the number of lifts (NLB) (R. Roman, A. Falameev, 1969; A. Chernyak, 1978). Counting the volume based on NLB only reflects that the bar is lifted a certain amount of times. However, the NLB parameter does not include the average weight of the bar, unlike the tonnage method. «It does not depend on an athlete's qualifications, age, nor height. This parameter only reflects the fact that the bar has been lifted a certain number of times. It is simple and convenient to record loads in weekly and monthly cycles ... By using this parameter, it is easy to judge how many times a bar should be lifted in order to develop the necessary strength and master a technique...».

It has been suggested that training volume be assessed by the number of lifts (S. Filanovsky, A. Falameyev, 1968; A. Chernyak, 1970, R. Roman, 1970), and the intensity by the average bar weight (L. Matveyev, 1960). Some others have also proposed to determine the intensity of training volume by pulse rate (L. Akyamov, 1959). In contrast to the planning of rated athletes, where training volume is determined by the number of lifts per workout, week, month and year, in the group of more advanced athletes this is not enough. Coaches should know the total amount of kilograms lifted (tonnage) as well as the average weight of the bar in every competitive exercise and the total amount per workout, week, month and year. Therefore, the volume of the special powerlifting load, as well as in weightlifting, is considered to be the amount of kilograms (tons), lifted during one exercise or exercises per workout, week, month and year. The volume of training load in kilograms is determined by the weight lifted multiplying by the number of lifts. For example:

150 kilos 3 times X 5 = 2250 kilos

If you combine the training volume of every single exercise done at a training, you will get the total training volume.

6.7. Load distribution by zones of intensity

In powerlifting as well as in weightlifting it is important to know NLB and NLB of particular weight. Depending on the weight lifted, different motor abilities improve. Nowadays, three parameters are used to determine training intensity by specialists, i.e. average bar weight,

number of lifts at 90-100% in bench press and distribution of lifts in intensity areas. It should be noted that the difficulty of an exercise is determined not only by the amount of work done, but also its intensity. An exercise is considered more difficult if it is performed with a bigger weight.

In weightlifting, an intensity measure is thought to be an average bar weight. Training intensity (average lifted weight) is counted in kilograms and is determined by dividing the volume in kilos by the number of lifts (for instance: 2250 kilos: 15 lifts = 150 kilos).

In order to improve strength in people of different weight, a concept of relative strength has been introduced, i.e. the ratio of maximum strength and weight (S. Yermoayev, 1937; A. Kryestovnikov, 1951; V. Chudinov, 1961; V. Zatsiorskiy, 1966, etc.). Muscle strength depends on many factors. Under equal conditions, it is proportional to the cross section of the muscle (Weber's Law). The degree of muscle contraction/activation, under equal conditions, is proportional to the length of muscle fibers (Bernoulli's Law).

According to the data collected by A. Chernyak (1978), based on the average bar weight it is impossible to compare the training intensity in particular exercises done by athletes of different qualifications and weight categories. This weight characterizes only the general level of strength qualities development. The relative intensity indicator, which is determined by the ratio (in %) of the average weight to the best result in a particular exercise, seems to be more informative. It should be noted that the complexity of the exercise is determined not only by the amount of work done, but also by its intensity (stress). The exercise is more difficult, if executed with more weight. There are absolute intensity (in kilograms) and relative (percentage).

As absolute intensity is the average bar weight in kilos, when planning training, one should take into account NLB as well as number of lifts of certain weight. Depending on the weight lifted, different motor skills are developed.

Relative intensity (RI) means the ratio of the average bar weight (in %) to the best result in a given exercise. This relative parameter characterizes the true work of the body when exercising, regardless of an athlete's weight, qualification or strength. Relative intensity has its steady individual expression. This intensity criteria is knows as intensity coefficient (A. Medvedev).

Chernyakov's research has shown that an average relative intensity (ARI) can also be a criteria of training intensity, it may be applied to a single exercise as well as to the whole training program. In order to make the training parameters more universal, Chernyak A., Medvedev A., and Matveyev L. (1972) suggested that ARI should be taken into consideration. Carl G. (1967) proposed to divide all the weights used in the practice into zones of intensity (20% and 10% intervals). Roman R. (1968) considered it appropriate to divide the training bar weight into 5% interval. Chernyak A., Medvedev A. (1972), Roman R., Chernyak A. (1974) for the planning and evaluation of training recommended an interval of 10%, starting with 50% bar weight. For example:

I zone - over 50% - 60%;
II zone - 61% - 70%;
III zone - 71% - 80%;
IV zone –81% - 90%;
V zone - 91% - 100%

When distributing the training volume of highly qualified sportsmen into intensity areas, R. Roman (1970) discovered that a certain proportion of training (25%) was performed with a bar weighing 70% of maximum weight, and another one (30%) with 70-80%. The main

training (40%) was performed with a bar weighing 80% and an insignificant part (5%) with over 100%.

Based on the USSR national team training program, Vorobyov A. (1971) recommended that the plan in the first zone be 14% lifts, in the second - 15.5% in the third - 18.5%, in the fourth - 30.5%, in the fifth - 17.5%, and in the sixth - 4 %.

Regarding the number of lifts distributed by intensity zones, expert opinions are divided. According to the R. Roman (1970), A. Vorobyov (1972), and V. Zakorko (1977), an athlete should lift large weight (IV area) to 30-40%, submaximal and maximal weight (V area) - 17-22%, however, V. Agudina (1973), A. Kalinichenko (1974), Y. Yablonovsky (1975), think that the intensity in zone IV and V should be moderate: in the fourth zone - 17-26%, in the fifth - 5-12%. I. Kudyukov (1976) and V. Alekseev (1977) propose even fewer lifts in these intensity zones (14-16%, 3-4%, 0-3.7% and 0-1%). It is obvious that with the decrease of lifts with submaximal and maximal weight, the number of lifts with small and average weight increases.

Unlike in weightlifting, where during intensity analysis five intensity zones are taken into account, in powerlifting B. Sheiko (2005) proposes the use of seven intensity zones:

I zone - from 50% to 60%;
II zone - over 61% to 70%;
III zone - over 71% to 80%;
IV zone - over 81% to 85%;
V zone - over 86% to 90%;
VI zone - over 91% to 95%
VII zone - over 96% to 100% and above

Dividing the training into 7 intensity zones seems to be more informative than into 5 zones.

7. Powerlifting Programs

Long-term powerlifting training is a complex process of sports mastery development, every level of which is characterized by its own goals, tasks, means, and organization of the training. What is a plan? Why do you need the ability and knowledge on how to plan? A plan is a map or a step-by-step guide that leads you to your goal. If there is no plan, your goal is blurry, it is difficult to focus on it. If there is a goal and a precise plan on how to achieve it, we will gradually be able to get closer to it. Any plan, even an imperfect one, is better than an unplanned training dependent on one's condition. It helps an athlete to get to his highest sports form at the right time. However, it is impossible to prepare a single plan that will fit all athletes. Even if you make a plan for those who want to perfect their squat, another one for those who want to press more and the third one for the athletes who have troubles with the deadlift, in all of the above you will still need to adjust the plan according to your personal needs. The best plans are the ones tailored for a specific athlete when their coach knows all their strengths and weaknesses. Even then, while training, adjustments will be necessary.

You need to keep it in mind that the optimal number of workouts per week depends on the ability of the body to recover; no recommendations regarding sets and repetitions, no program can be ideal for all sportsmen. You need to accept the fact that during the training process, methods of strength development, first of all, depend on the amount and intensity of sports training and athletes' ability to withstand it. You need to be very careful when working on a plan for a champion, you cannot just copy them. Therefore, training process planning and the workouts themselves should be designed based on the individual abilities of every trainee. The latter can be established by means of questionnaires, conversation, tests and analyzing the training process. Do not forget that a plan is not a dogma and you are not a robot. You can change your exercise routine depending on the weaknesses of any particular muscles. It is crucial not to think about the percentage of the weight you plan to lift. You should take into account the percentage of the real results. Along with planning, another way to control the process is the analysis that defines the effectiveness of training. It is used to assess training means and methods, to identify the dynamics of an athlete's fitness level and prognosis of sports achievements as well as competition results.

I am often asked what is the difference between my universal programs and my individual programs.

When I write an individual program for a specific lifter I pay attention to his/her technique. If I find mistakes in the execution of the lifts I plan specific preparatory exercises to correct these mistakes. Furthermore, I look at which muscles of the lifter are weak. I plan specific exercises and appropriate volume to increase the strength of these muscles. And the most important thing is that each individual athlete has different recovery abilities, living conditions, training conditions, working conditions, etc.

Each scheduled week is analyzed twice. The first time - I analyze the written program by volume and intensity in accordance with the previous weeks, the second time - for the follow up I analyze what actually transpired during that training week. If the lifter didn't complete all of the week's program or vice versa - made it very easy, I take this into account when

planning the following week. I can make either upwards or downwards corrections to the volume or intensity for example.

When I write individual programs there is always very close contact with the lifter.

Every Saturday/Sunday I receive a detailed report of the week where I can read the comments about every lift and to watch videos.

At the end of each week a student and I discuss the number of competitions he/she will take part in. Usually we choose 3-5 meets, although it depends on the level of an athlete. Nobody can maintain peak condition throughout the year. That is why we choose 1-2 of the most important competitions among those already chosen. All training is then planned with regard to the fact that twice a year the lifter must reach his/her peak strength.

Also of great importance is the gym equipment available to the lifter. I have to know if there are chains, bands, boards, boxes/blocks, types of machines, etc. These will affect the development of strength in the competition movements.

All of the above I do not have when writing universal programs for everyone. This means that my universal programs should be taken as a starting point and adding corrections for yourself due to your own capabilities and circumstances. Even perfect programs need to be changed for every single lifter according to his/her ability to recover, level of experience, age, gender, etc. Otherwise the program will not give you any progress at all.

Here are some rules which will help you to understand how to work with my programs.

1. The most important rule is to use only real maxes not desirable ones.
2. If the weight is heavy for you on a particular workout then you can reduce it by 5-10% and leave number of reps and sets as planned.
3. If the weight is easy then you can increase the intensity by 5-7.5kg (no more). Do this no more than once every 2 weeks. Also, you can leave weights as planned but increase the number of reps in each set by 1-2, no more.
4. If you improved your maxes on the test days, then start using the new results from the next week. If the test was closer than 6 weeks before the meet, keep using your old maxes. You should recalculate your weights only after the competition.
5. 100% x 1rep x 1-2sets - means that you should lift 100% for 1 rep. If you succeed, then add 2.5-5% and lift the barbell for 1 rep. If 100% was lifted but with rules violation, then go for 100% again and try to show proper technique.
6. When you decide what competition you want to compete in, then you need to count the weeks from the competition day in reverse order and start doing my program from that week you stopped counting. If you land on a high intensity week then you can decrease weights by 5-10%. The next week should be done as planned.
7. I recommend to take one week of active rest after the meet or the final test (if you don't compete). Then you can repeat the same program. You can skip Day 1 after the meet and continue training from Day 2. Intensity shouldn't be higher than 65-70% and the number of reps should be 3-4 in each set.
8. If you compete on Saturday then you rest (don't have workouts) on Thursday and Friday. If you compete on Sunday then you rest on Thursday, Friday, Saturday.

9. Use medium weight for GPP (bodybuilding) exercises. For example if it is written 8 reps in one set then you should take the weight that allows you to have strength for 3-4 reps more after the set. You can choose any lift for GPP exercise. It is often written the following way: pecs, delts, abs, lats, triceps, etc. Try to change the exercise for one group of muscles each week.

10. If you are able to train 4 times a week then do it. Doing so will give you better results.

11. I recommend you have a body weight no more than 1.5-2 kg over the category you will to compete in.

12. When doing the bench press with bands or chains, I prescribe the weight of the barbell only. Band and chains are on top of prescribed weights. The chains' weight is about 4-10 kg on each side (depending on your personal best). Bands should be slightly stretched when the barbell is on the chest.

If you train equipped.

1. You have to use very tough (latest models) equip only if you are a national team member, in the other cases - there is no need. Tough shirts and suits will destroy your technique if you are not experienced enough.

2. You have to have two shirts: one shirt (less tight) for 75-90% and another (more tight) for 95%+ and for competitions as well. You will not lift 95-110% very often. So your second shirt will not stretch too much and it will be in good condition for the competition.

3. You have to change your competition shirt approximately every 2 years. If you feel that you have a poor advantage from your shirt - change it.

4. You have two options for working with gear during a preparation for the meet.

- You train raw using your raw 1RMs, and 2-3 months before the meet you start working with equip, using your equip 1RMs. If you have just one set of equip, then you go for 50, 60, and 70% without a suit but with knee wraps. For 80-85% - you put on the suit, with straps off. For 90% and over, wear the suit with straps on. Use the same scheme for the deadlift. If you have only one pair of knee wraps, then adjust the tightness of the wrapping according to the weight on the barbell. If you have two pairs of knee wraps, then use the first one (looser) for intensities of 70-85%, and the second pair (tighter) for 90% and above. The same idea is used for the bench press. If you have only one shirt then wear it from 75% and higher.

- If you have two or more sets of equip then this is good for you. You can train equipped all the time. Use loose equip on weights at 70-85%, and tight (competition) suit or shirt for 90% and over. The same for the knee wraps.

Exercises replacement

1. Bench press with bands can be replaced with bench press with chains using the same load, and vice versa.

2. Board press - If you don't have a board you can replace it with a bottle filled with water or buy a special bench block.

3. Deficit deadlift - If you don't have a block you can put two 25kg plates under your feet.

4. Deadlift off blocks - If you don't have blocks, you can use 25kg plates under the barbell to raise it. Use 2-3 plates on each side.

Exercises library.

Squat with 2sec pause on the halfway up	https://youtu.be/jG8sxQ6J7iM
2sec paused squat	https://youtu.be/oHhE2qKm904
Squat with 2sec pause on the halfway down	https://youtu.be/B0b_tOBhfMg
Squat with chains	https://youtu.be/LRA8JqyXzWw
2sec paused bench press	https://youtu.be/WNo198jz3QQ
3sec paused bench press	https://youtu.be/wWCi4ZXj8h0
Bench press with bands	https://youtu.be/hCyBZHiE08Y
Bench press with chains	https://youtu.be/gaAZQtUDyJM
Board press	https://youtu.be/9_8fblBbZm0
Bench press with a slingshot	https://youtu.be/kkmPW1Fx41Y
Decline bench press	https://youtu.be/kIPFiWH0stw
Speed bench press	https://youtu.be/o6nLSuEF35o
Any deadlift exercise can be done either sumo or conventional	
Deadlift with bands	https://youtu.be/HyYeiE6gPBs
Deadlift with chains	https://youtu.be/fxkoix1xXCw
Deadlift off blocks, bar at 10-15cm below the knees	https://youtu.be/SXZj8zEXuRs
Deadlift off blocks with chains	https://youtu.be/KtLHLyu-Khg
Deficit deadlift (sumo)	https://youtu.be/rfprlfQCoik
Deficit deadlift (conventional)	https://youtu.be/wVg-MWtNxuo
Deadlift to the knees with 1sec pause	https://youtu.be/RO31qV_fc50
Deadlift to the knees + full deadlift	https://youtu.be/R_SlaZ8OY00
Deadlift with snatch grip, slow descent	https://youtu.be/I69DsTxPKs8
Deadlift above knees off blocks	https://youtu.be/eEKcyuF2xAE
Deadlift pause below & above knees	https://youtu.be/Tv31HHL1juc
Deadlift + deadlift from below the knees	https://youtu.be/NN8IgQwQrn0
Deadlift to the knees + deadlift from below the knees	https://youtu.be/rknYOthrBwE
Deadlift to the knees + full deadlift	https://youtu.be/R_SlaZ8OY00
Inclined shoulder press seated	https://youtu.be/ryFlnVqZV8U
Goodmorning seated	https://youtu.be/axWEJc7p1qM

In the following text, you can find five training programs. Four of them are written specifically for this book. Three of these programs are designed for athletes who have one weak competition movement and want to improve it, while continuing to train and maintain the other two lifts.

You can also find a program for benchers only. Finally, I decided to share the most popular program among powerlifters in recent years, which was published in my apps for iOS and Android. It is the medium volume program for advanced athletes. All five programs presented herein are 4 workouts per week. Enjoy, and be strong and patient so you can cross the finish line!

7.1. Three lifts: squat emphasis

Week 1			
Day 1	%	reps	sets
1.Squat	50	5	1
	60	4	1
	70	3	1
	75	3	4
2.Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3.Pecs		8	5
4.Medial delts		8	5
5.Goodmorning		5	5
Total: 44 reps			
Day 2	%	reps	sets
1.2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3.Deadlift off blocks, bar at 10-15cm below the knees	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	3
4.Leg curls 8reps x 5sets			
5.Abs 10reps x 3sets			
Total: 37 reps			
Day 3	%	reps	sets
1.Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	5
2.Bench press with bands	50	4	1
	60	4	1
	65	4	5
3.Pecs 8reps x 5sets			
4.Triceps pushdowns 8reps x 5sets			
5.Hyperextensions 8reps x 4sets			
Total: 49 reps			
Day 4	%	reps	sets
1.Inclined shoulder press seated		3	5
2.Deadlift with bands	50	3	1
	60	3	1
	70	3	4
3.Dumbbell bench press		6	5
4.Leg extensions		8	5
5.Reverse hyperextensions 8reps x 4sets		8	4
Total: 18 reps			
TOTAL A WEEK:148 reps			

Week 2			
Day 1	%	reps	sets
1.Squat	50	5	1
	60	4	1
	70	3	1
	80	2	5
2.Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
3.Pecs		8	5
4.Front delts		6	5
5.Hyperextensions		8	4
Total: 46 reps			
Day 2	%	reps	sets
1.Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	3
2.Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+3	1
	75	1+2	4
3.Lats		6	5
4.Triceps standing		6	5
5.Leg press		5	5
Total: 40 reps			
Day 3	%	reps	sets
1.Squat with chains	50	4	1
	60	4	1
	70	4	4
2.3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3.Pecs 8reps x 5sets		8	5
4.Medial delts 6reps x 5sets		6	5
5.Abs 10reps x 3sets		10	3
Total: 41 reps			
Day 4	%	reps	sets
1.Bench press with bands	50	4	1
	60	4	1
	65	4	4
2.Deadlift off blocks, bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	3	4
3.Lats 8reps x 5sets		6	5
4.Leg extensions 8reps x 5sets		8	5
5.Reverse hyperextensions 10reps x 4sets Total: 45 reps		8	4
TOTAL A WEEK:172 reps			

Week 3			
Day 1	%	reps	sets
1.Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2.Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3.Squat	55	3	1
	65	3	1
	75	3	4
4.Triceps lying 8reps x 4sets		8	5
5.Goodmorning 5reps x 4sets		8	5
Total: 55 reps		5	5
Day 2	%	reps	sets
1.3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2.Deadlift to the knees + full deadlift	50	3+1	1
	60	3+1	1
	70	3+1	1
	75	2+1	4
3.Lats		8	5
4.Medial delts		6	5
5.Leg press		5	5
Total: 41 reps			
Day 3	%	reps	sets
1.Squat	50	5	1
	60	5	1
	70	5	4
2.Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	2
	80	2	2
3.Pecs		8	5
4.Biceps		8	5
5.Hyperextensions		8	4
Total: 52 reps			
Day 4	%	reps	sets
1.Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	3
2.Inclined shoulder press seated		4	5
3.Rear delts		8	5
4.Leg curls		8	4
5.Abs		10	3
Total: 13 reps			
TOTAL A WEEK:161 reps			

Week 4			
Day 1	%	reps	sets
1.Squat with chains	50	4	1
	60	4	1
	70	4	5
2.Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
3.Pecs		8	5
5.Triceps pushdowns		8	5
6.Hyperextensions		8	4
Total: 52 reps			
Day 2	%	reps	sets
1.3,2,1 sec paused bench press	50	3	1
	60	3	1
	70	3	5
2.Deadlift with bands	50	4	1
	60	4	1
	70	4	4
3.Front delts		6	5
4.Lats		8	5
5.Abs		10	3
Total: 45 reps			
Day 3	%	reps	sets
1.Squat	50	5	1
	60	5	1
	70	4+7+3+8+2+6	6
			pyramid of 6 sets with the rest after each
2.Board press	55	3	1
	65	3	1
	75	3	1
	85	3	2
	90	2	3
3.Pecs		8	5
4.Triceps lying		6	5
5.Goodmorning seated		5	5
Total: 61 reps			
Day 4	%	reps	sets
1.Deadlift off blocks, bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
2.Board press	55	3	1
	65	3	1
	75	3	1
	85	3	3
3.Leg extensions		8	5
4.Reverse hyperextensions		10	4
Total: 34 reps			
TOTAL A WEEK:192 reps			

Week 5			
Day 1	%	reps	sets
1.Squat	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	2
	80	3	2
2.Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	2	2
	95	1	2
3.Pecs		8	5
4.Triceps standing		6	5
5.Goodmorning		5	5
Total: 47 reps			
Day 2	%	reps	sets
1.Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	3
2.Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	1
	85	2	4
3.Lats		8	5
4.Leg press		4	5
5.Abs		10	3
Total: 36 reps			
Day 3	%	reps	sets
1.Squat with chains	50	5	1
	60	5	1
	70	5	4
2.3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3.Pecs		8	5
4.Triceps pushdowns		8	5
5.Leg extensions		8	5
Total: 47 reps			
Day 4	%	reps	sets
1.Board press	55	3	1
	65	3	1
	75	3	1
	85	3	4
2.Deadlift off blocks	60	4	1
	70	4	1
	80	4	4
3.Dumbbell bench press 6reps x 5sets		6	5
4.Leg press 5reps x 5sets		8	5
5.Reverse hyperextensions 8reps x 4sets			
Total: 45 reps		8	4
TOTAL A WEEK:175 reps			

Week 6			
Day 1	%	reps	sets
1.Squat	50	5	1
	60	4	1
	70	3	1
	80	3	5
2.Bench press with chains	50	5	1
	60	5	1
	70	5	4
3.Pecs		8	5
4.Front delts		8	5
5.Good morning		5	5
Total: 57 reps			
Day 2	%	reps	sets
1.Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
2.Deadlift with bands	50	4	1
	60	4	1
	70	4	4
3.Medial delts		6	5
4.Triceps standing		6	5
5.Leg press		5	5
Total: 44 reps			
Day 3	%	reps	sets
1.Squat	50	5	1
	60	5	1
	70	3+5+7+6+5+4	
2.3 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3.Pecs		8	5
4.Abs		10	4
Total: 57 reps			
Day 4	%	reps	sets
1.Bench press	50	4	1
	60	4	1
	70	4	4
2.Deadlift with 1 sec pause 5-10 cm below the knees	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Lats		8	5
4.Leg curls		8	5
5.Reverse hyperextensions		8	4
Total: 41 reps			
TOTAL A WEEK: 199			

Week 7			
Day 1	%	reps	sets
1. 2 sec paused squat	50	3	1
	60	3	1
	65	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	5
3. Pecs		8	5
4. Triceps pushdowns		8	5
5. Hyperextensions		8	4
Total: 40 reps			
Day 2	%	reps	sets
1. 2 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
2. Deadlift to the knees + full deadlift	50	3+1	1
	60	3+1	1
	70	2+1	1
	75	2+1	4
3. Lats		8	5
4. Front delts		6	5
5. Leg		5	5
Total: 44 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	5
2. Bench press with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	4
Total: 39 reps			
Day 4	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	70	2	4
2. Bench press with chains	50	3	1
	60	3	1
	70	3	2
	75	2	3
3. Dumbbell bench press		6	5
4. Medial delts		6	5
5. Abs		10	3
Total: 32 reps			
TOTAL A WEEK: 155 reps			

Week 8			
Day 1	%	reps	sets
1. Squat	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	1
	85	2	4
3. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		8	5
5. Hyperextesions		8	5
Total: 60 reps			
Day 2	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	2
2. Deadlift + deadlift below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+2	4
3. Lats		8	5
4. Leg curls		8	5
5. Abs		10	3
Total: 36 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	5	1
	70	4+7+3+8+2+5	
2. 3 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Triceps lying		6	4
5. Good morning		5	4
Total: 56 reps			
Day 4	%	reps	sets
1. Deadlift off blocks	60	3	1
	70	3	1
	80	3	2
	85	2	3
2. Bench press with bands	50	4	1
	60	4	1
	70	4	4
3. Front delts		6	4
4. Reverse hyperextensions		6	4
Total: 42 reps			
TOTAL A WEEK:194 reps			

Week 9			
Day 1	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	3
3. Pecs		8	5
4. Triceps standing		6	5
5. Goodmorning seated		5	4
Total: 36 reps			
Day 2	%	reps	sets
1. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	2
	95	1	2
2. Deadlift with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Medial delts		6	5
4. Leg press		5	5
5. Abs		10	3
Total: 34 reps			
Day 3	%	reps	sets
1. Squat	50	4	1
	60	4	1
	70	3	1
	75	3	4
2. Bench press with bands	50	4	1
	60	4	1
	70	4	4
3. Pecs		8	5
4. Triceps lying		6	5
5. Hyperextensions		8	4
Total: 48 reps			
Day 4	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+2	1
	80	1+1	4
2. Inclined shoulder press seated		3	5
3. Front delts		6	5
4. Reverse hyperextensions		8	4
Total: 19 reps			
TOTAL A WEEK:137 reps			

Week 10			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	75	2	3
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	2
	80	2	2
3. Pecs		6	5
4. Triceps standing		6	5
5. Goodmorning		4	4
Total: 40 reps			
Day 2	%	reps	sets
1. Deadlift with 1sec pause 7-10cm below the knees	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Board press	60	3	1
	70	3	1
	80	3	2
	90	2	3
3. Lats		6	5
4. Abs		10	3
Total: 31 reps			
Day 3 TEST	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	1
	85	1	1
	90	1	1
	95	1	1
	100	1	1
	103-105	1	1
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		6	5
4. Front delts		6	5
5. Hyperextensions		8	4
Total: 39 reps			
Day 4	%	reps	sets
1. Deadlift off blocks, bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	2
	90	2	3
2. Bench press	50	5	1
	60	5	1
	70	4	4
3. Dumbbell bench press		6	5
4. Leg press		5	5
Total: 44 reps			
Total a week:154 reps			

Week 11			
Day 1	%	reps	sets
1. 2sec paused squat	50	3	1
	60	3	1
	65	3	1
	70	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	3
3. Pecs		8	5
4. Triceps standing		6	5
5. Goodmorning seated		5	4
Total: 40 reps			
Day 2	%	reps	sets
1. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	1
	90	1	2
	95	1	2
2. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+3	1
	75	1+2	4
3. Medial delts		8	5
4. Leg press		5	5
5. Abs		10	3
Total: 39 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
2. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Triceps lying		6	5
5. Hyperextensions		8	4
Total: 41 reps			
Day 4	%	reps	sets
1. Bench press	50	4	1
	60	4	1
	70	4	1
	75	4	4
2. Deadlift with chains	50	3	1
	60	3	1
	70	3	4
3. Leg extensions		8	5
4. Reverse hyperextensions		10	4
Total: 46 reps			
TOTAL A WEEK:166 reps			

Week 12			
Day 1	%	reps	sets
1. Squat	55	3	1
	65	3	1
	75	3	1
	85	2	3
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	3
4. Pecs		6	4
5. Hyperextensions		8	4
Total: 47 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	3
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Lats		6	4
4. Abs		8	3
Total: 31 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	4
4. Goodmorning		4	4
Total: 34 reps			
Day 4	%	reps	sets
1. Deadlift off blocks	60	3	1
	70	3	1
	75	2	2
	80	1	2
2. Inclined shoulder press seated		2	5
3. Lats		6	4
4. Abs		8	3
Total: 12 reps			
Total a week:124 reps			

Week 13			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Pecs		6	4
4. Triceps		6	4
5. Goodmorning seated		4	4
Total: 30 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Lats		6	4
3. Abs		10	3
Total: 34 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	75	2	2
	80	1	2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	2
	80	1	3
3. Pecs		8	4
4. Abs		8	3
Total: 31 reps			
Day 4	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	2	4
	75	1	3
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 13 reps			
TOTAL A WEEK:108 reps			

Week 14			
Day 1	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	75	2	3
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	3
3. Pecs		6	4
4. Abs		10	3
Total: 30 reps			
Day 2	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	2
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	1
	85	1	1
	90	1	2
	95	1	1
	100	1	1-2
3. Lats		8	4
4. Hyperextensions		5	4
*- If an athlete lifted 100%, then he should go for 103% or 105%.			
Total: 33 reps			
Day 3	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	4
4. Abs		8	3
Total: 31 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		3	5
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 00 reps			
TOTAL A WEEK: 94 reps			

Week 15			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	3
4. Goodmorning		4	4
Total: 34 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	2
	80	1	3
2. Deadlift	50	3	1
	60	3	1
	70	2	1
	75	1	3
3. Lats		6	4
4. Abs		8	3
Total: 27 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Bench press	50	3	1
	60	3	1
	70	2	1
	75	1	4
3. Hyperextensions		6	3
Total: 25 reps			
Day 4	%	reps	sets
REST			
Total: 0 reps			
Total a week: 86 reps			

Week 16			
Day 1	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Deadlift	50	3	1
	60	2	2
	70	1	3
3. Abs		8	3
Total: 23 reps			
Day 2	%	reps	sets
1. Squat	50	3	1
	60	2	2
	70	1	2
2. Bench press	50	3	1
	60	2	2
	70	1	3
Total: 19 reps			
Total a week: 42 reps			
Thursday-Friday			
REST			
COMPETITION !!!			

7.2. Three lifts: bench press emphasis

Week 1			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	75	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		8	5
4. Front delts		6	5
5. Triceps standing		6	5
Total: 44 reps			
Day 2	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	4
4. Lats		8	5
5. Abs		8	4
Total: 51 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press	50	6	1
	60	5	1
	70	4	1
	75	3	1
	80	2	2
	85	1	2
	75	3	1
	65	5	1
	55	7	1
3. Pecs		8	5
4. Triceps pushdowns		8	5
Hyperextensions		8	4
Total: 59 reps			
Day 4	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	3	4
2. Inclined shoulder press seated		4	5
3. Dumbbell bench press		6	5
4. Reverse hyperextensions		8	4
Total: 18 reps			
TOTAL A WEEK:172 reps			

Week 2			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Bench press with bands	50	5	1
	60	5	1
	65	5	4
4. Pecs		8	5
5. Hyperextensions		8	4
Total: 70 reps			
Day 2	%	reps	sets
1. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	3
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Medial delts		6	5
4. Triceps standing		6	5
5. Lats		8	5
Total: 33 reps			
Day 3	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
2. Squat	50	5	1
	60	5	1
	70	4	4
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
4. Pecs		8	5
5. Abs		10	3
Total: 59 reps			
Day 4	%	reps	sets
1. Decline bench press	50	4	1
	60	4	1
	70	4	4
2. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+2	1
	75	1+2	4
3. Dumbbell bench press		6	5
4. Rear delts		8	5
5. Reverse hyperextensions		8	5
Total: 47 reps			
TOTAL A WEEK:209 reps			

Week 3			
Day 1	%	reps	sets
1. 3,2,1 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
2. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	4
3. Board press	60	3	1
	70	3	1
	80	3	1
	90	2	3
4. Pecs		8	5
5. Triceps lying		8	5
Total: 54 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	5
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Lats		8	5
4. Front delts		6	5
5. Leg curls		8	5
Total: 39 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	5	1
	70	4	4
2. Bench press	50	6	1
	60	5	1
	70	4	1
	75	3	1
	80	2	2
	85	1	2
	75	3	1
	65	5	1
	55	7	1
3. Pecs		8	5
4. Biceps		8	5
5. Goodmorning		4	5
Total: 65 reps			
Day 4	%	reps	sets
1. Deadlift off blocks, bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	3	4
2. Speed bench press	40	4	1
	50	4	4
3. Dumbbell bench press		6	5
4. Leg extensions		8	5
5. Abs		8	4
Total: 41 reps			
TOTAL A WEEK:199 reps			

Week 4			
Day 1	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. 2sec paused squat	50	3	1
	60	3	1
	65	3	4
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		8	5
5. Hyperextensions		8	4
Total: 58 reps			
Day 2	%	reps	sets
1. 4,3,2 sec paused bench press	50	3	1
	60	3	1
	70	3	5
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Front delts		6	5
4. Lats		8	5
5. Abs		10	3
Total: 38 reps			
Day 3	%	reps	sets
1. Board press	60	3	1
	70	3	1
	80	3	1
	85	3	2
	90	2	3
2. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Bench press with chains	50	4	1
	60	4	1
	70	4	4
4. Pecs		6	5
5. Goodmorning seated		5	5
Total: 65 reps			
Day 4	%	reps	sets
1. Deadlift off blocks bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	2	1
	90	1	3
2. Decline bench press	50	5	1
	60	5	1
	70	5	4
3. Inclined shoulder press seated		4	5
4. Biceps		8	5
Total: 46 reps			
TOTAL A WEEK:207 reps			

Week 5			
Day 1	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	1
	90	1	2
	95	1	2
2. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	4
3. Bench press with bands	50	4	1
	60	4	1
	65	4	4
4. Pecs		6	5
5. Hyperextensions		8	4
Total: 57 reps			
Day 2	%	reps	sets
1. Bench press	50	6	1
	60	5	1
	70	4	1
	75	3	1
	80	2	2
	85	1	2
	75	3	1
	65	5	1
	55	7	1
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	3
3. Rear delts		8	5
4. Lats		6	5
5. Abs		10	3
Total: 55 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	3
2. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	5
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	4
Total: 38 reps			
Day 4	%	reps	sets
1. Deadlift off blocks with chains	55	3	1
	65	3	1
	75	3	1
	85	2	4
2. Inclined shoulder press seated		4	5
3. Dumbbell bench press		6	5
4. Medial delts		6	5
5. Reverse hyperextensions		8	4
Total: 17 reps			
TOTAL A WEEK:167 reps			

Week 6			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
3. Bench press with chains	50	4	1
	60	4	1
	70	4	4
4. Pecs		8	5
5. Goodmorning		5	4
Total: 68 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	3	4
3. Medial delts		6	5
4. Triceps standing		6	5
5. Leg curls		8	5
Total: 45 reps			
Day 3	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
2. Squat	50	5	1
	60	5	1
	70	5	4
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
4. Pecs		8	5
5. Abs		10	3
Total: 67 reps			
Day 4	%	reps	sets
1. Bench press	50	5	1
	60	5	1
	70	4	5
2. Deadlift off blocks, bar at 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	3	4
3. Lats		8	5
4. Front delts		6	5
5. Reverse hyperextensions		8	4
Total: 51 reps			
TOTAL A WEEK:231 reps			

Week 7			
Day 1	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	65	3	1
	70	3	1
	75	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	5
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	4
Total: 47 reps			
Day 2	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+1	4
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Lats		8	5
5. Front delts		6	5
Total: 56 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. Bench press with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Biceps		8	5
5. Hyperextensions		8	5
Total: 41 reps			
Day 4	%	reps	sets
1. Deadlift with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Speed bench press	40	4	1
	50	4	4
3. Dumbbell bench press		6	5
4. Medial delts		6	5
5. Abs		10	3
Total: 37 reps			
TOTAL A WEEK:181 reps			

Week 8			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	4
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	5
Total: 52 reps			
Day 2	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	4
2. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+3	1
	75	1+2	4
3. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	2
4. Lats		8	5
5. Abs		10	3
Total: 57 reps			
Day 3	%	reps	sets
1. Squat with chains	50	4	1
	60	4	1
	70	4	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		6	5
4. Triceps lying		6	4
5. Goodmorning		5	4
Total: 44 reps			
Day 4	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
2. Inclined shoulder press seated (with light weight)		3	4
3. Front delts (with light weight)		6	4
4. Reverse hyperextensions		6	4
Total: 15 reps			
TOTAL A WEEK:168 reps			

Week 9			
Day 1	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		6	5
5. Goodmorning		5	4
Total: 57 reps			
Day 2	%	reps	sets
1. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	2
2. Deadlift to the knees + full deadlift	50	3+1	1
	60	3+1	1
	70	2+1	1
	75	2+1	4
3. Medial delt		6	5
4. Lats		8	5
5. Abs		10	3
Total: 38 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	75	2	3
3. Pecs		6	5
4. Triceps lying		6	5
5. Hyperextensions		8	4
Total: 38 reps			
Day 4	%	reps	sets
1. Deadlift off blocks	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
2. Inclined shoulder press seated (with light weight)		3	4
3. Rear delts		6	4
4. Reverse hyperextensions		8	4
Total: 16 reps			
TOTAL A WEEK:149 reps			

Week 10			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press	50	4	1
	60	3	1
	70	2	2
	75	1	3
3. Pecs with light weight		6	4
4. Medial delts with light weight		6	4
5. Hyperextensions with no weight		6	4
Total: 34 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	1
	85	1	1
	90	1	1
	95	1	1
	100	1	1
	103	1	1
2. Deadlift to the knees	50	3	1
	60	3	1
	70	2	2
	80	1	3
3. Lats		6	5
4. Abs		10	3
Total: 33 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		8	5
4. Front delts		6	5
5. Hyperextensions		8	4
Total: 44 reps			
Day 4	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	3	4
2. Speed bench press	40	4	1
	45	4	1
	50	4	4
3. Dumbbell bench press		6	5
4. Leg press		6	5
5. Abs		8	3
Total: 42 reps			
Total a week:153 reps			

Week 11			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	1
	85	2	4
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	5
Total: 47 reps			
Day 2	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	4
2. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+3	1
	75	1+2	4
	100	1	1
	103	1	1
2. Deadlift to the knees	50	3	1
	60	3	1
	70	2	2
4. Lats		8	5
5. Abs		10	3
Total: 57 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
3. Pecs		6	5
4. Triceps lying		6	4
5. Goodmorning		5	4
Total: 44 reps			
Day 4	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Inclined shoulder press seated (with light weight)		3	4
3. Front delts 6reps x 4sets (with light weight)		6	4
4. Reverse hyperextensions		6	4
Total: 17 reps			
TOTAL A WEEK:165 reps			

Week 12			
Day 1	%	reps	sets
1. Bench press	55	3	1
	65	3	1
	75	2	2
	85	1	3
2. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		6	4
5. Hyperextensions		8	4
Total: 47 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	3	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Dumbbell bench press		6	5
4. Lats		8	4
5. Abs		10	3
Total: 36 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	4
4. Goodmorning		4	4
Total: 34 reps			
Day 4	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+2	4
2. Inclined shoulder press seated		2	5
3. Lats		6	4
4. Abs		8	3
Total: 21 reps			
Total a week:138 reps			

Week 13			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Pecs		6	4
4. Triceps		6	4
5. Goodmorning seated		4	4
Total: 30 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Lats		6	4
4. Abs		10	3
Total: 34 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	75	2	1
	80	1	2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	1
	80	1	3
3. Pecs		8	4
4. Abs		8	3
Total: 27 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	1
	45	4	1
	50	4	3
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 20 reps			
TOTAL A WEEK:111 reps			

Week 14			
Day 1	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	75	2	3
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	3
3. Pecs		6	4
4. Abs		10	3
Total: 30 reps			
Day 2	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	2
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	1
	85	1	1
	90	1	2
	95	1	1
	100	1	1-2
3. Lats		8	4
4. Hyperextensions		5	4
Total: 33 reps			
Day 3	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
3. Pecs		8	4
4. Abs		8	3
Total: 35 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		3	5
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 00 reps			
TOTAL A WEEK: 98 reps			

Week 15			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	3
4. Goodmorning		4	4
Total: 34 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	1
	80	1	3
2. Deadlift	50	3	1
	60	3	1
	70	2	1
	75	1	3
3. Lats		6	4
4. Abs		8	3
Total: 25 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Bench press	50	3	1
	60	3	1
	70	2	1
	75	1	3
3. Hyperextensions		6	3
Total: 24 reps			
Day 4	%	reps	sets
REST			
Total a week: 83 reps			

Week 16			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	2	3
2. Bench press	50	3	1
	60	3	1
	70	2	2
	75	1	3
3. Abs (with no weight)		8	3
Total: 25 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	2
	70	2	3
Total: 15 reps			
Total a week: 40 reps			
Thursday-Friday			
REST			
COMPETITION!!!			

7.3. Three lifts: deadlift emphasis

Week 1			
Day 1	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	65	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	5
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	4
Total: 40 reps			
Day 2	%	reps	sets
1. Deficit deadlift, box height is 8-10cm	50	3	1
	60	3	1
	65	2	4
2. Bench press with bands	50	4	1
	60	4	1
	70	4	4
3. Deadlift off blocks, bar at 10-15cm below the knees **	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
4. Medial delts		6	5
5. Lats		8	5
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	4
Pecs		8	5
Triceps pushdowns		8	5
Goodmorning		5	5
Total: 37 reps			
Day 4	%	reps	sets
1. Deadlift with chains	50	4	1
	60	4	1
	70	4	4
2. Inclined shoulder press seated		4	5
3. Dumbbell bench press		6	5
4. Deadlift with snatch grip, slow descent	30	5	4
5. Reverse hyperextensions		8	4
Total: 24 reps			
TOTAL A WEEK:155 reps			

Week 2			
Day 1	%	reps	sets
1. Squat with chains	50	5	1
	60	5	1
	70	4	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
3. Pecs		8	5
4. Front delts		6	5
5. Hyperextensions		8	5
Total: 50 reps			
Day 2	%	reps	sets
1. Deadlift with bands	50	4	1
	60	4	1
	70	4	4
2. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+2	4
4. Lats		8	
5. Triceps standing		6	5
6. Leg press		6	5
Total: 62 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press with a slingshot	55	5	1
	65	4	1
	75	3	1
	85	2	2
	95	1	3
3. Pecs		8	5
4. Deadlift with snatch grip, slow descent	30	5	4
5. Abs		8	3
Total: 46 reps			
Day 4	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	3
2. Decline bench press	50	4	1
	60	4	1
	70	4	4
3. Deadlift off blocks, bar at 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	2
4. Rear delts		8	5
5. Reverse hyperextensions		8	5
Total: 52 reps			
TOTAL A WEEK:210 reps			

Week 3			
Day 1	%	reps	sets
1. 2sec paused squat	50	3	1
	60	3	1
	65	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
	80	3	2
3. Pecs		8	5
4. Triceps lying		8	5
5. Goodmorning		5	5
Total: 48reps			
Day 2	%	reps	sets
1. 3,2,1 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
2. Deadlift with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Lats		8	5
4. Front delts		6	5
Total: 38 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	2
	80	3	2
2. Board press	60	3	1
	70	3	1
	80	3	2
	90	2	3
3. Pecs		8	5
4. Lats		6	5
5. Hyperextensions		8	5
Total: 46 reps			
Day 4	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	4
2. Speed bench press	40	4	1
	45	4	4
3. Deadlift off blocks, bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	90	2	2
	95	1	3
4.Dumbbell bench press		6	5
5. Abs		8	4
Total: 50 reps			
TOTAL A WEEK:182 reps			

Week 4			
Day 1	%	reps	sets
1. Squat with chains	50	5	1
	60	5	1
	70	4	5
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	4
3. Pecs		8	5
4. Triceps pushdowns		8	5
5. Hyperextensions		8	4
Total: 54 reps			
Day 2	%	reps	sets
1. Deadlift to the knees with 1sec pause	50	3	1
	60	3	1
	70	3	4
2. 4,3,2 sec paused bench press	50	3	1
	60	3	1
	70	3	4
3. Deadlift off blocks with chains, bar at 15-20 cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	3	3
4. Front delts		6	5
5. Lats		8	5
6. Abs		10	3
Total: 54 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. Bench press	50	6	1
	60	5	1
	70	4	1
	75	3	1
	80	2	2
	85	1	2
	75	3	1
	65	5	1
	55	7	1
3. Pecs		8	5
4. Triceps standing		6	5
5. Goodmorning seated		5	5
Total: 59 reps			
Day 4	%	reps	sets
1. Deadlift with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Decline bench press	50	4	1
	60	4	1
	70	4	4
3. Dumbbell bench press		6	5
4. Biceps		10	5
5. Deadlift with snatch grip, slow descent		5	4
Total: 48 reps			
TOTAL A WEEK:215 reps			

Week 5			
Day 1	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	2
	95	1	2
3. Pecs		8	5
4. Triceps lying		8	5
5. Hyperextensions		8	5
Total: 34 reps			
Day 2	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	3
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
	80	3	2
3. Deadlift off blocks, bat at 15-20cm below the knees	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	2
	95	1	2
4. Medial delts		6	5
5. Abs		10	3
Total: 58 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	5
3. Pecs		8	5
4. Triceps pushdowns		8	5
5. Goodmorning seated		5	5
Total: 43 reps			
Day 4	%	reps	sets
1. Full deadlift with 2sec pause at 10-15cm below the knees	50	3	1
	60	3	1
	70	3	4
2. Inclined shoulder press seated		4	6
3. Dumbbell bench press		6	5
4. Leg curls		8	5
5. Reverse hyperextensions		8	4
Total: 18 reps			
TOTAL A WEEK:153 reps			

Week 6			
Day 1	%	reps	sets
1. 2sec paused squat	50	3	1
	60	3	1
	65	3	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	5
3. Pecs		8	5
4. Triceps standing		6	5
5. Goodmorning		8	5
Total: 45 reps			
Day 2	%	reps	sets
1. Full deadlift with 1sec pause at 5-10cm below the knees	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Board press	60	3	1
	70	3	1
	80	3	1
	90	2	4
3. Deadlift off blocks, bar at 5-10cm below the knees	60	4	1
	70	4	1
	80	4	1
	85	4	4
4. Front delts		6	5
5. Leg extensions		8	5
Total: 62 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	5	1
	70	4	5
2. Bench press with bands	50	5	1
	60	5	1
	65	5	4
3. Pecs		8	5
4. Triceps pushdowns		8	5
5. Abs		10	4
Total: 60 reps			
Day 4	%	reps	sets
1. Bench press with a slingshot	60	3	1
	70	3	1
	80	3	1
	85	3	4
2. Deadlift to the knees + full deadlift	50	3+1	1
	60	3+1	1
	70	3+1	4
3. Lats		8	5
4. Deadlift with snatch grip, slow descent	40	5	5
5. Reverse hyperextensions		8	4
Total: 54 reps			
TOTAL A WEEK:221 reps			

Week 7			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	5
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
	80	2	2
3. Pecs		6	5
4. Triceps standing		6	5
5. Hyperextensions		8	4
Total: 50 reps			
Day 2	%	reps	sets
1. Deficit deadlift	50	3	1
	60	3	1
	65	2	2
	70	1	3
2. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
3. Deadlift off blocks bar is 10-15cm below the knees	60	3	1
	70	3	1
	80	3	1
	90	2	4
4. Lats		8	5
5. Medial delts		6	5
Total: 51 reps			
Day 3	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	1
	85	2	4
3. Pecs		8	5
4. Biceps		8	4
5. Abs		10	4
Total: 40 reps			
Day 4	%	reps	sets
1. Deadlift with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Inclined shoulder press seated		4	5
3. Dumbbell bench press		6	5
4. Goodmorning seated		5	5
Total: 17 reps			
TOTAL A WEEK:158 reps			

Week 8			
Day 1	%	reps	sets
1. Squat with chains	50	5	1
	60	5	1
	70	5	4
2. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Triceps lying		8	5
5. Hyperextensions		8	5
Total: 47 reps			
Day 2	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+3	1
	60	1+3	1
	70	1+3	1
	75	1+2	4
2. Board press	60	3	1
	70	3	1
	80	3	2
	90	2	3
3. Deadlift off blocks, bar at 15-20cm below the knees	60	4	1
	70	4	1
	80	4	2
	85	4	3
4. Lats		8	5
5. Leg press		5	4
6. Abs		10	3
Total: 72 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		8	5
4. Triceps pushdowns		6	4
5. Goodmorning		5	4
Total: 40 reps			
Day 4	%	reps	sets
1. Deadlift to the knees + full deadlift	50	2+2	1
	60	2+2	1
	70	2+2	1
	75	2+1	4
2. Speed bench press	40	6	2
	45	6	3
3. Medial delts (with light weight)		6	4
4. Reverse hyperextensions		6	4
Total: 54 reps			
TOTAL A WEEK:213 reps			

Week 9			
Day 1	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	3	1
	85	2	4
3. Pecs		8	5
4. Triceps standing		6	5
5. Goodmorning seated		5	4
Total: 40 reps			
Day 2	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	3	2
	85	3	4
3. Medial delts		6	5
4. Pecs		6	5
5. Abs		10	3
Total: 40 reps			
Day 3	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. Board press	60	3	1
	70	3	1
	80	3	1
	90	2	4
3. Triceps lying		6	5
4. Lats		8	5
5. Hyperextensions		8	4
Total: 41 reps			
Day 4	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+1	4
2. Inclined shoulder press seated		4	5
3. Dumbbell bench press		6	5
4. Reverse hyperextensions		8	4
Total: 17 reps			
TOTAL A WEEK:138 reps			

Week 10			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	2
	75	2	3
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
3. Pecs		6	5
4. Front delts		6	5
5. Hyperextensions (with no weight)		6	4
Total: 42 reps			
Day 2 TEST	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	1
	90	1	1
	95	1	1
	100	1	1
	103	1	1
3. Lats		6	5
4. Abs		8	3
Total: 46 reps			
Day 3	%	reps	sets
1. Squat with chains	50	4	1
	60	4	1
	70	4	5
2. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	5
3. Pecs		8	5
4. Medial delts		6	5
5. Hyperextensions		8	4
Total: 50 reps			
Day 4	%	reps	sets
1. Bench press with chains	50	4	1
	60	4	1
	70	4	5
2. Deadlift off blocks	55	3	1
	65	3	1
	75	3	1
	85	3	4
3. Triceps lying		6	5
4. Lats		6	5
5. Deadlift with snatch grip, slow descent	35	5	5
Total: 49 reps			
Total a week:189 reps			

Week 11			
Day 1	%	reps	sets
1. Squat	50	5	1
	60	4	1
	70	3	1
	80	3	4
2. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
3. Pecs		8	5
4. Triceps standing		6	5
5. Hyperextensions		8	5
Total: 41 reps			
Day 2	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+2	2
	80	1+1	3
2. 3sec paused bench press	50	3	1
	60	3	1
	70	3	4
3. Deadlift off blocks, bar at 15-20cm below the knees	60	3	1
	70	3	1
	80	3	1
	90	2	2
	95	1	2
4. Lats		8	5
5. Abs		10	3
Total: 54 reps			
Day 3	%	reps	sets
1. Squat with 2sec pause on the halfway down	50	5	1
	60	4	1
	70	3	1
	75	2	4
2. Bench press with bands	50	4	1
	60	4	1
	70	4	4
3. Pecs		8	5
4. Triceps lying		6	5
5. Reverse hyperextensions		8	4
Total: 44 reps			
Day 4	%	reps	sets
1. Deadlift + deadlift from below the knees	50	1+2	1
	60	1+2	1
	70	1+2	1
	75	1+2	4
2. Speed bench press	40	4	1
	45	4	4
3. Rear delts		8	4
4. Reverse hyperextensions		6	4
Total: 41 reps			
TOTAL A WEEK:180 reps			

Week 12			
Day 1	%	reps	sets
1. Squat	55	3	1
	65	3	1
	75	3	1
	85	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	3	4
3. Pecs		6	4
4. Front delts		6	4
5. Hyperextensions		8	4
Total: 38 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	3	1
	85	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	3	2
	85	2	3
3. Lats		8	4
4. Abs		8	3
Total: 31 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	4
4. Goodmorning		4	4
Total: 34 reps			
Day 4	%	reps	sets
1. Deadlift off blocks	60	3	1
	70	3	1
	80	3	2
	90	2	3
2. Speed bench press	40	4	2
	45	4	3
3. Lats		6	4
4. Abs		8	3
Total: 38 reps			
TOTAL A WEEK:141 reps			

Week 13			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Pecs		6	4
4. Triceps		6	4
5. Goodmorning seated		4	4
Total: 30 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	3
3. Lats		6	4
4. Abs		10	3
Total: 33 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
3. Pecs		8	4
4. Abs		8	3
Total: 30 reps			
Day 4	%	reps	sets
1. Decline bench press	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 13 reps			
TOTAL A WEEK:106 reps			

Week 14			
Day 1	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	75	2	3
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	3
3. Pecs		6	4
4. Abs		10	3
Total: 30 reps			
Day 2	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	1
	85	1	1
	90	1	1
	95	1	1
	100	1	1-2
3. Lats		8	4
4. Hyperextensions		5	4
Total: 33 reps			
Day 3	%	reps	sets
1. Deadlift	50	3	1
	60	3	1
	70	3	1
	80	2	1
	90	1	1
	95	1	1
	100	1	1-2
2. Bench press	50	3	1
	60	3	1
	70	3	1
	75	3	4
3. Pecs		8	4
4. Abs		8	3
Total: 35 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		3	5
2. Pecs		6	4
3. Hyperextensions		8	4
Total: 00 reps			
TOTAL A WEEK: 98 reps			

Week 15			
Day 1	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
3. Pecs		6	3
4. Goodmorning		4	4
Total: 34 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	2
	80	1	3
2. Deadlift	50	3	1
	60	3	1
	70	2	1
	75	1	3
3. Abs		8	3
Total: 27 reps			
Day 3	%	reps	sets
1. Squat	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Bench press	50	3	1
	60	3	1
	70	2	1
	75	1	4
3. Hyperextensions		6	3
Total: 25 reps			
Day 4	%	reps	sets
REST			
Total a week: 86 reps			

Week 16			
Day 1	%	reps	sets
1.Bench press	50	3	1
	60	3	1
	70	2	2
	75	1	2
2. Deadlift	50	3	1
	60	2	2
	65	1	3
3. Abs		8	3
Total: 22 reps			
Day 2	%	reps	sets
1. Squat	50	3	1
	60	2	1
	70	1	2
2. Bench press	50	3	1
	60	2	2
	70	1	3
Total: 19 reps			
Total a week: 41 reps			
Thursday-Friday			
REST			
COMPETITION!!!			

7.4. Bench press only

Week 1			
Day 1	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	5
2. Pecs		8	5
3. Front delts		6	5
4. Triceps standing		6	5
5. Abs		10	3
Total: 21 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	5
2. Lats		8	5
3. Biceps		8	5
4. Abs		10	4
Total: 22 reps			
Day 3	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	2
	90	1	3
2. Dumbbell bench press		6	5
3. Triceps lying		8	5
4. Leg press		5	4
Total: 16 reps			
Day 4	%	reps	sets
1. Bench press with bands	50	5	1
	60	5	1
	65	5	5
2. Pecs		8	5
3. Medial delts		8	5
4. Biceps		8	4
Total: 35 reps			
TOTAL A WEEK:101 reps			

Week 2			
Day 1	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Pecs		8	4
3. Bench press with a slingshot	60	3	1
	70	3	1
	80	3	1
	85	3	4
4. Triceps pushdowns		8	5
5. Hyperextensions		8	4
Total: 38 reps			
Day 2	%	reps	sets
1. Bench press with bands	50	4	1
	60	4	1
	65-70	4	4
2. Lats		6	5
3. Medial delts		6	5
4. Biceps		8	5
5. Leg press		5	5
Total: 24 reps			
Day 3	%	reps	sets
1. 3,2,1 sec paused bench press	50	3	1
	60	3	1
	70	3	4
2. Front delts		6	5
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	3	1
	90	2	3
3. Triceps standing		6	5
4. Abs		8	4
Total: 36 reps			
Day 4	%	reps	sets
1. Bench press	50	5	1
	60	5	1
	70	4	4
2. Lats		8	5
3. Biceps		8	5
4. Goodmorning seated		5	5
Total: 26 reps			
TOTAL A WEEK:124 reps			

Week 3			
Day 1	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Pecs		8	5
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Triceps lying		8	5
5. Abs		8	4
Total: 37 reps			
Day 2	%	reps	sets
1. Bench press with chains	50	4	1
	60	4	1
	70	3	5
2. Lats		8	5
3. Medial delts		6	5
4. Hyperextensions		8	4
Total: 23 reps			
Day 3	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	5
2. Leg press		5	5
3. Dumbbell bench press		6	4
4. Triceps pushdowns		8	
Total: 19 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		4	6
2. Medial delts		6	5
3. Dips		5	5
4. Leg press		5	5
Total: 00 reps			
TOTAL A WEEK: 79 reps			

Week 4			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	2
	80	3	2
2. Pecs		8	5
3. Rear delts		6	5
4. Triceps standing		6	5
Total: 28 reps			
Day 2	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Lats		6	5
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
4. Hyperextensions		8	4
Total: 33 reps			
Day 3	%	reps	sets
1. Bench press	50	5	1
	60	5	1
	70	5	5
2. Pecs		8	5
3. Front delts		6	5
4. Biceps		8	5
Total: 35 reps			
Day 4	%	reps	sets
1. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	3	4
2. Lats		8	5
3. Triceps lying		8	5
4. Abs		8	3
Total: 21 reps			
TOTAL A WEEK:117 reps			

Week 5			
Day 1	%	reps	sets
1. Bench press	55	3	1
	65	3	1
	75	3	1
	85	2	4
2. Squat (medium weight)		4	5
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		6	5
5. Triceps lying		6	5
Total: 34 reps			
Day 2	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	5
2. Dumbbell bench press		8	5
3. Medial delts		6	5
4. Abs		10	4
Total: 19 reps			
Day 3	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	80	3	1
	85	2	2
	90	1	3
2. Pecs		6	5
3. Triceps standing		6	5
4. Hyperextensions		8	4
Total: 19 reps			
Day 4	%	reps	sets
1. Speed bench press	40	5	2
	50	4	3
2. Dips		5	5
3. Medial delts		6	5
4. Biceps		8	4
Total: 22 reps			
TOTAL A WEEK: 94 reps			

Week 6			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	4
2. Pecs		8	5
3. Inclined shoulder press seated		4	5
4. Triceps standing		6	5
Total: 26 reps			
Day 2	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	3	2
	90	2	2
	95	1	2
2. Squat in depth		5	4
3. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	3
4. Rear delts		8	5
5. Hyperextensions		8	4
Total: 36 reps			
Day 3	%	reps	sets
1. Bench press	50	6	1
	60	6	1
	70	6	5
2. Pecs		8	5
3. Front delts		6	5
4. Biceps		8	5
Total: 42 reps			
Day 4	%	reps	sets
1. Bench press with a slingshot	60	3	1
	70	3	1
	80	2	2
	90	1	3
2. Dumbbell bench press		6	5
3. Dips		4	5
4. Abs		10	3
Total: 13 reps			
TOTAL A WEEK:117 reps			

Week 7			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	75	3	5
2. Pecs		6	5
3. Triceps pushdowns		6	5
4. Front delts		6	5
5. Hyperextensions		10	3
Total: 27 reps			
Day 2	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	5
2. Dumbbell bench press		6	5
3. Lats		8	5
4. Triceps lying		6	5
Total: 19 reps			
Day 3	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Squat (medium weight)			
3. Bench press with a slingshot	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
4. Pecs		6	5
5. Abs		10	3
Total: 36 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		4	5
2. Dumbbell bench press		6	5
3. Triceps pushdowns		6	5
4. Hyperextensions		8	4
Total: 00 reps			
TOTAL A WEEK: 82 reps			

Week 8			
Day 1	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	5
2. Pecs		6	5
3. Front delts		6	5
4. Triceps lying		8	5
Total: 24 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	5	1
	70	4	5
2. Dumbbell bench press		6	5
3. Triceps pushdowns		8	5
4. Hyperextensions		8	4
Total: 30 reps			
Day 3	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	4
2. Squat		4	5
3. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	1
	90	1	3
4. Lats		8	5
5. Leg press		4	5
Total: 34 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	2
	50	4	3
2. Medial delts (with light weight)		6	4
3. Triceps standing (with light weight)		6	4
4. Abs		10	3
Total: 20 reps			
TOTAL A WEEK:108 reps			

Week 9 TEST			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	75	2	3
2. Lats (with light weight)		6	4
3. Medial delts (with light weight)		6	4
4. Leg extensions		6	4
5. Abs		8	4
Total: 18 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	1
	85	1	1
	90	1	1
	95	1	1
	100-105	1	1-2
2. Pecs		6	4
3. Triceps lying		6	4
4. Goodmorning seated		4	4
Total: 19 reps			
Day 3	%	reps	sets
1. Bench press with a slingshot	55	3	1
	65	3	1
	75	3	1
	85	2	4
2. Dumbbell bench press		6	5
3. Pecs		6	5
4. Front delts		6	5
5. Squat		4	5
Total: 17 reps			
Day 4	%	reps	sets
1. Speed bench press	40	5	2
	50	5	3
2. Triceps pushdowns		8	5
3. Lats		8	5
4. Reverse hyperextensions		8	4
Total: 25 reps			
TOTAL A WEEK: 79 reps			

Week 10			
Day 1	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Squat (medium weight)		4	5
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		8	5
5. Triceps lying		8	5
Total: 37 reps			
Day 2	%	reps	sets
1. 4,3,2sec paused bench press	50	3	1
	60	3	1
	70	3	4
2. Dumbbell bench press		6	5
3. Medial delts		6	5
4. Lats		8	5
Total: 18 reps			
Day 3	%	reps	sets
1. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	2
2. Leg press		5	4
3. Bench press with bands	50	4	1
	60	4	1
	70	4	4
4. Pecs		6	5
5. Abs		10	3
Total: 39 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	1
	50	4	4
2. Inclined shoulder press seated		3	5
3. Seated rowing		8	5
4. Goodmorning seated		5	4
Total: 20 reps			
TOTAL A WEEK:114 reps			

Week 11			
Day 1	%	reps	sets
1. Bench press	50	6	1
	60	5	1
	70	4	1
	75	3	1
	80	2	2
	85	1	2
	75	3	1
	65	5	1
	55	7	1
2. Pecs		8	5
3. Triceps standing		6	5
4. Hyperextensions		8	4
Total: 39 reps			
Day 2	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Leg extensions		8	4
3. Bench press with chains	50	4	1
	60	4	1
	70	4	4
4. Lats		8	5
5. Abs		10	4
Total: 44 reps			
Day 3	%	reps	sets
1. Board press	60	3	1
	70	3	1
	80	3	1
	85	2	2
	90	1	3
2. Pecs		8	5
3. Triceps lying		6	5
4. Goodmorning			
Total: 19 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	1
	45	4	1
	50	4	4
2. Medial delts		6	5
3. Lats		8	5
4. Abs		10	3
Total: 24 reps			
TOTAL A WEEK:126 reps			

Week 12			
Day 1	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Squat (medium weight)		4	5
3. Bench press with bands	50	4	1
	60	4	1
	65	4	4
4. Triceps standing		6	5
5. Abs		10	3
Total: 41 reps			
Day 2	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	3	2
	85	2	3
	80	3	2
2. Pecs		8	5
3. Dumbbell bench press		6	5
4. Medial delts		6	5
5. Hyperextensions		10	4
Total: 30 reps			
Day 3	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	3	2
	90	2	2
	95	1	2
2. Lats		8	5
3. Triceps pushdowns		8	5
4. Abs		10	3
Total: 21 reps			
Day 4	%	reps	sets
1. Inclined shoulder press seated		4	5
2. Dumbbell bench press		6	5
3. Biceps		8	5
4. Goodmorning seated		5	4
Total: 00 reps			
TOTAL A WEEK: 91 reps			

Week 13			
Day 1	%	reps	sets
1. Bench press	50	5	1
	60	4	1
	70	3	1
	80	2	2
	85	1	3
2. Pecs		6	5
3. Front delts		6	5
4. Triceps pushdowns		8	5
5. Leg press		4	4
Total: 19 reps			
Day 2	%	reps	sets
1. Board press	55	3	1
	65	3	1
	75	3	1
	85	2	1
	90	2	1
	95	1	1
	100	1	2
2. Lats		8	5
3. Biceps		8	4
4. Abs		10	3
Total: 16 reps			
Day 3	%	reps	sets
1. 3sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Pecs		6	5
3. Medial delts		6	4
4. Triceps lying		6	5
5. Reverse hyperextensions		10	4
Total: 17 reps			
Day 4	%	reps	sets
1. Bench press with bands	50	4	1
	60	4	1
	65	4	4
2. Dumbbell bench press		6	5
3. Lats		8	4
4. Leg extensions		10	5
Total: 24 reps			
TOTAL A WEEK: 76 reps			

Week 14			
Day 1	%	reps	sets
1. Bench press	55	5	1
	65	4	1
	75	3	1
	85	2	4
2. Squat (medium weight)		4	5
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		8	5
5. Abs		10	3
Total: 37 reps			
Day 2	%	reps	sets
1. 3,2,1 sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	3	5
2. Dumbbell bench press		6	4
3. Lats		6	5
4. Front delts		6	4
Total: 24 reps			
Day 3	%	reps	sets
1. Bench press with chains	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Leg press		5	5
3. Bench press with a slingshot	60	3	1
	70	3	1
	80	3	1
	85	2	1
	90	1	2
	95	1	3
4. Triceps standing		6	5
5. Abs		10	3
Total: 33 reps			
Day 4	%	reps	sets
1. 2sec paused bench press	50	3	1
	60	3	1
	70	3	1
	75	2	4
2. Lats		6	5
3. Triceps pushdowns		6	4
4. Hyperextensions		6	3
Total: 17 reps			
TOTAL A WEEK:111 reps			

Week 15			
Day 1	%	reps	sets
1. Bench press	55	3	1
	65	3	1
	75	2	1
	85	1	4
2. Squat		4	4
3. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
4. Pecs		6	5
5. Abs		8	3
Total: 29 reps			
Day 2	%	reps	sets
1. Board press	50	3	1
	60	3	1
	70	3	1
	80	3	1
	85	2	1
	90	1	1
	95	1	3
2. Leg extensions		6	4
3. Triceps pushdowns		6	5
4. Hyperextensions		8	4
Total: 18 reps			
Day 3	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	5
2. Squat		4	5
3. Lats		6	5
4. Medial delts		6	5
Total: 19 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	4
2. Medial delts		6	4
3. Triceps lying		6	4
4. Abs		8	3
Total: 16 reps			
TOTAL A WEEK: 82 reps			

Week 16			
Day 1	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	3
2. Pecs		6	4
3. Front delts (with light weight)		6	5
4. Abs		8	3
Total: 15 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	1
	85	1	1
	90	1	1
	95	1	1
	100	1	1
	105	1	1
2. Lats		6	4
3. Triceps lying		6	4
4. Hyperextensions		8	3
Total: 16 reps			
Day 3	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	4
2. Pecs		6	4
3. Medial delts		6	5
4. Abs		8	3
Total: 17 reps			
Day 4	%	reps	sets
1. Speed bench press	40	4	2
	45	4	3
2. Lats		6	4
3. Leg extensions		8	4
4. Reverse hyperextensions		8	3
Total: 20 reps			
TOTAL A WEEK: 68 reps			

Week 17			
Day 1	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	80	2	2
	85	1	2
2. Pecs		6	4
3. Front delts (with light weight)		6	4
4. Abs		8	3
Total: 15 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	3	1
	75	2	2
	80	1	3
2. Triceps pushdowns 6reps x 4sets (with light weight)		6	4
3. Hyperextensions		6	3
Total: 16 reps			
Day 3	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	2	2
	75	2	3
2. Pecs (with light weight)		6	4
3. Abs		8	3
Total: 16 reps			
Day 4	%	reps	sets
REST			
TOTAL A WEEK: 47 reps			

Week 18			
Day 1	%	reps	sets
1. Bench press	50	3	1
	60	3	1
	70	2	2
	75	1	3
2. Pecs (with light weight)		6	3
3. Abs		8	2
Total: 13 reps			
Day 2	%	reps	sets
1. Bench press	50	3	1
	60	2	2
	70	1	3
Total: 10 reps			
TOTAL A WEEK: 23 reps			
COMPETITION !!!			

7.5. Three lifts. Medium volume for experienced athletes

Prep Cycle I

Week 1				
Day 1		%	reps	sets
1	Front squat	30	4	1
		40	4	1
		50	4	3
2	Bench press	50	5	1
		60	4	1
		70	3	1
		75	2	4
3	Squat	55	3	1
		65	3	1
		75	2	4
4	Chest muscles		8	4
5	Abs		10	3
	4. Abs		10	4
	Total: 22 reps			
Day 2		%	reps	sets
1	Deadlift to knees w/2s pause	50	3	1
		60	3	1
		70	3	4
2	Bench press	50	5	1
		60	5	1
		70	3	1
		70	5	1
		70	7	1
		70	4	1
		70	6	1
		70	8	1
3	Deadlift w/pause 5-7cm above knees	55	3	1
		65	3	1
		75	2	4
4	Lat muscles		6	4
5	Good morning		5	5
Day 3		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		75	3	4
2	Squat w/pause at half squat position on the way down	50	3	1
		60	3	1
		70	3	4
3	Board bench press	60	3	1
		70	3	1
		80	3	2
		85	2	3
4	Chest muscles		8	4
5	Hyperextension		8	4
Day 4		%	reps	sets
1	Deficit deadlift	50	3	1
		60	2	4
2	Incline Bench press		3	5
3	Deadlift off boxes, bar is 5-7cm below knees	55	3	1
		65	3	1
		75	3	2
		85	2	3
4	Dips (with weight)		4	5
5	Leg press (80-90% of squat 1RM)		5	5
6	Reverse hyperextension		8	4

Week 2				
Day 1		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		75	3	1
		80	2	5
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	5
3	French press lying		6	5
4	Chest muscles		8	4
5	Hyperextension		8	4
Day 2		%	reps	sets
1	Bench press w/chains plus 8-10 kg to each side in addition to this Weight	50	3	1
		60	3	1
		70	3	5
2	Deadlift w/pause 5-7cm below knees	50	3	1
		60	3	1
		70	3	1
		75	2	5
3	Delts		6	5
4	Leg extension (1 sec pause every rep at the top)		6	5
5	Abs		8	4
Day 3		%	reps	sets
1	Front squat	30	4	1
		40	4	1
		50	4	4
2	Board bench press	60	3	1
		70	3	1
		80	3	1
		85	2	2
		90	1	3
3	Dips (with Weights)		6	5
4	Lat muscles		8	4
5	Good morning		5	5
Day 4		%	reps	sets
1	Bench press narrow grip	50	4	1
		60	3	1
		65	3	4
2	Deadlift w/chains, 8-10kg each side plus this Weight	50	3	1
		60	3	1
		70	2	2
		75	1	3
3	Dumbbells press		6	5
4	Leg press		4	5
5	Reverse hyperextension		8	4

Week 3				
Day 1		%	reps	sets
1	Front squat	40	4	1
		50	4	1
		55	3	3
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	2
		85	1	2
		80	2	2
3	Squat w/pause at half squat position on the way down	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Lat muscles		6	5
5	Goodmorning		5	4
Day 2		%	reps	sets
1	Deadlift to knees with 2 sec pause	50	3	1
		60	3	1
		70	3	1
		75	2	4
2	Bench press	50	5	1
		60	5	1
		70	3	1
		70	7	1
		70	4	1
		70	8	1
3	Deadlift off boxes, bar is 5-7cm below knees	60	4	1
		70	4	1
		80	4	4
4	Chest muscles		6	4
5	Abs		10	3
Day 3		%	reps	sets
1	Board bench press	55	3	1
		65	3	1
		75	3	1
		85	2	2
		90	1	3
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	5
3	Bench press w/chains plus 8-10 kg to each side in addition to this Weight	50	4	1
		60	4	1
		65	4	4
4	Lat muscles		8	4
5	Goodmorning seated		4	4
Day 4		%	reps	sets
1	Deadlift w/pause 5-7cm above knees	50	3	1
		60	3	1
		70	3	1
		75	2	4
2	Bench press with middle grip	50	4	1
		60	4	1
		65	4	4
3	Deadlift w/ chains	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Triceps		6	5
5	Reverse hyperextension		8	4

Week 4				
Day 1		%	reps	sets
1	Front squat	40	3	1
		45	3	1
		50	3	2
		55	3	3
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	2
		85	2	3
		80	3	2
3	Squat	50	3	1
		60	3	1
		70	3	1
		80	3	4
4	Dips		6	5
5	Lat muscles		6	4
6	Hyperextension		8	4
Day 2		%	reps	sets
1	Deficit deadlift	50	3	1
		60	3	1
		70	2	4
2	Board bench press	60	3	1
		70	3	1
		80	3	1
		85	3	1
		90	2	2
		95	1	2
3	Deadlift off boxes, 5-7cm below knees	60	3	1
		70	3	1
		80	3	2
		90	2	2
		95	1	2
4	Chest muscles		8	4
5	French press		6	5
Day 3		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
3	Bench press w/chains plus 8-10 kg to each side in addition to this Weight	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Delts		6	4
5	Abs		10	3
Day 4		%	reps	sets
1	Deadlift w/pause 5-7cm below knees	50	3	1
		60	3	1
		70	3	1
		75	2	2
		80	1	3
2	Bench press	50	6	1
		60	6	1
		65	6	4
3	Deadlift w/chains	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Leg press		5	5
5	Reverse hyperextension		8	4

Prep Cycle II

Week 1				
Day 1		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	4
2	Squat	50	5	1
		60	5	1
		70	3	1
		70	7	1
		70	4	1
		70	8	1
		70	2	1
		70	5	1
3	Board press	60	3	1
		70	3	1
		80	3	2
		85	2	3
4	Chest Muscles		8	4
5	Abs		8	3
Day 2		%	reps	sets
1	Deficit deadlift	50	3	1
		60	3	1
		70	2	4
2	Bench press with chains	50	3	1
		60	3	1
		70	3	1
		75	2	4
3	Deadlift off boxes, bar is at 5 cm below knees	60	3	1
		70	3	1
		80	3	2
		85	3	3
4	Lat Muscles		6	5
5	Dips		5	5
6	Goodmorning		5	5
Day 3		%	reps	sets
1	Bench press	55	5	1
		65	4	1
		75	3	1
		85	2	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
3	Bench press	50	4	1
		60	4	1
		70	4	4
4	Chest Muscles		6	5
5	Abs		10	3
Day 4		%	reps	sets
1	Deadlift off boxes, bar is at 5 cm below knees	60	3	1
		70	3	1
		80	3	2
		85	3	4
2	Incline Shoulder Press		3	5
3	Delts		6	4
4	Lat Muscles		6	5
5	Hyperextension		8	4

Week 2				
Day 1		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
3	Bench press w/chains	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Chest Muscles		8	5
5	Good morning		5	5
Day 2		%	reps	sets
1	Deadlift to knees	50	3	1
		60	3	1
		70	2	4
2	Board press	55	5	1
		65	4	1
		75	4	2
		85	3	2
		90	2	3
3	Deadlift w/chains	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Triceps		6	5
5	Abs		10	4
Day 3		%	reps	sets
1	Squat	55	5	1
		65	4	1
		75	3	1
		85	2	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	5
3	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	3
4	Chest Muscle		6	5
5	Hyperextension		8	4
Day 4		%	reps	sets
1	Deadlift w/ 2 pauses, 5-7cm above and below knees	50	3	1
		60	3	1
		70	3	1
		75	2	4
2	Incline shoulder press		3	5
3	Dips		4	5
4	Lat Muscles		6	5

Week 3				
Day 1		%	reps	sets
1	Bench press	55	5	1
		65	4	1
		75	3	1
		85	2	4
2	Squat	50	5	1
		60	5	1
		70	3	1
		70	7	1
		70	4	1
		70	8	1
		70	2	1
		70	6	1
3	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	4
4	Chest Muscles		6	4
5	Good Morning		5	5
Day 2		%	reps	sets
1	Deficit Deadlift	50	3	1
		60	3	1
		70	2	4
2	Bench press	50	6	1
		60	5	1
		70	4	1
		75	3	1
		80	2	2
3	Deadlift from boxes bar is 5cm below knees	60	4	1
	Deadlift from boxes bar is 5cm below knees	70	4	1
		80	4	4
4	Triceps		6	5
5	Abs		10	3
Day 3		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
2	Bench press w/chains	50	4	1
		60	4	1
		70	4	5
3	Squat	50	4	1
		60	4	1
		70	4	4
4	Lat Muscles		6	5
5	Hyperextension		8	4
Day 4		%	reps	sets
1	1 + 1/2 (x2) Deadlift	50	3	1
	2 + 1/2 (x2) Deadlift	60	3	1
	3 + 1/2 (x2) Deadlift	70	3	1
	4 + 1/2 (x2) Deadlift	75	3	4
2	Close Grip Bench Press	50	3	1
		60	3	1
		70	3	4
3	Delts		6	5
4	Triceps		6	5
5	Reverse Hyperextension		8	4

Week 4				
Day 1		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	5
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	5
3	Chest Muscles		6	5
4	Dips		4	5
5	Good Morning		5	5
Day 2		%	reps	sets
1	Deadlift w/pause 5-7cm above knees	50	3	1
		60	3	1
		70	3	1
		80	2	4
2	Board press	55	3	1
		65	3	1
		75	3	1
		85	3	2
		90	2	2
		95	1	3
3	Deadlift up to knees	50	3	1
		60	3	1
		70	3	1
		75	2	4
4	Triceps		6	5
5	Lat Muscles		6	5
6	Abs		8	4
Day 3		%	reps	sets
1	Bench press	55	5	1
		65	4	1
		75	3	1
		85	2	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
3	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	3
4	Chest Muscles		6	5
5	Hyperextension		8	4
Day 4		%	reps	sets
1	Deficit Deadlift	50	3	1
		60	3	1
		70	2	4
4	Chest Muscles		6	5
3	Deadlift off boxes bar is 5cm below knees	60	3	1
		70	3	1
		80	3	1
		85	2	3
5	Lat Muscles		6	5
6	Leg Press		5	5
5	Reverse Hyperextension		8	4

Week 5				
Day 1		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	4
2	Squat	50	5	1
		60	5	1
		70	3	1
		70	5	1
		70	7	1
		70	4	1
		70	6	1
		70	8	1
3	Bench press	50	4	1
		60	4	1
		70	4	4
4	Chest Muscles		6	5
5	Good morning		5	5
Day 2		%	reps	sets
1	Deadlift up to knees	50	4	1
		60	4	1
		70	4	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	5
3	Deadlift off boxes, bar is 5cm below knees	55	3	1
		65	3	1
		75	3	1
		85	3	4
4	Triceps		6	5
5	Hyperextension		8	4
Day 3		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		75	3	4
2	Bench press	50	6	1
		60	5	1
		70	4	1
		75	3	1
		80	2	2
		85	1	2
		80	2	1
		75	3	1
		70	4	1
		65	5	1
		60	6	1
		55	7	1
		50	8	1
3	Squat	50	4	1
		60	4	1
		70	4	4
4	Chest Muscles		6	5
5	Abs		10	3
Day 4		%	reps	sets
1	Deadlift w/pause 5cm above knees	50	3	1
		60	3	1
		70	3	1
		75	2	4
2	DB Bench press		6	5
3	Deadlift w/chains off boxes, bar is 10-12cm below knees	50	4	1
		60	4	1
		70	4	1
		75	4	4
4	Lat Muscles		6	5
5	Hyperextension		8	4

Week 6				
Day 1		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	5
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	5
3	Chest Muscles		6	5
4	Hyperextension		8	4
Day 2		%	reps	sets
1	Deficit Deadlift	50	4	1
		60	4	1
		65	3	2
2	Bench press	50	5	1
		60	5	1
		70	3	1
		70	6	1
		70	4	1
		70	7	1
		70	2	1
		70	8	1
		70	5	1
3	Deadlift	50	3	1
		60	3	1
		70	3	1
		80	2	5
4	Delts		6	5
5	Lat Muscles		8	4
Day 3		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		75	3	5
3	Squat	50	5	1
		60	5	1
		70	5	4
4	Chest Muscles		8	5
5	Abs		8	4
Day 4		%	reps	sets
1	Deadlift up to knees	50	3	1
		60	3	1
		70	3	1
		75	3	4
2	Bench press	50	4	1
		60	4	1
		70	4	4
3	Deadlift w/chains	50	3	1
		60	3	1
		70	3	1
		75	3	4
4	Dips		4	5
5	Lat Muscles		6	5
6	Hyperextension		8	4

Prep Cycle III

Week 1				
Day 1		%	reps	sets
1	Squat	55	5	1
		65	4	1
		75	3	1
		85	2	3
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	4
4	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	4
3	Chest Muscles		6	4
5	Goodmorning		5	5
Day 2		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	2
		85	1	2
		80	2	2
2	Deadlift off boxes, bar is 5cm below knees	65	3	1
		75	3	1
		85	3	2
		90	2	3
3	Triceps		6	5
4	Lat Muscles		6	5
5	Abs		8	4
Day 3		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		75	2	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		75	2	4
3	Chest Muscles		6	5
4	Dips		5	5
5	Goodmorning		5	5
Day 4		%	reps	sets
1	Incline shoulder press		3	5
2	Deadlift w/pause 5cm above knees	50	3	1
		60	3	1
		70	3	1
		75	2	4
2	Lat Muscles		6	4
4	Hyperextension		8	4

Week 2				
Day 1		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	1
		90	1	1
		95	1	1
		100-105	1	2-3
2	Bench press	50	5	1
		60	4	1
		70	3	1
		75	2	3
3	Chest Muscles		5	3
4	Hyperextension		8	4
Day 2		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	1
		90	1	1
		95	1	1
		100-105	1	2-3
2	Deadlift	50	5	1
		60	4	1
		70	3	1
		80	2	1
		90	1	1
		95	1	1
		100-105	1	2-3
3	Delts		6	4
4	Lat Muscles		6	4
Day 3		%	reps	sets
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
1	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
4	Squat	50	5	1
		60	5	1
		70	5	4
3	Chest Muscles		8	5
4	Abs		8	4
Day 4		%	reps	sets
1	Incline shoulder press		3	5
1	Deadlift up to knees	50	3	1
		60	3	1
		70	3	1
		75	3	4
2	Bench Press	50	5	1
		60	5	1
		70	5	4
3	Deadlift off boxes, bar is 5cm below knees	60	4	1
		70	4	1
		80	4	4
4	Dips		4	5
5	Lat Muscles		6	5
6	Hyperextension		8	4

Week 3				
Day 1		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	3	5
3	Board press	60	4	1
		70	4	1
		80	4	4
4	Chest Muscles		6	4
5	Goodmorning		5	5
Day 2		%	reps	sets
1	Deadlift up to knees w/pause	50	4	1
		60	4	1
		70	4	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	2
		85	2	2
		80	3	1
		70	5	1
		60	7	1
		50	9	1
3	1(x2) + 1/2 (x2) Deadlift, alternating	50	4	1
	1(x2) + 1/2 (x2) Deadlift, alternating	60	4	1
	1(x2) + 1/2 (x2) Deadlift, alternating	70	4	1
	1 + 1/2 (x2) Deadlift, alternating	75	3	4
4	Triceps		6	5
5	Lat Muscles		6	5
6	Abs		10	4
Day 3		%	reps	sets
1	Squat	50	5	1
		60	5	1
		65	3	1
		65	7	1
		65	4	1
		65	8	1
		65	2	1
		65	6	1
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	5
3	Dips		4	5
4	Chest Muscles		6	4
5	Hyperextension		8	4
Day 4		%	reps	sets
3	Dips		4	5
4	Chest Muscles		6	4
5	Hyperextension		8	4
		75	3	4
2	Bench press w/chains	50	4	1
		60	4	1
		70	4	4
2	Bench press w/chains	50	4	1
		60	4	1

Week 4				
Day 1		%	reps	sets
1	Board press	55	3	1
		65	3	1
		75	3	1
		85	3	2
		90	2	3
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	3	2
		85	2	2
		80	3	2
3	Decline Bench Press	50	4	1
		60	4	1
		70	4	5
4	Chest Muscles		6	4
5	Abs		10	3
Day 2		%	reps	sets
1	Deadlift w/pause 5-7cm below knees	50	3	1
		60	3	1
		70	3	1
		80	2	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	2
		85	2	3
		80	3	2
3	Deadlift off boxes, bar is 5cm below knees	65	3	1
		75	3	1
		85	3	2
		95	2	3
4	Lat Muscles		6	5
5	Goodmorning		5	5
Day 3		%	reps	sets
1	Bench press w/chains	50	3	1
		60	3	1
		70	3	1
		75	3	4
2	Squat	50	5	1
		60	5	1
		70	3	1
		70	5	1
		70	7	1
		70	4	1
		70	6	1
		70	8	1
3	Board press	50	4	1
		60	4	1
		70	4	1
		75	4	4
4	Chest Muscles		6	5
Day 4		%	reps	sets
1	1(x2) + 1/2 (x2) Deadlift, alternating	50	4	1
	1(x2) + 1/2 (x2) Deadlift, alternating	60	4	1
	1 + 1/2 (x2) Deadlift	70	3	1
	1 + 1/2 Deadlift	80	2	4
2	Incline shoulder press		3	5
3	Lat Muscles		6	5
4	Hyperextension		8	4

Week 5				
Day 1		%	reps	sets
1	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	4
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	2	4
3	Bench press	50	4	1
		60	4	1
		70	4	4
4	Chest Muscles		6	4
5	Goodmorning		5	5
Day 2		%	reps	sets
1	Deficit deadlift	50	3	1
		60	3	1
		65	3	1
		70	2	4
2	Bench press	50	5	1
		60	4	1
		70	3	1
		75	2	4
3	Deadlift from boxes, bar is 5cm below knees	70	3	1
		80	3	1
		90	2	2
		95	1	3
4	Triceps		6	5
5	Goodmorning		5	4
Day 3		%	reps	sets
4	Triceps		6	5
5	Goodmorning		5	4
		70	3	1
		70	5	1
2	Bench press	50	6	1
		60	6	1
		70	6	5
4	Squat	55	3	1
		65	3	1
		75	3	4
3	Chest Muscles		6	5
5	Lat Muscles		6	5
Day 4		%	reps	sets
1	Deadlift up to knees w/pause	50	3	1
		60	3	1
		70	3	1
		75	3	4
2	Close Grip Bench Press	50	4	1
		60	4	1
		70	4	4
3	Deadlift off boxes, bar is 5cm below knees	55	3	1
		65	3	1
		75	3	1
		85	3	4
5	Triceps		6	5

Comp Cycle

Week 1				
Day 1		%	reps	sets
1	Bench press	55	5	1
		65	4	1
		75	3	1
		85	2	3
2	Squat	50	5	1
		60	4	1
		70	3	1
		80	3	4
3	Bench press	50	5	1
		60	4	1
		70	3	1
		80	2	4
4	Chest Muscles		8	4
5	Hyperextension		8	4
Day 2		%	reps	sets
1	Deficit Deadlift	50	3	1
		60	3	1
		70	2	2
2	Bench press	50	5	1
		60	4	1
		70	3	1
		80	3	4
3	Deadlift of boxes, bar is 5cm below knees	60	3	1
		70	3	1
		80	2	2
		90	1	3
4	Lat Muscles		6	4
5	Abs		8	4
Day 3		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	2
		85	1	3
2	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	2
		85	2	3
		80	2	2
3	Chest Muscles		6	4
4	Triceps		6	4
5	Goodmorning		5	4
Day 4		%	reps	sets
1	Incline Shoulder press		3	5
2	Deadlift off boxes, bar is 5cm below knees	55	3	1
		65	3	1
		75	3	1
		85	2	3
3	Delts		6	4
4	Lat Muscles		6	4
5	Reverse Hyperextension		8	4

Week 2				
Day 1		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	3	1
		75	2	3
2	Bench press	50	3	1
		60	3	1
		70	3	1
		75	2	3
3	Chest Muscles		6	4
4	Hyperextension		8	3
Day 2		%	reps	sets
1	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	1
		90	1	1
		95	1	1
		100	1	1-2
2	Deadlift	50	3	1
		60	3	1
		70	2	1
		80	2	1
		90	1	1
		95	1	1
		100	1	1-2
3	Triceps		6	4
4	Abs		8	3
Day 3		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	1
		90	1	1
		95	1	1
		100	1	1-2
2	Bench press	50	3	1
		60	3	1
		70	3	1
		75	2	4
3	Chest Muscles		6	4
4	Goodmorning		4	4
Day 4		%	reps	sets
1	DB Bench Press		5	5
2	Deadlift up to knees	50	3	1
		60	3	1
		70	3	1
		75	2	3
3	Lat Muscles		6	4

Week 3				
Day 1		%	reps	sets
1	Squat	55	3	1
		65	3	1
		75	2	2
		85	1	4
2	Bench press	50	3	1
		60	3	1
		70	3	1
		80	3	4
3	Chest Muscles		6	4
4	Hyperextension		8	4
Day 2		%	reps	sets
1	Deadlift up to knees	50	3	1
		60	3	1
		70	3	1
		75	2	3
2	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	2
		85	1	3
3	Deadlift	50	3	1
		60	3	1
		70	3	1
		80	2	4
4	Lat Muscles		6	4
5	Abs		8	3
Day 3		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	4
2	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	5
3	Delts		6	4
4	Triceps		6	4
5	Goodmorning		4	4
Day 4		%	reps	sets
1	Incline Shoulder Press		3	5
2	Deadlift off boxes, bar is 5cm below knees	55	3	1
		65	3	1
		75	3	1
		85	2	3
3	Lat Muscles		6	4
4	Reverse Hyperextension		8	4

Week 4				
Day 1		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	3	1
		80	2	4
2	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	2
		85	1	3
3	Chest Muscles		6	4
4	Goodmorning		4	4
Day 2		%	reps	sets
1	Bench press	50	3	1
		60	3	1
		70	3	1
		80	2	2
2	Deadlift	50	2	1
		60	2	1
		70	2	3
3	Lat Muscles		6	3
4	Abs		8	3
Day 3		%	reps	sets
1	Squat	50	3	1
		60	3	1
		70	2	2
		75	1	3
2	Bench press	50	3	1
		60	3	1
		70	3	1
		75	2	3
3	Hyperextension		6	3
Day 4		%	reps	sets
Rest				

Week 5				
Day 1		%	reps	sets
1	Bench press	50	3	1
		60	3	1
		70	2	2
		75	1	3
2	Deadlift	50	3	1
		60	3	1
		70	2	3
3	Abs		8	3
Day 2		%	reps	sets
1	Squat	50	3	1
		60	2	1
		70	1	3
2	Bench press	50	3	1
		60	3	1
		70	2	3
Competition				

Литература / Bibliography

Аксенов М.О. Управление тренировочным процессом спортсменов старших разрядов в пауэрлифтинге на основе современных методов моделирования // Молодежь Забайкалья: интеллект и здоровье: материалы VII междунар. науч. конф. - Чита, 2003. - С.48-51.

Аксенов М.О. Кластерный анализ тренировочной нагрузки пауэрлифтеров / М.О. Аксенов, А.В. Гаськов // Интеллектуальный потенциал вузов на развитие Дальневосточного региона России: материалы VII Междунар. конф. студентов, аспирантов и молодых ученых (25-26 мая 2005 г.) Кн. 4 / Институт сервиса, моды и дизайна. - Владивосток: Изд-во ВГУЭС, 2005. - С. 10-12.

Аксенов М.О. Управление тренировочным процессом в пауэрлифтинге на основе современных информационных технологий. Дис. ... канд. пед. наук: 13.00.04, Улан-Удэ, Изд-во БГУ, 2006. - 200 с.

Бельский И.В. Модель специальной силовой подготовленности пауэрлифтеров // Теория и практика физической культуры. - 2000. - №1. - С. 33-35.

Бельский И.В. Системы эффективной тренировки: Армрестлинг. Бодибилдинг. Бенчпресс. Пауэрлифтинг. Минск: Вида-Н, 2003. - 352 с.

Берштейн, Н.А. Биомеханика и физиология движений / Н.А. Берштейн. - Воронеж: МПО «МОДЭК», 1997. 608 с.

Бондарчук А.П. Периодизация спортивной тренировки. - К., 2000. - 558с.

Вайцеховский С.М. Книга тренера. - М.: ФиС. - 1971-312 с.

Вейдер Д. Принципы Уайдера – основа современного бодибилдинга. «Muscle and Fitness», июль 1987 года.

Вейдер Д. Строительство тела по системе Джо Вейдера (перевод с английского) / Д. Вейдер - М.: ФиС, 1992. -112 с.

Верхошанский Ю.В. Некоторые предпосылки к оптимальному управлению процессом становления спортивного мастерства. - Теория и практика физической культуры, 1966, №4, С.21-23.

Верхошанский Ю.В. Основы специально силовой подготовки в спорте. - М.: ФиС, 1977. - 216 с.

Верхошанский Ю.В. Методика оценки скоростно-силовых способностей спортсменов // Теория и практика физической культуры. - 1979. - №2 - С. 7-11.

Верхошанский Ю.В. Программирование тренировочного процесса квалифицированных спортсменов. - М.: ФиС, 1985.

Верхошанский Ю.В. Основы специальной физической подготовки спортсменов. - М.: ФиС, 1988. - 322 с.

Верхошанский Ю.В. Актуальные проблемы современной теории и методики спортивной тренировки // Теория и практика физической культуры. 1993. № 11-12. С. 21-24.

Верхошанский Ю.В. Основы методики специально силовой подготовки тяжелоатлетов / Ю.В. Верхошанский, А.С. Медведев. - Методическая разработка для слушателей Высшей школы тренеров и студентов Академии. М.: - 1997. 35 с.

Верхошанский Ю.В. Горизонты научной теории и методологии спортивной тренировки // Теория и практика физической культуры. 1998. №7. С. 41-54.

Воробьев А.Н. Тяжелоатлетический спорт. Очерки по физиологии и спортивной тренировке. М.: ФиС, 1971. - 224с.

Воробьев А.Н. Физиологические и гигиенические основы тяжелой атлетики. / А.Н. Воробьев, Н.И. Саксонов // Тяжелая атлетика. М.: ФиС, 1972. - С. 243-274.

Воробьев А.Н. Вариативность нагрузки важнейший фактор рационального построения тренировки / А.Н. Воробьев, А.Д. Ермаков // Теория и практика физической культуры. - 1972.-№6.- С. 8-10

Воробьев А.Н. Тяжелоатлетический спорт. Очерки по физиологии и спортивной тренировке.2-е доп. М.: ФиС, 1977. - 255с.

Воробьев А.Н. Сила как физическое качество и методы ее развития // Тяжелая атлетика: Ежегодник-1981. М.: ФиС, 1981. С. 117-131.

Воробьев А.Н. Методика тренировки. Тактика соревнований / А.Н. Воробьев, В.И. Родионов // Тяжелая атлетика. М.: ФиС, 1981. С. 131-190.

Воробьев А. Н., Роман Р. А. Методика тренировки / Тяжелая атлетика: Учеб. для ИФК, под ред. А. Н. Воробьева. - М., ФиС, 1988.

Воробьев А.Н. Тренировка. Работоспособность. Реабилитация. М.: ФиС, 1989. - 272 с., ил.

Глядя С.А. Стань сильным / С.А. Глядя, М.А. Старов, Ю.В. Батыгин. - Харьков. 1998. - 43 с.

Глядя С.А. Стань сильным. – 2. / С.А. Глядя, М.А. Старов, Ю.В. Батыгин. - Харьков. 1999, 71 с.

Глядя С.А. Стань сильным – 3. / С.А. Глядя, М.А. Старов, Ю.В. Батыгин. - Харьков. 2000. - 108 с.

Годик М.А. Контроль тренировочных и соревновательных нагрузок. М.: ФиС, 1980, 136 с.

Делавье Ф. Анатомия силовых упражнений для мужчин и женщин / Пер. с фр. О.Е. Ивановой. - М.: РИПОЛ классик, 2006. 144 с.: ил.

Донской Д.Д. Биомеханика с основами спортивной техники: учебник для ин-тов физкультуры. - М.: ФиС, 1971. - 288 с.

Донской Д.Д. Биомеханика. - М., 1975. - 239 с.

Дылык Т.Н. Структура подготовительного периода в пауэрлифтинге // Физическое воспитание студентов творческих специальностей / ХГАДИ (ХХПИ). - Харьков, 2003. - №1. - С. 40-46.

Завьялов И.В. Жим лежа // Мир силы, №1. 2000. - С. 22-25.

Завьялов И. В. Применение специальной экипировки в пауэрлифтинге // Мир силы, №2. 2000. - С. 25-27.

Зациорский В.М. Вопросы методики воспитания физических качеств. - М.: ФиС, 1961. - 175 с.

Зациорский В.М. Годик М.А. Методика и первые результаты исследования «взрывной» силы спортсмена.//Теория и практика физической культуры. 1965. - № 7. - С. 22-24.

Зациорский В.М. Двигательные качества спортсменов. (Исследования по теории и методике воспитания): Автореферат дис. ... д-ра пед. наук. - М., 1969. - 72 с.

Зациорский В.М. Физические качества спортсмена. М.: ФиС, 1970. - 200 с.

Зимкин Н.В. Физиологическая характеристика силы, быстроты и выносливости. - М.: ФиС, 1956. - 35 с.

Ибель Д.В. Терминология атлетических упражнений в бодибилдинге. М.: Олимпия Пресс, 2006. - С. 80.

Ингерлейб М.Б. Анатомия физических упражнений. - Изд. 2-е. - Ростов на Дону: Феникс, 2009. - 187 [1] с.: ил. - (Феникс-Фитнес).

Кичайкина, Н.Б. Оценка техники приседания со штангой на плечах в пауэрлифтинге с точки зрения мышечного обеспечения движения // Труды кафедры биомеханики: сборник статей / Н.Б. Кичайкина, Г.А. Самсонов. - НГУ им. П.Ф. Лесгафта, СПб / Под ред. А.В. Самсоновой, В.Н. Томилова. - СПб, 2010. - 79-103.

Книпст И.Н. Влияние величины тренировочного груза на увеличение силы мышц. Сб. Проблемы физиологии спорта. 1. - М.: ФиС, 1958.

Кострюков В.В. Совершенствование специальной силовой подготовки квалифицированных пауэрлифтеров на основе применения упражнений с переменными отягощениями. Дис. ... канд. пед. наук: Чебоксары, 2011. - 189 с.

Крестовников, А.Н. Очерки по физиологии физических упражнений. -М.: ФиС, 1951. 168 с.

Лукьянов Б.Г. Кинематические характеристики модели техники жима лежа. / Б.Г. Лукьянов, Б.И. Шейко, Г.А. Смоляков, И.С. Фролов. - Российская научная конференция “Инновационные подходы в организации и содержании образовательного процесса в спортивной и оздоровительной деятельности” УГАТУ 2008 133-135 с.

Лукьянов Б.Г. Структурная модель техники жима лежа. / Б.Г. Лукьянов, Б.И. Шейко, А.Б. Лукьянов. - Российская научная конференция “Инновационные подходы в организации и содержании образовательного процесса в спортивной и оздоровительной деятельности” УГАТУ 2008 141-146с

Лэмберт Майкл. Как приседать? // Muscle and Fitness,1988, март. С.12-13

Лэмберт Майкл. Приседания в силовом троеборье // Атлетизм, 1990, № 10. - С. 18.

Лэмберт Майкл. Как правильно тренировать становую тягу. //Олимп, 1993, № 2. - С.- 19-21.

Матвеев Л.П. Основы спортивной тренировки / Л.П. Матвеев. - М.: ФиС, 1977. - 297 с.

Матвеев Л.П. Величина тренировочной нагрузки и рост спортивных достижений / Л.П. Матвеев, Р.А. Роман, А.В. Черняк // Тяжелая атлетика. Сборник статей. Ежегодник 1973 / Сост. Р.А. Роман. - М.: ФиС, 1973. - С.13-19. 48.

Матвеев Л.П. Теория и методика физической культуры. - М.: ФиС, 1991. - 542 с.

Медведев А.С. Распределение тренировочной нагрузки тяжелоатлетов между недельными циклами за месяц до соревнования. // Теория и практика физической культуры. - 1966. № 5. С. 31-32.

Медведев А.С. Объем и интенсивность тренировочных нагрузок в соревновательный период у сильнейших тяжелоатлетов СССР: Автореф. дис. канд. пед. наук / Медведев Алексей Сидорович; [Гос. Центр. Ордена Ленина ин-т физ. культуры] - М.,1967. - 21 с.

Медведев А.С. Объем и интенсивность тренировочных нагрузок в предсоревновательном периоде сильнейших тяжелоатлетов // Теория и практика физической культуры. - 1969. - 2. - С. 21-23.

111.Медведев А.С., Черняк А.В. Некоторые варианты распределения объема и интенсивности суммарной тренировочной нагрузки тяжелоатлетов между недельными циклами за месяц до соревнований // Трибуна мастеров тяжелой атлетики. В помощь тренеру / Сост. сборника А.Н. Воробьев. - М.: ФиС, 1969. - С. 20-26.

Медведев А.С. Методика тренировки тяжелоатлета. Тяжелая атлетика. Учебник для студентов инст. физич. культ. М.: ФиС, 1972. - С. 160-220.

Медведев А.С. К вопросу о классификации и ранжировании тяжелоатлетических упражнений // Тяжелая атлетика. Ежегодник. - М.: ФиС, 1982. - С.19-23.

Медведев А.С. Классификация упражнений, применяемых в тренировочном процессе в силовом троеборье / А. С. Медведев, Я.Э. Якубенко // Олимп, 1997, №2. - С. 26-27.

121.Муминов, В.И. Развитие силы ног у пауэрлифтеров // Пауэрлифтинг. - № 6.- 2008. - С. 24-26.

Муравьев В.Л. Пауэрлифтинг. Путь к силе. - М.: Светлана П, 1998. - 32 с.

Муравьев В.Л. Жми лежа. Начинающим с «нуля». М.: Лана, 2001. - 32 с.

Назаренко Ю.Ф. Методика обучения соревновательным упражнениям в силовом троеборье. (пауэрлифтинг) / Ю.Ф. Назаренко, С.Ю. Те, С.В. Матук; Сибирский ГУФК. - Омск, 2003. - 44 с.

Назаренко Ю.Ф. Техника соревновательных упражнений в силовом троеборье / Ю.Ф. Назаренко, С.Ю. Те; Сибирский ГУФК. - Омск 2001, 27 с.

Наталов Г.Г. Проблема классификации физических упражнений // Теория и практика физической культуры. - 1968. - №8. - С. 64-67.

Объемно-силовая система тренировок / авт.- сост. Л.В. Аксенова. - М.: АСТ; Донецк: Сталкер, 2006.

Озолин Н.Г. Современная система спортивной тренировки. - М.: ФиС, 1970. - 480 с.

Озолин Н.Г. Проблемы совершенствования советской системы подготовки спортсменов // Теория и практика физической культуры. - 1984. - №10. С.48-50.

Озолин Н.Г. Настольная книга тренера: Наука побеждать /Н.Г. Озолин. - М.: ООО «Изд. Астрель»: ООО «Издательство АСТ», 2002. - 864 с.: ил. - (Профессия тренер).

Остапенко Л. А. Терминология бодибилдинга. // IronMan, 2000, №7.

Остапенко Л.А. Пауэрлифтинг. //Теория и практика телостроительства, 1994, №5. - С. 112-148.

Остапенко Л.А. Особенности тренировочного процесса в силовом троеборье на этапе отбора и начальной подготовки: Автореф. дисс. .канд. пед. наук / Л.А. Остапенко. - М., 2002. - 25 с.

Паков А.В. Оптимальные тренировочные нагрузки в полугодичном цикле у тяжелоатлетов-разрядников различной технической подготовленности: Автореф. дис…. канд. пед. наук. М., 1960. - 19 с.

Паков А.В. Особенности тренировки спортсменов различной технической подготовленности / А.В. Паков, А.В. Черняк // Тяжелая атлетика. М.: ФиС, 1977. С. 48-52.

Паков А.В. Экспериментальное обоснование оптимальных тренировочных нагрузок тяжелоатлетов-разрядников. /А.В. Паков, В.С. Аванесов // Тяжелая атлетика: Ежегодник, 1980 / Сост. Ю.А. Сандалов - М.: ФиС, 1980.- С.15-20.

Паков А.В. Оптимизация тренировочной нагрузки в тяжелой атлетике. / А.В. Паков, В.С. Аванесов, В.А. Нижегородов, В.В. Новиков. - Т.: Медицина, 1985. - 134 с.

Перов, П.В. Пути повышения эффективности тренировки в жиме лежа / П.В. Перов // Санкт-Петербург - родина отечественного атлетизма: Международный сб. научно-метод. трудов / СПб ГАФК им. П.Ф. Лесгафта.- СПб., 2004. - 39-40.139140.

Перов, П.В. Содержание физической подготовки на начальном этапе занятий пауэрлифтингом: Автореф. дис. .канд. пед. наук/ П.В. Перов; СПб ГУФК им. П. Ф. Лесгафта. СПб., 2005. - 24 с.

Петров В.К. Ваш помощник тренажер. М.: Советский спорт, 1991. - С.45.

Платонов В.Н. Современная спортивная тренировка. - Киев: Изд-во Здоровья, 1980. - 336 с.

Платонов В.Н. Подготовка квалифицированных спортсменов - М.: ФиС, 1986. - 286 с.

Платонов В.Н. Адаптация в спорте. - Киев: Здоровье,1988.-216 с.

Платонов В. Н. Система подготовки спортсменов в олимпийском спорте. Общая теория и ее практические приложения. - К.: Изд-во Олимпийская литература, 2004. - 808 с.

Плехов В.Н. Возьми в спутники силу. - М.: ФиС, 1988 - 240 с.

Полетаев П.А. О необходимости нового подхода к планированию, контролю и анализу тренировок тяжелоатлетов с целью разработки эффективных тренировочных программ Часть третья (окончание): Основные параметры тренировочной нагрузки (методологический анализ) // Олимп. № 2-3, 1999. С. 18-20.

Пономарев В.Н. Атлетизм. /В.Н. Пономарев, Ю.А. Богащенко. - Красноярск, 1991. - 254 с.

Попов Г.И. Биомеханика двигательной деятельности: учеб. для студ. учреждений высш. проф. образования / Г.И. Попов, А. В. Самсонова. - М.: Издательский центр «Академия», 2011. - 320 с. - (Сер. Бакалавриат).

Прилепин А.С. Экспериментальное определение оптимальной тренировочной нагрузки тяжелоатлетов-разрядников: Автореф. дис. канд. пед. наук. /Прилепин Александр Сергеевич; [Гос. Центр. Ордена Ленина ин-т физ. культуры]. - М.,1974. - 25

Прилепин А.С. Объем месячной нагрузки атлетов Ш и П разрядов при тренировке с отягощениями 70% предельного веса // Тяжелая атлетика: Ежегодник. - М,: ФиС,1974, с,22-24. 184

Прилепин А.С. Количество подъемов штанги весом 90% в тренировках тяжелоатлетов-разрядников 16-18 лет // Тяжелая атлетика. Ежегодник-1976. - М.: ФиС, 1976. С. 8-11

Пэйн, П. Чудовищный жим: руководство к действию. //Мир силы. - №1, 2001. - С. 32-33.

Пэйн, П. Становая тяга ваш главный козырь // Мир Силы. - №2, 2000. - С. 36-37.

Рогожников К.В. Школа чемпионов. Разговор о тренинге в современном пауэрлифтинге // Мир силы. №2, 2007. - С. 43-47.

Рогожников К.В. Как пожать 300? // Power life, №1, 2010. С. 30-32.

Родионов, В.И. Как развить силу / В.И. Родионов. - М.: ФиС, 1964. - 94 с.

Родионов В.И. Вариант планирования недельной тренировки // Тяжелая атлетика. М.: ФиС, 1977. С. 77-80.

Роман Р. А. Тренировка тяжелоатлета в двоеборье. - М.: ФиС,1974. - 151 с.

Роман Р.А. Тренировка начинающего тяжелоатлета в 17-18 лет // Тяжелая атлетика. - М.: ФиС,1975, С. 34-37.

Роман Р. А. Содержание тренировки спортсмена высокого класса // В дружбе - сила. М.: ФиС, 1978. С. 56-85.

Роман Р.А. Тренировка тяжелоатлета.- 2-е изд., перераб., доп. - М.: ФиС, 1986. - 175 с.

Рыбальский П.И. Жим в пауэрлифтинге. //Теория и практика физической культуры. - № 8, 1997. С. 58-60.

Рыбальский П.И. Становая тяга в пауэрлифтинге // Теория и практика физической культуры. № 10, 1997. - С .40-42.

Рыбальский П.И. Взрывная сила // Мир силы. № 4,1999. - С. 23-25.

Рыбальский П.И. Структура и содержание тренировочных микроциклов различной направленности в зависимости от характеристик соревновательных упражнений в пауэрлифтинге: автореф. дис. ... канд. пед. наук / П.И. Рыбальский. - М., 2000.

Сайпсон Майк. Моя русская программа жима. “Muscle and Fitness”, sept., 1987.

Самсонов, Г.А. Методика коррекции технической и специальной силовой подготовки пауэрлифтеров с целью преодоления “мертвых зон” в жиме штанги лежа / Г.А. Самсонов, Б.И. Шейко // Труды кафедры биомеханики университета имени П.Ф. Лесгафта. 2015. Вып.9. С. 46-51.

Самсонов, Г.А. Новый подход к определению понятия и выявлению “мертвой зоны” в жиме штанги лежа / Г.А. Самсонов // Российский журнал биомеханики, 2015. - Т. 19. - № 3. - С. 296-306.

Самсонов, Г.А. Преодоление “мертвых зон” при выполнении жима штанги лежа / Г.А. Самсонов, Н.Б. Кичайкина, Б.И. Шейко // Ученые записки университета имени П.Ф. Лесгафта, 2015. - № 10 (128). - С. 171-176.

Самсонов, Г.А. Устройство для фиксации отрыва таза при выполнении жима штанги лежа / Г.А. Самсонов, С.Н. Забавный // Труды кафедры биомеханики университета имени П.Ф. Лесгафта, 2015. - Вып.9. - С. 43-45.

Самсонов, Г.А. Электрическая активность широчайшей мышцы спины при жиме штанги лежа на горизонтальной скамье / Г.А. Самсонов, Д.Д. Дальский // Ученые записки университета имени П.Ф. Лесгафта, 2015.– № 8 (126). - С. 137-142.

Самсонов, Г.А. Коррекция техники жима штанги лежа пауэрлифтеров высокой квалификации с целью преодоления «мертвых зон», дис. ... канд. пед. наук / Г.А. Самсонов. - СПб:., 2016. 196 с.

Самсонова А.В. Биомеханика мышц: учебно-методическое пособие / А.В. Самсонова, Е.Н. Комиссарова / СПбГУФК им. П.Ф.Лесгафта, СПб.: [б.и.]. - 127 с.

Самсонова А.В. Гипертрофия скелетных мышц человека: монография / А.В. Самсонова; Национальный гос. Ун-т физ. культуры, спорта и здоровья им. П.Ф. Лесгафта. - СПб.: [б.и.], 2011. - 203 с. ил.

Самсонова, А.В. Электрическая активность мышц нижних конечностей при выполнении жима штанги лежа / А.В. Самсонова, Б.И. Шейко, Н.Б. Кичайкина, Г.А. Самсонов // Ученые записки университета имени П.Ф. Лесгафта, 2014. - № 5 (111). - С. 159-165.

Самсонова, А.В. Механизм передачи импульса от ног штанге при выпол- нении жима штанги лежа / А.В. Самсонова, Б.И. Шейко, Н.Б. Кичайкина, Г.А. Самсонов // Труды кафедры биомеханики университета имени П.Ф. Лесгафта, 2014. - Вып. 8. - С. 34-37.

Сандов Е. Сила и как сделаться сильным. - СПб: Изд-во Ф.И. Митюрникова, 1900. - 125 с.

Селуянов В. Н. Эмпирический и теоретический пути развития теории спортивной тренировки // Теория и практика физической культуры. 1998. № 3. С. 46-50.

Селуянов В.Н. Принципы построения силовой тренировки / В.Н. Селуянов, С.К. Сарсания // Юбилейный сборник трудов ученых РГАФК, посвященный 80-летию академии. - М.: 1998. - Т. 2. - С. 39-49.

Семенов Г.П. Исследование режимов работы мышц в связи с формированием силовых качеств человека. Автореф. дис. канд. биол. наук. Тарту, 1972. - С.27.

Силин, В.И. Интегральная подготовка мастеров пауэрлифтинга // Пауэрлифтинг. - 2009.- №7. - С. 12-14.

Симавский О. Вариативность нагрузки в предсоревновательный период // Мир силы. №3, 2002. С.36-39

Симмонс Луи Управление тренингом. //Железный мир. №5-6, 2003, С.182-183

Симмонс Луи Использование свободных отягощений. //Железный мир. №2, 2004, С.176-177.

Симпсон, Майкл. Моя русская программа жима / М. Симпсон: //Muscle and Fitness, sept., 1987

Слоан С.С. Жим лежа // IronMan. - №10. - 2000. - С. 34-37.

Смолов С.Ю. Тяга как одно из основных упражнений силового троеборья: краткий анализ и методика тренировки // Атлетизм, Инспорт, Малаховка, 1993. - №2. - С. 3-13.

Солоневич И. Тренировка тяжелоатлета. М.: ФиС, 1925, с. 53-57.

Солоневич И. Гиревой спорт. - М.: Книгоиздательство ВЦСПС, 1928. - 96 с.

Старов М.Д. Техника жима лежа в пауэрлифтинге. // IronMan. - №10, 2000. - С. 16-18.

Стеценко А.И. Теоретические и методические основы подготовки в пауэрлифтинге. Пауэрлифтинг Украины. - К.: № 1 (2), 1997. - 25 с.

Стеценко А.И. Построение тренировочного процесса в пауэрлифтинге на этапе непосредственной подготовки к соревнованиям: автореф. дис. ... канд. пед. наук / А.И. Стеценко. - Киев, 2000.

Стеценко А.И. Пауэрлифтинг. Теория и методика избранного вида спорта. Черкасси: ЧНУ Б. Хмельницкого, 2008.- 452с.

Стюарт Мак Роберт. Жим лежа = 180 кг! М.: Изд. «Сила и красота». 16 с.

Суровецкий А. Становая тяга // Мир силы.- № 4, 2000. - С. 25-28

Сухоцкий И.В. Тренировка в пауэрлифтинге. Методическая разработка. М.: МИИТ, 1999. 30 с.

Сухоцкий И.В. Пауэрлифтинг. Методика тренировки высококвалифицированных спортсменов. М.: МИИТ, 2000, 38 с.

Токарски Крэйг. Клуб 700 – фунтовиков // Muscle and Fitness, февраль 1993.

Терминология спорта. Толковый словарь спортивных терминов / Сост. Ф.П. Суслов, Д.А.Тышлер. - М.: СпортАкадемПресс, 2001. - 480 с.

Тяжелая атлетика: учеб. для ин-тов физ.культ. / А.Н. Воробьев. - М.: ФиС,1988. - 238 с., ил.

Уильяме, Дж. Вспомогательные упражнения для повышения результатов в жиме на скамье. // Muscular Development.- май, 1994. - С. 21-22.

Фаламеев А.И. Вариативность методики тренировки тяжелоатлета // Тяжелая атлетика: Ежегодник / Сост. Роман Р.А.- М.: ФиС, 1974.- С.17-20.

Фаламеев А.И. Систематика, классификация и терминология в тяжелой атлетике: метод. рекомендации / А.И. Фаламеев.- Л.: [б.и.], 1981.- 35 с.

Фаламеев А.И. Планирование тренировочной нагрузки новичков и разрядников // Тяжелая атлетика: Ежегодник. М.: ФиС, 1970, С. 36-43.

Фалеев А.В. Анти - МакРоберт: Думай! по-русски. Как тренироваться по циклам. - Ростов н/Д: Изд-во Феникс, 2006. - 285 с. - (Планета Спорт).

Фалеев А.В. Секреты силового тренинга. Как накачать силу и мышечную массу, занимаясь без тренера. - М.: Изд-во АСТ; Донецк: Сталкер, 2008. - 319 с.

Фарфель, В.С. Управление движениями в спорте. М.: ФиС, 1975. - 208 с.

Фетисов В.С. Информационно-измерительный комплекс для многопараметрической оценки состояния опорно-двигательного аппарата чело¬века. / В.С. Фетисов,

О.А. Дудов, П.С. Горулев, Е.П. Артеменко, А.Х. Кальметьев , Е.А. Константинов. // Медицинская техника. - №4, 2004. - С. 16-18.

Филипп Л. Некоторые принципы тренинга в силовом троеборье. Братислава. // Тренер. - №10, 1978. - С. 67-69.

Холодов Ж.К. Теория и методика физического воспитания и спорта: учеб. пособие для студ. высш. учеб. заведений /Ж.К. Холодов, В.С. Кузнецов. - 5-е изд., стер.- М.: Издательский центр «Академия», 2007.- 480 с.

Холопов В.А. Построение тренировочных нагрузок в микро и мезоциклах на базовом этапе годичного цикла подготовки пауэрлифтеров высших разрядов // Вестник спортивной науки. - Советский спорт. - №4. - 2007. - С. 62-65.

Холопов В.А. Построение нагрузок различной направленности в годичном цикле подготовки пауэрлифтеров старших разрядов.//Вестник читинского государственного университета. - Чита. - №1, 2008. - С. 103-109.

Холопов В.А. Построение и содержание тренировочных нагрузок в годичном цикле подготовки пауэрлифтеров старших разрядов.: дис. ... канд. пед. наук / В.А. Холопов. - М.: Изд-во ВНИФК, 2008. - 115 с.

Хэтфилд Ф.К. Присед. Техника и тренировочные методы. // Muscular Development, 1982, февраль. - С. 34-37.

Хэтфилд Ф.К. Всестороннее руководство по развитию силы. - Новый Орлеан, 1983. / Ретопринтное переводное издание Ассоциации федераций культуризма Восточной Сибири и Дальнего Востока. - Красноярск, 1992. - 288с.

Чередниченко С.А. Культуризм и пауэрлифтинг. - Краматорск: НПОНИИПТмаш, 1991. - 256 с.: ил.

Чернышев П.Н. Жим, ничего кроме жима // Мир силы. - июнь, 1999. - С. 34-37.

Черняк А.В. Распределение тренировочной нагрузки по неделям за два месяца до соревнований / А.В. Черняк, Н. Атанасов, А. Ермаков. // Тяжелая атлетика: Ежегодник / Сост. Р.А. Роман. - М.: ФиС, 1972. - С. 10-18.

Черняк А.В. О соотношении между объемом и интенсивностью тренировочной нагрузки. - В сб.: Тяжелая атлетика. - М.: ФиС, 1973, С. 23-26

Черняк А.В. Методика планирования тренировки тяжелоатлета. М.: ФиС., 1978. 136 с. ил.

Черняк А.В. Объем и интенсивность тренировочных нагрузок атлетов различных весовых категорий / А.В. Черняк, А.В. Чужин, М.С. Гисин, С.А. Качаев // Тяжелая атлетика: Ежегодник. М.: ФиС, 1976. - С. 19-24.

Черняк А.В. Распределение объема и интенсивности недельной нагрузки в течение года. / А.В.Черняк, Э.Ш. Каримов, Ж.Т. Бутинчинов // Тяжелая атлетика: Ежегодник. 1979. - М.: ФиС, С. 15-17.

Чудинов В.И. Абсолютная и относительная сила спортсмена // Теория и практика физической культуры. -1962. - № 3. - С. 34-36.

Шагапов Р.Х. Пауэрлифтинг – спорт высших достижений. Монография / Р.Х. Шагапов, О.П. Слива. Екатеринбург: УГТУ, 1998. - 154 с.

Шагеева Л.Г. О теории движений в силовых видах спорта. // Пауэрлифтинг. - №7, 2009. - С. 23-25.

Шварцнегер А. Новая энциклопедия бодибилдинга. - М.: изд-во ЭКСМО-Пресс, 2000, - 824с, ил.

Швуб Йожеф. Техника жима лежа. // Trener, 1980, ноябрь.

Шейко Б.И. Методика планирования для начинающих пауэрлифтеров // Мир силы. №4, 1999.- С. 28-31.

Шейко Б.И. Методика планирования спортсменов-разрядников // Мир силы. №1, 2000.- С.32-34

Шейко Б.И. Методика подготовки пауэрлифтеров в группе спортивного совершенствования (КМС-МС) // Олимп. №1, 2000. - С. 37-41

Шейко Б.И. Месячный план подготовки спортсмена – разрядника. Часть1. // Мир силы. №3, 2000. - С. 30-35.

Шейко Б.И. Месячный план подготовки разрядников (подготовительный период) // Мир силы. №4, 2000. - С. 22-25.

Шейко Б.И. Методика подготовки пауэрлифтеров в группе высшего спортивного мастерства (МС и МСМК) // Олимп. № 2-3, 2000. - С. 34-38.

Шейко Б.И. Методика достижения результатов в пауэрлифтинге: от начальной подготовки до спортивного совершенства. - Омск, 2000. - 136 с.

Шейко Б.И. Терминология упражнений применяемых в пауэрлифтинге. // Мир силы, №4, 2001. - С.20-27.

Шейко Б.И. Техника соревновательных упражнений. Тяга становая. // Мир силы. №1, 2002. - С.4-9.

Шейко Б.И. Техника выполнения тяги становой // Олимп. №1, 2002. - С. 33-37.

Шейко Б.И. Техника выполнения приседания // Олимп. №2, 2002. - С. 26-31.

Шейко Б.И. Техника выполнения жима лежа на скамье. // Олимп. №3-4, 2002. - С. 20-25.

Шейко Б.И. Краткий очерк истории развития пауэрлифтинга в России. // Железный мир. № 4, 2003. - С. 240-245

Шейко Б.И. Жим лежа для любителей пахать // Мир силы. №1, 2003. - С. 22-29

Шейко Б.И. Специальные упражнения в пауэрлифтинге (Методическое пособие). / Б.И. Шейко, П.С. Горулев. Уфа.: 2004.- 112 с.

Шейко Б.И. Секреты тренерской кухни Леонида Котенджи. // Железный мир. №3, 2005. - С. 168-171

Шейко Б.И. Пауэрлифтинг. (Учебное пособие для студентов высших учебных заведений) М.: ЗАО «ЕАМ СПОРТ СЕРВИС». 2005. – 544 с.

Шейко Б.И. Биомеханический анализ техники выполнения жима лежа. Первые результаты / Б.И. Шейко, Б.Г. Лукьянов, В.С. Фетисов // Железный мир. № 4, 2007. - С. 124-130.

Шейко Б.И. Пауэрлифтинг в цифрах и фактах 1980-2005. (книга 1)

Чемпионаты России, М.: ЗАО «ЕАМ СПОРТ СЕРВИС». 2007. - 720 с.

Шейко Б.И. Что дает спортсмену жимовая майка. / Б.И. Шейко, Б.Г. Лукьянов, В.С. Фетисов, О.А. Дудов, П.В. Репина. //Железный мир. №5, 2007. - С. 128-133

Шейко Б.И. Техника жима лежа. /Б.И. Шейко, Б.Г. Лукьянов, В.С. Фетисов, О.А. Дудов, П.В. Репина. //Железный мир. №6, 2007. - С. 128-133

Шейко Б.И. Видеотехнический комплекс для анализа движений спортсменов / Б.И. Шейко, Б.Г. Лукьянов, В.С. Фетисов, О.А Дудов // Вестник УГАТУ 2007. - Т. 9. - №5 (23). С. 64-71.

Шейко Б.И. Пауэрлифтинг в цифрах и фактах (книга 2) Чемпионаты.

Европы - EPF. М.: ЗАО «ЕАМ СПОРТ СЕРВИС». 2008. - 544 с.

Шейко Б.И. Методика обучения технике соревновательных упражнений в пауэрлифтинге. / Б.И. Шейко, Б.Г. Лукьянов. Международная научно-практическая конферен-

ция «Актуальные проблемы физической культуры, спорта и туризма». УГАТУ. 2009. - С. 88-90.

Шейко Б.И. Пауэрлифтинг России в цифрах и фактах (книга 3). Чемпионаты мира. IPF. (1971-2008). / Б.И. Шейко, Г.В. Ходосевич. Красноярск: ООО ИПЦ «КАСС». 2009. - 792 с.

Шейко Б.И. Техника приседания со штангой на спине Е. Ковалькова // Тяжелая атлетика и пауэрлифтинг. №1, 2009. - С. 26-28.

Шейко Б.И. Биомеханический анализ техники выполнения тяги становой стилем «сумо» трехкратного чемпиона мира Хупера Вэйда, США / Б.И. Шейко, В.С. Фетисов. IV Междунар. Науч.-практич. конф. «Актуальные проблемы физической культуры, спорта и туризма». - Уфа: УГАТУ, 2010. - С. 72-77

Шейко Б.И. Сравнительные характеристики биомеханического анализа техники тяги становой. / Б.И. Шейко, В.С. Фетисов. IV Междунар. Науч.-практич. конф. «Актуальные проблемы физической культуры, спорта и туризма».- Уфа: УГАТУ, 2010. - С.77-86

Шейко Б.И. Теоретические основы техники выполнения соревнователь-ного упражнения жим лежа. Учебное пособие / Б.И. Шейко, Е.В. Охлюев. - Уфа, Ред.-издат. центр Баш ИФК, 2010. – 190 с.: ил.

Шейко Б.И. Биомеханический анализ техники выполнения тяги Андрея Беляева. // Power Life. № 5, 2011. - С. 46-48.

Шейко Б.И. Биомеханический анализ техники выполнения тяги классическим стилем Баруздина Максима. // Power Life. №6, 2011. - С. 36-38.

Шейко Б.И. Ретроспективный взгляд на методологию увеличения силы. Башкирский институт физической культуры(филиал) ФГОУ ВПО «Уральский государственный университет физической культуры», г. Уфа.Актуальные проблемы физической культуры, спорта и туризма: мат. V Междунар. науч.-практич. конф. Уфимск. гос. авиац. техн. ун-т. - Уфа: УГАТУ, 2011 - С.163-167.

Шейко Б.И. Методика обучения студентов основам техники жима лежа. Актуальные проблемы физической культуры, спорта и туризма: Мат. VI Международной научно-практич. конференции, посвященной 80-летию УГАТУ/Уфимский гос. Авиационный техн. Ун-т. Уфа: УГАТУ, 2012. - С 219-221 с.

Шейко Б.И. Методика обучения студентов основам техники тяги. Актуальные проблемы физической культуры, спорта и туризма: Мат. VI Международной научно-практич. конференции, посвященной 80-летию УГАТУ/Уфимский гос. Авиационный техн. Ун-т. Уфа: УГАТУ, 2012. - С 224-228 с.

Шейко Б.И. Методика обучения студентов основам техники приседания со штангой на плечах. Актуальные проблемы физической культуры, спорта и туризма: Мат. VI Международной научно-практич. конференции, посвященной 80-летию УГАТУ / Уфимский гос. Авиационный техн. Ун-т. Уфа: УГАТУ, 2012. - С 221-224 с.

Шейко, Б.И. Пауэрлифтинг. От новичка до мастера. Монография / Б.И. Шейко, П.С. Горулев, Э.Р. Румянцева, Р.А. Цедов; под общ. ред. Б.И. Шейко Пауэрлифтинг. От новичка до мастера. - М: Медиагрупп «Актиформула». - 2013, С. 177-278.

Шейко Б.И. «Русская пирамида». Кто автор тренировочной системы «Пиковый цикл» и «Русская пирамида»? //Железный мир». №7 -2014. С.136-140

Шейко Б.И., Юламанова Г.М. Применение тренажеров в физкультурно-оздоровительной и спортивно-тренировочной деятельности. Учебное пособие/– Уфа, Редакционно-издательский центр Баш ИФК, 2015. - 121 с.: ил.

Шейко Б.И. Сарычев, К.И. Жим штанги лежа для спортсменов всех уровней подготовки и физических возможностей. - Москва, 2018. - 528 с.: ил.

Штольц А. Руководство к развитию силы и наращиванию мускулов посредством упражнений с гирями и другими приборами. - СПб.: Изд-во А.Ф. Сухановой, 1909. - 88 с.

Щетина Б.М. К вопросу о планировании тренировочной нагрузки в пауэрлифтинге. // Физическая культура, спорт и здоровье населения Дальнего Востока: материалы V Межрегион, науч. конф.-Хабаровск, 1997.-С.130-131.

Щетина Б.М. Двигательная структура соревновательных упражнений в пауэрлифтинге. Материалы межрегиональной науч. и науч.-практ. конф. Ч. III / Б.М. Щетина, Б.В. Райский. - Хабаровск, 1999. - С. 80-81.

Щетина Б.М. Двигательная структура соревновательного упражнения «приседание» в пауэрлифтинге. Материалы науч. конф. преподавателей ХГПУ / Б.М. Щетина, В.В. Ткачев, В.В. Мулин. - Хабаровск, 2000. - С. 91-92.

Щетина Б.М. Двигательная структура жима лежа. Материалы науч.- практ. конф. / Б.М. Щетина, В.В. Мулин, В.В. Ткачев. – Хабаровск, 2001. – С. 316–317.

Щетина Б.М. Силовое троеборье (теория и методика обучения и тренировки). Рекомендовано РОУМО ДВ РУМЦ в качестве учебного пособия для студентов высших учебных заведений. Хабаровск. Издательство ДВГУПС. 2008. 76 с.

Якубенко Я. Э. Сравнительный анализ объема тренировочной нагрузки в пауэрлифтинге у мужчин в зависимости от квалификации и массы тела Дис. ... канд. пед. наук : 13.00.04. - М., 2006.- 127 с.

Anderson, T., & Kearney, J.T. Effects of three resistance training programs on muscular strength and absolute and relative endurance. Res Q Exerc Sport, 53,1-7,1982,

Bacchle T.R. Essentials of Strength Training and Conditioning. Champaign IL: Human Kinetics, 1994.

Baker G. Exercise of the month. Strength Cond. J. 16:54-55.1994.

Berger R.A. The Effect of varied weight training programs on strength and endurance. Microfilm, University of Illinois, 1960.

Berger R.A. Effect of varied weight training programs on strength.// Res .Quart 1962. - Vol. 33.- No. 1 - P. 168-181.

Berger R.A. Repetitions for the Development of Strength // Res Quart-1962.-Vol. 33. - No. 3 - P. 334-338.

Berger R.A. Comparative effects of three weight training programs // Res Quart-1963.- Vol. 34 - P. 396-398.

Berger R. Comparison of the effect of various weight training loads on strength. //Res Quart-1965.-Vol. 36 - P. 141-146.

Berger R.A., and Hardage B. Effect of maximum loads for each of ten repetitions on strength improvement // Res Quart - 1967. - Vol. 38. - No. 4. - P. 715-718.

Borst S.E., DeHoyos D.V., Garzarella L., Vincent K., Pollock B.H., Lowenthal D.T. et al. Effects of resistance training on insulin like growth factor-1 and IGF binding proteins. // Med Sci Sports Exerc - 2001- Vol. 33 - P. 648-653.

Daniels D. Tips On Squatting. “Powerlifting USA”, January 1999.

Daniels D. Deadlift Details. “Powerlifting USA”, November 1999.

Daniels D. Close Grip Bench Press. “Powerlifting USA”, January 2000.

Daniels D. Deadlift 101, Part 1. Powerlifting USA. Vol 26. No.8. 2003

DeLorme T.L. Restoration of Muscle Power by Heavy-Resistance Exercises // J Bone Joint Surg. - 1945. - Vol. XXVII. - No 4. - P. 645-667.

DeLorme T.L. Heavy-Resistance Exercises // Arch. Phys Med. - 1946. - No. 27. - P. 607-630.

DeLorme T., Watkins A. Techniques of progressive resistance exercise // Arch. Phys Med. Rehabil. - 1948. - No. 29. - P. 263-273.

DeLorme T.L., West F.E. and Schreiber W.J. Influence of progressive-resistance exercises on knee function following femoral fractures // J Bone Joint Surg. - 1950. - Vol. 32A. - No.4. - P. 910-924.

DeLorme T.L., Watkins A.L. Progressive resistance exercises. - New York: Appleton-Century-Crofts. Division of the Meredith Publishing Co., 1951.

DeLorme, Thomas L., and Watkins A. L. Progressive resistance exercise: technic and medical application - New York: Appleton- Century-Crofts., Inc., Division of The Meredith Publishing Co., 1951.

DeLorme T.L. Effect of progressive Resistance exercise on Muscle Contraction // Arch Phys Med. Rehabil. - 1952. - No. 33. - P. 86.

Duclos M. (2008). A critical assessment of hormonal methods used in monitoring training status in athletes. Int. J. SportsMed (pp.56–66).

Farley K. Analysis of the Conventional Deadlift. Strength and Conditioning Journal. Vol 15. - No. 2. - 1995. - pp 55-58.

Fish D. E., Krabak B. J., Johnson-Greene D., DeLateur B.J. Optimal Resistance Training: Comparison of DeLorme with Oxford Techniques // Am J Phys Vaughan, ed. Boca Raton, FL: CRC Press, 1989. pp 169-211.

Grabiner M.D. and Garhammer, J. Analysis and assessment of human movement performance. In: Kinesiology and Applied Anatomy. P.J. Rasch, ed. Philadelphia: Lea & Febiger, 1989. pp 247-258.

Haff G.G., Hobbs R., Haff E., Sands W., Pierce K., Stone M. Cluster Training: A Novel Method for Introducing Training Program Variation. // Strength Cond J - 2008. - Vol. 30 - No. 1 - P. 67-76.

Harris T. The Squat. // Powerlifting USA, September 1998.

Hatfield F. Soviet System Of Peaking. “Muscle and Fitness”, June 1987.

Hatfield F. So, Do You Want To Be A Powerlifter? “Muscle & Fitness”, October 1985.

Hayes L.D., Bickerstaff G.F., Baker J.S. (2010). Interactions of cortisol, testosterone, and resistance training: influence of circadian rhythms. Chronobiol Int. (June 27(4), pp.675-705).

Hellebrandt F.A., Houtz S.J., and Eurbank R.N. Influence of alternate and reciprocal exercise on work capacity //Arch. Phys. Med. - 1951. - Vol. 32 - P. 766-776.

Henry C.G. A comparison of the effectiveness of two methods for the development of muscular strength. Unpublished master’s thesis, State University of Iowa, 1949.

Hettinger T., Muller E. Muskelleistung und Muskeltraining. “Arbeitsphysiologie”, 1953, 15.

Hoffman B. Deadlift. // Muscular Development, June 1985.

Horn T.S. A biomechanical comparison of sumo and conventional deadlifting techniques. Int. J. Sports Med. 9:150. 1988.

John S. O Brien Bench Assault: The shortest road to maximum power 1991. - 36 p.

John S. O Brien Squat and deadlift blitz: The shortest road to maximum power 1992 35p.

Kamand A. Sguatting For The Beginner and Intermediate Powerlifting. // Powerlifting USA, November 1994

Kuc J. Drug Free Squat Routine. “Powerlifting USA”, January 2000.

Krusen E.M. Functional improvement produced by resistance exercise of quadriceps muscles affected by poliomyelitis // Arch. Phys. Med. & Rehabilit. - 1949. - Vol. 30. - P. 271-277.

Landers J. Maximun based on repetitions // Strength Cond J - 1985. - No. 6. - P. 60-61.

Leighton J.R., Holmes D., Benson J., Wooton B., Schememer R. A study on the effectiveness of ten different methods of progressive resistance exercise on the development of strength, flexibility, girth and bodyweight. // J Assoc Phys Mental Rehabil 1967. - Vol.21- P. 78-81.

Marx J.O., Ratamees N.A., Nindl B.C., Gotshalk L.A., Volek J.S., Dohi, K. et al.Low volume circuit versus high volume periodised resistance training in women // Med Sci Sports Exerc - 2001. - Vol. 33. - P.635-643.

McGovern R.E., and Luscombe H.B. Useful modifications of progressive resistance exercise technique. //Arch. Phys. Med. - 1953. - Vol. 34 - P. 475-479.

McRobert S. Hargainer. “IRONMAN”, February 1998.

McRobert S. Sustaining Progress. “Hardgainer”, November 1991.

McGuigan, R.M. & Wilson B.D. Biomechanical Analysis of the Deadlift. Journal of Strength and Conditioning Research. 10(4), 250-255. 1996

McGovern R.E., and Luscombe H.B. Useful modifications of progressive resistance exercise technique. //Arch. Phys. Med. - 1953. - Vol. 34 - P. 475-479.

McLaughin T.M., Dillman C.J., and Lardner T.J. A Kinematic model of performance in the parallel squat by champion powerlifters. Med. Sci. Sports Exerc. 9:128-133. 1977.

McMorris R.O. and Elkins E.C. A study of production and evaluation of muscular hypertrophy // Arch. Phys. Med. - 1954. - Vol. 35 - P. 420-426.

Miller L. The Bench Press: Putting All The Pieces Together. “Powerlifting USA”, June 2000.

Ortiz Cervera V. Entrenamiento de fuerza y explosividad para la actividad fisica y el deporte de competition / Ortiz Cervera V., Gue N., Navarrro Puche J. A., Poletaev P., Rausel Peris L.: Ortiz Cervera V. (ed.) - Barcelona: INDE, 1996 (1a ed.), 1999 (2a ed.) - 397 p.

O’Shea J. P. Anabolic steroids, their effects on muscle function // Modern Med- 1969.- No. 12. - P. 64-70.

O’Shea J.P. Biochemical and physical effects of anabolic steroids in competitive swimmers and weightlifters // Nutrition Reports International. - 1970.- Vol. 2 - P. 351-362.

O’Shea J.P. Effects of anabolic steroids on dynamic strength levels of competitive weightlifters // Nutrition Reports Intern. - 1971.- Vol. 4 - P. 363-370.

O’Shea J.P. Biochemical evaluation of the effects of stanozolol on adrenal, liver, and muscle function in humans. // Nutrition Reports Intern. - 1974. - Vol.10-P.385-388.

Ostapenko, L.A. (2004). Nutrition in Powerlifting. Part 1: Is our nutrition correct? Zheleznyj mir (issues 5-6, pp. 158-162).

Ostapenko, L.A. (2005). Nutrition in Powerlifting. Part 2: How should you start to balance your diet? Zheleznyj mir (issue 1, pp. 155-160).

Ostapenko, L.A. (2005). Nutrition in Powerlifting. Part 3: How to balance your daily diet in practice? Zheleznyj mir (issue 2, pp. 161-165).

Ostapenko, L.A. (2001). How to reduce weight. EAM Sport Service.

Pacifico L. Add 44 Lbs. To Your Bench. “Powerlifting USA”, December 1999.

Palmieri G. Weight training and repetition speed // J Appl Sport Science Res. - 1987. - Vol. 1.- No 2. - P. 36-38.

Rippetoe, M. Popular Biomechanics / M.Rippetoe // CrossFit Journal, 2007, Issue55, March. - P. 1-5. Перевод А.В. Самсоновой.

Rooney, K. J., Herbert, R. D., & Balnave, R. J. Fatigue contributes to the strength of training stimulus. // Med Sci Sports Exerc - 1994. - Vol. 26. - P. 1160-1164.

Rick Weils The Bench Press 1986. - 52 p.

Rhea M.R., Alvar B.A., Burkett L.N. Single versus multiple sets for strength: a meta-analysis to address the controversy // Res Q Exerc Sport- 2002. - Vol. 73. - P. 485-488.

Rhea M.R., Alvar B.A., Burkett L.N., Ball S.D. A meta-analysis to determine the doseresponse for strength development // Med Sci Sports Exerc. - 2003. - Vol. 35. - No 3. - P. 456-464.

Sheiko B., Fetisov V., Lukyanov B. Bench press technique // Powerlifting USA. JAN/2010. P.12-13, 70-71

Sheiko B., Fetisov V., Lukyanov B. Bench press technique. Results of the Biomechanical Analysis of the BP technique // Powerlifting USA. MAR/2010. P.12-13, 74-75.

Simmons L. So you want to deadlift. Powerlifting USA. 17:34-35. 1994.

Simmons L. More Big Benches / L. Simmons // Powerlifting USA. - January, 2000. - P. 31-33.

Simmons L. Resistance Training. “Powerlifting USA”, October 1997.

Simmons L. The Dynamic Duo. “Powerlifting USA”, November 1999.

Simmons L. The Squat Workout. “Powerlifting USA”, December 1997.

Simmons L. Training: AN Overview. “Powerlifting USA”, April 2000.

Sloan C. S. Size, Strength and Power. “IRONMAN”, Winter 1999.

Starr B. Deadlift without deadlifting. Powerlifting USA. 18:10-11. 1995.

Wagman, Dan. The Bench Rules. “Muscle and Fitness”, August 1998.

Wagner, Robert. A Balding Man’s Advice on Powerlifting Technique. “Powerlifting USA”, September 1998.

Weider D. Training for competition. Welder’s Olympic Courses,London, I960, 29-31

Unitt Dennis. Fifty years ago there was no IPF – there was no powerlifting // The IPF Magazine 01 / 2010. - P. 18-27.

Yaglov V.V. (1986). Regulation Stages. Moscow: Soviet Russia (p.40).

Zatsiorsky, V.M. Intensity of Strength Training. Facts and Theory: Russian and Eastern European Approach. National Strength and Conditioning Association Journal. Vol. 14, Numb. 5, p. 46-57, 1992.

Zatsiorsky V.M. Science and Practice of Strength Training. Champaign IL. Human Kinetics. 1995.